PART I

ODOUR THRESHOLD VALUES IN AIR

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

A

abhexone⇒5-ETHYL-3-HYDROXY-4-METHYL-2(5*H*)-FURANONE
acetaldehyde⇒ETHANAL
acetamide⇒ETHANAMIDE

ACETANILIDE *[103-84-4]*

Laffort (1968a)		270

ACETIC ACID (ethanoic acid) *[64-19-7]*

Passy (1893b,1893c)	d	5 - 10
Grijns (1906)		49 - 76
Backman (1917)	r	4.8 - 5.0
Grijns (1919)		2
Mitsumoto (1926)	r	0.074 - 0.57
Hesse (1926)	r	0.6
Henning (1927)	d	3.6
Morimura (1934)	r	1.82 - 1.91
Jung (1936)	d	0.025
Jung (1936)	r	0.05
Balavoine (1943,1948)		300 - 500
Stone (1963c)	d	3.9
Stone & Bosley (1965)	d	4.2
Endo *et al.* (1967)		6.5
Takhirov (1969,1974)		0.60
Leonardos *et al.* (1969)	r	2.5
Homans *et al.* (1978)		0.37
Nauš (1982)	d	0.5
Nauš (1982)	r	25
Punter (1983)	d	0.09
Homans (1984)		0.93
Walker *et al.* (1990)		5.0
Nagy (1991)	d	0.37
Blank & Schieberle (1993)		0.03 - 0.09
Walker *et al.* (1996)		0.25 - 2.5
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	0.025
Nagata (2003)	d	0.015
Van Thriel *et al.* (2006)	d	1.45
Wise *et al.* (2007); Miyazawa *et al.* (2009a)	d	0.017 - 0.020
Miyazawa *et al.* (2009b)	d	0.001
Cain *et al.* (2010)	d	0.15
Cometto-Muñiz & Abraham (2010b)	d	0.013

ACETIC ANHYDRIDE *[108-24-7]*

Takhirov (1969)		0.49
Hellman & Small (1973,1974)	d	< 0.6
Hellman & Small (1973,1974)	r	1.5

acetone⇒PROPANONE
acetonitrile⇒METHYL CYANIDE

ACETOPHENONE *[98-86-2]*

Imasheva (1963)		0.01
Tkach (1965)		0.01

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Korneev (1965)		0.01
Gavaudan & Poussel (1966)		0.23
Hellman & Small (1973,1974)	d	1.5
Hellman & Small (1973,1974)	r	2.9
Sävenhed *et al.* (1985)	d	0.01 - 0.04
Randebrock (1986)		0.0012

acetyl chloride⇒ETHANOYL CHLORIDE

ACETYLDI-*tert*-BUTYL(METHYL)SILANE
Sunderkötter *et al.* (2010) > 0.5

ACETYLDICYCLOPROPYL(ETHYL)SILANE
Sunderkötter *et al.* (2010) 0.025

ACETYLDICYCLOPROPYL(METHYL)SILANE
Sunderkötter *et al.* (2010) 0.058

2-acetyl-3,4-dihydro-2*H*-azole⇒2-ACETYL-1-PYRROLINE

5-ACETYL-2,3-DIHYDRO-1,4-THIAZINE *[164524-93-0]*
Hofmann & Schieberle (1995,1997);
Hofmann *et al.* (1995) 0.000 02 - 0.000 08

2-ACETYL-5,6-DIHYDRO-4*H*-1,3-THIAZINE
Fuganti *et al.* (2007) 0.000 000 54

ACETYLDIISOPROPYL(METHYL)SILANE
Sunderkötter *et al.* (2010) 0.088

(-)-(9*R*,10*S*)-10-ACETYL-9,10-DIMETHYLBICYCLO[6.4.0]DODEC-1(8)-ENE
Kraft & Gallo (2004) 0.000 19

acetylene⇒ETHYNE
acetylene tetrabromide⇒ 1,1,2,2-TETRABROMOETHANE

7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE (fixolide, tonalide) *[1506-02-1]*

McGee *et al.* (1995)	d	0.000 5 - 0.005
Traynor (2001)		0.222

1-ACETYL-4-METHYLBENZENE (*p*-methyl acetophenone) *[122-00-9]*

Appell (1969)	0.000 15
Schieberle *et al.* (1988); Schieberle & Grosch (1988)	0.002 - 0.0108
Blank *et al.* (1989)	0.002 - 0.003

(±)-3α-ACETYL-2,3,4,4αβ,5,6,7,8-OCTAHYDRO-3β,4β,5,5-TETRAMETHYLNAPHTHALENE
Etzweiler *et al.* (1992) 0.5

ACETYLPYRAZINE *[22047-25-2]*
Hofmann *et al.* (1995);
Hofmann & Schieberle (1997) 0.000 4

2-ACETYLPYRROLE *[1072-83-9]*
Hofmann & Schieberle (1997) > 2

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-ACETYL-1-PYRROLINE (2-acetyl-3,4-dihydro-2*H*-azole) *[85213-22-5]*
Blank *et al.* (1989); Schieberle (1990,1991);
Schieberle & Grosch (1991,1994); Hofmann
et al. (1995); Hofmann & Schieberle (1997) 0.000 02 - 0.000 04

6-ACETYL-1,2,3,4/2,3,4,5-TETRAHYDROPYRIDINE
Schieberle (1990); Schieberle & Grosch (1991);
Hofmann *et al.* (1995);
Hofmann & Schieberle (1997) 0.000 06

2-ACETYL-2,3,8,8-TETRAMETHYL-1,2,3,4,5,6,7,8-OCTAHYDRONAPHTHALENE (cyclemone)
[68991-97-9]
Neuner-Jehle & Etzweiler (1991) 0.000 003

2-ACETYLTHIAZOLE *[24295-03-2]*
Hofmann *et al.* (1995);
Hofmann & Schieberle (1997) 0.004 - 0.0041

2-ACETYL-2-THIAZOLINE *[29926-41-8]*
Hofmann & Schieberle (1995,1997);
Hofmann *et al.* (1995) 0.000 02 - 0.000 08
Cerny (1998) 0.000 016 – 0.000 022

acrolein⇒2-PROPENAL
acrylic acid⇒PROPENOIC ACID
acrylonitrile⇒ETHENYL CYANIDE
active amyl acetate⇒2-METHYLBUTYL ACETATE
active amyl alcohol⇒2-METHYL-1-BUTANOL
1-adamantanaldehyde⇒TRICYCLO[3.3.1.1(3.7)]DECANE-1-CARBOXALDEHYDE
adamsite⇒10-CHLORO-5,10-DIHYDROPHENARSAZINE
aldehyde C16⇒ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE
allyl alcohol⇒2-PROPEN-1-OL
allylamine⇒2-PROPENYLAMINE

7-ALLYLBENZO[*b*][1,4]DIOXEPIN-3-ONE
Kraft & Eichenberger (2003) 0.000 051

allyl caproate⇒2-PROPENYL HEXANOATE
allyl chloride⇒3-CHLOROPROPENE

1-ALLYL-2,5-DIMETHOXY-3,4-METHYLENEDIOXYBENZENE (apiole) *[523-80-8]*
Tempelaar (1913); Zwaardemaker (1927) d 610

allyl isocyanide⇒2-PROPENYL ISOCYANIDE
allyl mercaptan⇒2-PROPENE-1-THIOL

1-ALLYL-4-METHOXYBENZENE (estragole, methylchavicol) *[140-67-0]*
Williams *et al.* (1977) d 0.000 13

5-ALLYL-1-METHOXY-2,3-METHYLENEDIOXYBENZENE (myristicin) *[607-91-0]*
Blank & Grosch (1991a) 0.0015

4-ALLYL-2-METHOXYPHENOL (eugenol) *[97-53-0]*
Tempelaar (1913); Zwaardemaker (1927) d 0.2 - 0.23
Ohma (1922) d 0.23
Van Anrooij (1931) d 0.2
Baldus (1936) d 0.005 - 0.0055
Baldus (1936) r 0.007 - 0.01
Jones (1955c) r 3.8

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Appell (1969)		0.000 008
Huber (1984)		0.000 076
Randebrock (1986)		0.000 022
De Wijk (1989)		0.11
Blank *et al.* (1989)		0.000 2 - 0.000 3
Ferreira *et al.* (1998)		0.000 61

4-ALLYL-1,2-(METHYLENEDIOXY)BENZENE (safrole) *[94-59-7]*

Tempelaar (1913); Zwaardemaker (1927)	d	0.1 - 2
Ohma (1922)	d	0.125
Van Anrooij (1931)	d	0.000 35
Baldus (1936)	d	0.005
Baldus (1936)	r	0.01
Jones (1955a)	r	0.08
Jones (1955b)	r	0.08 - 4.8
Jones (1955c)	r	3.0
Appell (1969)		0.000 07
Köster (1971)	d	16 - 30

allyl mustard oil⇒2-PROPENYL ISOTHIOCYANATE
ambretone⇒5-CYCLOHEXADECEN-1-ONE
ambrettolide⇒16-HYDROXY-7-HEXADECENOIC ACID LACTONE

2-AMINOETHANOL (ethanolamine) *[141-43-5]*

Weeks *et al.* (1960)	d	6.5
Weeks *et al.* (1960)	r	60

6-AMINOHEXANOIC ACID LACTAM (caprolactam) *[105-60-2]*

Krichevskaya (1968)	d	0.30

(*R*)-(+)-2-AMINO-3-MERCAPTOPROPANOIC ACID (L-cysteine) *[52-90-4]*

Laska (2010)	d	0.0045

(*S*)-(-)-2-AMINO-3-MERCAPTOPROPANOIC ACID (D-cysteine) *[921-01-7]*

Laska (2010)	d	0.0048

(*R*)-(-)-2-AMINO-4-(METHYLTHIO)BUTANOIC ACID (D-methionine) *[348-67-4]*

Laska (2010)	d	0.000 22

(*S*)-(+)-2-AMINO-4-(METHYLTHIO)BUTANOIC ACID (L-methionine) *[63-68-3]*

Laska (2010)	d	0.0018

1-AMINONAPHTHALENE (1-naphthylamine) *[134-32-7]*

Backman (1917)	r	0.14 - 0.29

2-AMINONAPHTHALENE (2-naphthylamine) *[91-59-8]*

Backman (1917)	r	1.4 - 1.9

AMMONIA *[7664-41-7]*

Valentin (1848,1850)		21
Grijns (1906)		21.6 - 42
Fieldner *et al.* (1921)		37
Smolczyk & Cobler (1930)		0.71 - 7.1
Geier (1936)	d	1.25
Geier (1936)	r	2.5
Carpenter *et al.* (1948)		0.7
Smyth (1956)	r	< 0.7

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Patty (1962a)		< 3.5
Saifutdinov (1966)		0.50 - 0.55
Endo *et al.* (1967)		37
Leonardos *et al.* (1969)	r	33
Hamanabe *et al.* (1969)		0.03
Stephens (1971)		2.7
Nishida *et al.* (1975)	d	1.8 - 37.8
Hill & Barth (1976)		21
Schoedder (1977)		5.0 - 7.6
Logtenberg (1978)	d	5.2
Nishida *et al.* (1979)	d	11.6
Anon. (1980)	d	0.1
Anon. (1980)	r	0.4
Nauš (1982)	d	1.5
Nauš (1982)	r	35
Nagy (1991)	d	3.7
Nagata (2003)	d	1.1
Van Thriel *et al.* (2006)	d	0.04
Smeets *et al.* (2007)	d	1.8 (static olfactometry)
Smeets *et al.* (2007)	d	1.8 (dynamic olfactometry)

amyl acetate⇒PENTYL ACETATE
amyl alcohol⇒1-PENTANOL
tert-amyl alcohol⇒2-METHYL-2-BUTANOL
amyl butyrate⇒PENTYL BUTANOATE
amyl isovalerate⇒PENTYL 3-METHYLBUTANOATE
amyl mercaptan⇒1-PENTANETHIOL
tert-amyl mercaptan⇒2-METHYL-2-BUTANETHIOL
amyl propionate⇒ PENTYL PROPANOATE
amyl salicylate⇒PENTYL SALICYLATE

4,16-ANDROSTADIEN-3-ONE ((8*S*,9*S*,10*R*,13*R*,14*S*)-10,13-dimethyl-1,2,6,7,8,9,11,12,14,15-decahydrocyclopenta[a]phenanthren-3-one) *[4075-07-4]*

Kraft & Popaj (2004)	d	0.000 002

5α-ANDROST-16-EN-3α-OL *[1153-51-1]*

Bajgrowicz & Fráter (2000)		0.000 02
Kraft & Popaj (2004)	d	0.000 000 67

5α-ANDROST-16-EN-3β-OL *[7148-51-8]*

Kraft & Popaj (2004)	d	0.0053

5α-ANDROST-16-EN-3-ONE *[18339-16-7]*

Amoore (1977)	d	0.0021
Baydar *et al.* (1992a,1993)	d	0.000 000 868
Kraft & Popaj (2004)	d	0.000 000 4

anethole⇒1-METHOXY-4-(1-PROPENYL)BENZENE

ANILINE *[62-53-3]*

Tempelaar (1913)	d	0.97
Huijer (1917); Zwaardemaker (1927)	d	46
Backman (1917)	r	5.0 - 5.8
Geier (1936)	d	1.2 - 1.5
Geier (1936)	r	2.0 - 2.5
Jacobson *et al.* (1958)		38
Tkachev (1963)		0.37
Leonardos *et al.* (1969)	r	3.8

Öztürk (1976)	d	2.21
Nauš (1982)	d	2
Nauš (1982)	r	20

anisaldehyde⇒4-METHOXYBENZALDEHYDE
anisole⇒METHOXYBENZENE
apiole⇒1-ALLYL-2,5-DIMETHOXY-3,4-METHYLENEDIOXYBENZENE

ARSINE *[7784-42-1]*

Patty (1962b)		< 3.2

artificial musk⇒1-*tert*-BUTYL-3-METHYL-2,4,6-TRINITROBENZENE
azine⇒PYRIDINE
aziridine⇒ETHYLENEIMINE
azolidine⇒PYRROLIDINE

B

BENZALDEHYDE *[100-52-7]*

Backman (1917)	r	0.33 - 0.50
Rocén (1920)	r	1.7
Ohma (1922)	d	0.44
Katz & Talbert (1930)		0.18
Jones (1955c)	r	4.1
Pliška & Janiček (1965)		13
Knuth (1973)		0.27
Laing (1975)	d	4.3
Nishida *et al.* (1979)	d	3,400
Randebrock (1986)		0.014
Stevens & Cain (1987a)	d	0.43 - 43
Khiari *et al.* (1992)	d	< 0.01
Von Ranson & Belitz (1992b)	d	0.61
Von Ranson & Belitz (1992b)	r	2.1
McGee *et al.* (1995)	d	0.1 - 1
Yang *et al.* (2008)		0.085

BENZENE *[71-43-2]*

Backman (1917)	r	6.6 - 6.9
Backman (1918); Zwaardemaker (1927)		5 - 5.3
Grijns (1919); Zwaardemaker (1927)		420
Schley (1934)	d	8.8
Schley (1934)	r	12
Jones (1954)	r	480 - 510
Jones (1955c)	r	180
Novikov (1957)		4.9
Deadman & Prigg (1959)	d	9
Gusev (1965)		2.8 - 4
Nauš (1962)	d	6
May (1966)	d	180
May (1966)	r	310
Elfimova (1966)		2.5
Schutte & Zubek (1967)	r	310
Leonardos *et al.* (1969)	r	15
Alibaev (1970)		2.9
Dravnieks (1971)		38
Köster (1971)	d	37
Dravnieks & Laffort (1972)		32.5

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Laffort & Dravnieks (1973)		14.5
Artho & Koch (1973)		100 - 1,000
Dravnieks (1974)	d	380
Nauš (1982)	d	1.5
Nauš (1982)	r	16
Punter (1983)	d	108
Nagata (2003)	d	8.6

BENZENETHIOL (thiophenol) *[108-98-5]*

Katz & Talbert (1930)		0.0012
Stuiver (1958)	d	0.000 14

(2*R*)-(+)-3-(1,3-benzodioxol-5-yl)-2-methylpropanal⇒(2*R*)-(+)-2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL
(2*S*)-(-)-3-(1,3-benzodioxol-5-yl)-2-methylpropanal⇒ (2*S*)-(-)-2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL

2,3-BENZOFURAN (cumarone, coumarone) *[271-89-6]*

Moriguchi *et al.* (1983)	d	0.048

1,4-BENZOQUINONE (quinone) *[106-51-4]*

Backman (1917)	r	0.047 - 0.050
Oglesby *et al.* (1947)		0.44

BENZOYL CHLORIDE *[98-88-4]*

Schley (1934)	d	0.012 - 0.024
Schley (1934)	r	0.012 - 0.036

BENZYL ACETATE *[140-11-4]*

Appell (1969)		0.001
Köster (1971)	d	85 - 135

benzylbenzene⇒DIPHENYLMETHANE

7-BENZYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.0068

BENZYLBROMO CYANIDE (bromobenzyl cyanide) *[5798-79-8]*

Prentiss (1937)		> 0.15

BENZYL CHLORIDE (α-chlorotoluene) *[100-44-7]*

Katz & Talbert (1930)		0.21
Leonardos *et al.* (1969)	r	0.24

benzyl mercaptan⇒α-TOLUENETHIOL

BENZYLTHIOTOLUENE (dibenzyl sulphide) *[538-74-9]*

Katz & Talbert (1930)		0.053
Leonardos *et al.* (1969)	r	0.018

BICYCLO[2.2.1]-5-HEPTENE-2(*exo*)-CARBOXALDEHYDE (5-norbornen-2(*exo*)aldehyde) *[5453-80-5]*

Von Ranson & Belitz (1992b)	d	0.34
Von Ranson & Belitz (1992b)	r	1.0

BICYCLO[2.2.1]-5-HEPTENE-2(*endo*)-CARBOXALDEHYDE (5-norbornen-2(*endo*)aldehyde) *[5453-80-5]*

Von Ranson & Belitz (1992b)	d	0.90
Von Ranson & Belitz (1992b)	r	2.1

biphenyl⇒DIPHENYL
bis(2-furfuryl) disulphide⇒2-FURFURYLDITHIO-2-FURYLMETHANE

bis(2-(1-mercaptoethyl)furane) disulphide⇒2-(1-MERCAPTOETHYL)FURYLDITHIO-2-(1-MERCAPTOETHYL)FURAN
bis(2-methyl-3-furyl) disulphide⇒3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN

BIS(METHYLTHIO)METHANE (2,4-dithiapentane, methylene bis(methyl sulphide)) *[1618-26-4]*

Gijs *et al.* (2000)		0.09

borneol⇒*endo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
bornyl acetate⇒*endo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL ACETATE
bornyl salicylate⇒*endo*-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL SALICYLATE

BORON TRIFLUORIDE *[7637-07-2]*

Torkelson *et al.* (1961)		4.2

bourgeonal⇒3-(4-*tert*-BUTYLPHENYL)PROPANAL

BROMINE *[7726-95-6]*

Valentin (1848,1850)		3
Henning (1924)	d	0.2
Rupp & Henschler (1967)	d	< 0.065
Rupp & Henschler (1967)	r	> 6.5
Leonardos *et al.* (1969)	r	0.3
Randebrock (1986)		0.90

α-bromoacetophenone⇒PHENACYL BROMIDE

3-BROMOBENZALDEHYDE *[3132-99-8]*

Von Ranson & Belitz (1992b)	d	1.0
Von Ranson & Belitz (1992b)	r	2.0

BROMOBENZENE *[108-86-1]*

Backman (1917)	r	1.7 - 2.1
Mateson (1955)		30.5
Punter (1983)	d	1.2

bromobenzyl cyanide⇒BENZYLBROMO CYANIDE

BROMOCHLOROIODOMETHANE *[34970-00-8]*

Cancho *et al.* (2001)		1.85

2-BROMO-2-CHLORO-1,1,1-TRIFLUOROETHANE (halothane) *[151-67-7]*

Flemming & Johnstone (1977)	r	267

6-BROMO-2,4-DICHLOROANISOLE (2,4-dichloro-6-bromoanisole)

Diaz *et al.* (2005)		0.0024

6-BROMO-2,5-DICHLOROANISOLE (2,5-dichloro-6-bromoanisole)

Diaz *et al.* (2005)		0.0016

3-BROMO-2,6-DICHLOROANISOLE (2,6-dichloro-3-bromoanisole)

Diaz *et al.* (2005)		0.0014

BROMODIIODOMETHANE *[557-95-9]*

Cancho *et al.* (2001)		0.52

BROMOETHANE *[74-96-4]*

Backman (1917)	r	12.1 - 16.0

bromoform⇒TRIBROMOMETHANE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1-BROMO-2-METHYLBENZENE (2-bromotoluene) *[95-46-5]*

Backman (1917)	r	0.10 - 0.12

1-(2-BROMO-3-METHYLBUTANOYL)UREA (bromural) *[496-67-3]*

Backman (1917)	r	2.9

2-BROMOPHENOL *[95-56-7]*

Kendall *et al.* (1968)	r	0.000 007

1-BROMO-2-PROPANONE (bromacetone) *[598-31-2]*

Prentiss (1937)		0.5

2-bromotoluene⇒1-BROMO-2-METHYLBENZENE
bromural⇒1-(2-BROMO-3-METHYLBUTANOYL)UREA

1,3-BUTADIENE *[106-99-0]*

Mullins (1955	r	169
Deadman & Prigg (1959)	d	2.1
Ripp (1968)		4
Laffort & Dravnieks (1973)		5.8
Hellman & Small (1974)	d	1.0
Hellman & Small (1974)	r	2.4
Jeltes (1975)		0.22
Nagata (2003)	d	0.51

BUTANAL (butyraldehyde) *[123-72-8]*

Backman (1917)	r	0.013 - 0.014
Pliška & Janiček (1965)		15
Hellman & Small (1973,1974)	d	< 0.013
Hellman & Small (1973,1974)	r	0.027
Teranishi *et al.* (1974)		0.042
Anon. (1980)	d	0.000 84
Anon. (1980)	r	0.011
Hall & Andersson (1983)	d	0.20
Cristoph (1983)	r	0.18 - 0.21
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	8.8
Nagata (2003)	d	0.0019
Cometto-Muñiz & Abraham (2010a)	d	0.0013
Laska & Ringh (2010)	d	0.10

BUTANE *[106-97-8]*

Patty & Yant (1929)		12,000
Mullins (1955)	r	6,160
Schneider *et al.* (1966)		8,700
Laffort & Dravnieks (1973)		3,000
Artho & Koch (1973)		1 - 10
Nagata (2003)	d	2,880

1,4-BUTANEDIAMINE (putrescine) *[110-60-1]*

El-Maaytah (1997)		0.088
Greenman *et al.* (2004)		0.080

2,3-BUTANEDIONE (diacetyl) *[431-03-8]*

Backman (1917)	r	0.003 - 0.006
Van Anrooij (1931)	d	0.0025

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Appell (1969)		0.0026
Artho & Koch (1973)		0.000 01
Punter (1975,1979)	d	0.000 007
Hall & Andersson (1983)	d	0.0050
Bahnmüller (1983)		0.000 7 - 0.087
Randebrock (1986)		10.2
Blank (1990)		0.015 - 0.030
Blank *et al.* (1992)		0.01 - 0.02
Nagata (2003)	d	0.000 18

1-BUTANETHIOL (butyl mercaptan) *[109-79-5]*

Allison & Katz (1919)		18
Katz & Talbert (1930)		0.0037
Deadman & Prigg (1959)	d	0.0015
Blinova (1965)		0.007 - 0.04
Kniebes *et al.* (1969)		0.003
Wilby (1969)	r	0.0027
Patte (1978); Patte & Punter (1979)	d	0.003
Nagata (2003)	d	0.000 010

2-BUTANETHIOL (*sec*-butyl mercaptan) *[513-53-1]*

Stuiver (1958)	d	0.000 002 - 0.000 2
Blanchard (1976)		0.007
Nagata (2003)	d	0.000 11

BUTANOIC ACID (butyric acid) *[107-92-6]*

Passy (1893b,1893c)	d	0.001
Backman (1917)	r	0.02
Allison & Katz (1919)		9
Mitsumoto (1926)	r	0.16 - 0.17
Hesse (1926)	r	0.1
Hesse (1927)	r	1.5
Tamman & Oelsen (1928)	d	0.09 - 0.12
Jones (1955a)	r	0.016
Jones (1955b)	r	0.000 5 - 3.6
Jones (1955c)	r	3.0
Kerka & Humphreys (1956)		0.002
Neuhaus (1957)	d	0.001
Neuhaus (1957)	r	0.025
Nader (1958)		0.058
Bocca & Battiston (1964)	d	0.007 - 0.14
Guadagni (1966)		0.24
Goldenberg (1967)	d	0.000 4
Leonardos *et al.* (1969)	r	0.004
Styazhkin & Khachaturyan (1973)		0.06
Cormack *et al.* (1974)		0.004
Punter (1975,1979)	d	0.0385
Bedborough & Trott (1979)	d	0.0086
Anon. (1980)	d	0.000 35
Anon. (1980)	r	0.0025
Punter (1983)	d	0.014 - 0.029
Dravnieks *et al.* (1986)	d	0.018
Cometto-Muñiz *et al.* (1998a); Cometto-Muñiz (1999)	d	0.013
Nagata (2003)	d	0.000 68
Greenman *et al.* (2004)		0.02
Wise *et al.* (2007); Miyazawa *et al.* (2009a)	d	0.000 35 - 0.000 86

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Cometto-Muñiz & Abraham (2010b)	d	0.000 94

1-BUTANOL (butyl alcohol) *[71-36-3]*

Passy (1892c)	d	1
Backman (1917)	r	0.35 - 0.6
Zwaardemaker (1927)		1
Jung (1936)	d	0.158 - 0.316
Jung (1936)	r	0.474 - 0.632
Gavaudan *et al.* (1948)		0.15
Mullins (1955)	r	37.2
Jones (1955a)	r	3.1
Jones (1955b)	r	110 - 285
Jones (1955c)	r	42
Scherberger *et al.* (1958)	r	45
Janiček *et al.* (1960)		20
Nauš (1962)	d	4
Pliška & Janiček (1965)		3,000
Gavaudan & Poussel (1966)		1.1
May (1966)	d	33
May (1966)	r	48
Dravnieks & Krotoszynski (1968)		1.35
Khachaturyan & Baikov (1969)		1.2 - 2
Cain (1969)	r	60
Corbit & Engen (1971)		13 - 20
Dravnieks & Laffort (1972)		10.0
Baikov & Khachaturyan (1973)		1.2
Laffort & Dravnieks (1973)		0.9
Hellman & Small (1974)	d	0.9
Hellman & Small (1974)	r	3.0
Moskowitz *et al.* (1974)		186
Jones *et al.* (1975)		< 132
Piggott & Harper (1975)		4 - 1,000
Dravnieks (1976)	d	0.36 - 10.2
Williams *et al.* (1977)	d	0.63 - 1.14
Amoore & Buttery (1978)	d	2.3
Homans *et al.* (1978)		13.94
Jones *et al.* (1978)	d	42 - 105
Laing *et al.* (1978)	r	10.5
Laing (1982)	d	3.0
Cain *et al.* (1983); Cain (1989)	d	< 4.2
Cristoph (1983)	r	0.7 - 0.9
Laing (1983)		6.0
Jensen & Flyger (1983)		0.10 - 2.40
Punter (1983)	d	2.6 - 5.3
Viswanathan *et al.* (1983)		1.26 - 2.4
Homans (1984)		21.5
Murphy & Cain (1985)	d	0.39 - 4.26
Roos *et al.* (1985)	d	0.101 - 0.136
Roos *et al.* (1985); Don (1986)	d	0.077
Ahlström *et al.* (1986a)	d	0.136 - 0.224
Dravnieks *et al.* (1986)	d	0.51 - 4.05
Hartigh (1986)	d	0.01 - 0.292
MacLeod *et al.* (1986)		0.69
Poostchi *et al.* (1986)	d	0.99 - 1.85
Poostchi *et al.* (1986)	r	3.72 - 4.02
Cain *et al.* (1988)		1.4
Dollnick *et al.* (1988)		0.384

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Stevens *et al.* (1988)	d	0.36 - 3.3
De Wijk (1989)		4.43
Hermans (1989)		0.015 - 0.214
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	5.4
Scharfenberger (1990)		0.5
Cain & Gent (1991)	d	3 - 9
Laska & Hudson (1991)	d	0.79
Lea & Ford (1991)		2
Nagy (1991)	d	3.1
Nagy (1991)	d	0.591
Cometto-Muñiz & Cain (1993)	d	162
Patterson *et al.* (1993)	d	5.4
Stevens & Dadarwala (1993)	d	0.48 - 38.4
Dalton *et al.* (1997a)	d	0.61 - 5.5
Dalton *et al.* (1997b)	d	8.2 - 15.8
Harreveld & Heeres (1997)		0.058 - 0.53
Wysocki *et al.* (1997)	d	0.48 - 9.6
Cometto-Muñiz *et al.* (1999)	d	1.7 - 3.8
Molhave *et al.* (2000)	d	11
Ziemer *et al.* (2000)	d	0.15
Feddes *et al.* (2001)	d	0.170
Mannebeck & Mannebeck (2002)	d	0.105 - 0.1739
Smeets & Dalton (2002)	d	42 - 54
Nagata (2003)	d	0.11
Cometto-Muñiz *et al.* (2004)	d	~ 0.97
Maxeiner & Mannebeck (2004)	d	0.1323 - 0.1957
Cometto-Muñiz & Abraham (2008)	d	0.024
Polednik *et al.* (2008)	d	0.2 - 0.4 (indoor air as medium)
Maxeiner (2006)	d	0.1071 - 0.1251
Maxeiner (2007)	d	0.111 - 0.130
Ueno *et al.* (2009)		0.051 (triangle odor bag method; forced-choice mode)
Ueno *et al.* (2009)		0.26 (triangle odor bag method; yes/no mode)
Ueno *et al.* (2009)		0.16 (dynamic olfactometer; forced-choice mode)
Ueno *et al.* (2009)		0.42 (dynamic olfactometer; yes/no mode)
Cain *et al.* (2010)	d	0.48
Nimmermark (2011)		0.078 - 1.4

2-BUTANOL (*sec*-butyl alcohol) *[78-92-2]*

Jung (1936)	d	7.4
Jung (1936)	r	14.4
Jones (1955c)	r	80
Laffort & Dravnieks (1973)		9
Hellman & Small (1974)	d	0.4
Hellman & Small (1974)	r	1.2
Bedborough & Trott (1979)	d	3.3
Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993)	d	285
Ziemer *et al.* (2000)	d	0.13
Nagata (2003)	d	0.66

(*S*)-(+)-2-BUTANOL ((*d*)-*sec*-butyl alcohol) *[4221-99-2]*

Punter (1983)	d	59.1

(*R*)-(-)-2-BUTANOL ((*l*)-*sec*-butyl alcohol) *[14898-79-4]*

Punter (1983)	d	41.8

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

tert-butanol⇒2-METHYL-2-PROPANOL

BUTANONE (methyl ethyl ketone) *[78-93-3]*

Backman (1917)	r	63 - 70
May (1966)	d	80
May (1966)	r	163
Leonardos *et al.* (1969)	r	29
Hartung *et al.* (1971)		7
Mukhitov & Azimbekov (1971)		0.75
Dravnieks & Laffort (1972)		22.0
Artho & Koch (1973)		100 - 1,000
Dravnieks (1974)	d	250
Hellman & Small (1974)	d	5.8
Hellman & Small (1974)	r	16
Anon. (1980)	d	8.4
Anon. (1980)	r	29
Hall & Andersson (1983)	d	61
Doty *et al.* (1988)	d	16.5 - 23.9
Scharfenberger (1990)		126
Doty (1991)	d	2.9 - 51.6
Nagy (1991)	d	4.8
Ziemer *et al.* (2000)	d	0.21
Nagata (2003)	d	1.3

2-BUTANOYL-2,3-DIHYDRO-1,4-THIAZINE

Hofmann *et al.* (1995)		> 2

2-BUTANOYL-1-PYRROLINE

Schieberle (1991)		> 2

(*E*)-2-BUTENAL (crotonaldehyde) *[123-73-9]*

Katz & Talbert (1930)		0.18 - 0.57
Teranishi *et al.* (1974)		0.420
Hall & Andersson (1983)	d	1.7
Nagata (2003)	d	0.067

1-BUTENE *[106-98-9]*

Katz & Talbert (1930)		2.1
Mullins (1955)	r	39.2
Krasovitskaya & Malyarova (1968)		15.4
Knuth (1973)		1.2
Nagata (2003)	d	0.83

2-BUTENE *[107-01-7]*

Katz & Talbert (1930)		4.8

(*E*)-2-BUTENE *[624-64-6]*

Mullins (1955)	r	2,700

(*Z*)-2-BUTENE *[590-18-1]*

Mullins (1955)	r	28.5

2-BUTENE-1-THIOL (crotyl mercaptan) *[5954-72-3]*

Katz & Talbert (1930)		0.000 43 - 0.0014

1-BUTEN-3-ONE (3-buten-2-one, methyl vinyl ketone) *[78-94-4]*

Martirosyan (1970)		0.5

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3-buten-2-one⇒1-BUTEN-3-ONE
butiphos⇒S,S,S-TRIBUTYL PHOSPHOROTRITHIOATE

BUTOXYBUTANE (butyl ether) *[142-96-1]*

Dravnieks & Laffort (1972)		0.03
Dravnieks (1974)	d	8
Hellman & Small (1974)	d	0.4
Hellman & Small (1974)	r	1.3

2-BUTOXYETHANOL (butyl cellosolve, ethylene glycol *n*-butyl ether) *[111-76-2]*

Hellman & Small (1973,1974)	d	0.5
Hellman & Small (1973,1974)	r	1.7
Nagy (1991)	d	1.9
Nagata (2003)	d	0.21

2-BUTOXYETHYL ACETATE (butyl cellosolve acetate) *112-07-2]*

Hellman & Small (1973,1974)	d	0.7
Hellman & Small (1973,1974)	r	1.3
Nagy (1991)	d	6.5

1-BUTOXY-2-PROPANOL (propylene glycol *n*-butyl ether) *[5131-66-8]*

Ziemer *et al.* (2000)	d	1.3
Nagata (2003)	d	0.87

BUTYL ACETATE *[123-86-4]*

Backman (1917)	r	1.3 - 1.7
Jung (1936)	d	0.044
Jung (1936)	r	0.044 - 0.13
Scherberger *et al.* (1958)	r	96
Gofmekler (1960)		0.6
Pliška & Janiček (1960)		190
Nauš (1962)	d	0.7
May (1966)	d	35
May (1966)	r	55
Köster (1971)	d	480 - 1,750
Dravnieks & Laffort (1972)		0.04
Dravnieks (1974)	d	3
Hellman & Small (1974)	d	0.03
Hellman & Small (1974)	r	0.18
Anon. (1980)	d	0.32
Anon. (1980)	r	2.4
Cristoph (1983)	r	0.46 - 0.55
Scharfenberger (1990)		4
Cometto-Muñiz & Cain (1991,1993);		
Cometto-Muñiz (1993)	d	11.5
Nagy (1991)	d	1
Nagy (1991)	d	0.521
Patterson *et al.* (1993)	d	7.7
Ziemer *et al.* (2000)	d	0.061
Cometto-Muñiz *et al.* (2002)		0.000 62
Cometto-Muñiz *et al.* (2003)		0.009
Nagata (2003)	d	0.077
Cometto-Muñiz *et al.* (2004)	d	~ 0.015
Komthomg *et al.* (2006)		165 - 1,570
Cometto-Muñiz *et al.* (2008)	d	0.020
Cain & Schmidt (2009)	d	0.010

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-BUTYL ACETATE (*sec*-butyl acetate) *[105-46-4]*

Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993)	d	22.6
Nagata (2003)	d	0.012

sec-butyl acetate⇒2-BUTYL ACETATE

tert-BUTYL ACETATE *[540-88-5]*

Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993)	d	6.2
Nagata (2003)	d	0.34
Cain & Schmidt (2009)	d	0.038

butyl acrylate⇒BUTYL 2-PROPENOATE
butyl alcohol⇒1-BUTANOL
sec-butyl alcohol⇒2-BUTANOL
(*d*)-*sec*-butyl alcohol⇒(*S*)-(+)-2-BUTANOL
(*l*)-*sec*-butyl alcohol⇒(*R*)-(-)-2-BUTANOL
tert-butyl alcohol⇒2-METHYL-2-PROPANOL

BUTYLAMINE *[109-73-9]*

Scherberger *et al.* (1960)		< 0.36
Sutton (1962a)		< 3
Hellman & Small (1973,1974)	d	0.24
Hellman & Small (1973,1974)	r	0.72
Laing *et al.* (1978)	r	41.7
Nagata (2003)	d	0.51

2-BUTYLAMINE (*sec*-butylamine) *[13952-84-6]*

Nagata (2003)	d	0.51

sec-butylamine⇒2-BUTYLAMINE
tert-butylamine⇒2-(2-METHYL)PROPYLAMINE

4-BUTYLBENZALDEHYDE *[1200-14-2]*

Von Ranson & Belitz (1992b)	d	0.058
Von Ranson & Belitz (1992b)	r	0.11

4-*tert*-BUTYLBENZALDEHYDE *[939-97-9]*

Von Ranson & Belitz (1992b)	d	0.065
Von Ranson & Belitz (1992b)	r	0.19

BUTYLBENZENE *[104-51-8]*

Cometto-Muñiz (1993);		
Cometto-Muñiz & Cain (1994)	d	23.4
Nagata (2003)	d	0.047
Cometto-Muñiz & Abraham (2009b)	d	0.014

7-BUTYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 26

BUTYL BUTANOATE (butyl butyrate) *[109-21-7]*

Nagata (2003)	d	0.028

butyl butyrate⇒BUTYL BUTANOATE
tert-butylcarbylamine⇒*tert*-BUTYL ISOCYANIDE
butyl cellosolve⇒2-BUTOXYETHANOL
butyl cellosolve acetate⇒2-BUTOXYETHYL ACETATE

4-*tert*-BUTYLCYCLOHEXYL ACETATE (oryclone) *[32210-23-4]*

Randebrock (1986)		0.000 69
Laska & Hudson (1991)	d	0.86

2-BUTYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE (2-butyl-3,5-dimethyl-5,6-dihydropyrazine)

Kurniadi *et al.* (2003)		0.0168
Kurniadi (2003)	r	0.336

3-BUTYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE (3-butyl-2,5-dimethyl-5,6-dihydropyrazine)

Kurniadi *et al.* (2003)		> 2
Kurniadi (2003)	r	> 40

3-BUTYL-5,6-DIHYDRO-4*H*-ISOBENZOFURAN-1-ONE *[141120-37-8]*

Nitz *et al.* (1992)		0.005

4-BUTYL-4,5-DIHYDROPHTHALIDE (sedanenolide) *[62006-39-7]*

Nitz *et al.* (1992)		5

5-BUTYLDIHYDRO-2(3*H*)-THIOPHENONE (γ-thiooctalactone)

Roling *et al.* (1998); Engel (1999)		0.000 3

2-butyl-3,5-dimethyl-5,6-dihydropyrazine⇒2-BUTYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE
3-butyl-2,5-dimethyl-5,6-dihydropyrazine⇒3-BUTYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE

4-*tert*-BUTYL-2,6-DIMETHYL-3,5-DINITROACETOPHENONE (musk ketone) *[81-14-1]*

Kraft & Fráter (2001)		0.000 1
Traynor (2001)		0.0012

5-BUTYL-2,3-DIMETHYLPYRAZINE *[15834-78-3]*

Wagner *et al.* (1999)		> 2

3-BUTYL-2,5-DIMETHYLPYRAZINE *[40790-29-2]*

Wagner *et al.* (1999)		> 2

2-BUTYL-3,5-DIMETHYLPYRAZINE *[50888-63-6]*

Wagner *et al.* (1999)		0.18

5-*sec*-BUTYL-2,3-DIMETHYLPYRAZINE *[3226-30-0]*

Wagner *et al.* (1999)		> 2

3-*sec*-BUTYL-2,5-DIMETHYLPYRAZINE *[70303-41-2]*

Wagner *et al.* (1999)		> 2

2-*sec*-BUTYL-3,5-DIMETHYLPYRAZINE *[70303-43-4]*

Wagner *et al.* (1999)		> 2

2-butyldimethyltetrahydropyrazine⇒2-BUTYLTETRAHYDRODIMETHYLPYRAZINE

1-*tert*-BUTYL-3,5-DIMETHYL-2,4,6,-TRINITROBENZENE (musk xylene) *[81-15-2]*

Backman (1917)	r	0.009
Allison & Katz (1919)		0.04
Katz & Talbert (1930)		0.000 59

butyl ether⇒BUTOXYBUTANE

BUTYL FORMATE *[592-84-7]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

May (1966)	d	70
May (1966)	r	90
Nagata (2003)	d	0.37

(3*S*,3a*S*,7a*R*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) 0.0013

(3*R*,3a*R*,7a*S*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) > 1.25

(3*S*,3a*S*,7a*S*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) 0.25

(3*R*,3a*R*,7a*R*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) > 1.25

(3*S*,3a*R*,7a*S*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) 0.25

(3*R*,3a*S*,7a*R*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) 0.125

(3*S*,3a*R*,7a*R*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) > 1.25

(3*R*,3a*S*,7a*S*)-3-BUTYLHEXAHYDROPHTHALIDE
Bartschat *et al.* (1997b) > 1.25

tert-BUTYL(1-HYDROXY-1-METHYLETHYL)DIMETHYLSILANE
Sunderkötter *et al.* (2010) 0.001

BUTYL 2-HYDROXYPROPANOATE (butyl lactate) *[138-22-7]*
Ziemer *et al.* (2000) d 0.000 000 029

butyl iodide⇒1-IODOBUTANE
butyl isobutyrate⇒BUTYL 2-METHYLPROPANOATE

tert-BUTYL ISOCYANIDE (*tert*-butylcarbylamine) *[7188-38-7]*

Stone & Pryor (1967)	d	1.67
Stone & Pryor (1967)	r	1.67

butyl isovalerate⇒BUTYL 3-METHYLBUTANOATE
butyl lactate⇒ BUTYL 2-HYDROXYPROPANOATE
butyl mercaptan⇒1-BUTANETHIOL
sec-butyl mercaptan⇒2-BUTANETHIOL
tert-butyl mercaptan⇒2-METHYL-2-PROPANETHIOL

2-BUTYL-3-MERCAPTOPROPANAL (3-mercapto-2-butylpropanal)
Vermeulen & Collin (2002) 0.004

2-BUTYL-3-MERCAPTO-1-PROPANOL (3-mercapto-2-butyl-1-propanol)
Vermeulen *et al.* (2003) 0.000 05

2-BUTYL-3-MERCAPTOPROPYL ACETATE (3-mercapto-2-butylpropyl acetate)
Vermeulen & Collin (2003) ` 0.01

butyl methacrylate⇒BUTYL 2-METHYL-2-PROPENOATE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1-*tert*-BUTYL-2-METHOXY-4-METHYL-3,5-DINITROBENZENE (musk ambrette) *[83-66-9]*

Appell (1969)		0.000 000 09
Etzweiler *et al.* (1980)		0.000 19
Randebrock (1986)		0.000 045

2-*sec*-BUTYL-3-METHOXYPYRAZINE *[24168-70-5]*

Wagner *et al.* (1999)		0.000 003

1-*tert*-BUTYL-4-METHYLBENZENE (4-*tert*-butyltoluene) *[98-51-1]*

Hine *et al.* (1954)	r	< 30.5

BUTYL 3-METHYLBUTANOATE (butyl isovalerate) *[109-19-3]*

Nagata (2003)	d	0.078

4-*tert*-butyl-α-methyldihydrocinnamaldehyde⇒3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL
(2*R*)-4-*tert*-butyl-α-methyldihydrocinnamaldehyde⇒(2*R*)-3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL
(2*S*)-4-*tert*-butyl-α-methyldihydrocinnamaldehyde⇒(2*S*)-3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL

BUTYL 2-METHYLPROPANOATE (butyl isobutyrate) *[97-87-0]*

Nagata (2003)	d	0.13
Komthong *et al.* (2006)		6.07

BUTYL 2-METHYL-2-PROPENOATE (butyl methacrylate) *[97-88-1]*

Piringer & Granzer (1984)		0.1

2-BUTYL-3-METHYLPYRAZINE *[15987-00-5]*

Wagner *et al.* (1999)		0.095 - 0.26

5-BUTYL-2-METHYLPYRAZINE *[29461-04-9]*

Wagner *et al.* (1999)		0.095 - 0.26

6-BUTYL-2-METHYLPYRAZINE *[32184-46-6]*

Wagner *et al.* (1999)		0.095 - 0.26

1-*tert*-BUTYL-3-METHYL-2,4,6-TRINITROBENZENE (artificial musk) *[547-94-4]*

Passy (1892a)	d	0.000 000 05 - 0.000 001
Passy (1892b)	d	0.000 005 - 0.000 01
Zwaardemaker (1914)	d	0.001

3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL (4-*tert*-butyl-α-methyldihydrocinnamaldehyde, lily aldehyde, lilial) *[80-54-6]*

Doszczak *et al.* (2007)		0.000 1

(2*R*)-3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL ((2*R*)-4-*tert*-butyl-α-methyldihydrocinnamaldehyde)

Bartschat *et al.* (1997a)		0.01

(2*S*)-3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL ((2*S*)-4-*tert*-butyl-α-methyldihydrocinnamaldehyde)

Bartschat *et al.* (1997a)		> 2.5

3-(4-*tert*-BUTYLPHENYL)PROPANAL (bourgeonal) *[18127-01-0]*

Doszczak *et al.* (2007)		0.000 16
Olsson & Laska (2010)	d	0.1 - 0.2

3-BUTYLPHTHALIDE *[6066-49-5]*

Nitz *et al.* (1992)		7

BUTYL PROPANOATE (butyl propionate) *[590-01-2]*

Nagata (2003)	d	0.19

BUTYL 2-PROPENOATE (butyl acrylate) *[141-32-2]*

Anon. (1969)		0.53
Van Gemert (1973)	d	0.005 - 0.01
Anon. (1980)	d	0.0015
Anon. (1980)	r	0.014
Piringer & Granzer (1984)		0.01
Nagata (2003)	d	0.0029

butyl propionate⇒BUTYL PROPANOATE

2-BUTYLPYRIDINE *[5058-19-5]*

Schieberle (1993)		0.0018

BUTYL SALICYLATE *[2052-14-4]*

Baldus (1936)	d	0.005 - 0.007
Baldus (1936)	r	0.01

2-BUTYLTETRAHYDRODIMETHYLPYRAZINE (2-butyldimethyltetrahydropyrazine)

Kurniadi (2003)	r	> 40

trans-3-BUTYLTETRAHYDROPHTHALIDE (*trans*-sedanolide) *[3553-29-5]*

Nitz *et al.* (1992)		2

S-BUTYL THIOACETATE

Gijs *et al.* (2000)		0.12

BUTYLTHIOBUTANE (dibutyl sulphide) *[544-40-1]*

Katz & Talbert (1930)		0.090
Fuller *et al.* (1964)	r	0.000 88

S-BUTYL THIOBUTANOATE (thiobutyl butanoate)

Cavaillé-Lefebvre *et al.* (1998)	d	1.12

tert-BUTYLTHIO-2-METHYLPROPANE (di-*tert*-butyl sulphide) *[107-47-1]*

Gijs *et al*. (2000)		0.32

S-BUTYL THIOPENTANOATE (thiobutyl pentanoate)

Cavaillé-Lefebvre *et al.* (1998)	d	1.88

S-BUTYL THIOPROPANOATE (thiobutyl propanoate)

Cavaillé-Lefebvre *et al.* (1998)	d	0.45

4-*tert*-butyltoluene⇒1-*tert*-BUTYL-4-METHYLBENZENE

2-BUTYNE *[503-17-3]*

Mullins (1955)	r	70.3

butyraldehyde⇒BUTANAL
butyric acid⇒BUTANOIC ACID

C

camphene⇒2,2-DIMETHYL-3-METHYLENEBICYCLO[2.2.1]HEPTANE
camphor⇒ 1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
(*S*)-(-)-camphor⇒(*S*)-(-)-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE

capric acid⇒DECANOIC ACID
caproic acid⇒HEXANOIC ACID
caprolactam⇒6-AMINOHEXANOIC ACID LACTAM
caprylaldehyde⇒OCTANAL
caprylic acid⇒OCTANOIC ACID
carbitol⇒3,6-DIOXA-1-OCTANOL
carbitol acetate⇒3,6-DIOXAOCTYL-1 ACETATE

CARBON DIOXIDE *[124-38-9]*

Lötsch *et al.* (1997)		540,000 - 1,080,000
Shusterman & Balmes (1997a,1997b)		486,000
Melzner *et al.* (2011)	d	95,400 (orthonasal stimulation, nasal breathing)
Melzner *et al.* (2011)	d	81,000 (orthonasal stimulation, oral breathing)
Melzner *et al.* (2011)	d	75,600 (retronasal stimulation, nasal breathing)
Melzner *et al.* (2011)	d	70,200 (retronasal stimulation, oral breathing)

CARBON DISULPHIDE *[75-15-0]*

Deadman & Prigg (1959)	d	0.07
Hildenskiold (1959)		0.05
Frantikova (1962)		1.3
Baikov (1963)		0.08 - 0.5
Leonardos *et al.* (1969)	r	0.65
Nauš (1982)	d	0.1
Nauš (1982)	r	1
Kleinschmidt (1983)	r	98.9
Moriguchi *et al.* (1983)	d	0.11
Don (1986)	d	0.18
Nagy (1991)	d	3.9
Nagy (1991)	d	1.269
Nagata (2003)	d	0.65

carbonic chloride⇒CARBONYL CHLORIDE
carbon tetrachloride⇒TETRACHLOROMETHANE

CARBONYL CHLORIDE (carbonic chloride, phosgene) *[75-44-5]*

Fieldner *et al.* (1921)		23
Suchier (1929)		4
Schley (1934)	d	0.5
Schley (1934)	r	0.5 - 1
Prentiss (1937)		4.4
Patty (1962c)		2.0
Leonardos *et al.* (1969)	r	4.0

CARBONYL SULPHIDE *[463-58-1]*

Polgar *et al.* (1975)		0.25
Nagata (2003)	d	0.14

3-carene⇒3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE
(+)-3-carene⇒(1*S*)-(+)-3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE
δ3-carene⇒3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE
carvacrol⇒5-ISOPROPYL-2-METHYLPHENOL
carvone⇒5-ISOPROPENYL-2-METHYL-2-CYCLOHEXEN-1-ONE
(*d*)-carvone⇒(*S*)-(+)-5-ISOPROPENYL-1-METHYL-2-CYCLOHEXEN-1-ONE
(*l*)-carvone⇒(*R*)-(-)-5-ISOPROPENYL-1-METHYL-2-CYCLOHEXEN-1-ONE
β-caryophyllene⇒4,11,11-TRIMETHYL-8-METHYLENEBICYCLO[7.2.0]-4-UNDECENE
cellosolve⇒2-ETHOXYETHANOL
cellosolve acetate⇒2-ETHOXYETHYL ACETATE
cervolide⇒16-HYDROXY-12-OXAHEXADECANOIC ACID LACTONE
chloral⇒TRICHLOROETHANAL
chloral hydrate⇒TRICHLOROETHANAL HYDRATE

CHLORINE *[7782-50-5]*

Fieldner *et al.* (1921); Prentiss (1937)		10
Smolczyk & Cobler (1930)		1.43 - 14.3
Takhiroff (1957)		0.8
Beck (1959)		0.15 - 0.3
Styazhkin (1963)		0.7
Rupp & Henschler (1967)	d	0.06 - 0.15
Rupp & Henschler (1967)	r	0.3
Leonardos *et al.* (1969)	r	0.6
Kramer (1974)		3.2 - 7.8
Dixon & Ikels (1977)	d	0.23
Nauš (1982)	d	3
Nauš (1982)	r	10
Randebrock (1986)		0.18
Nagata (2003)	d	0.14

CHLORINE DIOXIDE *[7758-19-2]*

Vincent *et al.* (1946)		42

CHLOROACETIC ACID *[79-11-8]*

Backman (1917)	r	0.6
Smith & Hochstettler (1969)	r	0.05

α-chloroacetophenone⇒PHENACYL CHLORIDE

2-CHLOROANILINE *[95-51-2]*

Punter (1983)	d	0.69

3-CHLOROANILINE *[108-42-9]*

Punter (1983)	d	1.11

4-CHLOROANILINE *[106-47-8]*

Kazakova (1968)		0.09
Punter (1983)	d	1.58

2-CHLOROANISOLE *[766-51-8]*

Strube & Buettner (2010)		0.01433

3-CHLOROANISOLE *[2845-89-8]*

Strube & Buettner (2010)		0.03445

4-CHLOROANISOLE *[623-12-1]*

Strube & Buettner (2010)		0.0029

10-CHLORO-5,10-DIHYDROPHENARSAZINE (adamsite) *[578-94-9]*

Prentiss (1937)		2.5

CHLOROBENZENE *[108-90-7]*

Backman (1917)	r	7.5 - 8.1
Mateson (1955)		21.6
Tarkhova (1965)		0.4
Leonardos *et al.* (1969)	r	0.97
Smith & Hochstettler (1969)	r	3
Punter (1983)	d	5.9
Don (1986)	d	1.0
Nagy (1991)	d	4.5

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Cometto-Muñiz (1993);
Cometto-Muñiz & Cain (1994) d 59.3

2-CHLORO-1,3-BUTADIENE (chloroprene) *[126-99-8]*
Mnatsakanyan (1962) 0.4 - 2.0
Nyström (1948) r 500 - 1,000

1-CHLOROBUTANE *[109-69-3]*
Mullins (1955) r 450
Hellman & Small (1973,1974) d 33
Hellman & Small (1973,1974) r 51

2-CHLORO-2-BUTENE *[4461-41-0]*
McCawley (1942) 1,110 - 3,700

1-CHLORO-1-(CHLOROETHYLTHIO)ETHANE (1,1'-dichlorodiethyl sulphide)
Katz & Talbert (1930) 0.015

2-CHLORO-1-(2-CHLOROETHYLTHIO)ETHANE (β,β'-dichlorethyl sulphide, mustard gas) *[505-60-2]*
Prentiss (1937) 1.3

1-CHLORO-2-(1-CHLOROISOPROPOXY)PROPANE (dichloroisopropylether) *[108-60-1]*
Punter (1983) d 0.19 - 0.36

1-CHLORODECANE (decyl chloride) *[1002-69-3]*
Mullins (1955) r 21.7

CHLORODIFLUOROMETHANE *[75-45-6]*
Braker & Mossman (1980) 708,000

CHLORODIIODOMETHANE *[638-73-3]*
Cancho *et al.* (2001) 0.77

CHLORODIPHENYLARSINE *[712-48-1]*
Prentiss (1937) 0.3

1-CHLORO-2,3-EPOXYPROPANE (epichlorohydrin) *[106-89-8]*
Toxicity data sheet (1959) 38 - 46
Fomin (1966) 0.3

CHLOROETHANE (ethyl chloride) *[75-00-3]*
Backman (1917) r 10 – 12
Nagy (1991) d > 1,000

2-CHLOROETHANOL (ethylene chlorohydrin) *[107-07-3]*
Semenova *et al.* (1980) 1.2

CHLOROETHENE (vinyl chloride) *[75-01-4]*
Hori *et al.* (1972) 520 - 910

chloroform⇒TRICHLOROMETHANE

1-CHLOROHEPTANE (heptyl chloride) *[629-06-1]*
Mullins (1955) r 157

CHLOROIODOMETHANE *[593-71-5]*
Cancho *et al.* (2001) 1.4

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

CHLOROMETHANE (methyl chloride) *[74-87-3]*

Leonardos *et al.* (1969)	r	> 21

5-CHLORO-2-METHYLANILINE (5-chloro-*o*-toluidine) *[95-79-4]*

Punter (1983)	d	0.10

1-CHLORO-2-METHYLBENZENE (2-chlorotoluene) *[95-49-8]*

Backman (1917)	r	0.95 - 1.4

1-CHLORO-3-NITROBENZE *[121-73-3]*

Andreescheva (1968)		0.02

2-CHLOROPHENOL *[95-57-8]*

Katz & Talbert (1930)		0.019
Kendall *et al.* (1968)	r	0.000 5

4-CHLOROPHENOL *[106-48-9]*

Kendall *et al.* (1968)	r	0.001

3-CHLOROPHENYL ISOCYANATE *[2909-38-8]*

Zibireva (1967)		0.015

4-CHLOROPHENYL ISOCYANATE *[104-12-1]*

Zibireva (1967)		0.010

chloropicrin⇒TRICHLORONITROMETHANE
chloroprene⇒2-CHLORO-1,3-BUTADIENE

3-CHLOROPROPENE (allyl chloride) *[107-05-1]*

Toxicity data sheet (1958a)		9.3 - 18.6
Torkelson *et al.* (1959)		3 - 9
Leonardos *et al.* (1969)	r	1.5

α-chlorotoluene⇒BENZYL CHLORIDE
2-chlorotoluene⇒1-CHLORO-2-METHYLBENZENE
5-chloro-*o*-toluidine⇒5-CHLORO-2-METHYLANILINE

CHLOROTRIFLUOROMETHANE *[75-72-9]*

Braker & Mossman (1980)		854,000

2-CHLOROVINYLDICHLOROARSINE (lewisite) *[541-25-3]*

Prentiss (1937)		14

1,8-cineole⇒1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE
cinnamaldehyde⇒3-PHENYL-2-PROPENAL
(*Z*)-cinnamaldehyde⇒3-PHENYL-(*Z*)-2-PROPENAL
cinnamic alcohol⇒3-PHENYL-2-PROPEN-1-OL
cinnamyl alcohol⇒3-PHENYL-2-PROPEN-1-OL
citral⇒3,7-DIMETHYL-2,6-OCTADIENAL
citral a⇒3,7-DIMETHYL-(*E*)-2,6-OCTADIENAL
citral b⇒3,7-DIMETHYL-(*Z*)-2,6-OCTADIENAL
citronellal⇒3,7-DIMETHYL-6-OCTENAL
citronellol⇒3,7-DIMETHYL-6-OCTEN-1-OL
β-citronellol⇒3,7-DIMETHYL-6-OCTEN-1-OL
(*R*)-(+)-citronellol⇒(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-OL
(*S*)-(-)-citronellol⇒(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-OL
citronellyl acetate⇒3,7-DIMETHYL-6-OCTEN-1-YL ACETATE

COUMARIN *[91-64-5]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Passy (1892a,1892b)	d	0.01 - 0.05
Tempelaar (1913)	d	0.000 78
Backman (1917)	r	0.015 - 0.044
Katz & Talbert (1930)		0.020
Appell (1969)		0.000 007
Randebrock (1986)		0.000 083

coumarone⇒2,3-BENZOFURAN
m-cresol⇒3-METHYLPHENOL
o-cresol⇒2-METHYLPHENOL
p-cresol⇒4-METHYLPHENOL
p-cresyl acetate⇒4-METHYLPHENYL ACETATE
crotonaldehyde⇒(*E*)-2-BUTENAL
crotyl mercaptan⇒2-BUTENE-1-THIOL
cumarone⇒2,3-BENZOFURAN
cumene⇒ISOPROPYLBENZENE
cumene hydroperoxide⇒ISOPROPYLBENZENE HYDROPEROXIDE
cuminaldehyde⇒4-ISOPROPYLBENZALDEHYDE

CYANODIPHENYLARSINE *[23525-22-6]*

Flury (1921)		< 0.005
Prentiss (1937)		0.3

CYANOGEN *[460-19-5]*

Braker & Mossman (1980)		> 533

CYANOGEN CHLORIDE *[506-77-4]*

Prentiss (1937)		2.5

cyclamen aldehyde⇒2-METHYL-3-(4-ISOPROPYLPHENYL)PROPANAL
cyclemone⇒2-ACETYL-2,3,8,8-TETRAMETHYL-1,2,3,4,5,6,7,8-OCTAHYDRONAPHTHALENE

5-CYCLOBUTYLPENTANAL

Von Ranson & Belitz (1992b)	d	0.0042
Von Ranson & Belitz (1992b)	r	0.0070

CYCLO-1,3-ETHYLENEDIOXY-1,3-TRIDECANEDIONE (ethylene brassylate, musk T) *[105-95-3]*

Traynor (2001)		0.066

CYCLOHEPTADECANONE *[3661-77-6]*

Eykman (1927)	d	0.0017 - 0.0035

CYCLOHEXADECANONE *[2550-52-9]*

Eykman (1927)	d	0.0025 - 0.0066

5-CYCLOHEXADECEN-1-ONE (ambretone) *[37609-25-9]*

Traynor (2001)		0.116

CYCLOHEXANE *[110-82-7]*

Schley (1934)	d	39
Schley (1934)	r	120
Jones (1955c)	r	900
Alibaev (1970)		1.8
Stone *et al.* (1972)	d	35.6
Dravnieks & Laffort (1972)		315
Laffort & Dravnieks (1973)		165
Dravnieks (1974)	d	2,700
Nagata (2003)	d	8.5

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

CYCLOHEXANECARBOXALDEHYDE *[2043-61-0]*

Von Ranson & Belitz (1992b)	d	0.0020
Von Ranson & Belitz (1992b)	r	0.025

CYCLOHEXANOL *[108-93-0]*

Dobrinsky (1964)		0.24
Punter (1983)	d	0.64
Van Thriel *et al.* (2006)	d	2.01

CYCLOHEXANONE *[108-94-1]*

Dobrinsky (1964)		0.21
Stone *et al.* (1967)	d	1.15
Köster (1971)	d	790 - 880
Stone *et al.* (1972)	d	1.6
Hellman & Small (1973,1974)	d	0.48
Hellman & Small (1973,1974)	r	0.48
Davis (1973)	d	2
Laing (1975)	d	40
Laing (1983)		1.0 - 2.4
Laska & Hudson (1991)	d	0.88 - 1.2
Ziemer *et al.* (2000)	d	1.1
Van Thriel *et al.* (2006)	d	5.27

CYCLOHEXENE *[110-83-8]*

Deadman & Prigg (1959)	d	0.6

1-CYCLOHEXENE-1-CARBOXALDEHYDE *[1192-88-7]*

Von Ranson & Belitz (1992b)	d	0.88
Von Ranson & Belitz (1992b)	r	5.3

2-CYCLOHEXENE-1-CARBOXALDEHYDE

Von Ranson & Belitz (1992b)	d	0.75
Von Ranson & Belitz (1992b)	r	13

3-CYCLOHEXENE-1-CARBOXALDEHYDE (1,2,3,6-tetrahydrobenzaldehyde) *[100-50-5]*

Von Ranson & Belitz (1992b)	d	0.015
Von Ranson & Belitz (1992b)	r	0.35

CYCLOHEXYLAMINE *[108-91-8]*

Van Thriel *et al.* (2006)	d	9.83

4-CYCLOHEXYLBUTANAL

Von Ranson & Belitz (1992b)	d	0.20
Von Ranson & Belitz (1992b)	r	0.32

3-CYCLOHEXYLENEPROPANAL

Von Ranson & Belitz (1992b)	d	0.000 19
Von Ranson & Belitz (1992b)	r	0.061

CYCLOHEXYLETHANAL

Von Ranson & Belitz (1992b)	d	0.16
Von Ranson & Belitz (1992b)	r	0.25

3-CYCLOHEXYLPROPANAL *[4361-28-8]*

Von Ranson & Belitz (1992b)	d	0.21
Von Ranson & Belitz (1992b)	r	0.28

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3-CYCLOHEXYL-(*E*)-2-PROPENAL
Von Ranson & Belitz (1992b) d 0.025
Von Ranson & Belitz (1992b) r 0.044

3-CYCLOHEXYL-(*Z*)-2-PROPENAL
Von Ranson & Belitz (1992b) d 0.0014
Von Ranson & Belitz (1992b) r 0.097

CYCLOMUSK *[84012-64-6]*
Kraft & Fráter (2001) 0.0045

CYCLOOCTANE *[292-64-8]*
Stone *et al.* (1972) d 3.6

cyclopentadecanolide⇒ 15-HYDROXYPENTADECANOIC ACID LACTONE

CYCLOPENTADECANONE (exaltone) *[502-72-7]*
Eykman (1927) d 0.0133 - 0.0236
Traynor (2001) 0.165

CYCLOPENTADIENE *[542-92-7]*
Deadman & Prigg (1959) d 5

CYCLOPENTANOL *[96-41-3]*
Köster (1971) d 4,200 - 8,700

CYCLOPENTANONE *[120-92-3]*
Köster (1968a,1971) d 31 - 1,120

4-CYCLOPENTYLBUTANAL
Von Ranson & Belitz (1992b) d 0.0070
Von Ranson & Belitz (1992b) r 0.070

3-CYCLOPENTYLPROPANAL
Von Ranson & Belitz (1992b) d 0.0056
Von Ranson & Belitz (1992b) r 0.033

7-CYCLOPROPYLHEPTANAL
Von Ranson & Belitz (1992b) d 0.0019
Von Ranson & Belitz (1992b) r 0.0062

6-CYCLOPROPYLHEXANAL
Von Ranson & Belitz (1992b) d 0.000 28
Von Ranson & Belitz (1992b) r 0.0028

CYCLOPROPYL(1-HYDROXY-1-METHYLETHYL)DIISOPROPYLSILANE
Sunderkötter *et al.* (2010) 0.0013

5-CYCLOPROPYLPENTANAL
Von Ranson & Belitz (1992b) d 0.013
Von Ranson & Belitz (1992b) r 0.015

cyclotene ⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE

CYCLOTETRADECANONE *[3603-99-4]*
Eykman (1927) d 0.0241 - 0.0315

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

o-cymene⇒1-ISOPROPYL-2-METHYLBENZENE
p-cymene⇒1-ISOPROPYL-4-METHYLBENZENE

D

β-damascenone⇒1-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-(*E*)-2-BUTEN-1-ONE

DECABORANE *[17702-41-9]*

Krackow (1953)		0.3

(*E*,*E*)-2,4-DECADIENAL *[25152-84-5]*

Teranishi *et al.* (1974)		0.000 18
Gasser & Grosch (1990a)		0.000 04 - 0.000 16
Yang *et al.* (2008)		0.0023

(*E*,*Z*)-2,4-DECADIENAL *[25152-83-4]*

Lin *et al.* (2001)		0.000 01

(*Z*)-1,5-DECADIEN-3-OL

Schieberle & Buettner (2001)		0.001 - 0.01

(*Z*)-1,5-DECADIEN-3-ONE

Schieberle & Buettner (2001)		0.000 1 - 0.001

(*R*)-γ-decalactone⇒(*R*)-4-HYDROXYDECANOIC ACID LACTONE
(*S*)-γ-decalactone⇒(*S*)-4-HYDROXYDECANOIC ACID LACTONE
(*R*)-δ-decalactone⇒(*R*)-5-HYDROXYDECANOIC ACID LACTONE
(*S*)-δ-decalactone⇒(*S*)-5-HYDROXYDECANOIC ACID LACTONE

DECANAL (decyl aldehyde) *[112-31-2]*

Pliška & Janiček (1965)		3.8
Guadagni (1966)		0.001
Randebrock (1971)		0.000 25
Teranishi *et al.* (1974)		0.005
Cristoph (1983)	r	0.03 - 0.04
Randebrock (1986)		0.000 62
Lindell (1991)	d	0.051
Von Ranson & Belitz (1992a)	d	0.063
Von Ranson & Belitz (1992a)	r	0.094
Nagata (2003)	d	0.0026
Yang *et al.* (2008)		0.0026

DECANE *[124-18-5]*

Laffort & Dravnieks (1973)		11.3
Nagy (1991)	d	160
Nagata (2003)	d	3.6

2,5-DECANEDIONE *[51575-16-7]*

Guth & Grosch (1989)		0.65 - 1.3

3,5-DECANEDIONE *[5336-67-4]*

Guth & Grosch (1989)		0.0075 - 0.0149

4,5-DECANEDIONE *[122404-66-4]*

Guth & Grosch (1989)		> 17.0

DECANOIC ACID (capric acid) *[334-48-5]*

Passy (1892b,1892c)	d	0.05

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Backman (1917)	r	0.08 - 0.09
1-DECANOL (decyl alcohol) *[112-30-1]*		
Backman (1917)	r	0.10 - 0.17
Mullins (1955)	r	19.4
Pliška (1962)		2.6
Punter (1983)	d	0.44
Cristoph (1983)	r	0.07 - 0.09
Nagata (2003)	d	0.0050
2-DECANONE (methyl octyl ketone) *[693-54-9]*		
Teranishi *et al.* (1974)		0.060
Hall & Andersson (1983)	d	0.11
(*E*)-2-DECENAL *[3913-81-3]*		
Teranishi *et al.* (1974)		0.0027
Hall & Andersson (1983)	d	0.0049
Von Ranson & Belitz (1992a)	d	0.077
Von Ranson & Belitz (1992a)	r	0.077
Yang *et al.* (2008)		0.0027
(*Z*)-2-DECENAL *[2497-25-8]*		
Von Ranson & Belitz (1992a)	d	0.0022
Von Ranson & Belitz (1992a)	r	0.0031
(*E*)-3-DECENAL		
Von Ranson & Belitz (1992a)	d	0.025
Von Ranson & Belitz (1992a)	r	0.16
(*Z*)-3-DECENAL *[69891-94-7]*		
Von Ranson & Belitz (1992a)	d	0.0062
Von Ranson & Belitz (1992a)	r	0.025
(*E*)-4-DECENAL *[65405-70-1]*		
Von Ranson & Belitz (1992a)	d	0.025
Von Ranson & Belitz (1992a)	r	0.071
(*Z*)-4-DECENAL *[21662-09-9]*		
Von Ranson & Belitz (1992a)	d	0.0031
Von Ranson & Belitz (1992a)	r	0.022
(*E*)-5-DECENAL *[60671-72-9]*		
Von Ranson & Belitz (1992a)	d	0.034
Von Ranson & Belitz (1992a)	r	0.068
(*Z*)-5-DECENAL *[21662-11-3]*		
Von Ranson & Belitz (1992a)	d	0.056
Von Ranson & Belitz (1992a)	r	0.10
(*E*)-6-DECENAL *[147159-48-6]*		
Jung *et al.* (1992)		0.003 - 0.005
Von Ranson & Belitz (1992a)	d	0.0019
Von Ranson & Belitz (1992a)	r	0.0019
(*Z*)-6-DECENAL *[105683-99-6]*		
Jung *et al.* (1992)		0.000 02 - 0.000 04

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(*E*)-7-DECENAL

Von Ranson & Belitz (1992a)	d	0.019
Von Ranson & Belitz (1992a)	r	0.022

(*Z*)-7-DECENAL *[21661-97-2]*

Von Ranson & Belitz (1992a)	d	0.0022
Von Ranson & Belitz (1992a)	r	0.0025

9-DECENAL *[39770-05-3]*

Von Ranson & Belitz (1992a)	d	0.0022
Von Ranson & Belitz (1992a)	r	0.0093

1-DECENE *[872-05-9]*

Koszinowski & Piringer (1983)		37

1-DECEN-3-OL *[51100-54-0]*

Koszinowski & Piringer (1983)		0.37
Schieberle & Buettner (2001)		0.001 - 0.01

1-DECEN-3-ONE *[56606-79-2]*

Koszinowski & Piringer (1983)		0.001
Schieberle & Buettner (2001)		ca. 0.000 1

DECYL ACETATE *[112-17-4]*

Laska & Hudson (1991)	d	47.6
Cometto-Muñiz & Cain (1991); Cometto-Muñiz (1993)	d	2.6

decyl alcohol⇒1-DECANOL
decyl aldehyde⇒DECANAL
decyl chloride⇒1-CHLORODECANE
(-)-dehydrogeosmin⇒(-)-(4*R*,4a*R*,8a*R*)-1,2,3,4,4a,5,8,8a-OCTAHYDRO-4,8a-DIMETHYLNAPHTHALEN-4a-OL
(+)-dehydrogeosmin⇒(+)-(4*S*,4a*S*,8a*S*)-1,2,3,4,4a,5,8,8a-OCTAHYDRO-4,8a-DIMETHYLNAPHTHALEN-4a-OL
(+)-(6*R*)-dehydromintlactone⇒(6*R*)-5,6-DIHYDRO-3,6-DIMETHYL-2(4*H*)-BENZOFURANONE

(-)-(*R*)-8,9-DEHYDROTHEASPIRONE

Serra *et al.* (2007)	0.0137

(+)-(*S*)-8,9-DEHYDROTHEASPIRONE

Serra *et al.* (2007)	0.0098

diacetone alcohol⇒4-HYDROXY-4-METHYL-2-PENTANONE
diacetyl⇒2,3-BUTANEDIONE
diallylamine⇒DI-2-PROPENYLAMINE
diallyl disulphide⇒2-PROPENYLDITHIO-2-PROPENE
dially sulphide⇒2-PROPENYLTHIO-2-PROPENE
diallyl tetrasulphide⇒2-PROPENYLTETRATHIO-2-PROPENE
1,2-diaminopropane⇒1,2-PROPANEDIAMINE
diamyl sulphide⇒PENTYLTHIOPENTANE
dibenzyl sulphide⇒BENZYLTHIOTOLUENE

DIBORANE *[19287-45-7]*

Krackow (1953)	2 - 4

1,2-DIBROMOBENZENE *[583-53-9]*

Backman (1917)	r	0.024 - 0.033

2,6-DIBROMO-3-CHLOROANISOLE

Diaz *et al.* (2005)	0.0018

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2,6-DIBROMO-4-CHLOROANISOLE

Diaz *et al.* (2005)		0.0018

2,3-DIBROMO-1-CHLOROPROPANE *[96-12-8]*

Torkelson & Rowe (1981)		0.1 - 0.3

1,2-DIBROMOETHANE (ethylene bromide, ethylene dibromide) *[106-93-4]*

Olmstead (1972)	r	< 77

DIBROMOIODOMETHANE *[593-94-2]*

Cancho *et al.* (2001)		0.8

DIBUTYLAMINE *[111-92-2]*

Hellman & Small (1973,1974)	d	0.42
Hellman & Small (1973,1974)	r	1.4
Laing *et al.* (1978)	r	2.76
Bahnmüller (1984)		0.44 - 4.069

DIBUTYL PHTHALATE *[84-74-2]*

Menshikova (1972)		0.26

dibutyl sulphide⇒BUTYLTHIOBUTANE
di-*tert*-butyl sulphide⇒*tert*-BUTYLTHIO-2-METHYLPROPANE

DICHLOROACETIC ACID *[79-43-6]*

Backman (1917)	r	0.23

3,4-DICHLOROANILINE *[95-76-1]*

Andreescheva (1968)		0.047
Punter (1983)	d	0.29

2,3-DICHLOROANISOLE *[1984-59-4]*

Strube & Buettner (2010)		0.0168

2,4-DICHLOROANISOLE *[553-82-2]*

Nijssen (1988)	d	0.0066
Strube & Buettner (2010)		0.01853

2,5-DICHLOROANISOLE *[1984-58-3]*

Strube & Buettner (2010)		0.0444

2,6-DICHLOROANISOLE *[1984-65-2]*

Sävenhed *et al.* (1985)	d	0.001 - 0.004
Strube & Buettner (2010)		0.000 6

3,4-DICHLOROANISOLE *[36404-30-5]*

Strube & Buettner (2010)		0.01763

3,5-DICHLOROANISOLE *[33719-74-3]*

Strube & Buettner (2010)		0.46738

1,2-DICHLOROBENZENE *[95-50-1]*

Backman (1917)	r	0.12
Hollingsworth *et al.* (1958)		< 300
Punter (1983)	d	4.2

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1,4-DICHLOROBENZENE *[106-46-7]*

Hollingsworth *et al.* (1956)		< 90
Punter (1983)	d	0.73

2,4-dichloro-6-bromoanisole⇒6-BROMO-2,4-DICHLOROANISOLE
2,5-dichloro-6-bromoanisole⇒6-BROMO-2,5-DICHLOROANISOLE
2,6-dichloro-3-bromoanisole⇒3-BROMO-2,6-DICHLOROANISOLE
1,1'-dichlorodiethyl sulphide⇒1-CHLORO-1-(CHLOROETHYLTHIO)ETHANE

DICHLORODIFLUOROMETHANE *[75-71-8]*

Braker & Mossman (1980)		988,000

1,2-DICHLOROETHANE (ethylene chloride, ethylene dichloride) *[107-06-2]*

McCawley (1942)		1,200 - 4,000
Jones (1955c)	r	1,500
Borisova (1957)		17.5 - 23.2
Scherberger *et al.* (1958)	r	820
Irish (1962)		200
May (1966)	d	450
May (1966)	r	750
Dravnieks & O'Donnell (1971)		190
Hellman & Small (1974)	d	25
Hellman & Small (1974)	r	165
Kleinschmidt (1983)	r	350

1,1-DICHLOROETHENE (vinylidene chloride) *[75-35-4]*

Rylova (1953)		200
Janiček *et al.* (1960)		5,500
Irish (1962)		2,000 - 4,000

(*E*)-1,2-DICHLOROETHENE *[156-60-5]*

Lehmann & Schmidt-Kehl (1936)		1,100

DICHLOROETHYLARSINE (ethyl dichloroarsine) *[598-14-1]*

Flury (1921)		0.17 - 0.85
Prentiss (1937)		1

β,β'-dichlorethyl sulphide⇒2-CHLORO-1-(2-CHLOROETHYLTHIO)ETHANE
dichloroisopropylether⇒1-CHLORO-2-(1-CHLOROISOPROPOXY)PROPANE

DICHLOROMETHANE (methylene chloride) *[75-09-2]*

Lehmann & Schmidt-Kehl (1936)		1,100
Scherberger *et al.* (1958)	r	1,530
May (1966)	d	550
May (1966)	r	790
Leonardos *et al.* (1969)	r	730
Basmadshijewa *et al.* (1970)	d	4.1 - 33.2
Don (1986)	d	640
Nagata (2003)	d	560

DICHLOROMETHYLARSINE (methyl dichloroarsine) *[593-89-5]*

Prentiss (1937)		0.8

2,4-DICHLOROPHENOL *[120-83-2]*

Punter (1983)	d	0.000 27

1,2-DICHLOROPROPANE (propylene chloride, propylene dichloride) *[78-87-5]*

Hellman & Small (1974)	d	1.2

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Hellman & Small (1974)	r	2.4
Nagy (1991)	d	40

1,3-DICHLOROPROPENE *[542-75-6]*

Torkelson & Oyen (1977)		< 4.5

dicyclopentadiene⇒3a,4,7,7a-TETRAHYDRO-4,7-METHANOINDENE
α-dicyclopentadiene⇒3a,4,7,7a-TETRAHYDRO-4,7-METHANOINDENE

DICYCLOPROPYL(ETHYL)(1-HYDROXY-1-METHYLETHYL)SILANE

Sunder kötter *et al.* (2010)		0.0021

DICYCLOPROPYL(1-ETHYL-1-HYDROXYPROPYL)METHYLSILANE

Sunderkötter *et al.* (2010)		0.040

DICYCLOPROPYL(1-HYDROXY-1-METHYLETHYL)ISOPROPYLSILANE

Sunderkötter *et al.* (2010)		0.000 40

DICYCLOPROPYL(1-HYDROXY-1-METHYLETHYL)METHYLSILANE

Sunderkötter *et al.* (2010)		0.0016

rac-DICYCLOPROPYL(-HYDROXY-1-METHYLPROPYL)METHYLSILANE

Sunderkötter *et al.* (2010)		0.021

DICYCLOPROPYL(METHYL)SILANOL

Sunderkötter *et al.* (2010)		0.125

(-)-(*R*)-3,4-didehydro-7,8-dihydro-γ-ionone⇒(*R*)-(-)-4-(6,6-DIMETHYL-2-METHYLENE-3-CYCLOHEXEN-1-YL)-2-BUTANONE
(+)-(*S*)-3,4-didehydro-7,8-dihydro-γ-ionone⇒(*S*)-(+)-4-(6,6-DIMETHYL-2-METHYLENE-3-CYCLOHEXEN-1-YL)-2-BUTANONE
(-)-(*R*)-3,4-didehydro-γ-ionone⇒(*R*)-(-)-(3*E*)-4-(2,2-DIMETHYL-6-METHYLENE-4-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(+)-(*S*)-3,4-didehydro-γ-ionone⇒(*S*)-(+)-(3*E*)-4-(2,2-DIMETHYL-6-METHYLENE-4-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE

DIETHANOLAMINE *[111-42-2]*

England *et al.* (1978)	r	1.2

DIETHYLAMINE *[109-89-7]*

Geier (1936)	d	0.01 - 0.1
Geier (1936)	r	2.25 - 5
Kosiborod (1968)		0.084
Hellman & Small (1973)	d	0.42
Hellman & Small (1973)	r	1.5
Hellman & Small (1974)	d	0.06
Hellman & Small (1974)	r	0.18
Cormack *et al.* (1974)		0.09
Laing *et al.* (1978)	r	42.9
Tkachev (1978)		0.044 - 0.558
Anon. (1980)	d	0.09
Anon. (1980)	r	0.9
Laing (1982)	d	4.0
Nagata (2003)	d	0.14

2-DIETHYLAMINOETHANOL (N,N-diethylethanolamine) *[100-37-8]*

Hellman & Small (1973,1974)	d	0.05
Hellman & Small (1973,1974)	r	0.19

1,2-DIETHYLBENZENE (*o*-diethylbenzene) *[135-01-3]*

Nagata (2003)	d	0.052

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1,3-DIETHYLBENZENE (*m*-diethylbenzene) *[141-93-5]*
Nagata (2003) d 0.39

1,4-DIETHYLBENZENE (*p*-diethylbenzene) *[105-05-5]*
Nagata (2003) d 0.0021

m-diethylbenzene⇒1,3-DIETHYLBENZENE
o-diethylbenzene⇒1,2-DIETHYLBENZENE
p-diethylbenzene⇒1,4-DIETHYLBENZENE

2,3-DIETHYL-5,6-DIHYDRO-5-METHYLPYRAZINE (2,3-diethyl-5-methyl-5,6-dihydropyrazine)
Kurniadi *et al.* (2003) 0.000 004
Kurniadi (2003) r 0.000 08

DIETHYL 2,3-DIHYDROXYBUTANEDIOATE (diethyl tartrate) *[13811-71-7]*
Backman (1917) r 0.078

diethyl disulphide⇒ETHYLDITHIOETHANE
diethylene glycol ethyl ether⇒3,6-DIOXA-1-OCTANOL
N,N-diethylethanolamine⇒2-DIETHYLAMINOETHANOL
diethyl ether⇒ETHOXYETHANE
diethyl ketone⇒3-PENTANONE

2,3-DIETHYL-5-METHYLPYRAZINE *[18138-04-0]*
Fors (1984); Fors & Olofsson (1985) d 0.023
Cerny (1992) 0.000 009 - 0.000 018
Czerny *et al.* (1996); Grosch *et al.* (1996);
Wagner *et al.* (1999) 0.000 01 - 0.000 014

2,3-diethyl-5-methyl-5,6-dihydropyrazine⇒2,3-DIETHYL-5,6-DIHYDRO-5-METHYLPYRAZINE

DIETHYL OXALATE *[95-92-1]*
Backman (1917) r 0.5

DIETHYL PHTHALATE *[84-66-2]*
Wünsche *et al.* (1995) d 0.33 - 3.3

2,3-DIETHYLPYRAZINE *[15707-24-1]*
Fors (1984); Fors & Olofsson (1985) d 0.050
Wagner *et al.* (1999) 0.0066

2,5-DIETHYLPYRAZINE *[13238-84-1]*
Wagner *et al.* (1999) 0.0017 - 0.074

2,6-DIETHYLPYRAZINE *[13067-27-1]*
Wagner *et al.* (1999) 0.0017 - 0.074

DIETHYL SELENIDE *[627-53-2]*
Katz & Talbert (1930) 0.0067

DIETHYL SUCCINATE *[123-25-1]*
Tempelaar (1913) d 0.79

diethyl sulphide⇒ETHYLTHIOETHANE
diethyl tartrate⇒DIETHYL 2,3-DIHYDROXYBUTANEDIOATE

2,3-DIETHYLTETRAHYDRO-5-METHYLPYRAZINE
Kurniadi *et al.* (2003) 0.0043

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Kurniadi (2003) r 0.086

O,S-DIETHYL THIOCARBONATE
Breme *et al.* (2009) 0.002

diethyl trisulphide⇒ETHYLTRITHIOETHANE
difurfuryl disulphide⇒FURFURYLDITHIOFURYLMETHANE
3,4-dihydro-2H-azole⇒1-PYRROLINE

(6*R*)-5,6-DIHYDRO-3,6-DIMETHYL-2(4*H*)-BENZOFURANONE ((+)-(6*R*)-dehydromintlactone) *[75640-26-5]*
Güntert *et al.* (2001) 0.0031
Buhr (2006) 0.001

5,6-DIHYDRO-2,5-DIMETHYL-3-PENTYLPYRAZINE (3-pentyl-2,5-dimethyl-5,6-dihydropyrazine)
Kurniadi *et al.* (2003) > 2
Kurniadi (2003) r > 40

5,6-DIHYDRO-3,5-DIMETHYL-2-PENTYLPYRAZINE (2-pentyl-3,5-dimethyl-5,6-dihydropyrazine)
Kurniadi *et al.* (2003) 0.0032
Kurniadi (2003) r 0.064

5,6-DIHYDRO-2,5-DIMETHYL-3-PROPYLPYRAZINE (3-propyl-2,5-dimethyl-5,6-dihydropyrazine)
Kurniadi *et al.* (2003) > 2
Kurniadi (2003) r > 40

5,6-DIHYDRO-3,5-DIMETHYL-2-PENTYLPYRAZINE (2-propyl-3,5-dimethyl-5,6-dihydropyrazine)
Kurniadi *et al.* (2003) 0.0025
Kurniadi (2003) r 0.05

2,5-DIHYDROFURAN *[1191-99-7]*
Nagata (2003) d 0.27

(*R*)-(+)-dihydro-α-ionone⇒(1*R*)-(+)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTANONE
(*S*)-(-)-dihydro-α-ionone⇒(1*S*)-(-)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTANONE
dihydro-β-ionone⇒4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-2-BUTANONE
(*R*)-(-)-dihydro-γ-ionone⇒(1*R*)-(-)-4-(2,2-DIMETHYL-6-METHYLENE-1-CYCLOHEXYL)-2-BUTANONE
(*S*)-(+)-dihydro-γ-ionone⇒(1*S*)-(+)-4-(2,2-DIMETHYL-6-METHYLENE-1-CYCLOHEXYL)-2-BUTANONE

6,7-DIHYDRO-1,1,2,3,3-PENTAMETHYL-5-(4*H*)-INDANONE
Bachman n & Pesaro (1990);
Bachmann *et al.* (1992) 0.0025

5,6-dihydro-2,3,5-trimethylpyrazine⇒2,3-DIHYDRO-2,5,6-TRIMETHYLPYRAZINE

2,3-DIHYDRO-2,5,6-TRIMETHYLPYRAZINE (5,6-dihydro-2,3,5-trimethylpyrazine) *[65826-70-2]*
Kurniadi (2003) r 0.2

1,2-DIHYDROXYETHANE DIACETATE (glycol diacetate) *[111-55-7]*
Hellman & Small (1974) d 0.6
Hellman & Small (1974) r 1.8

2,4-dihydroxy-3-methyl-2-hexenoic acid lactone⇒5-ETHYL-3-HYDROXY-4-METHYL-2(5H)-FURANONE
α,2-dihydroxytoluene⇒SALICYL ALCOHOL
diisoamyl sulphide⇒3-METHYLBUTYLTHIO-3-METHYLBUTANE
diisobutyl carbinol⇒2,6-DIMETHYL-4-HEPTANOL
diisobutyl ketone⇒2,6-DIMETHYL-4-HEPTANONE

1,6-DIISOCYANATOHEXANE *[822-06-0]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Kimmerle (1971) 0.035 - 0.07

2,4-DIISOCYANATO-1-METHYLBENZENE (4-methyl-1,3-phenylenediisocyanate, tolylenediisocyanate) *[584-84-9]*

Zapp (1957)		2.8
Henschler *et al.* (1962)		0.14 - 0.35
Chizhikov (1963)		0.2
Leonardos *et al.* (1969)	r	15

2,6-DIISOCYANATO-1-METHYLBENZENE (2-methyl-1,3-phenylenediisocyanate) *[91-08-7]*

Henschler *et al.* (1962)		0.14 - 0.35

DIISOPROPANOLAMINE *[110-97-4]*

England *et al.* (1978)	r	0.07

diisopropyl⇒2,3-DIMETHYLBUTANE

DIISOPROPYLAMINE *[108-18-9]*

Hellman & Small (1974)	d	0.56
Hellman & Small (1974)	r	1.6
England *et al.* (1978)	r	17.4

diisopropyl sulphide⇒2-(ISOPROPYLTHIO)PROPANE
diketene⇒3-HYDROXYBUT-3-ENOIC ACID LACTONE
dill ether⇒(3*S*,3aS,7aR)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN
dimethoate⇒O,O-DIMETHYL S(N-METHYLCARBAMOYLMETHYL)PHOSPHORODITHIOATE

(2,2-DIMETHOXYETHYL)BENZENE (2-phenylacetaldehyde dimethylacetal) *[101-48-4]*

Appell (1969)		0.000 6

DI-2-METHOXYPHENYL CARBONATE (duotal, guaiacol carbonate) *[553-17-3]*

Backman (1917)	r	0.0011 - 0.0012

N,N-DIMETHYLACETAMIDE *[127-19-5]*

Leonardos *et al.* (1969)	r	170

DIMETHYLAMINE *[124-40-3]*

Geier (1936)	d	0.65 - 1.0
Geier (1936)	r	2.2 - 3.0
Taylor & Bodurtha (1960)		1.1
Leonardos *et al.* (1969)	r	0.085
Stephens (1971)		0.16
Prusakov *et al.* (1976)		0.01 - 0.03
Tkachev (1978)		0.030
Anon. (1980)	d	0.0014
Anon. (1980)	r	0.023
Nagata (2003)	d	0.059
Van Thriel *et al.* (2006)	d	7.75

2-DIMETHYLAMINOETHANOL (N,N-dimethylethanolamine) *[108-01-0]*

Hellman & Small (1974)	d	0.05
Hellman & Small (1974)	r	0.16
England *et al.* (1978)	r	2.0

5β,10-DIMETHYL-*des*-A-18-*nor*-ANDROSTAN-13β-OL (limdrostanol)

Kraft & Popaj (2004)	d	0.090

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

5β,10-DIMETHYL-*des*-A-18-*nor*-ANDROSTAN-14α-OL

Kraft & Popaj (2004)	d	0.046

2,5-DIMETHYLANILINE *[95-78-3]*

Backman (1917)	r	0.25 - 0.30

N,N-DIMETHYLANILINE *[121-69-7]*

Backman (1917)	r	0.8 - 1.0
Geier (1936)	d	0.005 - 0.1
Geier (1936)	r	0.05 - 0.25
Deadman & Prigg (1959)	d	0.012

1,2-DIMETHYLBENZENE (*o*-xylene) *[95-47-6]*

Backman (1917)	r	1.0 - 1.2
Backman (1918)		0.8
Stuiver (1958)	d	2.1
Nauš (1962)	d	1
Köster (1968a,1971)	d	11
Anon. (1980)	d	0.77
Anon. (1980)	r	3.1
Punter (1983)	d	23.6
Nagata (2003)	d	1.6

1,3-DIMETHYLBENZENE (*m*-xylene) *[108-38-3]*

Backman (1917)	r	1.1 - 1.3
Stuiver (1958)	d	0.35
Gusev (1965)		0.6 - 1.9
Dravnieks & O'Donnell (1971)		1.3
Köster (1965,1968a,b,1971)	d	0.6 - 86
Anon. (1980)	d	0.52
Anon. (1980)	r	2.4
Punter (1983)	d	1.5 - 4.9
Don (1986); Hoshika *et al.* (1993)	d	0.52 - 0.54
Hoshika *et al.* (1993)	d	0.052
Nagata (2003)	d	0.18

1,4-DIMETHYLBENZENE (*p*-xylene) *[106-42-3]*

Backman (1917)	r	1.4 - 1.5
Stuiver (1958)	d	0.6
Leonardos *et al.* (1969)	r	2
Köster (1968a,1971)	d	8
Knuth (1973)		0.8
Anon. (1980)	d	0.52
Anon. (1980)	r	2.2
Punter (1983)	d	9.1
Nagata (2003)	d	0.25

3,6-DIMETHYLBENZO-[b]-FURAN-2(3*H*)-ONE (furamintone)

Güntert *et al.* (2001)	0.000 8

(3*R*,6)-DIMETHYLBENZO-[b]-FURAN-2(3*H*)-ONE ((*R*)-furamintone)

Güntert *et al.* (2001)	0.000 8

(3*S*,6)-DIMETHYLBENZO-[b]-FURAN-2(3*H*)-ONE ((*S*)-furamintone)

Güntert *et al.* (2001)	0.000 8

(±)-(1*R**,9'*R**,10'*S**)-1-(9',10'-DIMETHYLBICYCLO[6.4.0]DODEC-1'(8)-EN-10'-YL)ETHAN-1-OL

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Kraft & Gallo (2004) 0.305

(±)-(1*R**,9'*S**,10'*R**)-1-(9',10'-DIMETHYLBICYCLO[6.4.0]DODEC-1'(8)-EN-10'-YL)ETHAN-1-OL
Kraft & Gallo (2004) 0.0081

2,2-DIMETHYLBUTANE (neohexane) *[75-83-2]*
Nagata (2003) d 70

2,3-DIMETHYLBUTANE (diisopropyl) *[79-29-8]*
Nagata (2003) d 1.5

2,2-DIMETHYL-3-BUTANONE (methyl *tert*-butyl ketone, pinacolone) *[75-97-8]*
Backman (1917) r 0.65 - 0.95
Nagata (2003) d 0.18

5,5-DIMETHYL-1,3-CYCLOHEXANEDIONE (methone) *[126-81-8]*
Stone *et al.* (1967); Stone (1979) d 42.9

2,4-DIMETHYL-3-CYCLOHEXENECARBOXALDEHYDE *[68039-49-6]*
Wünsche *et al.* (1995) d 0.000 76

1''-(3''',3'''-DIMETHYLCYCLOHEX-1'''-ENYL)ETHOXYCARBONYLMETHYL PROPANOATE
Kraft & Eichenberger (2004) 0.000 6

1''-(5''',5'''-DIMETHYLCYCLOHEX-1'''-ENYL)ETHOXYCARBONYLMETHYL PROPANOATE
Kraft & Eichenberger (2004) 0.011

2'-[1''-(3''',3'''-DIMETHYLCYCLOHEX-1'''-ENYL)ETHOXY]-2'-METHYLPROPYL PROPANOATE
Kraft & Eichenberger (2004) 0.000 2

2'-[1''-(5''',5'''-DIMETHYLCYCLOHEX-1'''-ENYL)ETHOXY]-2'-METHYLPROPYL PROPANOATE
Kraft & Eichenberger (2004) 0.000 9

3,4-dimethylcyclopentenolone⇒2-HYDROXY-3,4-DIMETHYL-2-CYCLOPENTEN-1-ONE
(8*S*,9*S*,10*R*,13*R*,14*S*)-10,13-dimethyl-1,2,6,7,8,9,11,12,14,15-decahydrocyclopenta[a]phenanthren-3-one⇒4,16-ANDROSTADIEN-3-ONE

trans-1,10-DIMETHYL-*trans*-9-DECALOL (geosmin) *[19700-21-1]*
Sävenhed *et al.* (1985) d 0.000 05 - 0.000 2
Khiari *et al.* (1992) < 0.000 5
Nagata (2003) d 0.000 049
Ömür-Özbek *et al.* (2007) 0.000 21

1,1-DIMETHYL-2,3-DIHYDRO-1*H*-5,9-DIOXACYCLOHEPTA[*f*]INDEN-7-ONE
Kraft & Eichenberger (2003) 0.000 55

(*E/Z*)-1,2-DIMETHYL-2,3-DIHYDRO-1*H*-5,9-DIOXACYCLOHEPTA[*f*]INDEN-7-ONE
Kraft & Eichenberger (2003) 0.00116

dimethyl disulphide⇒METHYLDITHIOMETHANE
N,N-dimethylethanolamine⇒2-DIMETHYLAMINOETHANOL
dimethyl ether⇒METHOXYMETHANE

(*S*)-4-(6,6-DIMETHYL-2-ETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
Luparia *et al.* (2008) 0.000 085

N,N-DIMETHYLFORMAMIDE *[68-12-2]*
Odoshashvili (1962) 0.14

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Leonardos *et al.* (1969)	r	300

2,5-DIMETHYLFURAN *[625-86-5]*

Artho & Koch (1973)		100 - 1,000

2,5-DIMETHYL-3-FURANTHIOL *[55764-23-3]*

Gasser & Grosch (1990a)		0.000 0035 - 0.000 014

3-(2,5-DIMETHYL)FURYLDITHIO-3-(2,5-DIMETHYL)FURAN (3,3'-dithiobis(2,5-dimethylfuran)) *[28588-73-0]*

Etzweiler (1993)		0.000 05 - 0.000 16

2,6-DIMETHYL-4-HEPTANOL (diisobutyl carbinol) *[108-82-7]*

Hellman & Small (1974)	d	0.19
Hellman & Small (1974)	r	0.28

2,6-DIMETHYL-4-HEPTANONE (diisobutyl ketone) *[108-83-8]*

Hellman & Small (1973,1974)	d	< 0.6
Hellman & Small (1973,1974)	r	1.8
Nagy (1991)	d	9.3

(3*R*,3a*R*,7a*S*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN ((3*S*,4*R*,8*R*)-3,9-epoxy-1-*p*-menthene)

Blank *et al.* (1989)		0.000 8 - 0.0016

(3*S*,3a*S*,7a*R*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN ((3*R*,4*S*,8S)-3,9-epoxy-1-*p*-menthene, dill ether) *[70786-44-6]*

Blank *et al.* (1989); Blank & Grosch (1991a)		0.020 - 0.040

(3*R*,3a*S*,7a*R*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN ((3*R*,4*S*,8*R*)-3,9-epoxy-1-*p*-menthene)

Blank *et al.* (1989)		0.020 - 0.040

(1*S**)-1-[(1*S**)-1,5-DIMETHYL-5-HEXENYL]-4-METHYL-3-CYCLOHEXENYL-1-OL

Braun *et al.* (2003)	d	0.081

1,1-DIMETHYLHYDRAZINE *[57-14-7]*

Jacobson *et al.* (1955)		15 - 35
Rumsey & Cesta (1970)		< 0.75

O,O-DIMETHYL (1-HYDROXY-2,2,2-TRICHLOROETHYL)PHOSPHONATE (trichlorfon) *[52-68-6]*

Kaloyanova *et al.* (1967)		28
Tabokova (1968,1969)		0.2

O,O-DIMETHYL S-[2-(1-METHYLCARBAMOYLETHYLTHIO)ETHYL] PHOSPHORODITHIOATE (kilwal, kilval) *[2275-23-2]*

Ubaidullaev & Madzhidov (1978)		0.026

O,O-DIMETHYL S(N-METHYLCARBAMOYLMETHYL)PHOSPHORODITHIOATE (dimethoate) *[60-51-5]*

Kaloyanova *et al.* (1968)		0.01

2,5-DIMETHYL-3-(METHYLDITHIO)FURAN (methyl 2,5-dimethylfur-3-yl disulphide) *[61197-06-6]*

Etzweiler (1993)		0.000 57

2,2-DIMETHYL-3-METHYLENEBICYCLO[2.2.1]HEPTANE (camphene) *[79-92-5]*

Cristoph (1983)	r	26 - 30

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE (β-pinene) *[127-91-3]*

Cristoph (1983)	r	35 - 38
Lindell (1991)	d	8.9
Cometto-Muñiz *et al.* (1998b); Cometto-Muñiz (1999)	d	65
Nagata (2003)	d	0.18

(1*S*)-(-)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE ((-)-β-pinene) *[18172-67-3]*

Jagella & Grosch (1998)		> 2

(*R*)-(-)-4-(6,6-DIMETHYL-2-METHYLENE-3-CYCLOHEXEN-1-YL)-2-BUTANONE ((-)-(*R*)-3,4-didehydro-7,8-dihydro-γ-ionone)

Serra *et al.* (2006)		0.0994

(*S*)-(+)-4-(6,6-DIMETHYL-2-METHYLENE-3-CYCLOHEXEN-1-YL)-2-BUTANONE ((+)-(*S*)-3,4-didehydro-7,8-dihydro-γ-ionone)

Serra *et al.* (2006)		0.092

(*R*)-(-)-(3*E*)-4-(2,2-DIMETHYL-6-METHYLENE-4-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE ((-)-(*R*)-3,4-didehydro-γ-ionone)

Serra *et al.* (2006)		0.000 25

(*S*)-(+)-(3*E*)-4-(2,2-DIMETHYL-6-METHYLENE-4-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE ((+)-(*S*)-3,4-didehydro-γ-ionone)

Serra *et al.* (2006)		0.000 08

(1*R*)-(-)-4-(2,2-DIMETHYL-6-METHYLENE-1-CYCLOHEXYL)-2-BUTANONE ((*R*)-(-)-dihydro-γ-ionone)

Brenna *et al.* (2002)		0.0062

(1*S*)-(+)-4-(2,2-DIMETHYL-6-METHYLENE-1-CYCLOHEXYL)-2-BUTANONE ((*S*)-(+)-dihydro-γ-ionone)

Brenna *et al.* (2002)		0.039

(1*R*)-(-)-4-(2,2-DIMETHYL-6-METHYLENE-1-CYCLOHEXYL)-3-BUTEN-2-ONE ((*R*)-(-)-γ-ionone)

Fuganti *et al.* (2000); Brenna *et al.* (2002)		0.0108 - 0.011

(1*S*)-(+)-4-(2,2-DIMETHYL-6-METHYLENE-1-CYCLOHEXYL)-3-BUTEN-2-ONE ((*S*)-(+)-γ-ionone)

Fuganti *et al.* (2000); Brenna *et al.* (2002)		0.000 07

2,5-DIMETHYL-3-(METHYLTHIO)FURAN *[63359-63-7]*

Etzweiler (1993)		0.0107 - 0.0245

O,O-DIMETHYL O-(p-NITROPHENYL)PHOSPHOROTHIOATE (methyl parathion) *[298-00-0]*

Akhmedov (1968)		0.0125

3,7-DIMETHYL-2,6-OCTADIENAL (citral) *[5392-40-5]*

Passy (1892a,1892b)	d	0.1 - 0.5
Tempelaar (1913); Zwaardemaker (1927)	d	0.062 - 0.1
Backman (1917)	r	0.06 - 0.09
Ohma (1922)	d	0.13
Schneider & Wolf (1955)		0.027
Schneider *et al.* (1958)		0.12
Appell (1969)		0.000 5
Köster (1971)	d	0.17 - 0.19
Etzweiler *et al.* (1980)		0.020
Randebrock (1986)		0.000 15

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3,7-DIMETHYL-(*E*)-2,6-OCTADIENAL (citral a, geranial) *[141-27-5]*

Bocca & Battiston (1964)	d	0.04
Cristoph (1983)	r	0.055 - 0.07
Schieberle & Grosch (1988)		0.012

3,7-DIMETHYL-(*Z*)-2,6-OCTADIENAL (citral b, neral) *[106-26-3]*

Cristoph (1983)	r	0.03 - 0.05
Schieberle & Grosch (1988)		0.0088

3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-OL (geraniol) *[106-24-1]*

Cristoph (1983)	r	0.15 - 0.20
Randebrock (1986)		0.000 012
De Wijk (1989)		1.29
Wünsche *et al.* (1995)	d	0.000 71
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	0.6

3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-OL (nerol) *[106-25-2]*

Schwarz (1955)		0.0015
Cristoph (1983)	r	3.2 - 3.7
Ferreira *et al.* (1998)		0.049

3,7-DIMETHYL-1,6-OCTADIEN-3-OL (linalool) *[78-70-6]*

Fuller *et al.* (1964)	r	0.11
Appell (1969)		0.0014
Etzweiler *et al.* (1980)		0.0033
Schieberle & Grosch (1988);		
Blank *et al.* (1989)		0.000 4 - 0.0012
Laska & Hudson (1991)	d	0.044 - 0.090
Müller (1992)		0.000 5
Wünsche *et al.* (1995)	d	0.0013
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	6.0
Ferreira *et al.* (1998)		0.0044
Güntert *et al.* (2001)		0.002
Tacke & Metz (2008)		0.0024

(*S*)-(+)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL ((*d*)-linalool, *S*-(+)-linalool) *[126-90-9]*

Cristoph (1983)	r	0.035 - 0.040
Jagella & Grosch (1998)		0.0029
Maas *et al.* (1993)		0.14

(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL ((*l*)-linalool, *R*-(-)-linalool) *[126-91-0]*

Cristoph (1983)	r	0.009 - 0.011
Jagella & Grosch (1998)		0.000 036
Maas *et al.* (1993)		0.006 - 0.01

3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-YL ACETATE (geranyl acetate) *[105-87-3]*

Cristoph (1983)	r	12.0 - 18.0

3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-YL ACETATE (neryl acetate) *[141-12-8]*

Cristoph (1983)	r	20.0 - 25.0

3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE (linalyl acetate) *[115-95-7]*

Tempelaar (1913)	d	0.093
Van Anrooij (1931)	d	0.04
Cristoph (1983)	r	4.0 - 6.0

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE ((*R*)-(-)-linayl acetate) *[16509-46-9]*
Maas *et al.* (1993) > 3

(*S*)-(+)-3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE ((*S*)-(+)-linayl acetate)
Maas *et al.* (1993) > 3

3,7-DIMETYL-2,4-OCTANEDIONE
Guth & Grosch (1989) 0.0010 - 0.0020

3,7-DIMETHYL-1-OCTANOL (tetrahydrogeraniol) *[106-21-8]*
Randebrock (1986) 0.000 79

(*E*)-3,7-DIMETHYL-1,3,6-OCTATRIENE ((*E*)-β-ocimene) *[3779-61-1]*
Güntert *et al.* (2001) 0.0187

(*Z*)-3,7-DIMETHYL-1,3,6-OCTATRIENE ((*Z*)-β-ocimene) *[3338-55-4]*
Güntert *et al.* (2001) 0.010

2,6-DIMETHYL-(*Z*,*E*)-2,5,7-OCTATRIENYL ACETATE ((*E*,*Z*)-8-ocimenyl acetate)
Güntert *et al.* (2001) 0.000 4

2,6-DIMETHYL-(*Z*,*Z*)-2,5,7-OCTATRIENYL ACETATE ((*Z*,*Z*)-8-ocimenyl acetate)
Güntert *et al.* (2001) 0.005

3,7-DIMETHYL-6-OCTENAL (citronellal) *[106-23-0]*
Fuller *et al.* (1964) r 0.0052
Appell (1969) 0.000 3
Cristoph (1983) r 0.05 - 0.065
Randebrock (1986) 0.000 44

3,7-DIMETHYL-6-OCTEN-1-OL (citronellol, β-citronellol) *[106-22-9]*
Tempelaar (1913); Zwaardemaker (1927) d 0.28 - 2.8
Appell (1969) 0.000 12
Cristoph (1983) r 0.8 - 1.0
Ferreira *et al.* (1998) 0.0465

(*R*)-(+)-3,7-DIMETHYL-6-OCTEN-1-OL ((*R*)-(+)-citronellol) *[1117-61-9]*
Maas *et al.* (1993) 0.04 - 0.05

(*S*)-(-)-3,7-DIMETHYL-6-OCTEN-1-OL ((*S*)-(-)-citronellol) *[7540-51-4]*
Maas *et al.* (1993) 0.04 - 0.05

3,7-DIMETHYL-6-OCTEN-1-YL ACETATE (citronellyl acetate) *[150-84-5]*
Cristoph (1983) r 1.0 - 1.1

4,8-DIMETHYL-2-OXABICYCLO[3.3.1]-7-NONEN-3-ONE
Buhr (2006) > 0.1

(3*R*,3a*R*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(3*R*,3a*R*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*R*,3a*R*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(-)-(3*R*,3a*R*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*R*,3a*R*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(+)-(3*R*,3a*R*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*R*,3a*R*,6*S*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(-)-(3*R*,3a*R*,6*S*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*R*,3a*S*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(+)-(3*R*,3a*S*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-

BENZO[b]FURANONE
(3*R*,3a*S*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(3*R*,3a*S*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*R*,3a*S*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(-)-(3*R*,3a*S*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*R*,3a*S*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(+)-(3*R*,3a*S*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*R*,3a*S*,6*S*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(3*R*,3a*S*,6*S*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*S*,3a*R*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(3*S*,3a*R*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*S*,3a*R*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(-)-(3*S*,3a*R*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*S*,3a*R*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(3*S*,3a*R*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*S*,3a*S*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(+)-(3*S*,3a*S*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*S*,3a*S*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(-)-(3*S*,3a*S*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*S*,3a*S*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone⇒(+)-(3*S*,3a*S*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE
(3*S*,3a*S*,6*S*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-hexahydro-2(3*H*)-benzofuranone⇒(3*S*,3a*S*,6*S*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE

2,2-DIMETHYLPENTANE *[590-35-2]*

Nagata (2003)	d	156

2,3-DIMETHYLPENTANE *[565-59-3]*

Nagata (2003)	d	18

2,4-DIMETHYLPENTANE *[108-08-7]*

Dravnieks (1974)	d	3,000
Dravnieks & Laffort (1972)		870
Nagata (2003)	d	3.9

3,5-DIMETHYL-2-PENTYLPYRAZINE *[50888-62-5]*

Wagner *et al.* (1999)		0.024

2,3-DIMETHYLPHENOL (2,3-xylenol) *[526-75-0]*

Stuiver (1958)	d	0.000 5

2,4-DIMETHYLPHENOL (2,4-xylenol) *[105-67-9]*

Backman (1917)	r	0.001
Stuiver (1958)	d	0.000 5 - 0.4

2,5-DIMETHYLPHENOL (2,5-xylenol) *[95-87-4]*

Backman (1917)	r	2.0 - 2.3
Stuiver (1958)	d	0.000 5

2,6-DIMETHYLPHENOL (2,6-xylenol) *[576-26-1]*

Stuiver (1958)	d	0.000 2

3,4-DIMETHYLPHENOL (3,4-xylenol) *[95-65-8]*

Stuiver (1958)	d	0.003

3,5-DIMETHYLPHENOL (3,5-xylenol) *[108-68-9]*

Stuiver (1958)	d	0.000 04

2,2-DIMETHYLPROPANAL (pivaldehyde) *[630-19-3]*

Von Ranson & Belitz (1992a)	d	6.9
Von Ranson & Belitz (1992a)	r	13

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

N,N-DIMETHYL-1,3-PROPANEDIAMINE *[111-33-1]*
Nagy (1991) d > 0.2

2,3-DIMETHYL-5-(*Z*-1-PROPENYL)PYRAZINE
Wagner *et al.* (1999) > 2

2,5-DIMETHYL-3-(*Z*-1-PROPENYL)PYRAZINE
Wagner *et al.* (1999) > 2

3,5-DIMETHYL-2-(*Z*-1-PROPENYL)PYRAZINE
Wagner *et al.* (1999) 0.000 057

2,3-DIMETHYL-5-(*E*-1-PROPENYL)PYRAZINE
Wagner *et al.* (1999) > 2

2,5-DIMETHYL-3-(*E*-1-PROPENYL)PYRAZINE
Wagner *et al.* (1999) > 2

3,5-DIMETHYL-2-(*E*-1-PROPENYL)PYRAZINE
Wagner *et al.* (1999) 0.0018

2,3-DIMETHYL-5-(2-PROPENYL)PYRAZINE
Wagner *et al.* (1999) > 2

2,5-DIMETHYL-3-(2-PROPENYL)PYRAZINE
Wagner *et al.* (1999) > 2

3,5-DIMETHYL-2-(2-PROPENYL)PYRAZINE
Wagner *et al.* (1999) 0.3

2,3-DIMETHYL-5-PROPYLPYRAZINE *[32262-98-9]*
Grosch *et al.* (1996); Wagner *et al.* (1999) > 1.2

2,5-DIMETHYL-3-PROPYLPYRAZINE *[18433-97-1]*
Grosch *et al.* (1996); Wagner *et al.* (1999) > 2

3,5-DIMETHYL-2-PROPYLPYRAZINE *[32350-16-6]*
Grosch *et al.* (1996); Wagner *et al.* (1999) 0.023 - 0.024

2,3-DIMETHYLPYRAZINE *[5910-89-4]*
Fors (1984); Fors & Olofsson (1985) d 0.90
Grosch *et al.* (1996); Wagner *et al.* (1999) 0.88

2,5-DIMETHYLPYRAZINE *[123-32-0]*
Fors (1984); Fors & Olofsson (1985) d 0.17
Grosch *et al.* (1996); Wagner *et al.* (1999) 1.82

2,6-DIMETHYLPYRAZINE *[108-50-9]*
Fors (1984); Fors & Olofsson (1985) d 0.25
Grosch *et al.* (1996); Wagner *et al.* (1999) 1.72

2,6-DIMETHYLPYRIDINE (2,6-lutidine) *[108-48-5]*
Moriguchi *et al.* (1983) d 0.003

3,4-DIMETHYLPYRIDINE (3,4-lutidine) *[583-58-4]*
Moriguchi *et al.* (1983) d 0.005

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3,5-DIMETHYLPYRIDINE (3,5-lutidine) *[591-22-0]*
Moriguchi *et al.* (1983) d 0.005

3,7-DIMETHYL-3-SILA-1,6-OCTADIEN-3-OL (sila-linalool)
Tacke & Metz (2008) 0.018

α,*p*-dimethylstyrene⇒ 1-ISOPROPENYL-4-METHYLBENZENE
dimethyl sulphide⇒METHYLTHIOMETHANE

3,6-DIMETHYL-2,4,5,7a-TETRAHYDRO-1-BENZOFURAN (linden ether, 3,9-epoxy-1,4(8)-*p*-menthadiene)
Blank *et al.* (1989) 0.001 - 0.002

3,6-DIMETHYL-4,5,6,7-TETRAHYDRO-1-BENZOFURAN (menthofuran,
(1*R*)-3,9-epoxy-3,8(9)-*p*-menthadiene) *[494-90-6]*
Blank *et al.* (1989) 0.000 1 - 0.000 2

(6*R*)-3,6-DIMETHYL-4,5,6,7-TETRAHYDRO-1-BENZOFURAN ((+)-(6*R*)-menthofuran) *[17957-94-7]*
Güntert *et al.* (2001) 0.000 4

(6*R*,7a*R*)-3,6-DIMETHYL-5,6,7,7a-TETRAHYDRO-2(4*H*)-BENZOFURANONE (mintlactone) *[38049-04-6]*
Güntert *et al.* (2001) 0.0028
Buhr (2006) 0.0045

(6*R*,7a*S*)-3,6-DIMETHYL-5,6,7,7a-TETRAHYDRO-2(4*H*)-BENZOFURANONE (isomintlactone)
[76684-66-1]
Güntert *et al.* (2001) 0.00125
Buhr (2006) 0.0045

(6*S*,7a*S*)-3,6-DIMETHYL-5,6,7,7a-TETRAHYDRO-2(4*H*)-BENZOFURANONE (E-mintlactone)
Buhr (2006) 0.006

(3*R*,3a*R*,7a*R*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) 0.014

(3*R*,3a*R*,7a*S*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) > 0.1

(3*R*,3a*S*,7a*R*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) 0.000 25

(3*R*,3a*S*,7a*S*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) 0.008

(3*S*,3a*R*,7a*R*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) 0.000 05

(3*S*,3a*R*,7a*S*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) 0.080

(3*S*,3a*S*,7a*R*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) 0.000 000 01

(3*S*,3a*S*,7a*S*)-3,6-DIMETHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE
Buhr (2006) 0.000 007

(2*R*,4*R*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Nätscher *et al.* (2010) 0.000 5

(2*S*,4*R*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.0031

(2*R*,4*S*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.001

(2*S*,4*S*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.010

(2*R*,4*R*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,8-DISILA-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE
Büttner *et al.* (2007b) 0.000 31

(2*R*,4*S*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,8-DISILA-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE
Büttner *et al.* (2007b) 0.000 31

(2*S*,4*R*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,8-DISILA-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE
Büttner *et al.* (2007b) 0.0068

(2*S*,4*S*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,8-DISILA-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE
Büttner *et al.* (2007b) 0.0063

(2*R*,4*R*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYLINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.0025

(2*S*,4*R*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYLINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.018

(2*R*,4*S*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYLINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.0015

(2*S*,4*S*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYLINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.020

(2*R*,4*R*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-2-OXA-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.0018

(2*S*,4*R*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-2-OXA-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.017

(2*R*,4*S*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-2-OXA-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.0029

(2*S*,4*S*)-2,4-DIMETHYL-2-(1,1,3,3-TETRAMETHYL-2-OXA-1,3-DISILAINDAN-5-YL)-1,3-DIOXOLANE
Nätscher *et al.* (2010) 0.022

(2*R*,4*R*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE
Büttner *et al.* (2007b) 0.000 4

(2*R*,4*S*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Büttner *et al.* (2007b) 0.000 54

(2*S*,4*R*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE
Büttner *et al.* (2007b) 0.0125

(2*S*,4*S*)-2,4-DIMETHYL-2-(5,5,8,8-TETRAMETHYL-5,6,7,8-TETRAHYDRO-2-NAPHTHYL)-1,3-DIOXOLANE
Büttner *et al.* (2007b) 0.002

(+)-(*Z*)-5-(2,3-DIMETHYLTRICYCLO[2.2.1.$0^{2,6}$]-3-HEPTYL)-2-METHYL-2-PENTEN-1-OL ((+)-(*Z*)-α-santalol)
Bajgrowicz *et al.* (1998) 0.0074

(±)-(2*R**,3*S**,7*S**)-2,3-DIMETHYLTRICYCLO[7.6.0.$0^{3,7}$]PENTADEC-1(9)-EN-4-ONE
Kraft & Gallo (2004) 0.0049

dimethyl tetrasulphide⇒METHYLTETRATHIOMETHANE
dimethyl trisulphide⇒METHYLTRITHIOMETHANE

DIMETHYL TRITHIOCARBONATE *[2314-48-9]*
Katz & Talbert (1930) 0.033

4,6-dinitro-*o*-cresol⇒2-METHYL-4,6-DINITROPHENOL
dioxane⇒1,4-DIOXANE

1,4-DIOXANE (dioxane) *[123-91-1]*

Wirth & Klimmer (1937)	d	10
May (1966)	d	620
May (1966)	r	1,000
Köster (1968a,1971)	d	45 - 9,400
Dravnieks & Laffort (1972)		30.6
Hellman & Small (1973,1974)	d	2.9
Hellman & Small (1973,1974)	r	6.5
Dravnieks (1974)	d	270
Nagy (1991)	d	46

3,6-DIOXA-1-OCTANOL (carbitol, diethylene glycol ethyl ether) *[111-90-0]*

Hellman & Small (1973,1974)	d	< 1.2
Hellman & Small (1973,1974)	r	6.0

3,6-DIOXAOCTYL-1 ACETATE (carbitol acetate) *[112-15-2]*

Hellman & Small (1974)	d	0.19
Hellman & Small (1974)	r	1.1

1,3-DIOXOLANE *[646-06-0]*

Hellman & Small (1974)	d	51
Hellman & Small (1974)	r	192

DIPHENYL (phenylbenzene, biphenyl) *[92-52-4]*

Solomin (1961)		0.06
Nagy (1991)	d	0.0033

DIPHENYLAMINE *[122-39-4]*

Backman (1917)	r	0.15 - 0.17
Nagy (1991)	d	1.3

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

diphenyl ether⇒PHENOXYBENZENE

DIPHENYLMETHANE (benzylbenzene, ditan) *[101-81-5]*

Punter (1983)	d	0.41

DIPHENYLMETHANE 4,4'-DIISOCYANATE *[101-68-8]*

Woolrich (1982)		4

diphenyl sulphide⇒PHENYLTHIOBENZENE
diphosgene⇒TRICHLOROMETHYL CHLOROFORMATE

DI-2-PROPENYLAMINE (diallylamine) *[124-02-7]*

Hine *et al.* (1960)		8

DIPROPYLAMINE *[142-84-7]*

Hellman & Small (1974)	d	0.08
Hellman & Small (1974)	r	0.40

dipropyl disulphide⇒PROPYLDITHIOPROPANE
dipropyl sulphide⇒PROPYLTHIOPROPANE
ditan⇒DIPHENYLMETHANE
3,4-dithiahexane⇒ETHYLDITHIOETHANE
4,5-dithiaoctane⇒PROPYLDITHIOPROPANE
2,3-dithiapentane⇒METHYLDITHIOETHANE
2,4-dithiapentane⇒BIS(METHYLTHIO)METHANE
3,3'-dithiobis(2,5-dimethylfuran)⇒3-(2,5-DIMETHYL)FURYLDITHIO-3-(2,5-DIMETHYL)FURAN
3,3'-dithiobis(2-methylfuran)⇒3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN
divinyl sulphide⇒ETHENYLTHIOETHENE
DMDS⇒METHYLDITHIOMETHANE

(*Z*)-1,5-DODECADIEN-3-OL

Schieberle & Buettner (2001)		0.1 - 1

(*Z*)-1,5-DODECADIEN-3-ONE

Schieberle & Buettner (2001)		0.0011

(*R*)-γ-dodecalactone⇒(*R*)-4-HYDROXYDODECANOIC ACID LACTONE
(*S*)-γ-dodecalactone⇒(*S*)-4-HYDROXYDODECANOIC ACID LACTONE
(*R*)-δ-dodecalactone⇒(*R*)-5-HYDROXYDODECANOIC ACID LACTONE
(*S*)-δ-dodecalactone⇒(*S*)-5-HYDROXYDODECANOIC ACID LACTONE

DODECANAL (laurylaldehyde) *[112-54-9]*

Cristoph (1983)	r	0.014 - 0.020
Von Ranson & Belitz (1992a)	d	0.033
Von Ranson & Belitz (1992a)	r	0.033

DODECANE *[112-40-3]*

Laffort & Dravnieks (1973)		37
Patte (1978); Patte & Punter (1979)	d	50
Lindell (1991)	d	11.8
Nagata (2003)	d	0.77

1-DODECANETHIOL *[112-55-0]*

Kendall *et al.* (1968)	r	0.000 8
Patte (1978); Patte & Punter (1979)	d	0.000 000 9

DODECANOIC ACID (lauric acid) *[143-07-7]*

Passy (1893b,1893c)	d	0.1
Backman (1917)	r	0.004 - 0.005

1-DODECANOL *[112-53-8]*

Backman (1917)	r	0.02
Mullins (1955)	r	25.5
Pliška (1962)		2.1
Punter (1983)	d	1.2

(*E*)-2-DODECENAL *[20407-84-5]*

Von Ranson & Belitz (1992a)	d	0.0073
Von Ranson & Belitz (1992a)	r	0.011

(*Z*)-2-DODECENAL *[81149-96-4]*

Von Ranson & Belitz (1992a)	d	0.047
Von Ranson & Belitz (1992a)	r	0.088

(*E*)-3-DODECENAL

Von Ranson & Belitz (1992a)	d	0.044
Von Ranson & Belitz (1992a)	r	0.044

(*Z*)-3-DODECENAL *[68141-15-1]*

Von Ranson & Belitz (1992a)	d	0.000 036
Von Ranson & Belitz (1992a)	r	0.0069

(*E*)-8-DODECENAL

Von Ranson & Belitz (1992a)	d	0.033
Von Ranson & Belitz (1992a)	r	0.036

1-DODECEN-3-OL *[4048-42-4]*

Koszinowski & Piringer (1983)		3.7
Schieberle & Buettner (2001)		0.1 - 1

(R)-*cis*-6-γ-dodecenolactone⇒(*R*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE
(*S*)-*cis*-6-γ-dodecenolactone⇒(*S*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE

1-DODECEN-3-ONE *[58879-39-3]*

Koszinowski & Piringer (1983)		0.18
Schieberle & Buettner (2001)		0.01 - 0.1

DODECYL ACETATE (lauryl acetate) *[112-53-8]*

Cometto-Muñiz & Cain (1991);		
Cometto-Muñiz (1993)	d	0.41

tert-dodecyl mercaptan⇒ 2-METHYL-2-UNDECANETHIOL
duotal⇒DI-2-METHOXYPHENYL CARBONATE
durene⇒1,2,4,5-TETRAMETHYLBENZENE

E

epichlorohydrin⇒1-CHLORO-2,3-EPOXYPROPANE

1,4-EPOXYBUTANE (tetrahydrofuran) *[109-99-9]*

May (1966)	d	90
May (1966)	r	180
Popov (1970)		0.27
Kendall *et al.* (1968)	r	7.3 - 10.2
Nagy (1991)	d	18

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

trans-2,3-EPOXYDECANAL *[102369-06-2]*
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.0037

4,5-EPOXY-(*E*)-2-DECENAL
Heiler & Schieberle (1996) 0.000 002

trans-4,5-EPOXY-(*E*)-2-DECENAL *[134454-31-2]*
Guth & Grosch (1990) 0.000 000 5 - 0.000 005
Schieberle & Grosch (1991);
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.000 000 6 - 0.000 0025

trans-2,3-EPOXYDODECANAL
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.0154

trans-4,5-EPOXY-(*E*)-2-DODECENAL
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.000 006

1,2-EPOXYETHANE (ethylene oxide) *[75-21-8]*
Jacobson *et al.* (1956) 1,260
Yuldashev (1965) 1.5
Hellman & Small (1974) d 470
Hellman & Small (1974) r 900

trans-2,3-EPOXYHEPTANAL
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.0069

trans-4,5-EPOXY-(*E*)-2-HEPTENAL
Schieberle & Grosch (1991);
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) > 1

trans-2,3-EPOXYHEXANAL
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.0063

3,9-epoxy-1,4(8)-*p*-menthadiene⇒3,6-DIMETHYL-2,4,5,7a-TETRAHYDRO-1-BENZOFURAN
(1*R*)-3,9-epoxy-3,8(9)-*p*-menthadiene⇒3,6-DIMETHYL-4,5,6,7-TETRAHYDRO-1-BENZOFURAN
(3*S*,4*R*,8*R*)-3,9-epoxy-1-*p*-menthene⇒(3*S*,4*R*,8*R*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN
(3*R*,4*S*,8*R*)-3,9-epoxy-1-*p*-menthene⇒(3*R*,4*S*,8*R*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN
(3*R*,4*S*,8*S*)-3,9-epoxy-1-*p*-menthene⇒(3*R*,4*S*,8*S*)-3,6-DIMETHYL-2,3,3a,4,5,7a-HEXAHYDROBENZOFURAN

trans-2,3-EPOXYNONANAL
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.0035

trans-4,5-EPOXY-(*E*)-2-NONENAL *[128386-31-2]*
Guth & Grosch (1990) 0.000 25 - 0.0010
Schieberle & Grosch (1991);
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.000 07 - 0.000 28

trans-2,3-EPOXYOCTANAL *[104595-96-2]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.0026

trans-4,5-EPOXY-(*E*)-2-OCTENAL *[134452-45-2]*
Schieberle & Grosch (1991);
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.007 - 0.028

2,3-EPOXYPROPANAL (glycidaldehyde) *[765-34-4]*
Hine *et al.* (1961) < 2.9

1,2-EPOXYPROPANE (propylene oxide) *[75-56-9]*

Jacobson *et al.* (1956)		473
Hellman & Small (1974)	d	24
Hellman & Small (1974)	r	84

trans-2,3-EPOXYUNDECANAL
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.0061

4,5-EPOXY-(*E*)-2-UNDECENAL
Heiler & Schieberle (1996) 0.000 04

trans-4,5-EPOXY-(*E*)-2-UNDECENAL *[134346-43-3]*
Schieberle & Grosch (1991);
Buettner & Schieberle (2001);
Schieberle & Buettner (2001) 0.000 015 - 0.000 06

estragole⇒1-ALLYL-4-METHOXYBENZENE

ESTRA-1,3,5(10),16-TETRAEN-3-OL *[1150-90-9]*
Kraft & Popaj (2004) d 0.008

ETHANAL (acetaldehyde) *[75-07-0]*

Zwaardemaker (1914)	d	0.7
Backman (1917)	r	0.062 - 0.075
Katz & Talbert (1930)		0.12
Balavoine (1943)		10
Pliška & Janiček (1965)		1,800
Gofmekler (1967,1968)	d	0.012
Leonardos *et al.* (1969)	r	0.38
Hartung *et al.* (1971)		0.005
Takhirov (1974)		0.49
Teranishi *et al.* (1974)		0.041
Anon. (1980)	d	0.0027
Anon. (1980)	r	0.027
Nauš (1982)	d	1
Nauš (1982)	r	10
Nagy (1991)	d	0.09
Nagata (2003)	d	0.0027

ETHANAMIDE (acetamide) *[60-35-5]*
Backman (1917) r 140 - 160

ETHANE *[74-84-0]*

Mullins (1955)	r	899,000
Laffort & Dravnieks (1973)		25,000

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1,2-ETHANEDIAMINE (ethylenediamine) *[107-15-3]*

Hellman & Small (1974)	d	3.2
Hellman & Small (1974)	r	11

1,2-ETHANEDIOL (ethylene glycol) *[107-21-1]*

Nagy (1991)	d	13

1,2-ETHANEDITHIOL *[540-63-6]*

Katz & Talbert (1930)		0.12

ETHANESELENOL (ethyl selenomercaptan) *[593-69-1]*

Katz & Talbert (1930)		0.00013

ETHANETHIOL (ethyl mercaptan) *[75-08-1]*

Allison & Katz (1919)		46
Katz & Talbert (1930)		0.000 66 - 0.0076
Thomas *et al.* (1943)		0.005
Stuiver (1958)	d	0.000 1
Sales (1958)		0.0025 - 0.0045
Blinova (1965)		0.006 - 0.03
Endo *et al.* (1967)		0.00065
Leonardos *et al.* (1969)	r	0.0025
Wilby (1969)	r	0.001
Blanchard (1976)		0.016
Selyuzhitskii *et al.* (1976)		0.000 095
Whisman *et al.* (1978)		0.000 25 - 0.000 5
Bedborough & Trott (1979)	d	0.000 33
Anon. (1980)	d	0.000 043
Anon. (1980)	r	0.000 73
Cristoph (1983)	r	0.000 8 - 0.000 9
Stevens *et al.* (1987);		
Stevens & Cain (1987b)	d	0.0019 - 0.021
Nagata (2003)	d	0.000 022

ethanoic acid ⇒ACETIC ACID

ETHANOL (ethyl alcohol) *[64-17-5]*

Passy (1892c)	d	250
Parker & Stabler (1913)	r	17
Backman (1917)	r	175 - 200
Grijns (1919); Zwaardemaker (1927)		2,600
Henning (1924)	d	183
Jung (1936)	d	7.8
Jung (1936)	r	11.7 - 14
Balavoine (1943)		10,000
Mullins (1955)	r	9,230
Scherberger *et al.* (1958)	r	665
Janiček *et al.* (1960)		884
Nauš (1962)	d	2
Pliška & Janiček (1965)		76,000
Ubaidullaev (1966b)		7.1
Guadagni (1966)		100
May (1966)	d	93
May (1966)	r	190
Leonardos *et al.* (1969)	r	19
Dravnieks & Laffort (1972)		640

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Dravnieks (1974)	d	1,350
Nishida *et al.* (1979)	d	302
Anon. (1980)	d	0.64
Anon. (1980)	r	11.6
Nauš (1982)	d	2
Nauš (1982)	r	20
Cristoph (1983)	r	8.7 - 9.2
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	154
Scharfenberger (1990)		988
Nagy (1991)	d	36
Nagata (2003)	d	0.99
Cain *et al.* (2006)	d	0.17
Cometto-Muñiz & Abraham (2008)	d	0.62

ethanolamine⇒2-AMINOETHANOL

ETHANOYL CHLORIDE (acetyl chloride) *[75-36-5]*

Schley (1934)	d	0.11
Schley (1934)	r	0.11

ETHENE (ethylene) *[74-85-1]*

Mullins (1955)	r	1,180
Deadman & Prigg (1959)	d	125
Krasovitskaya & Malyarova (1968)		20
Laffort & Dravnieks (1973)		1,100
Hellman & Small (1974)	d	310
Hellman & Small (1974)	r	480

ETHENYL ACETATE (vinyl acetate) *[108-05-4]*

Gofmekler (1960)		1
Deese & Joyner (1969)	r	< 1.4
Hellman & Small (1973,1974)	d	0.4
Hellman & Small (1973,1974)	r	1.4

ETHENYLBENZENE (styrene, vinylbenzene) *[100-42-5]*

Wolf *et al.* (1956)		43 - 258
Deadman & Prigg (1959)	d	0.11
Li-Sheng (1961)		0.02
Stalker (1963)	d	0.073
Mühlen (1968)	r	4.3
Leonardos *et al.* (1969)	r	0.2 - 0.4
Smith & Hochstettler (1969)	r	0.2
Dravnieks & Laffort (1972)		1.7
Hellman & Small (1973,1974)	d	0.22 - 0.64
Hellman & Small (1973,1974)	r	0.64
Dravnieks (1974)	d	8
Anon. (1980)	d	0.14
Anon. (1980)	r	0.73
Don (1986); Hoshika *et al.* (1993)	d	0.068
Randebrock (1986)		0.012
Nagy (1991)	d	1.3
Hoshika *et al.* (1993)	d	0.14
Nagata (2003)	d	0.15
Dalton *et al.* (2007)	d	26.4

ETHENYL CYANIDE (acrylonitrile, vinyl cyanide) *[107-13-1]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Stalker (1963)	d	3.4
Leonardos *et al.* (1969)	r	47
Nagata (2003)	d	19

2-ETHENYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE (2-ethenyl-3,5-dimethyl-5,6-dihydropyrazine)

Kurniadi (2003)	r	0.000 02

3-ETHENYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE (3-ethenyl-2,5-dimethyl-5,6-dihydropyrazine)

Kurniadi (2003)	r	0.16

2-ethenyl-3,5-dimethyl-5,6-dihydropyrazine⇒2-ETHENYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE
3-ethenyl-2,5-dimethyl-5,6-dihydropyrazine⇒3-ETHENYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE

5-ETHENYL-2,3-DIMETHYLPYRAZINE *[160818-30-4]*

Wagner *et al.* (1999)	> 2

3-ETHENYL-2,5-DIMETHYLPYRAZINE *[80935-98-4]*

Czerny *et al.* (1996); Wagner *et al.* (1999)	0.84 - 0.87

2-ETHENYL-3,5-DIMETHYLPYRAZINE *[157615-33-3]*

Czerny *et al.* (1996); Grosch *et al.* (1996); Wagner *et al.* (1999)	0.000 01 - 0.000 012

2-ETHENYL-3-ETHYL-5-METHYLPYRAZINE *[181589-32-2]*

Czerny *et al.* (1996); Grosch *et al.* (1996); Wagner *et al.* (1999)	0.000 01 - 0.000 014

3-ETHENYL-2-ETHYL-5-METHYLPYRAZINE

Czerny *et al.* (1996); Grosch *et al.* (1996); Wagner *et al.* (1999)	0.095 - 0.11

5-ETHENYL-2-ETHYL-3-METHYLPYRAZINE

Grosch *et al.* (1996)	> 0.8
Wagner *et al.* (1999)	1.04

5-ETHENYL-3-ETHYL-2-METHYLPYRAZINE

Grosch *et al.* (1996)	> 0.8
Wagner *et al.* (1999)	1.04

4-ETHENYL-2-METHOXYPHENOL (*p*-vinylguaiacol, 4-vinylguaiacol, 2-methoxy-4-vinylphenol) *[7786-61-0]*

Blank *et al.* (1989,1992)	0.000 4 - 0.000 8
Yang *et al.* (2008)	0.0028

2-ETHENYL-3-METHYLPYRAZINE *[25058-19-9]*

Wagner *et al.* (1999)	0.087

5-ETHENYL-4-METHYLTHIAZOLE (4-methyl-5-vinylthiazole) *[1759-28-0]*

Gijs *et al.* (2000)	0.3

ETHENYLPYRAZINE (vinylpyrazine) *[4177-16-6]*

Fors (1984); Fors & Olofsson (1985)	d	0.33
Wagner *et al.* (1999)		> 2

2-ETHENYLPYRIDINE (2-vinylpyridine) *[100-69-6]*

Sutton (1962b)	1.3

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

ether⇒ETHOXYETHANE

2-ETHOXY-3,4-DIHYDRO-1,2-PYRAN *[103-75-3]*

Hellman & Small (1974)	d	0.1
Hellman & Small (1974)	r	0.5

ETHOXYETHANE (ether, diethyl ether) *[60-29-7]*

Passy (1892a,b,d)	d	0.5 - 4
Allison & Katz (1919)		5,833
Grijns (1919)		< 50
Henning (1924)	d	0.75
Zwaardemaker (1927)		1
Jung (1936)	d	35
Jung (1936)	r	35
Scherberger *et al.* (1958)	r	210
Flemming & Johnstone (1977)	r	4.8
Nagy (1991)	d	0.95

2-ETHOXYETHANOL (cellosolve, ethyl glycol, ethylene glycol ethyl ether) *[110-80-5]*

May (1966)	d	90
May (1966)	r	180
Hellman & Small (1973,1974)	d	1.1
Hellman & Small (1973,1974)	r	2.0
Nagy (1991)	d	11.6
Nagata (2003)	d	2.1

2-ETHOXYETHYL ACETATE (cellosolve acetate) *[111-15-9]*

Hellman & Small (1973,1974)	d	0.3
Hellman & Small (1973,1974)	r	0.7
Nagy (1991)	d	0.48
Nagata (2003)	d	0.26

1-ETHOXY-2,4-HEXADIENE *[56752-54-6]*

Chisholm & Samuels (1992)	1.15

2-ETHOXY-3,5-HEXADIENE *[56752-55-7]*

Chisholm & Samuels (1992)	0.0013

3-ETHOXY-4-HYDROXYBENZALDEHYDE (ethylvanillin) *[121-32-4]*

Randebrock (1971)	0.000 007
Randebrock (1986)	0.000 027

1-ETHOXYNAPHTHALENE (ethyl α-naphthyl ether) *[5328-01-8]*

Backman (1917)	r	0.000 8

2-ETHOXYNAPHTHALENE (ethyl β-naphthyl ether) *[93-18-5]*

Backman (1917)	r	0.009 - 0.015

ETHYL ACETATE *[141-78-6]*

Backman (1917)	r	15 - 17.5
Allison & Katz (1919)		686
Jung (1936)	d	3.6
Jung (1936)	r	3.6 - 5.4
Jones (1955c)	r	155
Clausen *et al.* (1955)	d	4.8
Gofmekler (1960)		0.6
Janiček *et al.* (1960)		1,120

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

May (1966)	d	180
May (1966)	r	270
Laffort & Dravnieks (1973)		27
Hellman & Small (1973,1974)	d	23
Hellman & Small (1973,1974)	r	48
Anon. (1980)	d	0.9
Anon. (1980)	r	5.0
Bahnmüller (1983)		3.7 - 25
Cristoph (1983)	r	4.6 - 5.0
Randebrock (1986)		0.34
Scharfenberger (1990)		141
Cometto-Muñiz & Cain (1991);		
Cometto-Muñiz (1993)	d	623
Nagy (1991)	d	28
Ziemer *et al.* (2000)	d	4.6
Nagata (2003)	d	3.1
Higuchi & Masuda (2004)	d	2.0 - 3.0
Komthong *et al.* (2006)		1,030
Van Thriel *et al.* (2006)	d	5.36
Cometto-Muñiz *et al.* (2008)	d	0.88
Ueno *et al.* (2009)		1.3 (triangle odor bag method; forced-choice mode)
Ueno *et al.* (2009)		6.1 (triangle odor bag method; yes/no mode)
Ueno *et al.* (2009)		4.7 (dynamic olfactometer; forced-choice mode)
Ueno *et al.* (2009)		4.3 (dynamic olfactometer; yes/no mode)

ethyl acetoacetate⇒ETHYL 3-OXOBUTANOATE
ethyl acid succinate⇒MONETHYL SUCCINATE
ethyl acrylate⇒ETHYL 2-PROPENOATE
ethyl alcohol⇒ETHANOL

ETHYLAMINE *[75-04-7]*

Tkachev (1969)		0.05
Hellman & Small (1974)	d	0.5
Hellman & Small (1974)	r	1.5
Laing *et al.* (1978)	r	6.5
Nagata (2003)	d	0.083

ethyl amyl ketone⇒3-OCTANONE

N-ETHYLANILINE *[103-69-5]*

Backman (1917)	r	1.7 - 2.0

4-ETHYLBENZALDEHYDE *[4748-78-1]*

Von Ranson & Belitz (1992b)	d	0.013
Von Ranson & Belitz (1992b)	r	0.040

ETHYLBENZENE *[100-41-4]*

Ivanov (1964)		2 - 2.6
Köster (1971)	d	0.4
Nagy (1991)	d	1.9
Khiari *et al.* (1992)		< 0.01
Cometto-Muñiz (1993);		
Cometto-Muñiz & Cain (1994)	d	78.3
Nagata (2003)	d	0.73
Cometto-Muñiz & Abraham (2009b)	d	0.026

ETHYL BENZOATE *[93-89-0]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Jung (1936)	d	0.05 - 0.1
Jung (1936)	r	0.15 - 0.2
Janiček *et al.* (1960)		6.7
Sfiras & Demeiliers (1967)	r	0.62
Appell (1969)		0.000 6

7-ETHYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 11

ETHYL BUTANOATE (ethyl butyrate) *[105-54-4]*

Tempelaar (1913)	d	0.28
Backman (1917)	r	0.13 - 0.23
Cristoph (1983)	r	0.002 - 0.003
Roos *et al.* (1985)	d	0.000 060 - 0.000 090
Roos *et al.* (1985); Don (1986)	d	0.000 030
Hartigh (1986)	d	0.000 017 - 0.000 023
Randebrock (1986)		0.000 58
Marin *et al.* (1988)		0.02 - 0.11
Hermans (1989)		0.000 016 - 0.000 991
Lea & Ford (1991)		0.1
Guth (1997)		0.0025
Guth (1997)		0.2 with 55.6 mg ethanol/l air
Buettner & Schieberle (2001a,2001b)		0.0027
Nagata (2003)	d	0.000 19
Atanasova *et al.* (2005)	d	0.0036 - 0.0217
Atanasova *et al.* (2005)	r	0.0040 - 0.1261
Komthong *et al.* (2006)		22.9
Schmidt & Cain (2010)	d	0.000 053

2-ETHYL-1-BUTANOL *[97-95-0]*

Hellman & Small (1974)	d	0.3
Hellman & Small (1974)	r	3.2

7-(2'-ETHYLBUTYL)BENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.0047

ethyl butyrate⇒ETHYL BUTANOATE
ethyl caprate⇒ETHYL DECANOATE
ethyl caproate⇒ETHYL HEXANOATE
ethyl caprylate⇒ETHYL OCTANOATE
ethyl carbanilate⇒ETHYL N-PHENYLCARBAMATE
ethyl chloride⇒CHLOROETHANE
ethyl citronellyl oxalate⇒ETHYL 3,7-DIMETHYL-6-OCTEN-1-YL OXALATE

ETHYL CYCLOHEXANECARBOXYLATE (ethyl cyclohexanoate) *[3289-28-9]*

Sfiras & Demeiliers (1967)	r	0.85
Guth & Grosch (1991)		0.000 003 - 0.000 012

ethyl cyclohexanoate⇒ETHYL CYCLOHEXANECARBOXYLATE

ETHYL 3-CYCLOHEXENECARBOXYLATE *[15111-56-5]*

Sfiras & Demeiliers (1967)	r	0.77

ETHYL DECANOATE (ethyl caprate) *[110-38-3]*

Backman (1917)	r	0.0024 - 0.0032
Schwarz (1955)		0.0013
Goldenberg (1967)	d	0.0012
Ferreira *et al.* (1998)		0.53

ethyl dichloroarsine⇒DICHLOROETHYLARSINE

2-ETHYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE (2-ethyl-3,5-dimethyl-5,6-dihydropyrazine)

Kurniadi *et al.* (2003)		0.000 002
Kurniadi (2003)	r	0.000 04

3-ETHYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE (3-ethyl-2,5-dimethyl-5,6-dihydropyrazine)

Kurniadi *et al.* (2003)		0.000 227
Kurniadi (2003)	r	0.00454

5-ETHYLDIHYDRO-2(3*H*)-THIOPHENONE (γ-thiohexalactone)

Roling *et al.* (1998); Engel (1999)		0.011

ETHYLDIMETHYLAMINE *[598-56-1]*

Kohler & Homans (1980); Homans (1984)		1.4 - 1.9

2-ethyl-3,5-dimethyl-5,6-dihydropyrazine⇒2-ETHYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE
3-ethyl-2,5-dimethyl-5,6-dihydropyrazine⇒3-ETHYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE

ETHYL 3,7-DIMETHYL-6-OCTEN-1-YL OXALATE (ethyl citronellyl oxalate) *[60788-25-2]*

Kraft (2004a)		0.014

2-ETHYL-3,5-DIMETHYLPYRAZINE *[13925-07-0]*

Cerny (1992); Cerny & Grosch (1994)		0.000 007 - 0.000 014
Czerny *et al.* (1996); Grosch *et al.* (1996);		
Wagner *et al.* (1999)		0.000 01 – 0.000 011
Kurniadi (2003)	r	0.000 04

3-ETHYL-2,5-DIMETHYLPYRAZINE (2-ethyl-3,6-dimethylpyrazine) *[13360-65-1]*

Fors (1984); Fors & Olofsson (1985)	d	0.020
Cerny & Grosch (1994); Grosch *et al.* (1996)		0.00245 - 0.0025
Wagner *et al.* (1999)		0.0036

5-ETHYL-2,3-DIMETHYLPYRAZINE *[15707-34-3]*

Cerny & Grosch (1994)		> 0.095
Grosch *et al.* (1996); Wagner *et al.* (1999)		0.2

ETHYLDITHIOETHANE (diethyl disulphide, 3,4-dithiahexane) *[110-81-6]*

Hermanides (1909)	r	0.000 3
Zwaardemaker (1914)	d	0.000 3
Wilby (1969)	r	0.023
Gijs *et al.* (2000)		0.09
Nagata (2003)	d	0.01

ETHYL DODECANOATE (ethyl laurate) *[106-33-2]*

Backman (1917)	r	0.002

ethylene⇒ETHENE
ethylene brassylate⇒CYCLO-1,3-ETHYLENEDIOXY-1,3-TRIDECANEDIONE
ethylene bromide⇒1,2-DIBROMOETHANE
ethylene chloride⇒1,2-DICHLOROETHANE
ethylene chlorohydrine⇒2-CHLOROETHANAL
ethylenediamine⇒1,2-ETHANEDIAMINE
ethylene dibromide⇒1,2-DIBROMOETHANE
ethylene dichloride⇒1,2-DICHLOROETHANE

ETHYLENE 1,12-DODECANEDIOATE (musk C-14) *[54982-83-1]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Traynor (2001)		0.105

ethylene glycol⇒1,2-ETHANEDIOL
ethylene glycol *n*-butyl ether⇒2-BUTOXYETHANOL
ethylene glycol ethyl ether⇒2-ETHOXYETHANOL
ethylene glycol methyl ether⇒2-METHOXYETHANOL

ETHYLENEIMINE (aziridine) *[151-56-4]*

Carpenter *et al.* (1948)		3.6
Berzins (1967)		1.25 - 3.5

ethylene oxide⇒1,2-EPOXYETHANE
ethylene sulphide⇒THIIRANE

ETHYL ETHOXYDITHIOFORMATE (ethyl xanthate) *[151-01-9]*

Deadman & Prigg (1959)	d	0.03

ethyl ethoxy propionate⇒ ETHYL 3-ETHOXYPROPANOATE

ETHYL 3-ETHOXYPROPANOATE (ethyl ethoxy propionate) *[763-69-9]*

Nagy (1991)	d	0.11
Ziemer *et al.* (2000)	d	0.0011

ETHYL N-ETHYLCARBAMATE (ethylurethane) *[623-78-9]*

Backman (1917)	r	1.0 - 1.1

ETHYL FORMATE *[109-94-4]*

Backman (1917)	r	54 – 61
Nagata (2003)	d	8.1
Van Thriel *et al.* (2006)	d	90.9

ethyl glycol⇒2-ETHOXYETHANOL
p-ethylguaiacol⇒4-ETHYL-2-METHOXYPHENOL
4-ethylguaiacol⇒4-ETHYL-2-METHOXYPHENOL

ETHYL HEPTANOATE *[106-30-9]*

Backman (1917)	r	0.16 - 0.21
Ferreira *et al.* (1998)		0.039
Cometto-Muñiz *et al.* (2004)	d	0.21 - 0.65
Cometto-Muñiz *et al.* (2005)	d	0.24

(*R*)-2-ETHYLHEXANAL

Bartschat & Mosandl (1998)		> 2.5

(*S*)-2-ETHYLHEXANAL

Bartschat & Mosandl (1998)		> 2.5

ETHYL HEXANOATE (ethyl caproate) *[123-66-0]*

Backman (1917)	r	0.0043 - 0.0049	
Schwarz (1955)		0.03	
Cristoph (1983)	r	0.006 - 0.010	
Berger *et al.* (1985)	d	0.01 - 0.02	
Marin *et al.* (1988)		0.029 - 0.041	
Guth (1997)		0.009	
Guth (1997)		0.09	with 55.6 mg ethanol/l air
Ferreira *et al.* (1998)		0.0485	
Buettner & Schieberle (2001a,2001b)		0.003	
Komthong *et al.* (2006)		18.1 - 27.4	

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-ETHYL-1-HEXANOL *[104-76-7]*

Hellman & Small (1973,1974)	d	0.40
Hellman & Small (1973,1974)	r	0.74
Nagy (1991)	d	0.8

ETHYL (*E*)-2-HEXENOATE *[27829-72-7]*

Berger *et al.* (1985)	d	0.14 - 0.20

ETHYL (*E*)-3-HEXENOATE *[26553-46-8]*

Berger *et al.* (1985)	d	0.25 - 0.50

ETHYL (*Z*)-3-HEXENOATE *[64187-83-3]*

Berger *et al.* (1985)	d	0.01 - 0.02

2-ETHYLHEXYL ACETATE *[103-09-3]*

Hellman & Small (1974)	d	0.7
Hellman & Small (1974)	r	1.4

2-ethylhexyl acrylate⇒2-ETHYLHEXYL 2-PROPENOATE
2-ethylhexyl methacrylate⇒2-ETHYLHEXYL 2-METHYL-2-PROPENOATE

2-ETHYLHEXYL 2-METHYL-2-PROPENOATE (2-ethylhexyl methacrylate) *[688-84-6]*

Piringer & Granzer (1984)		0.8

2-ETHYLHEXYL 2-PROPENOATE (2-ethylhexyl acrylate) *[103-11-7]*

Hellman & Small (1974)	d	0.55
Hellman & Small (1974)	r	1.35
Piringer & Granzer (1984)		0.6

(*R*)-ETHYL 3-HYDROXYHEXANOATE

Buettner & Schieberle (2001a,2001b)		0.0021

(*S*)-ETHYL 3-HYDROXYHEXANOATE

Buettner & Schieberle (2001a,2001b)		0.264

5-ETHYL-3-HYDROXY-4-METHYL-2(5*H*)-FURANONE (abhexone, ethylsotolon, 2,4-dihydroxy-3-methyl-2-hexenoic acid lactone) *[698-10-2]*

Huber (1984)		0.000 000 2
Blank *et al.* (1992); Hofmann & Schieberle (1997);		
Blank (1997, 2002)		0.000 002 - 0.000 004 [GC-O on FFAP capillary column]
Blank (1997, 2002)		0.002 - 0.004 [GC-O on SE-54 capillary column]

2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE (sugarone, homofuraneol) *[27538-10-9]*

Blank & Schieberle (1993)		0.000 1 - 0.000 2

ETHYL 2-HYDROXYPROPANOATE (ethyl lactate) *[97-64-3]*

Backman (1917)	r	8

5-ETHYLIDENEBICYCLO[2.2.1]-2-HEPTENE (ethylidenenorbornene) *[16219-75-3]*

Kinkead *et al.* (1971a)		0.035 - 0.07
Hellman & Small (1974)	d	0.1
Hellman & Small (1974)	r	0.4

ethylidenenorbornene⇒5-ETHYLIDENEBICYCLO[2.2.1]-2-HEPTENE
ethyl iodide⇒IODOETHANE
ethyl isobutyrate⇒ETHYL 2-METHYLPROPANOATE

ETHYL ISOTHIOCYANATE (ethyl mustard oil) *[542-85-8]*

Katz & Talbert (1930)		6.1

ethyl isovalerate⇒ETHYL 3-METHYLBUTANOATE
ethyl lactate⇒ETHYL 2-HYDROXYPROPANOATE
ethyl laurate⇒ETHYL DODECANOATE
ethyl mercaptan⇒ETHANETHIOL

ETHYL 2-MERCAPTOACETATE (ethyl 2-thiolacetate, ethyl thioglycolate) *[623-51-8]*

Deadman & Prigg (1959)	d	0.008

2-ETHYL-3-MERCAPTOPROPANAL (3-mercapto-2-ethylpropanal)

Vermeulen & Collin (2002)		0.13

2-ETHYL-3-MERCAPTO-1-PROPANOL (3-mercapto-2-ethyl-1-propanol)

Vermeulen *et al.* (2003)		0.000 1

2-ETHYL-3-MERCAPTOPROPYL ACETATE (3-mercapto-2-ethylpropyl acetate)

Vermeulen & Collin (2003)		0.11

ethyl methacrylate⇒ETHYL 2-METHYL-2-PROPENOATE

4-ETHYL-2-METHOXYPHENOL (*p*-ethylguaiacol, 4-ethylguaiacol) *[2785-89-9]*

Blank *et al.* (1992)		0.000 01 - 0.000 03

N-ETHYL-2-METHYLANILINE (ethyl-*o*-toluidine) *[94-68-8]*

Nagata (2003)	d	0.14

1-ETHYL-2-METHYLBENZENE (*o*-ethyltoluene) *[611-14-3]*

Nagata (2003)	d	0.36

1-ETHYL-3-METHYLBENZENE (*m*-ethyltoluene) *[620-14-4]*

Nagata (2003)	d	0.088

1-ETHYL-4-METHYLBENZENE (*p*-ethyltoluene) *[622-96-8]*

Nagata (2003)	d	0.041

ETHYL 2-METHYLBUTANOATE (ethyl 2-methylbutyrate) *[7452-79-1]*

Cristoph (1983)	r	0.000 06 - 0.000 12
Marin *et al.* (1988)		0.0012 - 0.0028
Lea & Ford (1991)		0.01
Guth & Grosch (1991); Rychlik (1998)		0.000 06 - 0.000 24
Komthong *et al.* (2006)		0.968 - 11.7

ETHYL 3-METHYLBUTANOATE (ethyl isovalerate, ethyl 3-methylbutyrate) *[108-64-5]*

Tempelaar (1913)	d	4.6
Backman (1917)	r	0.4 - 0.5
Cristoph (1983)	r	0.000 07 - 0.000 1
Ferreira *et al.* (1998)		0.0084
Nagata (2003)	d	0.000 069

ethyl 2-methylbutyrate⇒ETHYL 2-METHYLBUTANOATE
ethyl 3-methylbutyrate⇒ETHYL 3-METHYLBUTANOATE
ethyl methyl disulphide⇒ METHYLDITHIOETHANE
ethyl methyl phenyl glycidate⇒ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE

ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE (aldehyde C16, strawberry aldehyde, ethyl

methyl phenyl glycidate) *[77-83-8]*

Balavoine (1948)		0.000 01	
Appell (1969)		0.000 000 7	

ETHYL 2-METHYLPROPANOATE (ethyl isobutyrate) *[97-62-1]*

Backman (1917)	r	0.86 - 1.06	
Appell (1969)		0.0035	
Cristoph (1983)	r	0.000 3 - 0.000 5	
Blank (1990)		0.000 1 - 0.000 2	
Guth (1997)		0.000 3	
Guth (1997)		0.038	with 55.6 mg ethanol/l air
Nagata (2003)	d	0.000 11	

ETHYL 2-METHYL-2-PROPENOATE (ethyl methacrylate) *[97-63-2]*

Piringer & Granzer (1984)		0.05	

2-ETHYL-3-METHYLPYRAZINE *[15707-23-0]*

Fors (1984); Fors & Olofsson (1985)	d	0.15	
Wagner *et al.* (1999)		0.035	

2-ETHYL-5-METHYLPYRAZINE *[13360-64-0]*

Fors (1984); Fors & Olofsson (1985)	d	0.036	

5-ETHYL-2-METHYLPYRIDINE *[104-90-5]*

Kendall *et al.* (1968)	r	0.013	
Hellman & Small (1974)	d	0.03	
Hellman & Small (1974)	r	0.04	
Dravnieks *et al.* (1986)	d	0.004	

ethyl methyl sulphide⇒METHYLTHIOETHANE

3-ETHYL-6-METHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE (3a,4,5,7a-tetrahydro-3-ethyl-6-methyl-2(3*H*)-benzofuranone)

Buhr (2006)		0.050	

ETHYL 2-(METHYLTHIO)ACETATE *[4455-13-4]*

Berger *et al.* (1985)	d	2.0 - 3.0	

ETHYL 3-(METHYLTHIO)PROPANOATE *[13327-56-5]*

Berger *et al.* (1985)	d	0.01 - 0.02	

N-ETHYLMORPHOLINE (4-ethylmorpholine) *[100-74-3]*

Hellman & Small (1974)	d	0.4	
Hellman & Small (1974)	r	1.2	

4-ethylmorpholine⇒N-ETHYLMORPHOLINE
ethyl mustard oil⇒ETHYL ISOTHIOCYANATE
ethyl α-naphthyl ether⇒1-ETHOXYNAPHTHALENE
ethyl β-naphthyl ether⇒2-ETHOXYNAPHTHALENE

ETHYL NONANOATE (ethyl pelargonate) *[123-29-5]*

Backman (1917)	r	0.009 -0.011	
Schwarz (1955)		0.01	

ETHYL OCTANOATE (ethyl caprylate) *[106-32-1]*

Backman (1917)	r	0.002	
Schwarz (1955)		0.01	

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Ferreira *et al.* (1998)		0.022
Rychlik (1998)		0.040

(*R*)-4-ETHYLOCTANOIC ACID

Karl *et al.* (1994)	r	< 0.026

(*S*)-4-ETHYLOCTANOIC ACID

Karl *et al.* (1994)	r	< 0.012

ETHYL 3-OXOBUTANOATE (ethyl acetoacetate) *[141-97-9]*

Backman (1917)	r	0.016 - 0.020

ethyl pelargonate⇒ETHYL NONANOATE

3-ETHYLPENTANE *[617-78-7]*

Nagata (2003)	d	1.5

ETHYL PENTANOATE (ethyl valerate) *[539-82-2]*

Backman (1917)	r	0.26 - 0.42
Stone (1963b,1963c)	d	0.33 - 0.39
Cristoph (1983)	r	0.000 3 - 0.000 4
Nagata (2003)	d	0.000 58

3-ETHYLPHENOL *[620-17-7]*

Blank (1990)		0.001 - 0.002

ETHYL PHENYLACETATE *[101-97-3]*

Schieberle (1998)		0.0033 - 0.0134

ETHYL N-PHENYLCARBAMATE (phenylurethane, ethyl carbanilate) *[1013-75-8]*

Backman (1917)	r	0.036 - 0.053

ETHYL PROPANOATE (ethyl propionate) *[105-37-3]*

Backman (1917)	r	0.3 - 0.5
Cristoph (1983)	r	0.8 - 1.0
Randebrock (1986)		0.20
Nagata (2003)	d	0.029
Cometto-Muñiz *et al.* (2004)	d	~ 1.3
Cometto-Muñiz *et al.* (2005)	d	1.1
Komthong *et al.* (2006)		15.1 - 40.3

ETHYL 2-PROPENOATE (ethyl acrylate) *[140-88-5]*

Leonardos *et al.* (1969)	r	0.0019
Hellman & Small (1973,1974)	d	0.0010
Hellman & Small (1973,1974)	r	0.0015
Anon. (1980)	d	0.000 82
Anon. (1980)	r	0.0053
Piringer & Granzer (1984)		0.001
Nagy (1991)	d	0.013
Nagata (2003)	d	0.0011
Van Thriel *et al.* (2006)	d	0.000 027

ethyl propionate⇒ETHYL PROPANOATE

ETHYLPYRAZINE *[13925-00-3]*

Fors (1984); Fors & Olofsson (1985)	d	0.25
Wagner *et al.* (1999)		> 2

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-ETHYLPYRIDINE *[100-71-0]*

Schieberle (1993)		0.057

ETHYL SALICYLATE *[118-61-6]*

Backman (1917)	r	0.12
Baldus (1936)	d	0.0035
Baldus (1936)	r	0.005

ethyl selenomercaptan⇒ETHANESELENOL

ETHYL SILICATE (tetraethyl orthosilicate) *[78-10-4]*

Smyth & Seaton (1940)	d	< 720
Hellman & Small (1974)	d	31
Hellman & Small (1974)	r	43

ethylsotolon⇒5-ETHYL-3-HYDROXY-4-METHYL-2(5*H*)-FURANONE

2-ETHYLTETRAHYDRO-3,5-DIMETHYLPYRAZINE

Kurniadi *et al.* (2003)		0.0019
Kurniadi (2003)	r	0.038

3-ETHYLTETRAHYDRO-2,5-DIMETHYLPYRAZINE

Kurniadi *et al.* (2003)		0.07
Kurniadi (2003)	r	1.4

S-ETHYL THIOACETATE *[625-60-5]*

Gijs *et al.* (2000)		0.14

S-ETHYL THIOBUTANOATE (thioethyl butanoate)

Cavaillé-Lefebvre *et al.* (1998)	d	1.16

ETHYLTHIOETHANE (diethyl sulphide) *[352-93-2]*

Allison & Katz (1919)		12
Katz & Talbert (1930)		0.010
Deadman & Prigg (1959)	d	0.0045
Wilby (1969)	r	0.022
Ifaedi (1972)		0.31
Anon. (1980)	d	0.0014
Anon. (1980)	r	0.02
Gijs *et al.* (2000)		0.022
Nagata (2003)	d	0.000 12

ethyl thioglycolate⇒ETHYL 2-MERCAPTOACETATE

S-ETHYL THIOHEXANOATE (thioethyl hexanoate)

Cavaillé-Lefebvre *et al.* (1998)	d	1.31

ethyl 2-thiolacetate⇒ETHYL 2-MERCAPTOACETATE

S-ETHYL THIO-3-METHYLBUTANOATE (thioethyl isopentanoate)

Cavaillé-Lefebvre *et al.* (1998)	d	3.56

S-ETHYL THIO-2-METHYLPROPANOATE (thioethyl isobutanoate)

Cavaillé-Lefebvre *et al.* (1998)	d	2.94

S-ETHYL THIOPROPANOATE

Gijs *et al.* (2000)		0.07

m-ethyltoluene⇒1-ETHYL-3-METHYLBENZENE
o-ethyltoluene⇒1-ETHYL-2-METHYLBENZENE
p-ethyltoluene⇒1-ETHYL-4-METHYLBENZENE
ethyl-*o*-toluidine⇒N-ETHYL-2-METHYLANILINE

(1*S*,2*S*,1'*S*,3'*R*,5'*R*)-1-ETHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]HEX-3-YLMETHYL) CYCLOPROPYLMETHANOL

Bajgrowicz & Fráter (2000)		0.000 04

(1*R*,2*R*,1'*S*,3'*R*,5'*R*)-1-ETHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]HEX-3-YLMETHYL) CYCLOPROPYLMETHANOL

Bajgrowicz & Fráter (2000)		> 0.25

(1*S*,2*S*,1'*R*,3'*S*,5'*R*)-1-ETHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]HEX-3-YLMETHYL) CYCLOPROPYLMETHANOL

Bajgrowicz & Fráter (2000)		0.099

(1*R*,2*R*,1'*R*,3'*S*,5'*R*)-1-ETHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]HEX-3-YLMETHYL) CYCLOPROPYLMETHANOL

Bajgrowicz & Fráter (2000)		0.0031

ETHYLTRITHIOETHANE (diethyl trisulphide, 3,4,5-trithiaheptane) *[3600-24-6]*

Wilby (1969)	r	0.0054

ETHYL UNDECANOATE *[627-90-7]*

Schwarz (1955)		0.005
Goldenberg (1967)	d	0.000 3

ethylurethane⇒ETHYL N-ETHYLCARBAMATE
ethyl valerate⇒ETHYL PENTANOATE
ethylvanillin⇒3-ETHOXY-4-HYDROXYBENZALDEHYDE
ethyl vinyl ketone⇒1-PENTEN-3-ONE
ethyl xanthate⇒ETHYL ETHOXYDITHIOFORMATE

ETHYNE (acetylene) *[74-86-2]*

Deadman & Prigg (1959)	d	240
Babin *et al.* (1965)		1,300 - 2,750
Nagy (1991)	d	510

eucalyptol⇒1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE
eugenol⇒4-ALLYL-2-METHOXYPHENOL
exaltolide⇒15-HYDROXYPENTADECANOIC ACID LACTONE
exaltone⇒CYCLOPENTADECANONE

F

fenchone⇒1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE
fenchyl alcohol⇒1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL
(*E*)-filbertone⇒5-METHYL-(*E*)-2-HEPTEN-4-ONE
fixolide⇒7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE
(+)-florhydral®⇒(+)-3-(3-ISOPROPYLPHENYL)BUTANAL
(-)-florhydral®⇒(-)-3-(3-ISOPROPYLPHENYL)BUTANAL
(-)-(2*R*,4*R*)-florol®⇒(-)-(2*R*,4*R*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL
(-)-(2*R*,4*S*)-florol®⇒(-)-(2*R*,4*S*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL
(+)-(2*S*,4*R*)-florol®⇒(+)-(2*S*,4*R*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL
(+)-(2*S*,4*S*)-florol®⇒(+)-(2*S*,4*S*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL

FLUORINE *[7782-41-4]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Belles (1965)		0.15 - 0.30

fluorotrichloromethane⇒TRICHLOROFLUOROMETHANE
formaldehyde⇒METHANAL

FORMIC ACID (methanoic acid) *[64-18-6]*

Passy (1893b,1893c)	d	25 - 50
Zwaardemaker (1914)	d	640
Backman (1917)	r	21 - 24
Schley (1934)		3.0 - 6.0
Guadagni (1966)		450
Nauš (1982)	d	2
Nauš (1982)	r	20
Kleinschmidt (1983)	r	453
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	14.5
Van Thriel *et al.* (2006)	d	12.4
Cometto-Muñiz & Abraham (2010b)	d	0.98

4-formyl-2-methoxyphenol⇒4-HYDROXY-3-METHOXYBENZALDEHYDE

2-FORMYL-5-METHYLTHIOPHENE (5-methyl-2-thiophenecarbaldehyde) *[13679-70-4]*

Gasser & Grosch (1990a)		0.00175 - 0.0074

furamintone⇒3,6-DIMETHYLBENZO-[b]-FURAN-2(3H)-ONE
(*R*)-furamintone⇒(3*R*,6)-DIMETHYLBENZO-[b]-FURAN-2(3H)-ONE
(*S*)-furamintone⇒(3*S*,6)-DIMETHYLBENZO-[b]-FURAN-2(3H)-ONE

FURAN *[110-00-9]*

Nagata (2003)	d	28

2,5-FURANDIONE (maleic anhydride) *[108-31-6]*

Grigorieva (1964)		1.0 - 1.3

furaneol⇒4-HYDROXY-2,5-DIMETHYL-3(2*H*)-FURANONE
2-furanmethanethiol⇒2-FURYLMETHANETHIOL

2-FURANMETHANOL (furfuryl alcohol) *[98-00-0]*

Jacobson *et al.* (1958)		32

FURFURAL *[98-01-1]*

Ubaidullaev (1961)		1.0
Appell (1969)		0.008
Makeicheva (1978)		0.98
Bedborough & Trott (1979)	d	0.25
Nagy (1991)	d	2.8

furfuryl alcohol⇒2-FURANMETHANOL

2-FURFURYLDITHIO-2-FURYLMETHANE (difurfuryl disulphide, bis(2-furfuryl) disulphide) *[4437-20-1]*

Gasser & Grosch (1990b)		0.000 000 15 - 0.000 000 6

2-FURFURYLDITHIOMETHANE (2-furfuryl methyl disulphide) *[57500-00-2]*

Gasser & Grosch (1990b)		0.000 04 - 0.000 17

furfuryl mercaptan⇒2-FURYLMETHANETHIOL
2-furfuryl methyl disulphide⇒2-FURFURYLDITHIOMETHANE

2-FURFURYL 2-METHYL-3-FURYL DISULPHIDE
Hofmann & Schieberle (1995) 0.000 000 4 - 0.000 0016

2-furfuryl methyl sulphide⇒2-FURFURYLTHIOMETHANE

2-FURFURYL 3-OXO-2-BUTYL DISULPHIDE
Hofmann & Schieberle (1995) 0.000 004 - 0.000 016

2-furfurylthiol⇒2-FURYLMETHANETHIOL

2-FURFURYLTHIOMETHANE (2-furfuryl methyl sulphide) *[1438-91-1]*
Gasser & Grosch (1990b) 0.000 4 - 0.0016

2-FURYLMETHANETHIOL (furfuryl mercaptan, 2-furfurylthiol, 2-furanmethanethiol) *[98-02-2]*
Gasser & Grosch (1990a,b) 0.000 004 - 0.000 02
Blank *et al.* (1992) 0.000 01 - 0.000 02
Miyazawa *et al.* (2009a) d 0.000 0049

G

galaxolide⇒1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN
(-)-(1*R*,2*S*)-georgywood®⇒(-)-(1*R*,2*S*)-1-(1,2,3,4,5,6,7,8-OCTAHYDRO-1,2,8,8-TETRAMETHYLNAPHTHALEN-2-YL)ETHANONE
(+)-(1*S*,2*R*)-georgywood®⇒(+)-(1*S*,2*R*)-1-(1,2,3,4,5,6,7,8-OCTAHYDRO-1,2,8,8-TETRAMETHYLNAPHTHALEN-2-YL)ETHANONE
geosmin⇒*trans*-1,10-DIMETHYL-*trans*-9-DECALOL
geranial⇒3,7-DIMETHYL-(*E*)-2,6-OCTADIENAL
geraniol⇒3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-OL
geranyl acetate⇒3,7-DIMETHYL-(*E*)-2,6-OCTADIEN-1-YL ACETATE
glutaraldehyde⇒1,5-PENTANEDIAL
glycidaldehyde⇒2,3-EPOXYPROPANAL
glycol diacetate⇒1,2-DIHYDROXYETHANE DIACETATE
guaiacol⇒2-METHOXYPHENOL
guaiacol carbonate⇒DI-2-METHOXYPHENYL CARBONATE

H

habanolide⇒15-HYDROXY-11(12)-PENTADECENOIC ACID LACTONE
halothane⇒2-BROMO-2-CHLORO-1,1,1-TRIFLUOROETHANE
helional⇒2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL
heliotropine⇒3,4-METHYLENEDIOXYBENZALDEHYDE

(*E*,*E*)-2,4-HEPTADIENAL *[4313-03-5]*
Hall & Andersson (1983) d 0.057

(*Z*)-1,5-HEPTADIEN-3-OL
Schieberle & Buettner (2001) 0.001 - 0.1

(*Z*)-1,5-HEPTADIEN-3-ONE
Schieberle & Buettner (2001) 0.000 000 6

(*R*)-γ-heptalactone⇒(*R*)-4-HYDROXYHEPTANOIC ACID LACTONE
(*S*)-γ-heptalactone⇒(*S*)-4-HYDROXYHEPTANOIC ACID LACTONE
heptaldehyde⇒HEPTANAL

HEPTANAL (heptaldehyde) *[111-71-7]*
Backman (1917) r 0.009 - 0.010
Deadman & Prigg (1959) d 0.006
Pliška & Janiček (1965) 9.5
Teranishi *et al.* (1974) 0.033

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Hall & Andersson (1983)	d	0.26
Cristoph (1983)	r	0.04 - 0.05
Lindell (1991)	d	0.061
Khiari *et al.* (1992)		0.02
Von Ranson & Belitz (1992a)	d	0.064
Von Ranson & Belitz (1992a)	r	0.15
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	0.14
Nagata (2003)	d	0.00085
Yang *et al.* (2008)		0.000 9
Laska & Ringh (2010)	d	0.26

HEPTANE *[142-82-5]*

Patty & Yant (1929)		410
Mullins (1955)	r	2,240
Jones (1955c)	r	750
May (1966)	d	930
May (1966)	r	1,350
Laffort & Dravnieks (1973)		165
Nagy (1991)	d	110
Nagata (2003)	d	2.7

HEPTANOIC ACID (oenanthic acid) *[111-14-8]*

Passy (1893b,1893c)	d	0.3
Backman (1917)	r	2.6 - 3.3
Hendriks (1979)	d	0.033
Punter (1983)	d	0.022

1-HEPTANOL *[111-70-6]*

Passy (1895)	d	1
Backman (1917)	r	0.1 - 0.2
Zwaardemaker (1927)		0.1
Pliška (1962)		17
Stone (1963b,1963c)	d	1.1
Stone *et al.* (1967)	d	0.05
Corbit & Engen (1971)		1
Laing (1975)	d	2.4
Punter (1983)	d	0.21
Cristoph (1983)	r	0.16 - 0.20
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	0.48 - 0.53
Lea & Ford (1991)		10
Nagy (1991)	d	1.6
Nagata (2003)	d	0.023

2-HEPTANOL *[543-49-7]*

Acree *et al.* (1976)		0.1

4-HEPTANOL *[589-55-9]*

Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993)	d	39.4

2-HEPTANONE (methyl amyl ketone) *[110-43-0]*

Stone *et al.* (1962)	d	0.90
Pangborn *et al.* (1964)	d	0.82
Teranishi *et al.* (1974)		0.840
Hall & Andersson (1983)	d	1.3

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Nagy (1991)	d	1.2
Nagy (1991)	d	0.398
Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993)	d	3.3
Cometto-Muñiz *et al.* (1999)	d	0.29 - 0.65
Ziemer *et al.* (2000)	d	0.045
Nagata (2003)	d	0.032
Cometto-Muñiz *et al.* (2004)	d	~ 0.47
Cain *et al.* (2008)	d	0.062
Cometto-Muñiz *et al.* (2008);		
Cometto-Muñiz & Abraham (2009a)	d	0.023
Yang *et al.* (2008)		0.0035
(*E*)-2-HEPTENAL *[18829-55-5]*		
Teranishi *et al.* (1974)		0.034
Eriksson *et al.* (1976)	d	0.14
Hall & Andersson (1983)	d	0.25
Guth (1991)		0.0525
Von Ranson & Belitz (1992a)	d	2.4
Von Ranson & Belitz (1992a)	r	2.8
(*Z*)-2-HEPTENAL *[57266-86-1]*		
Von Ranson & Belitz (1992a)	d	0.056
Von Ranson & Belitz (1992a)	r	0.22
(*E*)-3-HEPTENAL		
Von Ranson & Belitz (1992a)	d	0.16
Von Ranson & Belitz (1992a)	r	0.47
(*Z*)-3-HEPTENAL *[21662-18-0]*		
Von Ranson & Belitz (1992a)	d	0.0022
Von Ranson & Belitz (1992a)	r	0.016
(*E*)-4-HEPTENAL *[929-22-6]*		
Von Ranson & Belitz (1992a)	d	0.034
Von Ranson & Belitz (1992a)	r	0.034
(*Z*)-4-HEPTENAL *[6728-31-0]*		
Von Ranson & Belitz (1992a)	d	0.0034
Von Ranson & Belitz (1992a)	r	0.040
6-HEPTENAL		
Von Ranson & Belitz (1992a)	d	0.0090
Von Ranson & Belitz (1992a)	r	0.031
1-HEPTENE *[592-76-7]*		
Koszinowski & Piringer (1983)		37
Nagata (2003)	d	1.5
1-HEPTEN-3-OL *[4938-52-7]*		
Koszinowski & Piringer (1983)		0.07
Schieberle & Buettner (2001)		0.01 - 0.1
(*E*)-3-HEPTEN-2-ONE *[5609-09-6]*		
Teranishi *et al.* (1974)		0.060
1-HEPTEN-3-ONE *[2918-13-0]*		

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Koszinowski & Piringer (1983)		0.000 7
Buettner & Schieberle (2001a); Schieberle & Buettner (2001)		0.000 08

HEPTYL ACETATE *[112-06-1]*

Cristoph (1983)	r	1.9 - 2.8
Cometto-Muñiz & Cain (1991); Cometto-Muñiz (1993)	d	6.3

HEPTYLBENZENE *[1078-71-3]*

Cometto-Muñiz (1993); Cometto-Muñiz & Cain (1994)	d	4.0

7-HEPTYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.00165

heptyl chloride⇒1-CHLOROHEPTANE

2-HEPTYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE (2-heptyl-3,5-dimethyl-5,6-dihydropyrazine)

Kurniadi *et al.* (2003)		> 2
Kurniadi (2003)	r	> 40

3-HEPTYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE (3-heptyl-2,5-dimethyl-5,6-dihydropyrazine)

Kurniadi *et al.* (2003)		> 2
Kurniadi (2003)	r	> 40

2-heptyl-3,5-dimethyl-5,6-dihydropyrazine⇒2-HEPTYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE
3-heptyl-2,5-dimethyl-5,6-dihydropyrazine⇒3-HEPTYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE
2-heptyldimethyltetrahydropyrazine⇒2-HEPTYLTETRAHYDRODIMETHYLPYRAZINE

2-HEPTYLTETRAHYDRODIMETHYLPYRAZINE (2-heptyldimethyltetrahydropyrazine)

Kurniadi (2003)	r	> 40

6-HEPTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE (δ-thiododecalactone)

Roling *et al.* (1998); Engel (1999)		0.005

hexachloran⇒1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE

1,2,3,4,5,6-HEXACHLOROCYCLOHEXANE (hexachloran) *[319-84-6]*

Miryakubova (1970)		0.32

HEXACHLORO-1,3-CYCLOPENTADIENE *[77-47-4]*

Treon *et al.* (1955)		1.7

HEXADECANE *[544-76-3]*

Patte (1978); Patte & Punter (1979)	d	0.5

(*E*,*E*)-2,4-HEXADIENAL (sorbic aldehyde) *[142-83-6]*

Teranishi *et al.* (1974)		0.0018

2,4-HEXADIEN-1-OL *[111-28-4]*

Chisholm & Samuels (1992)		0.787

3,5-HEXADIEN-2-OL *[3280-51-1]*

Chisholm & Samuels (1992)		6.02

(*Z*)-1,5-HEXADIEN-3-OL

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Schieberle & Buettner (2001) 0.01 - 0.1

(*Z*)-1,5-HEXADIEN-3-ONE
Schieberle & Buettner (2001) ca. 0.000 01

hexahydroazine⇒PIPERIDINE

1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN (galaxolide) *[1222-05-5]*

Baydar *et al.* (1992a,1993)	d	0.000 489
Frater *et al.* (1997); Kraft & Fráter (2001)		0.000 9 - 0.0017
Traynor (2001)		0.042

(-)-(4*S*,7*R*)-1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN
Frater *et al.* (1997); Fráter *et al.* (1999) 0.000 3 - 0.000 63

(-)-(4*S*,7*S*)-1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN
Frater *et al.* (1997); Fráter *et al.* (1999) 0.000 6 - 0.001

(+)-(4*R*,7*S*)-1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN
Frater *et al.* (1997); Fráter *et al.* (1999) 0.13 - 0.2

(+)-(4*R*,7*R*)-1,3,4,6,7,8-HEXAHYDRO-4,6,6,7,8,8-HEXAMETHYLCYCLOPENTA[g]-2-BENZOPYRAN
Frater *et al.* (1997); Fráter *et al.* (1999) 0.4 - 0.44

(1*R**,1'*R**)-3',4',5',6',7',8'-HEXAHYDRO-3,3,1',8',8'-PENTAMETHYL-1'*H*-SPIRO(CYCLOPENTANE-1,2'-NAPHTHALEN)-2-ONE
Kraft *et al.* (2006a) 0.000 7

(1*R**,3*R**,1'*R**)-3',4',5',6',7',8'-HEXAHYDRO-3,1',8',8'-TETRAMETHYL-1'*H*-SPIRO(CYCLOPENTANE-1,2'-NAPHTHALEN)-2-ONE
Kraft *et al.* (2006a) 0.0018

(1*R**,3*S**,1'*R**)-3',4',5',6',7',8'-HEXAHYDRO-3,1',8',8'-TETRAMETHYL-1'*H*-SPIRO(CYCLOPENTANE-1,2'-NAPHTHALEN)-2-ONE
Kraft *et al.* (2006a) 0.000 094

(1*R**,1'*S**)-3',4',5',6',7',8'-HEXAHYDRO-1',8',8'-TRIMETHYL-1'*H*-SPIRO(CYCLOPENTANE-1,2'-NAPHTHALEN)-2-ONE
Kraft *et al.* (2006a) 0.0017

(*R*)-γ-hexalactone⇒(*R*)-4-HYDROXYHEXANOIC ACID LACTONE
(*S*)-γ-hexalactone⇒(*S*)-4-HYDROXYHEXANOIC ACID LACTONE

1-(1,1,2,3,3,6-HEXAMETHYL-1*H*-INDAN-5-YL)ETHANONE (phantolide) *[15323-35-0]*
Kraft & Fráter (2001) 0.0067

1,1,2,3,3,6-hexamethyl-4,5,6,7-tetrahydro-5-indanone⇒4,5,6,7-TETRAHYDRO-1,1,2,3,3,6-HEXAMETHYL-5-INDANONE
3,5,5,6,8,8-hexamethyl-1,2,3,4,5,6,7,8-octahydro-2-naphthalenone⇒1,2,3,4,5,6,7,8-OCTAHYDRO-3,5,5,6,8,8-HEXAMETHYL-2-NAPHTHALENONE

HEXANAL *[66-25-1]*

Dravnieks & Laffort (1972)		0.25
Teranishi *et al.* (1974)		0.039
Eriksson *et al.* (1976)	d	0.067
Logtenberg (1978)	d	0.028
Hendriks (1979)	d	0.029 - 0.054
Hall & Andersson (1983)	d	0.043

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Cristoph (1983)	r	0.02 - 0.04
Ulrich & Grosch (1988b); Grosch (1989)		0.065 - 0.098
Guth (1991)		0.030
Lindell (1991)	d	0.025
Khiari *et al.* (1992)		0.02
Von Ranson & Belitz (1992a)	d	0.034
Von Ranson & Belitz (1992a)	r	0.16
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	0.33
Nagata (2003)	d	0.0011
Komthong *et al.* (2006)		84.5
Yang *et al.* (2008)		0.0011
Ueno *et al.* (2009)		0.000 82 (triangle odor bag method; forced-choice mode)
Ueno *et al.* (2009)		0.0028 (triangle odor bag method; yes/no mode)
Ueno *et al.* (2009)		0.0039 (dynamic olfactometer; forced-choice mode)
Ueno *et al.* (2009)		0.0020 (dynamic olfactometer; yes/no mode)
Cometto-Muñiz & Abraham (2010a)	d	0.0014
Laska & Ringh (2010)	d	0.23

HEXANE *[110-54-3]*

Patty & Yant (1929)		875
Laffort & Dravnieks (1973)		230
De Wijk (1989)		107
Nagata (2003)	d	5.3

1,6-HEXANEDIAMINE *[124-09-4]*

Kulakov (1964)		0.0032

1-HEXANETHIOL (hexyl mercaptan) *[111-31-9]*

Kay *et al.* (1967)		11.8
Patte (1978); Patte & Punter (1979)	d	0.000 02
Nagata (2003)	d	0.000 072

HEXANOIC ACID (caproic acid) *[142-62-1]*

Passy (1893b,1893c)	d	0.04
Tempelaar (1913); Zwaardemaker (1927)	d	0.5 - 0.52
Backman (1917)	r	0.23
Zwaardemaker & Komuro (1921)	d	0.33
Hesse (1926)	r	2.3
Hesse (1927)	r	3.5
Dubrovskaya (1969)		0.07 - 0.1
Punter (1983)	d	0.020
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	0.012
Nagata (2003)	d	0.0029
Wise *et al.* (2007); Miyazawa *et al.* (2009a)	d	0.0047 - 0.010
Cometto-Muñiz & Abraham (2010b)	d	0.0048

1-HEXANOL (hexyl alcohol) *[111-27-3]*

Backman (1917)	r	1.0 - 1.3
Mullins (1955)	r	9.94
Pliška (1962)		65
Cain (1969)	r	3.5
Stone *et al.* (1972)	d	1.5
Dravnieks & Laffort (1972)		0.01
Dravnieks (1974)	d	0.3
Hellman & Small (1974)	d	0.04

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Hellman & Small (1974)	r	0.38
Punter (1983)	d	1.93
Cristoph (1983)	r	0.10 - 0.15
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	4.0
Ferreira *et al.* (1998)		0.74
Nagata (2003)	d	0.025
Komthong *et al.* (2006)		12.3
Cometto-Muñiz & Abraham (2008)	d	0.034
2-HEXANOL *[626-93-7]*		
Dravnieks (1974)	d	50
2-HEXANONE (methyl butyl ketone) *[591-78-6]*		
Backman (1917)	r	0.28 - 0.35
Hall & Andersson (1983)	d	4.7
Nagata (2003)	d	0.098
2-HEXANOYL-1-PYRROLINE		
Schieberle (1991)		> 2
(*E*)-2-HEXENAL (leaf aldehyde) *[6728-26-3]*		
Teranishi *et al.* (1974)		0.034
Eriksson *et al.* (1976)	d	0.63
Hall & Andersson (1983)	d	0.48
Lea & Ford (1991)		1
Guth & Grosch (1991)		0.05 - 0.2
Von Ranson & Belitz (1992a)	d	0.79
Von Ranson & Belitz (1992a)	r	1.8
Yang *et al.* (2008)		0.0031
(*Z*)-2-HEXENAL *[16635-54-4]*		
Von Ranson & Belitz (1992a)	d	0.088
Von Ranson & Belitz (1992a)	r	1.2
(*E*)-3-HEXENAL *[69112-21-6]*		
Von Ranson & Belitz (1992a)	d	> 0.79
Von Ranson & Belitz (1992a)	r	> 0.79
(*Z*)-3-HEXENAL *[6789-80-6]*		
Teranishi *et al.* (1974)		0.0022
Ulrich & Grosch (1988a)		0.0040 - 0.011
Ulrich & Grosch (1988b)		0.000 4 - 0.0011
Grosch (1989)		0.000 6 - 0.0016
Guth & Grosch (1991)		0.000 09 - 0.000 36
Von Ranson & Belitz (1992a)	d	< 0.0012
Von Ranson & Belitz (1992a)	r	0.0027
(*E*)-4-HEXENAL *[25166-87-4]*		
Teranishi *et al.* (1974)		0.0026
Von Ranson & Belitz (1992a)	d	0.0029
Von Ranson & Belitz (1992a)	r	0.014
5-HEXENAL *[764-59-0]*		
Von Ranson & Belitz (1992a)	d	0.012
Von Ranson & Belitz (1992a)	r	0.027

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1-HEXENE *[592-41-6]*

Nagata (2003)	d	0.48

(*Z*)-3-HEXEN-1-OL (leaf alcohol) *[928-96-1]*

Cain *et al.* (1987)	d	0.078
Guth & Grosch (1991)		0.004 - 0.016
Wünsche *et al.* (1995)	d	0.013

1-HEXEN-3-OL *[4798-44-1]*

Koszinowski & Piringer (1983)		0.07
Schieberle & Buettner (2001)		0.001 - 0.01

1-HEXEN-3-ONE *[1629-60-3]*

Koszinowski & Piringer (1983)		0.000 7
Blank *et al.* (1989)		0.000 02 - 0.000 04
Schieberle & Buettner (2001)		0.000 01 - 0.000 1

(*Z*)-3-HEXENYL ACETATE *[3681-71-8]*

Guth & Grosch (1991)		0.009 - 0.036

2-HEXOXYETHANOL (ethylene glycol hexyl ether) *[112-25-4]*

Nagy (1991)	d	2.5

HEXYL ACETATE *[142-92-7]*

Stone *et al.* (1972)	d	2.3
Acree *et al.* (1976)		0.1
Cristoph (1983)	r	0.07 - 0.09
Cometto-Muñiz & Cain (1991);		
Cometto-Muñiz (1993)	d	3.7
Nagata (2003)	d	0.011
Komthong *et al.* (2006)		73.7 – 179
Cometto-Muñiz *et al.* (2008)	d	0.017

sec-hexyl acetate⇒4-METHYLPENTYL-2 ACETATE
hexyl alcohol⇒1-HEXANOL

HEXYLAMINE *[111-26-2]*

Dravnieks *et al.* (1986)	d	0.062 - 0.13

4-HEXYLBENZALDEHYDE *[49763-69-1]*

Von Ranson & Belitz (1992b)	d	0.31
Von Ranson & Belitz (1992b)	r	0.80

HEXYLBENZENE (1-phenylhexane) *[1077-16-3]*

Cometto-Muñiz (1993);		
Cometto-Muñiz & Cain (1994)	d	4.2
Cometto-Muñiz & Abraham (2009b)	d	0.029

7-HEXYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 019

HEXYL BUTANOATE (hexyl butyrate) *[2639-63-6]*

Lea & Ford (1991)		5

hexyl butyrate⇒HEXYL BUTANOATE

2-HEXYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE (2-hexyl-3,5-dimethyl-5,6-dihydropyrazine)

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Kurniadi *et al.* (2003)		> 2
Kurniadi (2003)	r	> 40

3-HEXYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE (3-hexyl-2,5-dimethyl-5,6-dihydropyrazine)

Kurniadi *et al.* (2003)		> 2
Kurniadi (2003)	r	> 40

5-HEXYLDIHYDRO-2(3*H*)-THIOPHENONE (γ-thiodecalactone)

Roling *et al.* (1998); Engel (1999)		0.003

2-hexyl-3,5-dimethyl-5,6-dihydropyrazine⇒2-HEXYL-5,6-DIHYDRO-3,5-DIMETHYLPYRAZINE
3-hexyl-2,5-dimethyl-5,6-dihydropyrazine⇒3-HEXYL-5,6-DIHYDRO-2,5-DIMETHYLPYRAZINE

2-HEXYL-3,4-DIMETHYLPYRAZINE

Wagner *et al.* (1999)		> 2

2-hexyldimethyltetrahydropyrazine⇒2-HEXYLTETRAHYDRODIMETHYLPYRAZINE
hexylene glycol⇒2-METHYL-2,4-PENTANEDIOL
hexyl mercaptan⇒1-HEXANETHIOL

2-HEXYLPYRIDINE *[1129-69-7]*

Schieberle (1993)		0.001

2-HEXYLTETRAHYDRODIMETHYLPYRAZINE (2-hexyldimethyltetrahydropyrazine)

Kurniadi (2003)	r	> 40

homofuraneol⇒2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE

HYDRAZINE *[302-01-2]*

Jacobson *et al.* (1955)		3.9 - 5.2
Jacobson *et al.* (1958)		5.2

hydrocinnamaldehyde⇒3-PHENYLPROPANAL
hydrocinnamyl alcohol⇒3-PHENYL-1-PROPANOL

HYDROGEN CHLORIDE *[7647-01-0]*

Schley (1934)		4.5
Elfimova (1959)		0.1 - 0.2
Heyroth (1962)	d	1.5 -7.5
Styazhkin (1963)		0.2
Melekhina (1968)	d	0.39
Leonardos *et al.* (1969)	r	15
Takhirov (1974)		0.38
Nauš (1982)	d	7
Nauš (1982)	r	15
Van Thriel *et al.* (2006)	d	0.09

HYDROGEN CYANIDE (prussic acid) *[74-90-8]*

Sherrard (1928)		6
Smolczyk & Cobler (1930)		< 1.1
Prentiss (1937)		1
Artho & Koch (1973)		0.01 - 0.1
Braker & Mossman (1980)	r	2.2 - 5.6

HYDROGEN FLUORIDE *[7664-39-3]*

Sadilova (1968)		0.03

HYDROGEN SELENIDE *[7783-07-5]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Dudley & Miller (1941)		< 1
HYDROGEN SULPHIDE *[7783-06-4]*		
Valentin (1848,1850)		2
Lehmann (1897)		< 2
Kulka & Homma (1910)		0.2 - 0.3
Henderson & Haggard (1922)		< 0.001
Henning (1924)	d	0.000 1
Katz & Talbert (1930)		0.18
Thomas *et al.* (1943)		0.035
Loginova (1957)		0.040
Duan-Fen-Djuy (1959)		0.012 - 0.03
Sanders & Dechant (1961)		0.04 - 0.10
Baikov (1963)		0.014 - 0.03
Young & Adams (1966)		0.008 - 0.011
Cederlof *et al.* (1966)		0.01
Sakuma *et al.* (1967)		0.007
Endo *et al.* (1967)		1.4
Basmadzhieva & Argirova (1968)		0.012
Adams *et al.* (1968)		0.0047 - 0.0090
Leonardos *et al.* (1969)	r	0.000 66 - 0.0066
Pomeroy & Cruse (1969)		0.0042 - 0.042
Wilby (1969)	r	0.0063
Lindvall (1970)	d	0.000 21 - 0.0016
Stephens (1971)		0.067
Randebrock (1971)		0.012
Nishida *et al.* (1975)	d	0.0014 - 0.055
Winkler (1975)	d	0.003
Winkler (1975)	r	0.03
Hill & Barth (1976)		0.0007
Williams *et al.* (1977)	d	0.27
Logtenberg (1978)	d	0.002
Nishida *et al.* (1979)	d	0.074
Winneke *et al.* (1979)	d	0.00265
Thiele (1979)		0.0016
Bedborough & Trott (1979)	d	0.0036
Brunekreef & Harssema (1980)		0.0011 - 0.0024
Anon. (1980)	d	0.000 7
Anon. (1980)	r	0.0078
Thiele *et al.* (1981)		0.0013 - 0.0053
Thiele (1982)		0.0028
Nauš (1982)	d	0.1
Nauš (1982)	r	5
Jensen & Flyger (1983)		0.0038 - 0.0067
Kobal & Thiele (1983)		0.0022
Bahnmüller (1983)		0.0014 - 0.023
Moriguchi *et al.* (1983)	d	0.000 7
Bahnmüller (1984)		0.0012 - 0.0073
Thiele (1984)		0.0018
Roos *et al.* (1985)	d	0.000 85 -0.00105
Roos *et al.* (1985) & Don (1986);		
Hoshika *et al.* (1993)	d	0.000 4 - 0.000 43
Randebrock (1986)		0.0096
Heeres *et al.* (1986)		0.000 4 - 0.0052
Dollnick *et al.* (1988)		0.00166
Winneke *et al.* (1988)		0.0015 - 0.0026
Hermans (1989)		0.000 056 - 0.001545

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Nagy (1991)	d	0.0055
Hoshika *et al.* (1993)	d	0.000 7
Lötsch *et al.* (1997)		0.14 - 2.8
Mannebeck & Mannebeck (2002)	d	0.000 491 - 0.000 946
Nagata (2003)	d	0.000 57
Greenman *et al.* (2004)		0.022
McGinley & McGinley (2004)		0.000 70 - 0.003
McGinley & McGinley (2004)	r	0.000 64 - 0.0013
McGinley & McGinley (2004)	d	0.000 57 - 0.00142
McGinley & McGinley (2004)	r	0.000 71 - 0.0032
Glindemann *et al.* (2006)	d	0.001
Ueno *et al.* (2009)		0.000 45 (triangle odor bag method; forced-choice mode)
Ueno *et al.* (2009)		0.0018 (dynamic olfactometer; forced-choice mode)

2-HYDROXYBENZALDEHYDE (salicylaldehyde) *[90-02-8]*

Tempelaar (1913)	d	0.014
Backman (1917)	r	0.38 - 0.41

3-HYDROXYBUTANOIC ACID *[300-85-6]*

Backman (1917)	r	42

3-HYDROXY-3-BUTENOIC ACID LACTONE (diketene) *[674-82-8]*

Feldman (1967)	0.019

hydroxycitronellal⇒7-HYDROXY-3,7-DIMETHYLOCTANAL

(*R*)-4-HYDROXYDECANOIC ACID LACTONE ((*R*)-γ-decalactone)

Buhr (2006)	0.0125

(*S*)-4-HYDROXYDECANOIC ACID LACTONE ((*S*)-γ-decalactone)

Buhr (2006)	0.0125

(*R*)-5-HYDROXYDECANOIC ACID LACTONE ((*R*)-δ-decalactone)

Buhr (2006)	0.000 6

(*S*)-5-HYDROXYDECANOIC ACID LACTONE ((*S*)-δ-decalactone)

Buhr (2006)	0.0020

2-HYDROXY-3-DECANONE

Neuser *et al.* (2000)	r	1 - 1.2

3-HYDROXY-2-DECANONE

Neuser *et al.* (2000)	r	7 - 8

2-HYDROXY-3,4-DIMETHYL-2-CYCLOPENTEN-1-ONE (3,4-dimethylcyclopentenolone) *[21835-00-7]*

Blank *et al.* (1992); Blank (1997, 2002)	0.000 05 - 0.000 1 [GC-O on FFAP capillary column]
Blank (1997, 2002)	0.001 - 0.002 [GC-O on OV-1701 capillary column]

3-HYDROXY-4,5-DIMETHYL-2(5*H*)-FURANONE (sotolon) *[28664-35-9]*

Blank *et al.* (1992); Blank & Schieberle (1993); Hofmann & Schieberle (1997);	
Blank (1997, 2002)	0.000 01 - 0.000 02 [GC-O on FFAP capillary column]
Blank (1997, 2002)	0.000 6 – 0.0012 [GC-O on OV-1701 capillary column]

4-HYDROXY-2,5-DIMETHYL-3(2*H*)-FURANONE (furaneol) *[3658-77-3]*

Blank *et al.* (1992); Blank & Schieberle (1993); Schieberle & Grosch (1994); Blank (1997, 2002)	0.000 5 - 0.0015[GC-O on FFAP capillary column]

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Blank (1997, 2002) 0.001 - 0.002 [GC-O on OV-1701 capillary column]

7-HYDROXY-3,7-DIMETHYLOCTANAL (hydroxycitronellal) *[107-75-5]*
Randebrock (1986) 0.000 0069

(1*R**,4*S**)-1-HYDROXY-1,4-DIMETHYLSPIRO[4.6]-2-UNDECANONE
Kraft *et al.* (2005) 0.184

4-HYDROXY-2,5-DIMETHYL-3(2*H*)-THIOPHENE
Hofmann & Schieberle (1997) 0.0016

(±)-(1*R**,3*S**,6*S**,7*S**,8*S**)-3-HYDROXY-6,8-DIMETHYLTRICYCLO[5.3.1.$0^{3,8}$]UNDECAN-2-ONE
Kraft *et al.* (2006b) 0.0021

(±)-(1*R**,3*S**,7*S**,8*S**)-3-HYDROXY-6,8-DIMETHYLTRICYCLO[5.3.1.$0^{3,8}$]UNDEC-5-EN-2-ONE
Kraft *et al.* (2006b) 0.0014

(*R*)-4-HYDROXYDODECANOIC ACID LACTONE ((*R*)-γ-dodecalactone)
Buhr (2006) 0.000 8

(*S*)-4-HYDROXYDODECANOIC ACID LACTONE ((*S*)-γ-dodecalactone)
Buhr (2006) 0.0060

(*R*)-5-HYDROXYDODECANOIC ACID LACTONE ((*R*)-δ-dodecalactone)
Buhr (2006) 0.012

(*S*)-5-HYDROXYDODECANOIC ACID LACTONE ((*S*)-δ-dodecalactone)
Buhr (2006) 0.000 4

(*R*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE ((*R*)-*cis*-6-γ-dodecenolactone)
Guichard *et al.* (1991) 0.000 062 - 0.000 082

(*S*)-(*Z*)-4-HYDROXY-6-DODECENOIC ACID LACTONE ((*S*)-*cis*-6-γ-dodecenolactone)
Guichard *et al.* (1991) 0.000 002 - 0.000 052

rac-(1-HYDROXYETHYL)DIISOPROPYL(METHYL)SILANE
Sunderkötter *et al.* (2010) 0.026

(*R*)-4-HYDROXYHEPTANOIC ACID LACTONE ((*R*)-γ-heptalactone)
Buhr (2006) 0.0050

(*S*)-4-HYDROXYHEPTANOIC ACID LACTONE ((*S*)-γ-heptalactone)
Buhr (2006) 0.0050

2-HYDROXY-3-HEPTANONE *[71467-29-3]*
Neuser *et al.* (2000) r 1 - 1.5

3-HYDROXY-2-HEPTANONE
Neuser *et al.* (2000) r 4 - 5

2-HYDROXY-4*E*-HEPTEN-3-ONE
Neuser *et al.* (2000) r 0.5 - 0.8

3-HYDROXY-4*E*-HEPTEN-2-ONE
Neuser *et al.* (2000) r 2.5 – 3

16-HYDROXY-7-HEXADECENOIC ACID LACTONE (ambrettolide) *[28645-51-4]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Traynor (2001)		0.124

(1*R**,4*R**)-1-HYDROXY-1,4,7,7,9,9-HEXAMETHYLSPIRO[4.5]-2-DECANONE

Kraft *et al.* (2005)		0.051

(*R*)-4-HYDROXYHEXANOIC ACID LACTONE ((*R*)-γ-hexalactone)

Buhr (2006)		0.375

(*S*)-4-HYDROXYHEXANOIC ACID LACTONE ((*S*)-γ-hexalactone)

Buhr (2006)		0.375

(*R*)-5-HYDROXYHEXANOIC ACID LACTONE ((*R*)-δ-hexalactone)

Buhr (2006)		> 0.1

(*S*)-5-HYDROXYHEXANOIC ACID LACTONE ((*S*)-δ-hexalactone)

Buhr (2006)		> 0.1

2-HYDROXY-3-HEXANONE

Neuser *et al.* (2000)	r	5 - 6

3-HYDROXY-2-HEXANONE

Neuser *et al.* (2000)	r	9 - 10

4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE (α-terpineol, α-terpinol) *[98-55-5]*

Hermanides (1909)	r	0.185
Zwaardemaker (1914,1927)	d	0.18 - 0.2
Backman (1917)	r	0.15 - 0.23
Zwaardemaker & Komuro (1921)	d	0.39
Baldus (1936)	d	0.01
Baldus (1936)	r	0.0125
Cristoph (1983)	r	100 - 110
Ferreira *et al.* (1998)		0.0208
Molhave *et al.* (2000)	d	0.86

4-HYDROXY-3-METHOXYBENZALDEHYDE (vanillin, 4-formyl-2-methoxyphenol) *[121-33-5]*

Passy (1892a,b,d)	d	0.000 07 - 0.005
Tempelaar (1913); Zwaardemaker (1927)	d	0.000 18 - 0.000 2
Backman (1917)	r	0.0015 - 0.0020
Baldus (1936)	d	0.000 001
Baldus (1936)	r	0.000 004
Appell (1969)		0.000 001
Randebrock (1971)		0.000 006
Herrmann & Abd El Salam (1980a,b)		0.08 - 0.12
Kleinschmidt (1983)	r	0.578
Randebrock (1986)		0.000 033
Blank *et al.* (1989,1992)		0.000 6 - 0.0012

2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE (3-methyl-1,2-cyclopentanedione, cyclotene, methylcyclopentenolone, maple lactone) *[80-71-7, 765-70-8]*

Blank (1997, 2002)		0.01 – 0.02 [GC-O with SE-54 capillary column]
Blank (1997, 2002)		0.01 – 0.02 [GC-O with FFAP capillary column]
Miyazawa *et al.* (2009a)	d	0.026

cis-(5*R*)-HYDROXY-(4*R*)-METHYLDECANOIC ACID LACTONE

Bartschat *et al.* (1995)		0.3

cis-(5*S*)-HYDROXY-(4*S*)-METHYLDECANOIC ACID LACTONE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Bartschat *et al.* (1995) 0.0013

trans-(5*R*)-HYDROXY-(4*S*)-METHYLDECANOIC ACID LACTONE
Bartschat *et al.* (1995) 0.5

trans-(5*S*)-HYDROXY-(4*R*)-METHYLDECANOIC ACID LACTONE
Bartschat *et al.* (1995) 1

(1-HYDROXY-1-METHYLETHYL)DIISOPROPYL(METHYL)SILANE
Sunderkötter *et al.* (2010) 0.010

rac-(1-HYDROXY-1-METHYLETHYL)ISOBUTYL(ISOPROPYL)METHYLSILANE
Sunderkötter *et al.* (2010) 0.114

(1-HYDROXY-1-METHYLETHYL)TRIISOPROPYLSILANE
Sunderkötter *et al.* (2010) 0.0025

(1-HYDROXY-1-METHYLETHYL)TRIVINYLSILANE
Sunderkötter *et al.* (2010) 0.0045

3-HYDROXY-4-METHYLHEPTANOIC ACID
Natsch *et al.* (2006) 0.109 (GC-odor threshold)
Natsch *et al.* (2006) 0.000 17 (dynamic dilution olfactometry)

2-HYDROXY-4-METHYL-3-HEPTANONE
Neuser *et al.* (2000) r 0.8 - 1.2

3-HYDROXY-4-METHYL-2-HEPTANONE
Neuser *et al.* (2000) r 2 - 2.5

2-HYDROXY-5-METHYL-3-HEXANONE *[246511-74-0]*
Neuser *et al.* (2000) r 4 - 5

3-HYDROXY-5-METHYL-2-HEXANONE *[163038-04-8]*
Neuser *et al.* (2000) r 8 - 9

4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE (3-methyl-4-hydroxyoctanoic acid lactone, β-methyl-γ-octalactone, oak lactone, whisky lactone) *[39212-23-2]*
Atanasova *et al.* (2005) d 0.0031 - 0.0475
Atanasova *et al.* (2005) r 0.0049 - 0.0775

cis-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE (*cis*-β-methyl-γ-octalactone) *[55013-32-6]*
Abbott *et al.* (1995) 0.000 001

trans-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE (*trans*-β-methyl-γ-octalactone) *[39638-67-0]*
Abbott *et al.* (1995) 0.000 02

(12*R*)-(-)-14-HYDROXY-12-METHYL-9-OXATETRADECANOIC ACID LACTONE
((-)-(12*R*)-12-methyl-9-oxa-14-tetradecanolide)
Kraft & Fráter (2001) 0.000 09

4-HYDROXY-4-METHYL-2-PENTANONE (diacetone alcohol) *[123-42-2]*
Hellman & Small (1974) d 1.3
Hellman & Small (1974) r 5.2
Nagy (1991) d 60
Nagy (1991) d 37.418

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3-HYDROXY-6-METHYL-2(2*H*)-PYRANONE
Hofmann & Schieberle (1997) 0.000 76

1-HYDROXY-1-METHYLSPIRO[4.5]-2-DECANONE
Kraft *et al.* (2005) 0.0635

(12*R*)-14-HYDROXY-12-METHYL-9(*Z*)-TETRADECENOIC ACID LACTONE ((12*R*;9*Z*)-12-methyl-14-tetradeca-9-enolide)
Kraft & Fráter (2001) 0.000 05

2-HYDROXY-5-(METHYLTHIO)-3-PENTANONE
Neuser *et al.* (2000) r 0.15 - 0.2

3-HYDROXY-5-(METHYLTHIO)-2-PENTANONE
Neuser *et al.* (2000) r 0.05 - 0.1

1-HYDROXYNAPHTHALENE (1-naphthol) *[90-15-3]*
Backman (1917) r 0.0030 - 0.0052

2-HYDROXYNAPHTHALENE (2-naphthol) *[135-19-3]*
Backman (1917) r 0.23 - 0.30

4-HYDROXYNONANOIC ACID LACTONE (γ-nonalactone, 4-nonalide) *[104-61-0]*
Appell (1969) 0.000 000 8

(*R*)-4-HYDROXYNONANOIC ACID LACTONE ((*R*)-γ-nonalactone)
Buhr (2006) 0.0045

(*S*)-4-HYDROXYNONANOIC ACID LACTONE ((*S*)-γ-nonalactone)
Buhr (2006) 0.0045

(*R*)-5-HYDROXYNONANOIC ACID LACTONE ((*R*)-δ-nonalactone)
Buhr (2006) 0.0013

(*S*)-5-HYDROXYNONANOIC ACID LACTONE ((*S*)-δ-nonalactone)
Buhr (2006) 0.0020

2-HYDROXY-3-NONANONE
Neuser *et al.* (2000) r 1 - 1.2

3-HYDROXY-2-NONANONE
Neuser *et al.* (2000) r 5 - 6

(*R*)-4-HYDROXYOCTANOIC ACID LACTONE ((*R*)-γ-octalactone)
Buhr (2006) 0.0018

(*S*)-4-HYDROXYOCTANOIC ACID LACTONE ((*S*)-γ-octalactone)
Buhr (2006) 0.0018

(*R*)-5-HYDROXYOCTANOIC ACID LACTONE ((*R*)-δ-octalactone)
Buhr (2006) 0.0125

(*S*)-5-HYDROXYOCTANOIC ACID LACTONE ((*S*)-δ-octalactone)
Buhr (2006) 0.0125

2-HYDROXY-3-OCTANONE
Neuser *et al.* (2000) r 0.4 - 0.5

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3-HYDROXY-2-OCTANONE
Neuser *et al.* (2000) r 2.5 - 3

2-HYDROXY-4(*E*)-OCTEN-3-ONE
Neuser *et al.* (2000) r 0.2 - 0.3

3-HYDROXY-4(*E*)-OCTEN-2-ONE
Neuser *et al.* (2000) r 0.8 – 1

16-HYDROXY-10-OXAHEXADECANOIC ACID LACTONE (10-oxahexadecanolide, oxalide T) *[1725-01-5]*
Traynor (2001) 0.094

16-HYDROXY-11-OXAHEXADECANOIC ACID LACTONE (11-oxahexadecanolide, musk R-1) *[3391-83-1]*
Traynor (2001) 0.084

16-HYDROXY-12-OXAHEXADECANOIC ACID LACTONE (12-oxahexadecanolide, cervolide) *[6707-60-4]*
Traynor (2001) 0.168

15-HYDROXYPENTADECANOIC ACID LACTONE (exaltolide, pentadecalactone, cyclopentadecanolide, thibetolid, 15-pentadecanolide) *[106-02-5]*
Amoore (1977) d 0.18
Randebrock (1986) 0.000 013
Kraft & Fráter (2001) 0.0021
Traynor (2001) 0.098

15-HYDROXY-11(12)-PENTADECENOIC ACID LACTONE (habanolide) *[34902-57-3]*
Traynor (2001) 0.166

(1*R**,4*R**,5*S**,9*R**)-1-HYDROXY-1,4,7,7,9-PENTAMETHYLSPIRO[4.5]-2-DECANONE
Kraft *et al.* (2005) 0.018

(1*R**,4*S**,5*S**,9*R**)-1-HYDROXY-1,4,7,7,9-PENTAMETHYLSPIRO[4.5]-2-DECANONE
Kraft *et al.* (2005) 0.000 067

(+)-(1*S*,4*R*,5*R*,9*S*)-1-HYDROXY-1,4,7,7,9-PENTAMETHYLSPIRO[4.5]-2-DECANONE
Kraft *et al.* (2005) 0.000 027

3-HYDROXY-2-PENTANONE *[3142-66-3]*
Neuser *et al.* (2000) r 5 - 6

2-HYDROXY-3-PENTANONE *[5704-20-1]*
Neuser *et al.* (2000) r 2.5 - 3

3-HYDROXY-4-PHENYL-2-BUTANONE *[5355-63-5]*
Neuser *et al.* (2000) r 0.75 - 1

2-HYDROXYPROPANOIC ACID (lactic acid) *[50-21-5]*
Endo *et al.* (1967) 9

(*R*)-4-HYDROXYTETRADECANOIC ACID LACTONE ((*R*)-δ-tetradecalactone)
Buhr (2006) 0.188

(*S*)-4-HYDROXYTETRADECANOIC ACID LACTONE ((*S*)-δ-tetradecalactone)
Buhr (2006) 0.188

(1*R**,4*S**,5*R**,7*R**,9*S**)-1-HYDROXY-1,4,7,9-TETRAMETHYLSPIRO[4.5]-2-DECANONE

Kraft *et al.* (2005) 0.032

(1*R**,4*R**,5*R**,7*R**,9*S**)-1-HYDROXY-1,4,7,9-TETRAMETHYLSPIRO[4.5]-2-DECANONE
Kraft *et al.* (2005) 0.335

(±)-(5*R**,6*R**)-6-HYDROXY-1,1,6-TRIMETHYLSPIRO[4.5]DECAN-7-ONE
Kraft & Bruneau (2007) 0.0172

(±)-6-HYDROXY-2,2,6-TRIMETHYLSPIRO[4.5]DECAN-7-ONE
Kraft & Bruneau (2007) 0.233

4-HYDROXYUNDECANOIC ACID LACTONE (γ-undecalactone, 4-undecanolide) *[104-67-6]*
Appell (1969) 0.000 000 6
Huber (1984) 0.000 044

4-HYDROXYUNDECANOIC ACID LACTONE (γ-undecalactone)
Buhr (2006) 0.000 8

4-HYDROXYUNDECANOIC ACID LACTONE (γ-undecalactone)
Buhr (2006) 0.0045

5-HYDROXYUNDECANOIC ACID LACTONE (δ-undecalactone)
Buhr (2006) 0.060

5-HYDROXYUNDECANOIC ACID LACTONE (δ-undecalactone)
Buhr (2006) 0.000 8

I

INDENE *[95-13-6]*

Deadman & Prigg (1959)	d	0.02
Moriguchi *et al.* (1983)	d	0.013

INDOLE *[120-79-9]*

Tempelaar (1913)	d	0.000 6
Punter (1975,1979	d	0.0071
Etzweiler *et al.* (1980)		0.000 04
Randebrock (1986)		0.000 033
Nagata (2003)	d	0.0014
Yang *et al.* (2008)		0.0081

IODINE *[7553-56-2]*
Randebrock (1986) 10.1

IODOBENZENE *[591-50-4]*
Backman (1917) r 0.024 - 0.037

1-IODOBUTANE (butyl iodide) *[542-69-8]*

Dravnieks & Laffort (1972)		4.6
Dravnieks (1974)	d	60

IODOETHANE (ethyl iodide) *[75-03-6]*
Backman (1917) r 8 - 9

iodoform⇒TRIIODOMETHANE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1-IODO-2-METHYLBENZENE (2-iodotoluene) *[615-37-2]*

Backman (1917)	r	0.009

2-IODOPHENOL *[533-58-4]*

Kendall *et al.* (1968)	r	0.000 009

2-iodotoluene⇒1-IODO-2-METHYLBENZENE
α-ionone⇒4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(*R*)-(+)-α-ionone⇒(*R*)-(+)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(*S*)-(-)-α-ionone⇒(*S*)-(-)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(*d*)-α-ionone⇒(*R*)-(+)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(*l*)-α-ionone⇒(*S*)-(-)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
β-ionone⇒4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE
(+)-*cis*-γ-irone⇒(+)-4-((1*S*,5*R*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
(-)-*cis*-γ-irone⇒(-)-4-((1*R*,5*S*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
(+)-*trans*-γ-irone⇒(+)-4-((1*R*,5*R*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
(-)-*trans*-γ-irone⇒(-)-4-((1*S*,5*S*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
α-irone⇒4-(*cis*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
isoamyl acetate⇒ISOPENTYL ACETATE
isoamyl alcohol⇒3-METHYL-1-BUTANOL
isoamyl butyrate⇒ISOPENTYL BUTANOATE
isoamyl isovalerate⇒ISOPENTYL 3-METHYLBUTANOATE
isoamyl mercaptan⇒3-METHYL-1-BUTANETHIOL
isoamyl propylate⇒ISOPENTYL PROPANOATE
isobutane⇒2-METHYLPROPANE
isobutanol⇒2-METHYL-1-PROPANOL

2-ISOBUTOXYETHANOL (isobutyl cellosolve) *[4439-24-1]*

Hellman & Small (1974)	d	0.09
Hellman & Small (1974)	r	0.55

ISOBUTYL ACETATE *[110-19-0]*

Backman (1917)	r	1.9 - 2.1
May (1966)	d	17
May (1966)	r	34
Hellman & Small (1974)	d	1.7
Hellman & Small (1974)	r	2.4
Cristoph (1983)	r	0.42 - 0.52
Nagata (2003)	d	0.038
Komthong *et al.* (2006)		21.1 - 612

isobutyl acrylate⇒ISOBUTYL 2-PROPENOATE
isobutyl alcohol⇒2-METHYL-1-PROPANOL

ISOBUTYLAMINE *[78-81-9]*

Nagata (2003)	d	0.0045

ISOBUTYL BUTANOATE (isobutyl butyrate, 2-methylpropyl butanoate) *[539-90-2]*

Backman (1917)	r	0.06 - 0.10
Laing (1975)	d	18
Nagata (2003)	d	0.0094

isobutyl butyrate⇒ISOBUTYL BUTANOATE
isobutyl cellosolve⇒2-ISOBUTOXYETHANOL

4-ISOBUTYL-2,6-DIMETHYL-3,5-DINITROACETOPHENONE

Huber (1984)	0.000 1

5-ISOBUTYL-2,3-DIMETHYLPYRAZINE *[32263-00-6]*

Wagner *et al.* (1999)	> 2

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3-ISOBUTYL-2,5-DIMETHYLPYRAZINE *[32736-94-0]*

Wagner *et al.* (1999)		> 2

2-ISOBUTYL-3,5-DIMETHYLPYRAZINE *[70303-42-3]*

Wagner *et al.* (1999)		> 2

isobutylene⇒2-METHYLPROPENE

ISOBUTYL FORMATE *[542-55-2]*

Nagata (2003)	d	2.1

isobutyl isobutyrate⇒ISOBUTYL 2-METHYLPROPANOATE
isobutyl isovalerate⇒ISOBUTYL 3-METHYLBUTANOATE
isobutyl mercaptan⇒2-METHYL-1-PROPANETHIOL

2-ISOBUTYL-3-METHOXYPYRAZINE (pepper pyrazine) *[24683-00-9]*

Blank *et al.* (1992)		0.000 002 - 0.000 004
Khiari *et al.* (1992)		0.000 005
Wagner *et al.* (1999)		0.000 003

ISOBUTYL 3-METHYLBUTANOATE (isobutyl isovalerate) *[589-59-3]*

Backman (1917)	r	1.6
Nagata (2003)	d	0.034

ISOBUTYL 2-METHYLPROPANOATE (isobutyl isobutyrate) *[97-85-8]*

Punter (1983)	d	3.7
Nagy (1991)	d	2.9
Nagata (2003)	d	0.44

(-)-(2*R*,4*R*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL ((-)-(2*R*,4*R*)-florol®)

Abate *et al.* (2004)		0.00121

(-)-(2*R*,4*S*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL ((-)-(2*R*,4*S*)-florol®)

Abate *et al.* (2004)		0.52

(+)-(2*S*,4*R*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL ((+)-(2*S*,4*R*)-florol®)

Abate *et al.* (2004)		>0.6

(+)-(2*S*,4*S*)-2-ISOBUTYL-4-METHYLTETRAHYDRO-2*H*-PYRAN-4-OL ((+)-(2*S*,4*S*)-florol®)

Abate *et al.* (2004)		0.026

ISOBUTYL PROPANOATE *[540-42-1]*

Backman (1917)	r	0.39 - 0.43
Nagata (2003)	d	0.11

ISOBUTYL 2-PROPENOATE (isobutyl acrylate) *[106-63-8]*

Hellman & Small (1974)	d	0.01
Hellman & Small (1974)	r	0.05
Nagata (2003)	d	0.0047

isobutyraldehyde⇒2-METHYLPROPANAL
isobutyric acid⇒2-METHYLPROPANOIC ACID
isocaproic acid⇒4-METHYLPENTANOIC ACID
iso-α-irone⇒4-(*trans*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
(*d*)-*iso*-α-irone⇒(+)-4-(*trans*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
(*l*)-*iso*-α-irone⇒(-)-4-(*trans*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE
isomintlactone⇒(6*R*,7a*S*)-3,6-DIMETHYL-5,6,7-7a-TETRAHYDRO-2(4*H*)-BENZOFURANONE
isooctanol⇒2-METHYL-1-HEPTANOL

isooctylalcohol⇒2-METHYL-1-HEPTANOL
isopentane⇒2-METHYLBUTANE
isopentanol⇒3-METHYL-1-BUTANOL

ISOPENTYL ACETATE (isoamyl acetate, 3-methylbutyl acetate) *[123-92-2]*

Hermanides (1909)	r	0.09
Zwaardemaker (1914)	d	0.09
Backman (1917)	r	0.18 - 0.29
Katz & Talbert (1930)		0.018
Jung (1936)	d	0.2
Jung (1936)	r	0.2
Kerka & Humphreys (1956)		0.2
Pliška & Janiček (1965)		5
Appell (1969)		0.004
Nishida *et al.* (1979)	d	1,100
Punter (1983)	d	0.070 - 0.084
Cristoph (1983)	r	0.015 - 0.020
Don (1986)	d	0.075
Lea & Ford (1991)		0.5
Laska & Hudson (1991)	d	0.13 - 0.14
Hoshika *et al.* (1997)	r	8
Langridge (2004)		0.2259 orthonasal
Langridge (2004)		0.0107 retronasal
Atanasova *et al.* (2005)	d	0.018 - 0.919
Atanasova *et al.* (2005)	r	0.067 - 0.918
Komthong *et al.* (2006)		1,950

ISOPENTYL BUTANOATE (isoamyl butyrate) *[106-27-4]*

Backman (1917)	r	0.13 - 0.14
Cain & Gent (1991)	d	0.07 - 0.20

ISOPENTYL FORMATE *[110-45-2]*

Komthong *et al.* (2006)		149

ISOPENTYL 3-METHYLBUTANOATE (isoamyl isovalerate) *[659-70-1]*

Backman (1917)	r	0.64
Katz & Talbert (1930)		0.046

ISOPENTYL PROPANOATE (isoamyl propylate) *[105-68-0]*

Backman (1917)	r	0.17 - 0.23
Komthong *et al.* (2006)		31.3

isophorone⇒3,5,5-TRIMETHYL-2-CYCLOHEXEN-1-ONE
isopral⇒1,1,1-TRICHLORO-2-PROPANOL
isoprene⇒2-METHYL-1,3-BUTADIENE
isopropanol⇒2-PROPANOL

ISOPROPENYLBENZENE (α-methylstyrene) *[98-83-9]*

Wolf *et al.* (1956)		48 - 240
Minaev (1966)		0.1
Hellman & Small (1974)	d	0.25
Hellman & Small (1974)	r	0.75
Nagy (1991)	d	2.2

ISOPROPENYL CYANIDE (methacrylonitrile, 2-methyl-2-propenenitrile) *[126-98-7]*

Pozzani *et al.* (1968)	d	19
Nagata (2003)	d	8.1

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE (perillaldehyde) *[2111-75-3]*

Cristoph (1983)	r	0.26 - 0.34

1-ISOPROPENYL-4-METHYLBENZENE (α,*p*-dimethylstyrene) *[1195-32-0]*

Schieberle *et al.* (1988)		0.665 - 2.660

2-ISOPROPENYL-5-METHYL-1-CYCLOHEXANONE (isopulegone) *[529-00-0]*

Benzo *et al.* (2007)		0.00 076

4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE (limonene) *[138-86-3]*

Fuller *et al.* (1964)	r	0.058
Appell (1969)		0.01
Dravnieks *et al.* (1986)	d	1.7
Nagata (2003)	d	0.21
Langridge (2004)		1.6878 orthonasal
Langridge (2004)		0.0539 retronasal

(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE ((*d*)-limonene, (4*R*)-(+)-limonene) *[5989-27-5]*

Cristoph (1983)	r	2.2 - 2.3
Randebrock (1986)		0.0059
Stevens & Cain (1987a)	d	0.056 - 5.6
De Wijk (1989)		4.70
Blank & Grosch (1991a)		0.7
Jagella & Grosch (1998)		0.135
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	55
Buettner & Schieberle (2001a,2001b)		0.424
Benzo *et al.* (2007)		0.113 - 0.137
Cain *et al.* (2007a)	d	0.045

(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE ((*l*)-limonene, (4*S*)-(-)-limonene) *[5989-54-8]*

Randebrock (1986)		0.0084
Jagella & Grosch (1998)		0.270
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	25
Molhave *et al.* (2000)	d	12

5-ISOPROPENYL-1-METHYL-2-CYCLOHEXEN-1-ONE (carvone) *[99-49-0]*

Tempelaar (1913); Zwaardemaker (1927)	d	0.5 - 0.55
Langridge (2004)		0.0077 orthonasal
Langridge (2004)		0.000 2 retronasal

(*S*)-(+)-5-ISOPROPENYL-1-METHYL-2-CYCLOHEXEN-1-ONE ((*d*)-carvone) *[2244-16-8]*

Cristoph (1983)	r	0.30 - 0.40
Blank & Grosch (1991a)		0.03

(*R*)-(-)-5-ISOPROPENYL-1-METHYL-2-CYCLOHEXEN-1-ONE ((*l*)-carvone) *[6485-40-1]*

Amoore (1982)	d	0.034
Cristoph (1983)	r	0.10 - 0.12
Marin *et al.* (1988)		0.017 - 0.025
Schieberle & Grosch (1989);		
Rychlik *et al.* (1998)		0.0166
Laska & Hudson (1991)	d	0.085 - 0.15

3-(3-ISOPROPENYLPHENYL)BUTANAL (3-(3-propen-2-ylphenyl)butanal)

Chalk (1990)		0.000 7

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-ISOPROPOXYPROPANE (isopropyl ether) *[108-20-3]*

Hellman & Small (1974)	d	0.07
Hellman & Small (1974)	r	0.22

ISOPROPYL ACETATE *[108-21-4]*

Backman (1917)	r	27 - 33
Jung (1936)	d	1.9
Jung (1936)	r	1.9 - 2.9
May (1966)	d	140
May (1966)	r	170
Hellman & Small (1974)	d	2.1
Hellman & Small (1974)	r	3.8
Scharfenberger (1990)		68
Nagy (1991)	d	9.4
Nagata (2003)	d	0.67

ISOPROPYLAMINE *[75-31-0]*

Hellman & Small (1974)	d	0.5
Hellman & Small (1974)	r	1.7
Nagata (2003)	d	0.06

isopropyl alcohol⇒2-PROPANOL

2-ISOPROPYLBENZALDEHYDE

Von Ranson & Belitz (1992b)	d	0.018
Von Ranson & Belitz (1992b)	r	0.077

3-ISOPROPYLBENZALDEHYDE *[34246-57-6]*

Von Ranson & Belitz (1992b)	d	0.053
Von Ranson & Belitz (1992b)	r	0.59

4-ISOPROPYLBENZALDEHYDE (cuminaldehyde) *[122-03-2]*

Von Ranson & Belitz (1992b)	d	0.086
Von Ranson & Belitz (1992b)	r	0.14

ISOPROPYLBENZENE (cumene) *[98-82-8]*

Solomin (1964)		0.06
Elfimova (1966)		0.025
Köster (1971)	d	0.25
Turk (1973)	r	4.8 - 6.4
Hellman & Small (1974)	d	0.04
Hellman & Small (1974)	r	0.23
Anon. (1980)	d	0.074
Anon. (1980)	r	0.54
Punter (1983)	d	0.65
Bahnmüller (1983)		0.017 - 1.19
Nagy (1991)	d	0.6
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	5.3
Nagata (2003)	d	0.041

ISOPROPYLBENZENE HYDROPEROXIDE (cumene hydroperoxide) *[80-15-9]*

Solomin (1964)		0.03

ISOPROPYL BUTANOATE (isopropyl butyrate) *[638-11-9]*

Nagata (2003)	d	0.033

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

isopropyl butyrate⇒ISOPROPYL BUTANOATE

4-ISOPROPYL-1,3-CYCLOHEXADIEN-1-CARBOXALDEHYDE (1,3-*p*-menthadien-7-al) *[1197-15-5]*

Blank *et al.* (1989)		0.001 - 0.002

5-ISOPROPYL-2,3-DIMETHYLPYRAZINE *[40790-21-4]*

Wagner *et al.* (1999)		> 2

3-ISOPROPYL-2,5-DIMETHYLPYRAZINE *[40790-20-3]*

Wagner *et al.* (1999)		> 2

2-ISOPROPYL-3,5-DIMETHYLPYRAZINE *[101613-23-4]*

Wagner *et al.* (1999)		1.12

isopropyl ether⇒2-ISOPROPOXYPROPANE

ISOPROPYL FORMATE *[625-55-8]*

Nagata (2003)	d	1.0

4-ISOPROPYLIDENE-1-METHYL-3-CYCLOHEXANONE (pulegone) *[89-82-7]*

Tempelaar (1913); Zwaardemaker (1927)	d	0.084 - 0.1
Benzo *et al*. (2007)		0.000 884 (GC-O technique)
Benzo *et al*. (2007)		0.00187 (dynamic dilution olfactometry)

isopropyl isobutyrate⇒ISOPROPYL 2-METHYLPROPANOATE
isopropyl mercaptan⇒2-PROPANETHIOL

4-ISOPROPYL-8-MERCAPTO-1-METHYL-1-CYCLOHEXENE (1-*p*-menthene-8-thiol) *[71159-90-5]*

Huber (1984)		0.000 001

2-ISOPROPYL-3-METHOXYPYRAZINE *[25773-40-4]*

Buttery *et al.* (1969)		0.000 0036
Sävenhed *et al.* (1985)	d	< 0.000 003
Blank *et al.* (1992)		0.000 000 5 - 0.000 001
Khiari *et al.* (1992)		0.000 005
Wagner *et al.* (1999)		0.000 002

1-ISOPROPYL-2-METHYLBENZENE (*o*-cymene) *[527-84-4]*

Backman (1917)	r	0.004 - 0.005

1-ISOPROPYL-4-METHYLBENZENE (*p*-cymene) *[99-87-6]*

Backman (1917)	r	0.012
Cristoph (1983)	r	2.1 - 2.4
Schieberle *et al.* (1988);		
Schieberle & Grosch (1988)		0.196 - 0.785
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	7.2

trans-5-ISOPROPYL-2-METHYLBICYCLO[3.1.0]-2-HEXANOL (*trans*-sabinene hydrate) *[17699-16-0]*

Cristoph (1983)	r	9.2 - 10.0

1-ISOPROPYL-4-METHYLBICYCLO[3.1.0]-3-HEXANONE (α-thujone) *[546-80-5]*

Köster (1971)	d	19 - 35

4-ISOPROPYL-1-METHYL-1,3-CYCLOHEXADIENE (α-terpinene) *[99-86-5]*

Cristoph (1983)	r	2.35 - 2.45
Cometto-Muñiz *et al.* (1998b);		

Cometto-Muñiz (1999)	d	7.9

4-ISOPROPYL-1-METHYL-1,4-CYCLOHEXADIENE (γ-terpinene) *[99-85-4]*

Cristoph (1983)	r	1.4 - 1.6
Lindell (1991)	d	2.5
Cometto-Muñiz *et al.* (1998b); Cometto-Muñiz (1999)	d	55

4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE (α-phellandrene) *[99-83-2]*

Cristoph (1983)	r	2.9 - 3.9

(*R*)-(-)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE ((*l*)-α-phellandrene) *[4221-98-1]*

Jagella & Grosch (1998)		0.093

(*S*)-(+)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE ((*d*)-α-phellandrene) *[2243-33-6]*

Blank & Grosch (1991a)		0.7
Jagella & Grosch (1998)		0.040

(1*S*,2*R*,5*S*)-(+)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL ((*d*)-menthol) *[15356-60-2]*

Doll & Bournot (1949)	d	0.9
Doll & Bournot (1949)	r	2.1

(1*R*,2*S*,5R)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL ((*l*)-menthol) *[2216-51-5]*

Doll & Bournot (1949)	d	0.9
Doll & Bournot (1949)	r	2.1
Marin *et al.* (1988)		2.17 - 2.55

2-ISOPROPYL-5-METHYLCYCLOHEXANONE ((isomenthone) *[36977-92-1]*

Benzo *et al.* (2007)		0.040451

2-ISOPROPYL-5-METHYLCYCLOHEXANONE (menthone, *p*-menthan-3-one) *[10458-14-7]*

Marin *et al.* (1988)		0.021 - 2.2
Yang *et al.* (2008)		0.0047

4-ISOPROPYL-1-METHYL-1-CYCLOHEXENE (1-*p*-menthene) *[5502-88-5]*

Cristoph (1983)	r	7.4 - 7.7

1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL (1-terpinen-4-ol) *[562-74-3]*

Cristoph (1983)	r	11.7 - 12.2
Schieberle & Grosch (1988)		0.59

(*R*)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL ((*R*)-terpinen-4-ol) *[20126-76-5]*

Maas *et al.* (1993)		> 3

(*S*)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL ((*S*)-terpinen-4-ol) *[2438-10-0]*

Maas *et al.* (1993)		> 3

1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE (sabinene) *[3387-41-5]*

Cristoph (1983)	r	1.5 - 3.0
Jagella & Grosch (1998)		> 2

5-ISOPROPYL-2-METHYLPHENOL (carvacrol) *[499-75-2]*

Tempelaar (1913); Zwaardemaker (1927)	d	0.1
Backman (1917)	r	0.011 - 0.017
Van Anrooij (1931)	d	0.18

2-ISOPROPYL-5-METHYLPHENOL (thymol) *[89-83-8]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Tempelaar (1913); Zwaardemaker (1927)	d	0.038 - 0.04
Backman (1917)	r	0.056 - 0.075
Van Anrooij (1931)	d	0.04
Baldus (1936)	d	0.05
Baldus (1936)	r	0.1

ISOPROPYL 2-METHYLPROPANOATE (isopropyl isobutyrate) *[617-50-5]*

Nagata (2003)	d	0.19

3-(3-ISOPROPYLPHENYL)BUTANAL (3-(3-propan-2-ylphenyl)butanal)

Chalk (1990)	0.000 07

(+)-3-(3-ISOPROPYLPHENYL)BUTANAL ((+)-florhydral®)

Abate *et al.* (2002)	0.000 035

(-)-3-(3-ISOPROPYLPHENYL)BUTANAL ((-)-florhydral®)

Abate *et al.* (2002)	0.000 88

ISOPROPYL PROPANOATE (isopropyl propionate) *[637-78-5]*

Nagata (2003)	d	0.020

isopropyl propionate⇒ISOPROPYL PROPANOATE

2-(ISOPROPYLTHIO)PROPANE (diisopropyl sulphide) *[625-80-9]*

Wilby (1969)	r	0.018
Gijs *et al.* (2000)		0.007

(1*R*,2*S*,4*R*,2'*S*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)] ((1*R*,2*S*,4*R*,2'*S*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)])

Bajgrowicz & Frank (2001)	0.000 2

(1*R*,2*S*,4*R*,2'*R*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)] ((1*R*,2*S*,4*R*,2'*R*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)])

Bajgrowicz & Frank (2001)	0.040

(1*S*,2*R*,4*S*,2'*R*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)] ((1*S*,2*R*,4*S*,2'*R*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)])

Bajgrowicz & Frank (2001)	0.088

(1*S*,2*R*,4*S*,2'*S*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)] ((1*S*,2*R*,4*S*,2'*S*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)])

Bajgrowicz & Frank (2001)	0.040

(1*R*,2*R*,4*R*,2'*R*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)] ((1*R*,2*R*,4*R*,2'*R*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)])

Bajgrowicz & Frank (2001)	0.004

(1*R*,2*R*,4*R*,2'*S*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)] ((1*R*,2*R*,4*R*,2'*S*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)])

Bajgrowicz & Frank (2001)	0.250

isopulegone⇒2-ISOPROPENYL-5-METHYL-1-CYCLOHEXANONE
isovaleraldehyde⇒3-METHYLBUTANAL
isovaleric acid⇒3-METHYLBUTANOIC ACID

K-L

kilval⇒O,O-DIMETHYL S-[2-(1-METHYLCARBAMOYLETHYLTHIO)ETHYL]PHOSPHORODITHIOATE
kilwal⇒O,O-DIMETHYL S-[2-(1-METHYLCARBAMOYLETHYLTHIO)ETHYL]PHOSPHORODITHIOATE

KRYPTON *[7439-90-9]*
Laffort & Gortan (1987) 6,700,000

lactic acid⇒2-HYDROXYPROPANOIC ACID
lauric acid⇒DODECANOIC ACID
lauryl acetate⇒DODECYL ACETATE
laurylaldehyde⇒DODECANAL
(*S*)-lavandulol⇒(*S*)-5-METHYL-2-(1-METHYLETHENYL)-4-HEXEN-1-OL
(*R*)-lavandulol⇒(*R*)-5-METHYL-2-(1-METHYLETHENYL)-4-HEXEN-1-OL
leaf alcohol⇒(*Z*)-3-HEXEN-1-OL
leaf aldehyde⇒(*E*)-2-HEXENAL
(2*R*,2'*R*,5'*R*)-lilac alcohol⇒(2*R*,2'*R*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*R*,2'*R*,5'*S*)-lilac alcohol⇒(2*R*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*R*,2'*S*,5'*R*)-lilac alcohol⇒(2*R*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*R*,2'*S*,5'*S*)-lilac alcohol⇒(2*R*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*S*,2'*R*,5'*R*)-lilac alcohol⇒(2*S*,2'*R*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*S*,2'*R*,5'*S*)-lilac alcohol⇒(2*S*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*S*,2'*S*,5'*R*)-lilac alcohol⇒(2*S*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*S*,2'*S*,5'*S*)-lilac alcohol⇒(2*S*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL
(2*R*,2'*R*,5'*R*)-lilac aldehyde⇒(2*R*,2'*R*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
(2*R*,2'*R*,5'*S*)-lilac aldehyde⇒(2*R*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
(2*R*,2'*S*,5'*R*)-lilac aldehyde⇒(2*R*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
(2*R*,2'*S*,5'*S*)-lilac aldehyde⇒(2*R*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
(2*S*,2'*R*,5'*R*)-lilac aldehyde⇒(2*S*,2'*R*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
(2*S*,2'*R*,5'*S*)-lilac aldehyde⇒(2*S*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
(2*S*,2'*S*,5'*R*)-lilac aldehyde⇒(2*S*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
(2*S*,2'*S*,5'*S*)-lilac aldehyde⇒(2*S*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL
lilial⇒3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL
lily aldehyde⇒3-(4-*tert*-BUTYLPHENYL)-2-METHYLPROPANAL
limdrostanol⇒5β,10-DIMETHYL-*des*-A-18-*nor*-ANDROSTAN-13β-OL
limonene⇒4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(4*R*)-(+)-limonene⇒(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(4*S*)-(-)-limonene⇒(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(*d*)-limonene⇒(*R*)-(+)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
(*l*)-limonene⇒(*S*)-(-)-4-ISOPROPENYL-1-METHYL-1-CYCLOHEXENE
linalool⇒3,7-DIMETHYL-1,6-OCTADIEN-3-OL
(*d*)-linalool⇒(*S*)-(+)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
((*l*)-linalool⇒(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
R-(-)-linalool⇒(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
S-(+)-linalool⇒(*S*)-(+)-3,7-DIMETHYL-1,6-OCTADIEN-3-OL
linalyl acetate⇒3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE
(*R*)-(-)-linayl acetate⇒(*R*)-(-)-3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE
(*S*)-(+)-linayl acetate⇒(*S*)-(+)-3,7-DIMETHYL-1,6-OCTADIEN-3-YL ACETATE
linden ether⇒3,6-DIMETHYL-2,4,5,7a-TETRAHYDRO-1-BENZOFURAN
(+)-α-longipinene⇒(1*R*,2*S*,7*R*,8*R*)-(+)-2,6,9,9-TETRAMETHYLTRICYCLO[5.4.0.0(2.8)]-9-UNDECENE
2,6-lutidine⇒2,6-DIMETHYLPYRIDINE
3,4-lutidine⇒3,4-DIMETHYLPYRIDINE
3,5-lutidine⇒3,5-DIMETHYLPYRIDINE

M

trans-(2*R*,3*R*)-magnolione⇒*trans*-(2*R*,3*R*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE
cis-(-)-(2*R*,3*S*)-magnolione⇒*cis*-(-)-(2*R*,3*S*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE
cis-(+)-(2*S*,3*R*)-magnolione⇒*cis*-(+)-(2*S*,3*R*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE
trans-(2*S*,3*S*)-magnolione⇒*trans*-(2*S*,3*S*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE
maleic anhydride⇒2,5-FURANDIONE
maple lactone⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE
1,3-*p*-menthadien-7-al⇒4-ISOPROPYL-1,3-CYCLOHEXADIEN-1-CARBOXALDEHYDE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

p-menthan-3-one⇒2-ISOPROPYL-5-METHYLCYCLOHEXANONE
1-*p*-menthene⇒4-ISOPROPYL-1-METHYL-1-CYCLOHEXENE

p-MENTH-1-EN-3-[2-METHYL-1,3-BUTADIENYL]-8-OL

Krings *et al.* (2006)		< 1

1-*p*-menthene-8-thiol⇒4-ISOPROPYL-8-MERCAPTO-1-METHYL-1-CYCLOHEXENE
menthofuran⇒3,6-DIMETHYL-4,5,6,7-TETRAHYDRO-1-BENZOFURAN
(+)-(6*R*)-menthofuran ⇒(6*R*)-3,6-DIMETHYL-4,5,6,7-TETRAHYDRO-1-BENZOFURAN

MENTHOL *[89-78-1]*

Tempelaar (1913); Zwaardemaker (1927)	d	68
Backman (1917)	r	0.16 - 0.31
Baldus (1936)	d	0.000 4
Baldus (1936)	r	0.000 5
Doll & Bournot (1949)	d	0.9
Doll & Bournot (1949)	r	2.1
Murphy (1983)	d	1.7 - 3.3
Kleinschmidt (1983)	r	3.33
Randebrock (1986)		0.039
Cometto-Muñiz & Cain (1990); Cometto-Muñiz (1993)	d	0.13

(*d*)-menthol⇒(1*S*,2*R*,5*S*)-(+)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL
(*l*)-menthol⇒(1*R*,2*S*,5*R*)-(-)-2-ISOPROPYL-5-METHYLCYCLOHEXANOL
menthone⇒2-ISOPROPYL-5-METHYLCYCLOHEXANONE

MERCAPTOACETALDEHYDE *[4124-63-4]*

Vermeulen & Collin (2006)		0.000 06

MERCAPTOACETIC ACID (thioglycolic acid) *[68-11-1]*

Dravnieks *et al.* (1986)	d	0.000 8

mercaptoacetone⇒1-MERCAPTO-2-PROPANONE

1-MERCAPTO-3-BUTANOL

Vermeulen *et al.* (2003)		0.000 05

3-MERCAPTO-1-BUTANOL

Vermeulen *et al.* (2003)		0.001

1-MERCAPTO-2-BUTANONE

Vermeulen & Collin (2006)		0.004

3-MERCAPTO-2-BUTANONE *[40789-98-8]*

Hofmann & Schieberle (1995)		0.000 2 - 0.000 8
Vermeulen & Collin (2006)		0.006

1-MERCAPTO-3-BUTANONE

Vermeulen *et al.* (2003)		0.000 1

3-MERCAPTOBUTANAL

Vermeulen & Collin (2002)		0.04

3-MERCAPTOBUTYL ACETATE *[89534-38-3]*

Vermeulen & Collin (2003)		0.009

1-MERCAPTO-3-BUTYL ACETATE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Vermeulen & Collin (2003) 0.03

3-mercapto-2-butylpropanal⇒2-BUTYL-3-MERCAPTOPROPANAL
3-mercapto-2-butyl-1-propanol⇒2-BUTYL-3-MERCAPTO-1-PROPANOL
3-mercapto-2-butylpropyl acetate⇒2-BUTYL-3-MERCAPTOPROPYL ACETATE

2-MERCAPTO-1-CYCLOHEXANOL
Vermeulen & Collin (2006) 0.005/0.015 (different enantiomers)

1-MERCAPTO-3,3-DIMETHYL-2-BUTANONE (1-mercaptopinacolone)
Vermeulen & Collin (2006) 0.002

3-MERCAPTO-2,2-DIMETHYL-1-PROPANOL
Vermeulen & Collin (2006) 0.000 04

2-MERCAPTOETHANOL *[60-24-2]*
Vermeulen & Collin (2006) 0.24

2-(1-MERCAPTOETHYL)FURAN
Hofmann & Schieberle (1997) 0.000 009

2-(1-MERCAPTOETHYL)FURYLDITHIO-2-(1-MERCAPTOETHYL)FURAN (bis(2-(1-mercaptoethyl)furane) disulphide)
Hofmann & Schieberle (1997) 0.000 005

3-mercapto-2-ethylpropanal⇒2-ETHYL-3-MERCAPTOPROPANAL
3-mercapto-2-ethyl-1-propanol⇒2-ETHYL-3-MERCAPTO-1-PROPANOL
3-mercapto-2-ethylpropyl acetate⇒2-ETHYL-3-MERCAPTOPROPYL ACETATE

2-(1-MERCAPTOETHYL)THIOPHENE
Hofmann & Schieberle (1997) 0.000 016

N-(2-MERCAPTOETHYL)-1,3-THIAZOLIDINE
Engel & Schieberle (2002) 0.000 005

3-MERCAPTOHEPTANAL
Vermeulen & Collin (2002) 0.000 3

3-MERCAPTO-1-HEPTANOL
Vermeulen *et al.* (2003) 0.000 06

7-MERCAPTO-1-HEPTANOL
Vermeulen & Collin (2006) 0.012

3-MERCAPTOHEPTYL ACETATE *[548774-80-7]*
Vermeulen & Collin (2003) 0.000 6

3-MERCAPTOHEXANAL *[51755-72-7]*
Vermeulen & Collin (2002) 0.000 03

3-MERCAPTO-1-HEXANOL (3-sulfanylhexan-1-ol) *[51755-83-0]*
Vermeulen *et al.* (2003) 0.000 04

(*R*)-3-MERCAPTO-1-HEXANOL ((*R*)-3-sulfanylhexan-1-ol) *[90180-884-]*
Steinhaus *et al.* (2009) 0.000 08

(*S*)-3-MERCAPTO-1-HEXANOL ((*S*)-3-sulfanylhexan-1-ol) *[90180-90-8]*
Steinhaus *et al.* (2009) 0.000 07

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

6-MERCAPTO-1-HEXANOL *[1633-78-9]*
Vermeulen & Collin (2006) 0.019

4-MERCAPTO-2-HEXANOL (5-mercaptohexan-3-ol)
Vermeulen *et al.* (2001) 0.000 2/0.000 6 (different diastereoisomers)

5-mercaptohexan-3-ol⇒4-MERCAPTO-2-HEXANOL

2-MERCAPTO-4-HEXANONE (5-mercaptohexan-3-one)
Vermeulen *et al.* (2001) 0.000 2

5-mercaptohexan-3-one⇒2-MERCAPTO-4-HEXANONE
6-mercapto-2-hexanone⇒1-MERCAPTO-5-HEXANONE

1-MERCAPTO-5-HEXANONE (6-mercapto-2-hexanone)
Vermeulen & Collin (2006) 0.000 3

2-MERCAPTO-4-HEXYL ACETATE
Vermeulen & Collin (2003) 0.004/0.03 (different diastereoisomers)

3-MERCAPTOHEXYL ACETATE (3-sulfanylhexyl acetate) *[136954-20-6]*
Vermeulen & Collin (2003) 0.000 2

(*R*)-3-MERCAPTOHEXYL ACETATE ((*R*)-3-sulfanylhexyl acetate)
Steinhaus *et al.* (2009) 0.000 1

(*S*)-3-MERCAPTOHEXYL ACETATE ((*S*)-3-sulfanylhexyl acetate)
Steinhaus *et al.* (2009) 0.000 03

3-MERCAPTO-2-METHYLBUTANAL
Vermeulen & Collin (2002) 0.000 02

3-MERCAPTO-3-METHYLBUTANAL
Vermeulen & Collin (2002) 0.000 6

3-MERCAPTO-2-METHYL-1-BUTANOL *[227456-33-9]*
Vermeulen *et al.* (2003) 0.000 1

(*l*)-3-MERCAPTO-2-METHYL-1-BUTANOL
Acuna *et al.* (2003, 2004) 0.000 4

(*u*)-3-MERCAPTO-2-METHYL-1-BUTANOL
Acuna *et al.* (2003, 2004) 0.000 004

3-MERCAPTO-3-METHYL-1-BUTANOL *[34300-94-2]*
Vermeulen *et al.* (2003) 0.004

3-MERCAPTO-2-METHYLBUTYL ACETATE
Vermeulen & Collin (2003) 0.02/0.02 (different diastereoisomers)

3-MERCAPTO-3-METHYLBUTYL ACETATE (3-methyl-3-sulfanylbutyl acetate) *[50746-09-3]*
Vermeulen & Collin (2003) 0.003
Miyazawa *et al.* (2009a) d 0.000 10

3-MERCAPTO-3-METHYLBUTYL FORMATE *[50746-10-6]*
Blank & Grosch (1991b,1992);

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Blank *et al.* (1992) 0.000 000 2 - 0.000 000 4

3-MERCAPTO-2-METHYL-5-HEXANOL (5-methyl-4-mercaptohexan-2-ol)
Vermeulen *et al.* (2001) 0.000 02/0.000 02 (different diastereoisomers)

3-MERCAPTO-2-METHYL-5-HEXANONE (5-methyl-4-mercaptohexan-2-one) *[181356-02-5]*
Vermeulen *et al.* (2001) 0.000 3

4-MERCAPTO-5-METHYL-2-HEXYL ACETATE (5-methyl-4-mercaptohexyl-2-acetate)
Vermeulen & Collin (2003) 0.05/0.08 (different diastereoisomers)

3-MERCAPTO-2-METHYLPENTANAL *[227456-28-2]*
Vermeulen & Collin (2002) 0.001/0.000 03 (different diastereoisomers)

3-MERCAPTO-2-METHYL-1-PENTANOL *[227456-30-6]*
Vermeulen *et al.* (2003) 0.000 04/0.000 07 (different diastereoisomers)

(2*R*,3*S*)-3-MERCAPTO-2-METHYL-1-PENTANOL
Sabater-Lüntzel *et al.* (2000);
Widder *et al.* (2001, 2002) d 0.000 000 07 – 0.000 000 2

(2*S*,3*R*)-3-MERCAPTO-2-METHYL-1-PENTANOL
Sabater-Lüntzel *et al.* (2000) d 0.000 003 – 0.000 007

4-MERCAPTO-3-METHYL-2-PENTANOL
Vermeulen *et al.* (2001) 0.000 001
Vermeulen *et al.* (2003) 0.000 001/0.000 6 (different diastereoisomers)

4-MERCAPTO-4-METHYL-2-PENTANOL *[31539-84-1]*
Vermeulen *et al.* (2001) 0.000 09

4-MERCAPTO-3-METHYL-2-PENTANONE
Vermeulen *et al.* (2001) 0.000 2/0.004 (different diastereoisomers)
Vermeulen & Collin (2002) 0.000 98 (different diastereoisomers)

4-MERCAPTO-4-METHYL-2-PENTANONE (2-mercapto-2-methyl-4-pentanone) *[19872-52-7]*
Vermeulen *et al.* (2001); Vermeulen & Collin (2002) 0.000 04

2-mercapto-2-methyl-4-pentanone⇒4-MERCAPTO-4-METHYL-2-PENTANONE

3-MERCAPTO-2-METHYLPENTYL ACETATE
Vermeulen & Collin (2003) 0.000 7

4-MERCAPTO-3-METHYL-2-PENTYL ACETATE
Vermeulen & Collin (2003) 0.001/0.02 (different diastereoisomers)

4-MERCAPTO-4-METHYL-2-PENTYL ACETATE
Vermeulen & Collin (2003) 0.08

3-MERCAPTO-2-METHYLPROPANAL
Vermeulen & Collin (2002) 0.03

3-MERCAPTO-2-METHYL-1-PROPANOL *[56160-79-3]*
Vermeulen *et al.* (2003) 0.000 7

(3*R*)-3-MERCAPTO-2-METHYL-1-PROPANOL
Vermeulen & Collin (2006) 0.038

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(3*S*)-3-MERCAPTO-2-METHYL-1-PROPANOL
Vermeulen & Collin (2006) 0.006

1-MERCAPTO-2-METHYL-2-PROPANOL
Vermeulen & Collin (2006) 0.005

3-MERCAPTO-2-METHYLPROPYL ACETATE
Vermeulen & Collin (2003) 0.211

3-MERCAPTO-1-NONANOL
Vermeulen *et al.* (2003) 0.000 8

4-MERCAPTO-2-NONANOL
Vermeulen *et al.* (2003) 0.000 02/0.000 08 (different diastereoisomers)

4-MERCAPTO-2-NONANONE
Vermeulen *et al.* (2003) 0.000 2

3-MERCAPTONONYL ACETATE
Vermeulen & Collin (2003) 0.15

4-MERCAPTO-2-NONYL ACETATE
Vermeulen & Collin (2003) 0.44/1.15 (different diastereoisomers)

3-MERCAPTOOCTANAL
Vermeulen & Collin (2002) 0.000 01

3-MERCAPTO-1-OCTANOL
Vermeulen *et al.* (2003) 0.000 04

8-MERCAPTO-1-OCTANOL
Vermeulen & Collin (2006) 0.033

3-MERCAPTOOCTYL ACETATE
Vermeulen & Collin (2003) 0.009

3-MERCAPTOPENTANAL
Vermeulen & Collin (2002) 0.01

1-MERCAPTO-3-PENTANOL
Vermeulen *et al.* (2003) 0.000 5

3-MERCAPTO-1-PENTANOL
Vermeulen *et al.* (2003) 0.000 01

4-MERCAPTO-2-PENTANOL
Vermeulen *et al.* (2001) 0.000 02/0.000 2 (different diastereoisomers)

3-MERCAPTO-2-PENTANONE *[67633-97-0]*
Gasser & Grosch (1990a) 0.000 045 - 0.000 18
Hofmann & Schieberle (1995) 0.000 05 - 0.000 2
Kerscher & Grosch (1998) 0.000 15

4-MERCAPTO-2-PENTANONE *[92585-08-5]*
Vermeulen *et al.* (2001) 0.000 3
Vermeulen & Collin (2002) 0.000 69

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

5-mercapto-2-pentanone⇒1-MERCAPTO-4-PENTANONE

1-MERCAPTO-3-PENTANONE
Vermeulen *et al.* (2003);
Vermeulen & Collin (2006) 0.000 09

2-MERCAPTO-3-PENTANONE *[17042-24-9]*
Kerscher & Grosch (1998) 0.000 6

1-MERCAPTO-4-PENTANONE (5-mercapto-2-pentanone)
Vermeulen & Collin (2006) 0.000 02

1-MERCAPTO-3-PENTYL ACETATE
Vermeulen & Collin (2003) 0.04

3-MERCAPTOPENTYL ACETATE
Vermeulen & Collin (2003) 0.01

4-MERCAPTO-2-PENTYL ACETATE
Vermeulen & Collin (2003) 0.008/0.02 (different diastereoisomers)

1-mercaptopinacolone⇒1-MERCAPTO-3,3-DIMETHYL-2-BUTANONE

3-MERCAPTOPROPANAL
Vermeulen & Collin (2002) 0.000 6

(2*R*)-2-MERCAPTO-1-PROPANOL
Vermeulen & Collin (2006) 0.01

3-MERCAPTO-1-PROPANOL *[19721-22-3]*
Vermeulen *et al.* (2003); Vermeulen & Collin (2006) 0.002 - 0.003

1-MERCAPTO-2-PROPANOL *[1068-47-9]*
Vermeulen & Collin (2006) 0.046

1-MERCAPTO-2-PROPANONE (mercaptoacetone) *[24653-75-6]*
Hofmann & Schieberle (1995) 0.0017 - 0.0068
Vermeulen & Collin (2006) 0.019

3-MERCAPTOPROPYL ACETATE
Vermeulen & Collin (2003) 0.009

mesidine⇒2,4,6-TRIMETHYLANILINE
mesitylene⇒1,3,5-TRIMETHYLBENZENE
mesityl oxide⇒4-METHYL-3-PENTEN-2-ONE
mesotan⇒2-OXAPROPYL SALICYLATE
methacrolein⇒2-METHYL-2-PROPENAL
methacrylic acid⇒2-METHYLPROPENOIC ACID
methacrylonitrile⇒ISOPROPENYL CYANIDE

METHANAL (formaldehyde) *[50-00-0]*
Backman (1917) r 0.033 - 0.036
Melekhina (1958) 0.07
Buchberg *et al.* (1961) 1.1 - 2.2
Pliška & Janiček (1965) 12,000
Sgibnev (1968) 0.3 - 0.4
Leonardos *et al.* (1969) r 1.2

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Feldman & Bonashevskya (1971)		0.073
Takhirov (1974)		0.065
Makeicheva (1978)		0.077
Anon. (1980)	d	0.49
Anon. (1980)	r	2.3
Berglund *et al.* (1984);		
Ahlström *et al.* (1986b)	d	0.06
Berglund *et al.* (1987)	d	0.14 - 0.21
Winneke *et al.* (1988)		0.15 - 0.29
Nagy (1991)	d	2.2
Berglund & Nordin (1992)	d	0.066 - 0.11
Berglund & Esfandabad (1992)		0.18 (odor)
Berglund & Esfandabad (1992)		0.69 (irritation)
Nagata (2003)	d	0.6

METHANE *[74-82-8]*

Laffort & Gortan (1987)		1,900,000

METHANETHIOL (methyl mercaptan) *[74-93-1]*

Katz & Talbert (1930)		0.081
Bozza & Colombo (1949)		1
Freudenberg & Reichert (1955)		0.000 5
Guadagni (1966)		0.000 2
Endo *et al.* (1967)		1.1
Leonardos *et al.* (1969)	r	0.0042
Wilby (1969)	r	0.002
Hamanabe *et al.* (1969)		0.000 2
Sanders *et al.* (1970)		0.0019
Selyuzhitskii (1972)		0.000 5
Artho & Koch (1973)		0.000 000 000 001
Blanchard (1976)		0.003
Williams *et al.* (1977)	d	0.000 000 3
Nishida *et al.* (1979)	d	0.038
Bedborough & Trott (1979)	d	0.000 16
Anon. (1980)	d	0.000 24
Anon. (1980)	r	0.0013
Nagy (1991)	d	0.0024
Nagata (2003)	d	0.000 14
Greenman *et al.* (2004)		0.000 48
Glindemann *et al.* (2006)		d 0.001

methanoic acid⇒FORMIC ACID

METHANOL (methyl alcohol) *[67-56-1]*

Passy (1892c)	d	1,000
Zwaardemaker (1914)	d	600
Backman (1917)	r	900 - 1,000
Grijns (1919); Zwaardemaker (1927)		2,150
Jung (1936)	d	23.4 - 54.6
Jung (1936)	r	54.6 - 62.4
Gavaudan *et al.* (1948)		150
Mullins (1955)	r	19,300
Scherberger *et al.* (1958)		r 1,950
Chao-Chen-Tzi (1959)		4.3
Janiček *et al.* (1960)		4,000
Pliška & Janiček (1965)		260,000
May (1966)	d	7,800

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

May (1966)	r	11,700	
Ubaidullaev (1966a)		4.5	
Leonardos *et al.* (1969)	r	130	
Hellman & Small (1974)		d	5.5
Hellman & Small (1974)		r	69
Anon. (1980)	d	74	
Anon. (1980)	r	260	
Nauš (1982)	d	4	
Nauš (1982)	r	10	
Cometto-Muñiz & Cain (1990);			
Cometto-Muñiz (1993)	d	2,096	
Scharfenberger (1990)		1,975	
Nagata (2003)	d	43	

methional⇒3-(METHYLTHIO)PROPANAL
methionol⇒3-(METHYLTHIO)-1-PROPANOL
methone⇒5,5-DIMETHYL-1,3-CYCLOHEXANEDIONE

4-METHOXYBENZALDEHYDE (anisaldehyde) *[123-11-5]*

Van Anrooij (1931)	d	0.034
Blank *et al.* (1989)		0.000 1 - 0.000 2

METHOXYBENZENE (anisole) *[100-66-3]*

Hendriks (1979)	d	0.56
Punter (1983)	d	1.5

1-METHOXYBUTYL ACETATE

Nagy (1991)	d	0.68

6-METHOXY-2,6-DIMETHYLHEPTANAL (methoxy melonal) *[62439-41-2]*

Mueller (2007)		0.00192

6-METHOXY-2,6-DIMETHYLOCTANAL

Mueller (2007)		0.000 78

2-METHOXY-3,5-DIMETHYLPYRAZINE

Czerny & Grosch (2000)	d	0.000 000 001

3-METHOXY-2,5-DIMETHYLPYRAZINE *[19846-22-1]*

Czerny & Grosch (2000)	d	0.056

2-METHOXYETHANOL (methyl glycol, methyl cellosolve, ethylene glycol methyl ether) *[109-86-4]*

May (1966)	d	190
May (1966)	r	280
Hellman & Small (1973,1974)	d	< 0.3
Hellman & Small (1973,1974)	r	0.7

2-METHOXYETHYL ACETATE (methyl cellosolve acetate) *[110-49-6]*

Hellman & Small (1973,1974)	d	1.6
Hellman & Small (1973,1974)	r	3.1

1-METHOXY-3-HEPTANETHIOL

Van de Waal *et al.* (2001)		0.000 000 82

1-METHOXY-3-HEXANETHIOL

Van de Waal *et al.* (2001)		0.000 000 36

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(*R*)-1-METHOXY-3-HEXANETHIOL

Van de Waal *et al.* (2001)		0.000 00109

(*S*)-1-METHOXY-3-HEXANETHIOL

Van de Waal *et al.* (2001)		0.000 000 04

methoxy melonal⇒6-METHOXY-2,6-DIMETHYLHEPTANAL

METHOXYMETHANE (dimethyl ether) *[115-10-6]*

Nagy (1991)	d	430
Nagy (1991)	d	303.967

1-METHOXY-3-METHYL-3-BUTANETHIOL *[94087-83-9]*

Guth & Grosch (1991)		0.000 000 08 - 0.000 000 3

1-METHOXY-4-METHYL-3-PENTANETHIOL

Van de Waal *et al.* (2001)		0.000 043

2-METHOXY-2-METHYLPROPANE (MTBE, methyl-*tert*-butyl ether) *[1634-04-4]*

Smith & Duffy (1995)	d	0.11
Smith & Duffy (1995)	r	0.22
Prah *et al.* (1994); Schulman (2001)	d	0.63

methoxymethyl salicylate⇒2-OXAPROPYL SALICYLATE

1-METHOXYNAPHTHALENE (methyl α-naphthyl ether) *[2216-69-5]*

Backman (1917)	r	0.000 8 - 0.000 9

2-METHOXYNAPHTHALENE (methyl β-naphthyl ether) *[93-04-9]*

Passy (1892a)	d	0.001 - 0.005
Backman (1917)	r	0.008 - 0.016
Sävenhed *et al.* (1985)	d	0.001 - 0.004

2-METHOXYPHENOL (guaiacol) *[90-05-1]*

Hermanides (1909)	r	0.0037
Zwaardemaker (1914,1927)	d	0.0037 - 0.004
Backman (1917)	r	0.0025
Zwaardemaker & Komuro (1921)	d	0.64
Dravnieks *et al.* (1986)	d	0.0026
Guth & Grosch (1991)		0.000 1 - 0.000 8
Ferreira *et al.* (1998)		0.000 4
Atanasova *et al.* (2005)	d	0.000 20 - 0.000 62
Atanasova *et al.* (2005)	r	0.000 37 - 0.00452
Yang *et al.* (2008)		0.0015

1-METHOXY-2-PROPANOL (propylene glycol monomethyl ether) *[107-98-2]*

Stewart *et al.* (1970)	d	37
Nagy (1991)	d	121
Nagy (1991)	d	30.908

1-METHOXY-4-(1-PROPENYL)BENZENE (anethole) *[104-46-1]*

Tempelaar (1913)	d	0.14
Rocén (1920)	r	0.19
Laska & Hudson (1991)	d	0.057
Langridge (2004)		0.0218 orthonasal
Langridge (2004)		0.0018 retronasal

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1-METHOXY-2-PROPYL ACETATE (propylene glycol methyl ether acetate) *[108-65-6]*

Nagy (1991)	d	0.7
Ziemer *et al.* (2000)	d	0.016

2-methoxy-4-vinylphenol⇒4-ETHENYL-2-METHOXYPHENOL

N-METHYLACETANILIDE *[579-10-27]*

Backman (1917)	r	0.42 - 0.51

METHYL ACETATE *[79-20-9]*

Zwaardemaker (1914,1927)	d	2
Backman (1917)	r	67
Gofmekler (1960)		0.5
Janiček *et al.* (1960)		5,250
Nauš (1962)	d	0.7
May (1966)	d	550
May (1966)	r	900
Anon. (1980)	d	22
Anon. (1980)	r	63
Scharfenberger (1990)		579
Cometto-Muñiz & Cain (1991);		
Cometto-Muñiz (1993)	d	8,628
Nagata (2003)	d	5.1

p-methyl acetophenone⇒1-ACETYL-4-METHYLBENZENE
methyl acrylate⇒METHYL 2-PROPENOATE
methyl alcohol⇒METHANOL

METHYLAMINE *[74-89-5]*

Leonardos *et al.* (1969)	r	0.027
Nishida *et al.* (1975)	d	0.065
Nishida *et al.* (1979)	d	6.1
Anon. (1980)	d	0.0012
Anon. (1980)	r	0.012
Hill & Barth (1976)		0.027
Nagy (1991)	d	0.23
Nagata (2003)	d	0.046

METHYL 2-AMINOBENZOATE (methyl anthranilate) *[134-20-3]*

Passy (1892d)	d	0.005 - 0.4
Zwaardemaker (1914,1927)	d	0.006
Katz & Talbert (1930)		0.058
Appell (1969)		0.000 006

2-(METHYLAMINO)ETHANOL *[109-83-1]*

Hellman & Small (1974)	d	3.1
Hellman & Small (1974)	r	10.5

METHYL 3-AMINO-4-HYDROXYBENZOATE *[536-25-4]*

Backman (1917)	r	0.2 - 1.9

methyl amyl ketone⇒2-HEPTANONE

2-METHYLANILINE (*o*-toluidine) *[95-53-4]*

Huijer (1917)	d	29
Backman (1917)	r	4.0 - 5.4
Stuiver (1958)	d	0.11

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3-METHYLANILINE (*m*-toluidine) *[108-44-1]*

Huijer (1917)	d	26
Backman (1917)	r	3.0 - 3.9
Stuiver (1958)	d	2

4-METHYLANILINE (*p*-toluidine) *[106-49-0]*

Huijer (1917)	d	14
Backman (1917)	r	1.0 - 1.3
Stuiver (1958)	d	0.12

N-METHYLANILINE *[100-61-8]*

Backman (1917)	r	6.9 - 8.6

4-METHYLANISOLE (methyl *p*-cresol) *[104-93-8]*

Appell (1969)	0.000 2

methyl anthranilate⇒METHYL 2-AMINOBENZOATE

2-METHYLBENZALDEHYDE (*o*-tolualdehyde) *[529-20-4]*

Von Ranson & Belitz (1992b)	d	0.0096
Von Ranson & Belitz (1992b)	r	0.038

3-METHYLBENZALDEHYDE (*m*-tolualdehyde) *[620-23-5]*

Von Ranson & Belitz (1992b)	d	0.60
Von Ranson & Belitz (1992b)	r	0.89

4-METHYLBENZALDEHYDE (*p*-tolualdehyde) *[104-87-0]*

Von Ranson & Belitz (1992b)	d	0.0012
Von Ranson & Belitz (1992b)	r	0.0072

methylbenzene⇒TOLUENE

4-METHYLBENZENETHIOL (*p*-thiocresol) *[106-45-6]*

Katz & Talbert (1930)	0.014

METHYL BENZOATE *[93-58-3]*

Appell (1969)	0.0025
Randebrock (1986)	0.0015

7-METHYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)	0.000 031

α-methyl-1,3-benzodioxole-5-propanal⇒2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL

2-METHYL-1,3-BUTADIENE (isoprene) *[78-79-5]*

Artho & Koch (1973)		1 – 10
Nagata (2003)	d	0.13

(*R*)-2-METHYLBUTANAL

Bartschat & Mosandl (1998)	0.1

(*S*)-2-METHYLBUTANAL

Bartschat & Mosandl (1998)	0.01

3-METHYLBUTANAL (isovaleraldehyde) *[590-86-3]*

Bedborough & Trott (1979)	d	0.0016

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Blank (1990); Blank *et al.* (1992);
Schieberle & Grosch (1994) 0.002 - 0.006
Nagata (2003) d 0.000 35

(*E*)-2-METHYLBUTANAL OXIME O-METHYL ETHER
Nussbaumer & Hostettler (1996) 0.001

(*Z*)-2-METHYLBUTANAL OXIME O-METHYL ETHER
Nussbaumer & Hostettler (1996) 0.001

2-METHYLBUTANE (isopentane) *[78-78-4]*
Nagata (2003) d 3.8

3-METHYL-1-BUTANETHIOL (isoamyl mercaptan) *[541-31-1]*
Katz & Talbert (1930) 0.0018
Nagata (2003) d 0.000 0033

2-METHYL-2-BUTANETHIOL (*tert*-amyl mercaptan) *[1679-09-0]*
Deadman & Prigg (1959) d 0.0015

METHYL BUTANOATE (methyl butyrate) *[623-42-7]*
Tempelaar (1913); Zwaardemaker (1927) d 0.077 - 0.1
Nagata (2003) d 0.030

2-METHYLBUTANOIC ACID (α-methylbutyric acid) *[116-53-0]*
Punter (1983) d 0.020

3-METHYLBUTANOIC ACID (isovaleric acid) *[503-74-2]*
Backman (1917) r 0.07 - 0.13
Hesse (1926) r 3.0
Schwarz (1955) 0.008
Kerka & Humphreys (1956) 0.005
Amoore (1977); Amoore & Buttery (1978) d 0.0041 - 0.0042
Hendriks (1979) d 0.000 7 - 0.001
Anon. (1980) d 0.000 22
Anon. (1980) r 0.0018
Punter (1983) d 0.0052 - 0.0069
Dravnieks *et al.* (1986) d 0.014
Blank & Schieberle (1993) 0.001 - 0.002
Nagata (2003) d 0.000 33
Greenman *et al.* (2004) 0.0018

2-METHYL-1-BUTANOL (active amyl alcohol) *[137-32-6]*
Backman (1917) r 1.4 - 1.7
Hellman & Small (1973,1974) d 0.14
Hellman & Small (1973,1974) r 0.83
Cristoph (1983) r 0.9 - 1.0
Komthong *et al.* (2006) 329

(*S*)-(-)-2-METHYL-1-BUTANOL *[1565-80-6]*
Passy (1892c) d 0.6

3-METHYL-1-BUTANOL (isoamyl alcohol, isopentanol) *[123-51-3]*
Passy (1892c) d 0.1
Backman (1917) r 0.26
Jung (1936) d 0.08
Jung (1936) r 0.16

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Bahnmüller (1983)		0.019 - 0.547	
Bahnmüller (1984)		0.030 - 0.16	
Dollnick *et al.* (1988)		0.116	
Guth (1997)		0.125	
Guth (1997)		6.3	with 55.6 mg ethanol/l air
Ferreira *et al.* (1998)		2.8	
Nagata (2003)	d	0.0061	

2-METHYL-2-BUTANOL (*tert*-amyl alcohol, *tert*-pentanol) *[75-85-4]*

Passy (1892c)	d	20 - 40
Backman (1917)	r	2.0 - 3.0
Nagata (2003)	d	0.32

3-METHYL-2-BUTANONE (methyl isopropyl ketone) *[563-80-4]*

Backman (1917)	r	15 - 17
Nagata (2003)	d	1.8

3-METHYL-2-BUTENE-1-THIOL (prenyl mercaptan) *[5287-45-6]*

Mayer (1996)	0.000 013 - 0.000 026
Vermeulen & Collin (2006)	0.000 000 5

3-METHYL-3-BUTEN-2-ONE (methyl isopropenyl ketone) *[814-78-8]*

Rowe & Wolf (1962)	1

2-METHYLBUTYL ACETATE (active amyl acetate) *[624-41-9]*

Cristoph (1983)	r	0.14 - 0.21

3-methylbutyl acetate⇒ISOPENTYL ACETATE

7-(2'-METHYLBUTYL)BENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)	0.000 08

7-(3'-METHYLBUTYL)BENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)	0.000 014

methyl-*tert*-butyl ether⇒2-METHOXY-2-METHYLPROPANE
methyl butyl ketone⇒2-HEXANONE
methyl *sec*.butyl ketone⇒3-METHYL-2-PENTANONE
methyl *tert*-butyl ketone⇒2,2-DIMETHYL-3-BUTANONE

(3-METHYLBUTYLTHIO)-3-METHYLBUTANE (diisoamyl sulphide) *[544-02-5]*

Katz & Talbert (1930)	0.021

methyl butyrate⇒METHYL BUTANOATE
α-methylbutyric acid⇒2-METHYLBUTANOIC ACID
methyl cellosolve⇒2-METHOXYETHANOL
methyl cellosolve acetate⇒2-METHOXYETHYL ACETATE
methylchavicol⇒1-ALLYL-4-METHOXYBENZENE
methyl chloride⇒CHLOROMETHANE
methylchloroform⇒1,1,1-TRICHLOROETHANE
methyl *p*-cresol⇒4-METHYLANISOLE

METHYL CYANIDE (acetonitrile) *[75-05-8]*

Pozzani *et al.* (1959)		< 67
Dravnieks & Laffort (1972)		285
Dravnieks (1974)	d	1,950
Nagata (2003)	d	22

methyl 2-cyanoacrylate⇒METHYL 2-CYANO-2-PROPENOATE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

METHYL 2-CYANO-2-PROPENOATE (methyl 2-cyanoacrylate) *[137-05-3]*
McGee *et al.* (1968) 4.5 - 13.5

METHYLCYCLOHEXANE *[108-87-2]*
Nagata (2003) d 0.60

2-METHYLCYCLOHEXANONE *[583-60-8]*
Van Thriel *et al.* (2006) d 0.83

3-METHYLCYCLOPENTADECANONE (muscone) *[541-91-3]*
Traynor (2001) 0.029

(3*R*)-(-)-3-METHYLCYCLOPENTADECANONE ((3*R*)-(-)-muscone)
Kraft & Fráter (2001) 0.0021

3-METHYLCYCLOPENTADECENONE (muscenone) *[82356-51-2]*
Traynor (2001) 0.184

METHYLCYCLOPENTANE *[96-37-7]*
Nagata (2003) d 5.8

3-methyl-1,2-cyclopentanedione⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE
methylcyclopentenolone⇒2-HYDROXY-3-METHYL-2-CYCLOPENTEN-1-ONE

(*E*)-2-METHYLCYCLOPROPAN-1-CARBALDEHYDE
Von Ranson & Belitz (1992b) d 0.22
Von Ranson & Belitz (1992b) r 0.22

(*Z*)-2-METHYLCYCLOPROPAN-1-CARBALDEHYDE
Von Ranson & Belitz (1992b) d 0.34
Von Ranson & Belitz (1992b) r 0.34

6-((*E*)-2-METHYLCYCLOPROP-1-YL)HEXANAL
Von Ranson & Belitz (1992b) d 0.068
Von Ranson & Belitz (1992b) r 0.15

6-((*Z*)-2-METHYLCYCLOPROP-1-YL)HEXANAL
Von Ranson & Belitz (1992b) d 0.23
Von Ranson & Belitz (1992b) r 0.54

(*E*)-3-METHYLCYCLO-5-TETRADECEN-1-ONE
Helmlinger *et al.* (2001) 0.000 2

(*Z*)-3-METHYLCYCLO-5-TETRADECEN-1-ONE
Helmlinger *et al.* (2001) 0.0022

(+)-(5*R*)-4-METHYL-3(*E*)-DECEN-5-OL ((+)-(*R*)-undecavertol)
Abate *et al.* (2005) 0.000 31

(-)-(5*S*)-4-METHYL-3(*E*)-DECEN-5-OL ((-)-(*S*)-undecavertol)
Abate *et al.* (2005) 0.0047

methyl dichloroarsine⇒DICHLOROMETHYLARSINE

1-METHYL-2,3-DIHYDRO-1*H*-5,9-DIOXACYCLOHEPTA[*f*]INDEN-7-ONE
Kraft & Eichenberger (2003) 0.000 26

methyl 2,5-dimethylfur-3-yl disulphide⇒2,5-DIMETHYL-3-(METHYLDITHIO)FURAN

2-METHYL-4,6-DINITROPHENOL (4,6-dinitro-*o*-cresol) *[534-52-1]*

Kurtschatowa & Dawidkowa (1970)		0.004 - 0.021

METHYLDITHIOETHANE (ethyl methyl disulphide, methyl ethyl disulphide, 2,3-dithiapentane) *[20333-39-5]*

Wilby (1969)	r	0.062

METHYLDITHIOMETHANE (dimethyl disulphide, DMDS) *[624-92-0]*

Wilby (1969)	r	0.029
Lindvall (1970)	d	0.003 - 0.014
Selyuzhitskii (1972)		3.5
Bedborough & Trott (1979)	d	0.046
Anon. (1980)	d	0.0011 - 0.0020
Anon. (1980)	r	0.011 - 0.017
Moriguchi *et al.* (1983)	d	0.007
Ahlström *et al.* (1986a)	d	0.050 - 0.078
Nagy (1991)	d	0.066
Gijs *et al.* (2000)		0.82
Greenman *et al.* (2004)		5.6
Nagata (2003)	d	0.0084

(*E*)-2-METHYL-4-DODECENAL

Kaiser (2007)		0.064

(*E*)-3-METHYL-4-DODECENAL

Kaiser (2007)		0.02

(*E*)-4-METHYL-4-DODECENAL

Kaiser (2007)		0.001

METHYL DODECANOATE (methyl laurate) *[111-82-0]*

Backman (1917)	r	0.0015 - 0.0026

methylene bis(methyl sulphide)⇒BIS(METHYLTHIO)METHANE
methylene chloride⇒DICHLOROMETHANE

3,4-(METHYLENEDIXOY)BENZALDEHYDE (piperonal, heliotropine) *[120-57-0]*

Passy (1892a,1892b)	d	0.05 - 0.1
Backman (1917)	r	0.014 - 0.024
Henning (1924)	d	0.01
Zwaardemaker (1927)		0.000 1
Van Anrooij (1931)	d	0.001
Baldus (1936)	d	0.02 - 0.03
Baldus (1936)	r	0.04
Appell (1969)		0.000 03

METHYL ETHOXYDITHIOFORMATE (methyl xanthate)

Deadman & Prigg (1959)	d	0.008

methyl ethyl disulphide⇒METHYLDITHIOETHANE
methyl ethyl ketone⇒2-BUTANONE

METHYL FORMATE *[107-31-3]*

Backman (1917)	r	165 - 180
May (1966)	d	5,000

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

May (1966)	r	6,900
Nagata (2003)	d	325

2-METHYL-3-FURANTHIOL *[28588-74-1]*

Gasser & Grosch (1990a,b)	0.000 0025 - 0.000 01
Blank *et al.* (1992); Blank (1997, 2002)	0.000 001 - 0.000 002 [GC-O on SE-54 capillary column]
Blank (1997, 2002)	0.005 - 0.01 [GC-O on FFAP capillary column]

3-(2-METHYL)FURYLDITHIO-3-(2-METHYL)FURAN (bis(2-methyl-3-furyl) disulphide, 3,3'-dithiobis(2-methylfuran)) *[28588-75-2]*

Gasser & Grosch (1990a,b)	0.000 000 6 - 0.000 0028
Etzweiler (1993)	0.000 02 - 0.000 057

5-METHYL-2-FURYLMETHANETHIOL (5-methyl-2-furfurylthiol) *[59303-05-8]*

Hofmann & Schieberle (1997)	0.000 006

2-METHYL-3-FURYL 3-OXO-2-BUTYL DISULPHIDE

Hofmann & Schieberle (1995)	0.000 01 - 0.000 04

methyl glycol⇒2-METHOXYETHANOL

2-METHYLHEPTANE *[592-27-8]*

Nagata (2003)	d	0.52

3-METHYLHEPTANE *[589-81-1]*

Nagata (2003)	d	7.1

4-METHYLHEPTANE *[589-53-7]*

Nagata (2003)	d	8.0

METHYL HEPTANOATE *[106-73-0]*

Backman (1917)	r	0.29 - 0.59

2-METHYL-1-HEPTANOL (isooctanol, isooctylalcohol) *[60435-70-3]*

Tsulaya *et al.* (1972)		0.26
Nagata (2003)	d	0.049

5-METHYL-3-HEPTANONE *[541-85-5]*

Toxicity data sheet (1958b)	31

3-METHYL-4-HEPTANONE (5-methyl-4-heptanone) *[15726-15-5]*

Burdack-Freitag & Schieberle (2010)	0.0018

5-methyl-4-heptanone⇒3-METHYL-4-HEPTANONE
methylheptenone⇒6-METHYLHEPT-5-EN-2-ONE

6-METHYL-5-HEPTEN-2-ONE (methylheptenone) *[110-93-0]*

Tempelaar (1913); Zwaardemaker (1927)	d	0.3
Backman (1917)	r	0.51 - 0.54
Fuller *et al.* (1964)	r	0.0063
Cristoph (1983)	r	1.8 - 2.5
Benzo *et al*. (2007)		0.01889

2-METHYL-1-HEPTEN-3-ONE

Koszinowski & Piringer (1986)	0.3

5-METHYL-(*E*)-2-HEPTEN-4-ONE ((*E*)-filbertone) *[102322-83-8]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Burdack-Freitag & Schieberle (2010) 0.000 08

methyl heptine carbonate⇒METHYL 2-OCTYNOATE
methyl heptyl ketone⇒2-NONANONE

(*R*)-4-METHYLHEXANAL
Bartschat & Mosandl (1998) 0.03

(*S*)-4-METHYLHEXANAL
Bartschat & Mosandl (1998) 0.25

2-METHYLHEXANE *[591-76-4]*
Nagata (2003) d 1.7

3-METHYLHEXANE *[589-34-4]*
Nagata (2003) d 3.4

S-methyl hexanethioate⇒S-METHYL THIOHEXANOATE

5-METHYL-2-HEXANONE (methyl isoamyl ketone) *[110-12-3]*
Hellman & Small (1974) d 0.06
Hellman & Small (1974) r 0.23
Nagy (1991) d 0.63
Nagata (2003) d 0.0099

3-METHYL-2-HEXENOIC ACID
Natsch *et al.* (2006) 0.019

(*E*)-3-METHYL-2-HEXENOIC ACID *[35205-70-0]*
Baydar *et al.* (1992b) d 0.014

(*Z*)-3-METHYL-2-HEXENOIC ACID *[27960-21-0]*
Baydar *et al.* (1992b) d 0.279

2-METHYL-1-HEXEN-3-ONE
Koszinowski & Piringer (1986) 0.1

methyl hexyl ketone⇒2-OCTANONE

METHYLHYDRAZINE *[60-34-4]*
Jacobson *et al.* (1955) 1.9 - 5.7

(*R*)-METHYL 3-HYDROXYHEXANOATE
Buettner & Schieberle (2001b) 0.0021

(*S*)-METHYL 3-HYDROXYHEXANOATE
Buettner & Schieberle (2001b) 4.38

METHYL (2*S*,3*S*)-2-HYDROXY-3-METHYLPENTANOATE
Steinhaus *et al.* (2009) 0.01

METHYL (2*R*,3*R*)-2-HYDROXY-3-METHYLPENTANOATE
Steinhaus *et al.* (2009) >0.2

METHYL (2*S*,3*R*)-2-HYDROXY-3-METHYLPENTANOATE
Steinhaus *et al.* (2009) 0.11

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

METHYL (2*R*,3*S*)-2-HYDROXY-3-METHYLPENTANOATE

Steinhaus *et al.* (2009)		0.011

3-methyl-4-hydroxyoctanoic acid lactone⇒4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

METHYL 2-HYDROXYPROPANOATE (methyl lactate) *[547-64-8]*

Backman (1917)	r	21

3-METHYLINDOLE (skatole) *[83-34-1]*

Hermanides (1909)	r	0.000 35
Zwaardemaker (1914,1927)	d	0.000 4
Katz & Talbert (1930)		0.10
Van Anrooij (1931)	d	0.000 78
Punter (1975,1979)	d	0.000 35
Logtenberg (1978)	d	0.000 5
Greenman *et al.* (2004)		0.000 094
Nagata (2003)	d	0.000 030

methyl isoamyl ketone⇒5-METHYL-2-HEXANONE
2-methylisoborneol⇒1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL
(-)-2-methylisoborneol⇒(1*R*)-1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL
methyl isobutyl ketone⇒4-METHYL-2-PENTANONE
methyl isobutyrate⇒METHYL 2-METHYLPROPANOATE

METHYL ISOCYANATE *[624-83-9]*

Kimmerle & Eben (1964)		5

METHYL ISOCYANIDE *[593-75-9]*

Stone & Pryor (1967)	d	0.0069 - 0.0127
Stone & Pryor (1967)	r	0.069
Poziomek *et al.* (1971)	r	0.000 6 - 0.006

methyl isopropenyl ketone⇒3-METHYL-3-BUTEN-2-ONE
methyl isopropyl ketone⇒3-METHYL-2-BUTANONE

2-METHYL-3-(4-ISOPROPYLPHENYL)PROPANAL (cyclamen aldehyde) *[103-95-7]*

Randebrock (1986)		0.000 0031

METHYL ISOTHIOCYANATE (methyl mustard oil) *[556-61-6]*

Allison & Katz (1919)		15
Nesterova (1969)		0.3
Cain *et al.* (2010)	d	5.1

methyl isovalerate⇒METHYL 3-METHYLBUTANOATE
methyl lactate⇒METHYL 2-HYDROXYPROPANOATE
methyl laurate⇒METHYL DODECANOATE
methyl mercaptan⇒METHANETHIOL

METHYL 2-MERCAPTOACETATE (methyl 2-thiolacetate, methyl thioglycolate) *[2365-48-2]*

Deadman & Prigg (1959)	d	0.00

5-methyl-4-mercaptohexan-2-ol⇒3-MERCAPTO-2-METHYL-5-HEXANOL
5-methyl-4-mercaptohexan-2-one⇒3-MERCAPTO-2-METHYL-5-HEXANONE
5-methyl-4-mercaptohexyl-2-acetate⇒4-MERCAPTO-5-METHYL-2-HEXYL ACETATE
methyl methacrylate⇒METHYL 2-METHYL-2-PROPENOATE

METHYL 2-METHYLBUTANOATE *[868-57-5]*

Blank & Grosch (1991a)		0.000 15
Komthong *et al.* (2006)		1.18 - 8.76

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

METHYL 3-METHYLBUTANOATE (methyl isovalerate) *[556-24-1]*

Backman (1917)	r	0.12 - 0.20
Nagata (2003)	d	0.011

2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL (helional, α-methyl-1,3-benzodioxole-5-propanal) *[1205-17-0]*

Cometto-Muñiz & Abraham (2010a)	d	0.0011
Olsson & Laska (2010)	d	3.8 - 4.1

(2*R*)-(+)-2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL ((2*R*)-(+)-3-(1,3-benzodioxol-5-yl)-2-methylpropanal, (*R*)-tropional®)

Enders & Backes (2004)		0.00343

(2*S*)-(-)-2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL ((2*S*)-(-)-3-(1,3-benzodioxol-5-yl)-2-methylpropanal, (*S*)-tropional®)

Enders & Backes (2004)		0.000 64

2-METHYL-3-(METHYLDITHIO)FURAN (methyl 2-methylfur-3-yl disulphide) *[65505-17-1]*

Gasser & Grosch (1990b)		0.000 02 - 0.000 08
Etzweiler (1993)		0.000 8

7-METHYL-3-METHYLENE-1,6-OCTADIENE (myrcene) *[123-35-3]*

Cristoph (1983)	r	0.13 - 0.15
Jagella & Grosch (1998)		0.041
Buettner & Schieberle (2001a,2001b)		0.0445
Güntert *et al.* (2001)		0.1125

(*S*)-5-METHYL-2-(1-METHYLETHENYL)-4-HEXEN-1-OL ((*S*)-lavandulol)

Maas *et al.* (1993)		> 3

(*R*)-5-METHYL-2-(1-METHYLETHENYL)-4-HEXEN-1-OL ((*R*)-lavandulol) *[498-16-8]*

Maas *et al.* (1993)		> 3

methyl 2-methylfur-3-yl disulphide⇒2-METHYL-3-(METHYLDITHIO)FURAN

(-)-(Z)-2-METHYL-5-(2-METHYL-3-METHYLENEBICYCLO[2.2.1]-2-HEPTYL-2-PENTEN-1-OL ((-)-(Z)-β-santalol) *[77-42-9]*

Bajgrowicz *et al.* (1998)		0.000 54

METHYL 2-METHYLPROPANOATE (methyl isobutyrate) *547-63-7]*

Blank (1990)		0.000 1 - 0.000 2
Nagata (2003)	d	0.0080

METHYL 2-METHYL-2-PROPENOATE (methyl methacrylate) *[80-62-6]*

Filatova (1962)		0.2
Leonardos *et al.* (1969)	r	0.85
Hellman & Small (1973,1974)	d	0.2
Hellman & Small (1973,1974)	r	1.4
Holland (1974)		0.057
Anon. (1980)	d	0.62
Anon. (1980)	r	1.9
Piringer & Granzer (1984)		0.7
Nagy (1991)	d	2.7
Nagata (2003)	d	0.86

cis-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN (*cis*-rose oxide) *[876-17-5]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Blank *et al.* (1989) 0.000 1 - 0.000 2

(2*R*,4*S*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*R*,4*S*)-*cis*-rose oxide)
Maas *et al.* (1993) > 3

(2*S*,4*R*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*S*,4*R*)-*cis*-rose oxide) *[3033-23-6]*
Maas *et al.* (1993) 0.004

trans-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN (*trans*-rose oxide) *[876-18-6]*
Blank *et al.* (1989) 0.080 - 0.170

(2*R*,4*R*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*R*,4*R*)-*trans*-rose oxide) *[5258-11-7]*
Maas *et al.* (1993) > 3

(2*S*,4*S*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN ((2*S*,4*S*)-*trans*-rose oxide)
Maas *et al.* (1993) 0.004

S-METHYL 2-METHYLTHIOBUTANOATE *[51534-66-8]*
Gijs *et al.* (2000) 0.11

METHYL(METHYLTHIO)DITHIOMETHANE (methyl (methylthio)methyl disulphide, 2,3,5-trithiahexane) *[423474-44-2]*
Spadone *et al.* (2006) d 0.007

2-METHYL-3-(METHYLTHIO)FURAN *[63012-97-5]*
Gasser & Grosch (1990b) 0.0076 - 0.0304
Etzweiler (1993) 0.2

methyl (methylthio)methyl disulphide⇒METHYL(METHYLTHIO)DITHIOMETHANE
methyl mustard oil⇒METHYL ISOTHIOCYANATE

2-METHYLNAPHTHALENE *[91-57-6]*
Moriguchi *et al.* (1983) d 0.004

methyl α-naphthyl ether⇒1-METHOXYNAPHTHALENE
methyl β-naphthyl ether⇒2-METHOXYNAPHTHALENE

1-METHYL-3-NITROBENZENE (3-nitrotoluene) *[99-08-1]*
Punter (1983) d 0.25

4-METHYL-2-NITROPHENOL (2-nitro-*p*-cresol) *[119-33-5]*
Backman (1917) r 0.13 - 0.14

3-METHYL-2,4-NONANEDIONE *[113486-29-6]*
Guth & Grosch (1989) 0.000 007 - 0.000 014

METHYL NONANOATE (methyl pelargonate) *[1731-84-6]*
Backman (1917) r 0.040 - 0.048

2-METHYL-1-NONEN-3-ONE *[51756-19-5]*
Koszinowski & Piringer (1986) 0.001

methyl nonyl ketone⇒2-UNDECANONE
β-methyl-γ-octalactone⇒4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE
cis-β-methyl-γ-octalactone⇒*cis*-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE
trans-β-methyl-γ-octalactone⇒*trans*-4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(*R*)-4-METHYLOCTANOIC ACID
Karl *et al.* (1994) r 0.025

(*S*)-4-METHYLOCTANOIC ACID
Karl *et al.* (1994) r 0.013

4-METHYL-3-OCTENOIC ACID
Natsch *et al.* (2006) 0.022

2-METHYL-1-OCTEN-3-ONE
Koszinowski & Piringer (1986) 0.003

methyl octyl ketone⇒2-DECANONE

METHYL 2-OCTYNOATE (methyl heptine carbonate) *[111-12-6]*
Fuller *et al.* (1964) r 0.000 48
Appell (1969) 0.000 02

(-)-(12*R*)-12-methyl-9-oxa-14-tetradecanolide⇒(12*R*)-(-)-14-HYDROXY-12-METHYL-9-OXATETRADECANOIC ACID LACTONE
1-methyl-2-oxopyrrolidine⇒1-METHYL-2-PYRROLIDONE
methyl parathion⇒O,O-DIMETHYL O-(p-NITROPHENYL)PHOSPHOROTHIOATE
methyl pelargonate⇒METHYL NONANOATE

2-METHYLPENTANAL *[123-15-9]*
Hellman & Small (1974) d 0.4
Hellman & Small (1974) r 0.6

(*R*)-2-METHYLPENTANAL
Bartschat & Mosandl (1998) > 2.5

(*S*)-2-METHYLPENTANAL
Bartschat & Mosandl (1998) > 2.5

(*R*)-3-METHYLPENTANAL
Bartschat & Mosandl (1998) > 0.5

(*S*)-3-METHYLPENTANAL
Bartschat & Mosandl (1998) 0.03

2-METHYLPENTANE *[107-83-5]*
Nagata (2003) d 25

3-METHYLPENTANE *[96-14-0]*
Nagata (2003) d 31

2-METHYL-2,4-PENTANEDIOL (hexylene glycol) *[107-41-5]*
Nagy (1991) d 19

METHYL PENTANOATE (methyl n-valerate) *[624-24-8]*
Nagata (2003) d 0.011

4-METHYLPENTANOIC ACID (isocaproic acid) *[646-07-1]*
Backman (1917) r 0.35 - 0.42
Punter (1983) d 0.037
Nagata (2003) d 0.0019

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-METHYL-1-PENTANOL *[105-30-6]*

Hellman & Small (1974)	d	0.10
Hellman & Small (1974)	r	0.10

4-METHYL-2-PENTANOL *[108-11-2]*

Hellman & Small (1974)	d	1.4
Hellman & Small (1974)	r	2.2

3-METHYL-2-PENTANONE (methyl *sec*.butyl ketone) *[565-61-7]*

Nagata (2003)	d	0.098

4-METHYL-2-PENTANONE (methyl isobutyl ketone) *[108-10-1]*

Backman (1917)	r	0.6 - 0.8
Middleton (1956)	r	1.9
May (1966)	d	32
May (1966)	r	64
Stone *et al.* (1967,1972)	d	0.97/9.7
Steinmetz *et al.* (1969)	d	1.21
Leonardos *et al.* (1969)	r	1.9
Hellman & Small (1974)	d	0.4
Hellman & Small (1974)	r	1.1
Anon. (1980)	d	0.70
Anon. (1980)	r	2.8
Dravnieks *et al.* (1986)	d	0.14
Nagy (1991)	d	6.3
Dalton *et al.* (2000)	d	41
Ziemer *et al.* (2000)	d	1.1
Nagata (2003)	d	0.70

4-METHYL-3-PENTEN-2-ONE (mesityl oxide) *[141-79-7]*

Toxicity data sheet (1957)		48
Hellman & Small (1974)	d	0.07
Hellman & Small (1974)	r	0.20

4-METHYLPENTYL-2 ACETATE (*sec*-hexyl acetate) *[108-84-9]*

Hellman & Small (1974)	d	< 0.4
Hellman & Small (1974)	r	1.4

7-(2'-METHYLPENTYL)BENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 38

7-(3'-METHYLPENTYL)BENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 043

7-(4'-METHYLPENTYL)BENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 038

2-METHYLPHENOL (*o*-cresol) *[95-48-7]*

Backman (1917)	r	0.004
Stuiver (1958)	d	0.000 4
Kendall *et al.* (1968)	r	0.0028
Anon. (1980)	d	0.0017
Anon. (1980)	r	0.027
Moriguchi *et al.* (1983)	d	0.020
Schieberle *et al.* (1988)		0.000 7 - 0.0027
Nagata (2003)	d	0.0012

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

3-METHYLPHENOL (*m*-cresol) *[108-39-4]*

Backman (1917)	r	0.000 7 - 0.000 9
Stuiver (1958)	d	0.000 4
Nader (1958)		0.000 22 - 0.035
Anon. (1980)	d	0.000 57
Anon. (1980)	r	0.011
Nagata (2003)	d	0.000 44

4-METHYLPHENOL (*p*-cresol) *[106-44-5]*

Backman (1917)	r	0.03 - 0.04
Baldus (1936)	d	0.0125
Baldus (1936)	r	0.015
Stuiver (1958)	d	0.000 05
Leonardos *et al.* (1969)	r	0.0044
Punter (1975,1979)	d	0.024
Anon. (1980)	d	0.000 18
Anon. (1980)	r	0.0084
Schieberle *et al.* (1988); Schieberle & Grosch (1988); Blank *et al.* (1989); Blank (1990)		0.000 3 - 0.001
Nagata (2003)	d	0.000 24

4-METHYLPHENYL ACETATE (4-tolyl acetate, *p*-cresyl acetate) *[140-39-6]*

Appell (1969)		0.000 02

methyl phenyl carbinol⇒1-PHENYLETHANOL
4-methyl-1,3-phenylenediisocyanate⇒2,4-DIISOCYANATO-1-METHYLBENZENE
2-methyl-1,3-phenylenediisocyanate⇒2,6-DIISOCYANATO-1-METHYLBENZENE

3-METHYL-1-PHENYL-3-PENTANOL ((β)-phenylethylmethylethyl carbinol) *[10415-87-9]*

Cain & Gent (1991)	d	0.22

methyl phosphite⇒TRIMETHYL PHOSPHITE

2-METHYLPROPANAL (isobutyraldehyde) *[78-84-2]*

Hellman & Small (1973,1974)	d	0.14
Hellman & Small (1973,1974)	r	0.41
Amoore (1977)	d	0.015
Hendriks (1979)	d	0.022
Nagata (2003)	d	0.0010

2-METHYLPROPANE (isobutane) *[75-28-5]*

Mullins (1955)	r	1,370

2-METHYL-1-PROPANETHIOL (isobutyl mercaptan) *[513-44-0]*

Allison & Katz (1919)		8
Stuiver (1958)	d	0.000 006 - 0.000 3
Wilby (1969)	r	0.0036
Gijs *et al.* (2000)		0.002
Nagata (2003)	d	0.000 025

2-METHYL-2-PROPANETHIOL (*tert*-butyl mercaptan) *[75-66-1]*

Stuiver (1958)	d	0.000 09
Nevers & Oister (1965)		0.0019
Kniebes *et al.* (1969)		0.0019
Wilby (1969)	r	0.000 33
Blanchard (1976)		0.007
Moschandreas & Jones (1983)	d	0.0033

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Moschandreas & Jones (1983)	r	0.0081	
Nagata (2003)	d	0.000 11	

METHYL PROPANOATE (methyl propionate) *[554-12-1]*

Backman (1917)	r	10 - 12.5	
Nagata (2003)	d	0.35	

2-METHYLPROPANOIC ACID (isobutyric acid) *[79-31-2]*

Anon. (1980)	d	0.0050	
Anon. (1980)	r	0.025	
Punter (1983)	d	0.24 - 0.33	
Nagata (2003)	d	0.0054	
Ueno *et al.* (2009)		0.013 (triangle odor bag method; forced-choice mode)	
Ueno *et al.* (2009)		0.040 (dynamic olfactometer; forced-choice mode)	

2-METHYL-1-PROPANOL (isobutyl alcohol, isobutanol) *[78-83-1]*

Passy (1892c)	d	1	
Zwaardemaker (1914)	d	500	
Backman (1917)	r	0.2 - 0.4	
Jones (1955c)	r	31	
May (1966)	d	120	
May (1966)	r	160	
Laffort & Dravnieks (1973)		3	
Hellman & Small (1973,1974)	d	2.0	
Hellman & Small (1973,1974)	r	5.4	
Anon. (1980)	d	0.036	
Anon. (1980)	r	0.66	
Punter (1983)	d	3.8 - 8.1	
Cristoph (1983)	r	0.7 - 1.0	
Nagy (1991)	d	2.64	
Nagy (1991)	d	1.723	
Guth (1997)		0.64	
Guth (1997)		200	with 55.6 mg ethanol/l air
Nagata (2003)	d	0.033	

2-METHYL-2-PROPANOL (*tert*-butyl alcohol, *tert*-butanol) *[75-65-0]*

Passy (1892c)	d	10 - 20	
Backman (1917)	r	36 - 40	
Jones (1955c)	r	750	
Dravnieks & Laffort (1972)		71	
Dravnieks (1974)	d	2,900	
Nagy (1991)	d	42	
Cometto-Muñiz & Cain (1993);			
Cometto-Muñiz (1993)	d	1,827	
Ziemer *et al.* (2000)	d	24.2	
Nagata (2003)	d	14	

2-METHYL-2-PROPENAL (methacrolein) *[78-85-3]*

Nagata (2003)	d	0.025	

2-METHYLPROPENE (isobutylene) *[115-11-7]*

Katz & Talbert (1930)		3.0	
Mullins (1955)	r	4,880	
Anon. (1980)	d	15	
Anon. (1980)	r	46	
Nagata (2003)	d	23	

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-methyl-2-propenenitrile⇒ISOPROPENYL CYANIDE

METHYL 2-PROPENOATE (methyl acrylate) *[96-33-3]*

Bezpalkova (1967a,b)		0.017
Anon. (1980)	d	0.010
Anon. (1980)	r	0.060
Bahnmüller (1984)		0.015 - 0.088
Piringer & Granzer (1984)		0.05
Nagy (1991)	d	0.061
Nagata (2003)	d	0.012

2-METHYLPROPENOIC ACID (methacrylic acid) *[79-41-4]*

Piringer & Granzer (1984)		10
Nagy (1991)	d	1.9

methyl propionate⇒METHYL PROPANOATE

2-(2-METHYL)PROPYLAMINE (*tert*-butylamine) *[75-64-9]*

Nagata (2003)	d	0.51

7-(2'-METHYLPROPYL)BENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.00165

2-methylpropyl butanoate⇒ISOBUTYL BUTANOATE
methyl propyl ketone⇒2-PENTANONE

2-METHYL-3-PROPYLPYRAZINE *[15986-80-8]*

Wagner *et al.* (1999)		0.99 - 1.8

2-METHYL-5-PROPYLPYRAZINE *[29461-03-8]*

Wagner *et al.* (1999)		0.99 - 1.8

2-METHYL-6-PROPYLPYRAZINE *[29444-46-0]*

Wagner *et al.* (1999)		0.99 - 1.8

METHYLPYRAZINE *[109-08-0]*

Fors (1984); Fors & Olofsson (1985)	d	1.9
Grosch *et al.* (1996); Wagner *et al.* (1999)		> 2

2-METHYLPYRIDINE (2-picoline, α-picoline) *[109-06-8]*

Hellman & Small (1974)	d	0.05
Hellman & Small (1974)	r	0.09
Moriguchi *et al.* (1983)	d	0.010

(*S*)-(-)-3-(1-METHYL-2-PYRROLIDINYL)PYRIDINE (S-(-)-nicotine) *[54-11-5]*

Walker *et al.* (1996)		0.066

1-METHYL-2-PYRROLIDONE (1-methyl-2-oxopyrrolidine) *[872-50-4]*

Nagy (1991)	d	41
Nagy (1991)	d	17.113

METHYL SALICYLATE *[119-36-8]*

Tempelaar (1913); Zwaardemaker (1927)	d	0.1 - 0.97
Backman (1917)	r	0.8 - 0.9
Allison & Katz (1919)		100
Baldus (1936)	d	0.002
Baldus (1936)	r	0.0035

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Jung (1936)	d	0.023
Jung (1936)	r	0.023
Gundlach & Kenway (1939)	d	119
Jones (1954)	r	16.1
Jones (1955c)	r	22.3
Kerka & Humphreys (1956)		0.06
Nader (1958)		0.031 - 0.56
Köster (1971)	d	98 - 112
Etzweiler *et al.* (1980)		0.016

3-(4-(2-METHYL-2-SILAPROPANE)PHENYL)-2-METHYLPROPANAL (sila-lilial)
Doszczak *et al.* (2007) 0.0033

3-(4-(2-METHYL-2-SILAPROPANE)PHENYL)PROPANAL (sila-bourgeonal)
Doszczak *et al.* (2007) 0.000 55

α-methylstyrene⇒ISOPROPENYLBENZENE
3-methyl-3-sulfanylbutyl acetate⇒3-MERCAPTO-3-METHYLBUTYL ACETATE
3-(methylsulfanyl)propanal⇒3-(METHYLTHIO)PROPANAL
(12*R*;9*Z*)-12-methyl-14-tetradeca-9-enolide⇒(12*R*)-14-HYDROXY-12-METHYL-9(*Z*)-TETRADECENOIC ACID LACTONE

6-METHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE (3a,4,5,7a-tetrahydro-6-methyl-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 25

6-METHYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE (δ-thiohexalactone)
Roling *et al.* (1998); Engel (1999) 0.022

METHYLTETRATHIOMETHANE (dimethyl tetrasulphide) *[5756-24-1]*
Gijs *et al.* (2000) 0.13

5-METHYL-2-THENYLTHIOL
Hofmann & Schieberle (1997) 0.000 019

4-METHYL-2-THIAZOLINE
Schieberle *et al.* (1996) > 2

4-METHYL-3-THIAZOLINE *[52558-99-3]*
Schieberle *et al.* (1996) 0.000 1

S-METHYL THIOACETATE *[1534-08-3]*
Gijs *et al.* (2000) 0.23

S-METHYL THIOBUTANOATE *[2432-51-1]*
Berger *et al.* (1999) 0.06
Gijs *et al.* (2000) 0.14

METHYL THIOCYANATE *[556-64-9]*
Katz & Talbert (1930) 0.75

S-METHYL THIODECANOATE *[1680-29-1]*
Berger *et al.* (1999) 0.1 - 0.5

S-METHYL THIODODECANOATE
Berger *et al.* (1999) 0.05 - 0.1

METHYLTHIOETHANE (methyl ethyl sulphide, ethyl methyl sulphide) *[624-89-5]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Moschandreas & Jones (1983)	d	0.042
Moschandreas & Jones (1983)	r	0.124
Gijs *et al.* (2000)		0.022

methyl thioglycolate⇒METHYL 2-MERCAPTOACETATE

S-METHYL THIOHEXADECANOATE

Berger *et al.* (1999)		0.1 - 0.5

S-METHYL THIOHEXANOATE (S-methyl hexanethioate) *[20756-86-9]*

Berger *et al.* (1999)		0.1 - 0.5

methyl 2-thiolacetate⇒METHYL 2-MERCAPTOACETATE

METHYLTHIOMETHANE (dimethyl sulphide) *[75-18-3]*

Katz & Talbert (1930)		0.0094
Nevers & Oister (1965)		0.0035
Guadagni (1966)		0.003
Leonardos *et al.* (1969)	r	0.0025
Wilby (1969)	r	0.0063
Lindvall (1970)	d	0.002 - 0.03
Laffort (1968b); Laffort & Dravnieks (1973)		0.014
Hamanabe *et al.* (1969)		0.025
Selyuzhitskii (1972)		0.75
Ifeadi (1972)		0.65
Cormack *et al.* (1974)		0.0075
Nishida *et al.* (1975)	d	0.0025 - 0.065
Nishida *et al.* (1979)	d	0.16
Anon. (1980)	d	0.000 3
Anon. (1980)	r	0.0058
Moschandreas & Jones (1983)	d	0.027
Moschandreas & Jones (1983)	r	0.049
Randebrock (1986)		20.6
Nagy (1991)	d	0.051
Nagata (2003)	d	0.0075
Glindemann *et al.* (2006)	d	0.001

(methylthio)methyl disulphide⇒2,4,5,7-TETRATHIAOCTANE

S-METHYL THIOOCTADECANOATE (S-methyl thiostearate)

Berger *et al.* (1999)		0.9

S-METHYL THIOOCTANOATE *[2432-83-9]*

Berger *et al.* (1999)		0.1 - 0.5

S-METHYL THIOPENTANOATE *[42075-43-4]*

Berger *et al.* (1999)		0.1 - 0.5

5-methyl-2-thiophenecarbaldehyde⇒2-FORMYL-5-METHYLTHIOPHENE

2-METHYL-3-THIOPHENETHIOL *[2527-76-6]*

Hofmann & Schieberle (1995)		0.000 0032 - 0.000 0128

3-(METHYLTHIO)PROPANAL (methional, 3-(methylsulfanyl)propanal, 4-thiapentanal) *[3268-49-3]*

Blank *et al.* (1989,1992);		
Schieberle & Grosch (1994)		0.000 08 - 0.000 2
Von Ranson & Belitz (1992a)	d	0.000 063

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Von Ranson & Belitz (1992a) r 0.000 063
Gijs *et al.* (2000) 0.06

S-METHYL THIOPROPANOATE (S-methyl thiopropionate) *[5925-75-7]*
Berger *et al.* (1999) 0.05 - 0.1

3-(METHYLTHIO)-1-PROPANOL (methionol) *[505-10-2]*
Gijs *et al.* (2000) 1.71

METHYLTHIO-2-PROPENE (methyl allyl sulphide) *[10152-76-8]*
Nagata (2003) d 0.000 50

S-methyl thiopropionate⇒S-METHYL THIOPROPANOATE
S-methyl thiostearate⇒S-METHYL THIOOCTADECANOATE

S-METHYL TETRADECANOATE
Berger *et al.* (1999) 0.1 - 0.5

2-methyl-3,3,3-trichloro-1-propanol⇒1,1,1-TRICHLORO-2-METHYL-3-PROPANOL

(±)-(2*R**,3*S**,7*S**)-2-METHYLTRICYCLO[7.6.0.0^{3,7}]PENTADEC-1(9)-EN-4-ONE
Kraft & Gallo (2004) 0.124

(1*S*,2*S*,1'*S*,3'*R*,5'*R*)-1-METHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]-3-HEXYLMETHYL) CYCLOPROPYLMETHANOL
Bajgrowicz *et al.* (1998);
Bajgrowicz & Fráter (2000) 0.000 015

(1*R*,2*R*,1'*S*,3'*R*,5'*R*)-1-METHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]-3-HEXYLMETHYL) CYCLOPROPYLMETHANOL
Bajgrowicz *et al.* (1998);
Bajgrowicz & Fráter (2000) > 0.015

(1*S*,2*S*,1'*R*,3'*S*,5'*S*)-1-METHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]-3-HEXYLMETHYL) CYCLOPROPYLMETHANOL
Bajgrowicz *et al.* (1998);
Bajgrowicz & Fráter (2000) 0.000 51

(1*R*,2*R*,1'*R*,3'*S*,5'*R*)-1-METHYL-2-(1,2,2-TRIMETHYLBICYCLO[3.1.0]-3-HEXYLMETHYL) CYCLOPROPYLMETHANOL
Bajgrowicz *et al.* (1998);
Bajgrowicz & Fráter (2000) 0.0014

(+)-(*R*,*E*)-3-METHYL-4-(2,6,6-TRIMETHYLCYCLOHEX-2-ENYL)-3-BUTEN-2-ONE
Abate *et al.* (2007) 0.000 079

(-)-(*S*,*E*)-3-METHYL-4-(2,6,6-TRIMETHYLCYCLOHEX-2-ENYL)-3-BUTEN-2-ONE
Abate *et al.* (2007) 0.000 83

(+)-(*R*,*E*)-2-METHYL-4-(1,1,2-TRIMETHYL-2-PENTEN-5-YL)-2-BUTEN-1-OL
Bajgrowicz *et al.* (1998) 0.000 06

(-)-(*S*,*E*)-2-METHYL-4-(1,1,2-TRIMETHYL-2-PENTEN-5-YL)-2-BUTEN-1-OL
Bajgrowicz *et al.* (1998) 0.004

(2'*E*)-2'-METHYL-2'-(1'',2'',4''-TRIMETHYLPENT-2''-ENYLOXY)PROPYL ACETATE
Kraft & Eichenberger (2004) 0.012

(2’*Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL ACETATE
Kraft & Eichenberger (2004) 0.054

(2’*E/Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL BUTANOATE
Kraft & Eichenberger (2004) 0.0018

(2’’E/*Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL 3-BUTENOATE
Kraft & Eichenberger (2004) 0.0046

(2*E*,2’’E/*Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL 2-BUTENOATE
Kraft & Eichenberger (2004) 0.0085

(2’*E*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL CYCLOPROPANECARBOXYLATE
Kraft & Eichenberger (2004) 0.0057

(2’*Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL CYCLOPROPANECARBOXYKATE
Kraft & Eichenberger (2004) 0.0081

(2’*E/Z*)-6-METHYL-6-(1’,2’,4’-TRIMETHYLPENT-2-ENYLOXY)-3-HEPTANONE
Kraft & Eichenberger (2004) 0.000 55

(2’*E/Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL 2-METHYL-2-PROPENOATE
Kraft & Eichenberger (2004) 0.0037

(2’*E/Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL 3-METHYLPROPANOATE
Kraft & Eichenberger (2004) 0.0022

(2’*E*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL PROPANOATE
Kraft & Eichenberger (2004) 0.0012

(2’*Z*)-2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENT-2’’-ENYLOXY)PROPYL PROPANOATE
Kraft & Eichenberger (2004) 0.0054

2’-METHYL-2’-(1’’,2’’,4’’-TRIMETHYLPENTYLOXY)PROPYL BUTANOATE
Kraft & Eichenberger (2004) 0.001

6-METHYL-6-(1’,2’,4’-TRIMETHYLPENTYLOXY)-3-HEPTANONE
Kraft & Eichenberger (2004) 0.000 87

METHYLTRITHIOMETHANE (dimethyl trisulphide, 2,3,4-trithiapentane) *[3658-80-8]*
Wilby (1969) r 0.0073
Blank *et al.* (1989) 0.000 06 - 0.000 1
Khiari *et al.* (1992) 0.0075
Gijs *et al.* (2000); Vermeulen & Collin (2002) 0.014

2-METHYL-2-UNDECANETHIOL (*tert*-dodecyl mercaptan)
Blinova (1965) 4 – 9

methyl n-valerate⇒METHYL PENTANOATE
methyl vinyl ketone⇒1-BUTEN-3-ONE

(2*R*,2’*R*,5’*R*)-2-[(5’-METHYL-5’-VINYL)TETRAHYDROFURAN-2’-YL]PROPANAL ((2*R*,2’*R*,5’*R*)-lilac

aldehyde)
Kreck & Mosandl (2003) 0.22

(2*R*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL ((2*R*,2'*R*,5'*S*)-lilac aldehyde)
Kreck & Mosandl (2003) 0.004

(2*R*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL ((2*R*,2'*S*,5'*R*)-lilac aldehyde)
Kreck & Mosandl (2003) 0.18

(2*R*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL ((2*R*,2'*S*,5'*S*)-lilac aldehyde)
Kreck & Mosandl (2003) 0.003

(2*S*,2'*R*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL ((2*S*,2'*R*,5'*R*)-lilac aldehyde)
Kreck & Mosandl (2003) 0.20

(2*S*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL ((2*S*,2'*R*,5'*S*)-lilac aldehyde)
Kreck & Mosandl (2003) 0.003

(2*S*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL ((2*S*,2'*S*,5'*R*)-lilac aldehyde)
Kreck & Mosandl (2003) 0.04

(2*S*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANAL ((2*S*,2'*S*,5'*S*)-lilac aldehyde)
Kreck & Mosandl (2003) 0.002

(2*R*,2'*R*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*R*,2'*R*,5'*R*)-lilac alcohol)
Kreck & Mosandl (2003) > 1

(2*R*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*R*,2'*R*,5'*S*)-lilac alcohol)
Kreck & Mosandl (2003) 0.04

(2*R*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*R*,2'*S*,5'*R*)-lilac alcohol)
Kreck & Mosandl (2003) 0.22

(2*R*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*R*,2'*S*,5'*S*)-lilac alcohol)
Kreck & Mosandl (2003) 0.04

(2*S*,2'*R*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*S*,2'*R*,5'*R*)-lilac alcohol)
Kreck & Mosandl (2003) 0.80

(2*S*,2'*R*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*S*,2'*R*,5'*S*)-lilac alcohol)
Kreck & Mosandl (2003) 0.02

(2*S*,2'*S*,5'*R*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*S*,2'*S*,5'*R*)-lilac alcohol)

Kreck & Mosandl (2003)		0.74

(2*S*,2'*S*,5'*S*)-2-[(5'-METHYL-5'-VINYL)TETRAHYDROFURAN-2'-YL]PROPANOL ((2*S*,2'*S*,5'*S*)-lilac alcohol)

Kreck & Mosandl (2003)		0.02

4-methyl-5-vinylthiazole⇒5-ETHENYL-4-METHYLTHIAZOLE
methyl xanthate⇒METHYL ETHOXYDITHIOFORMATE
MIB⇒1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL
mintlactone⇒(6*R*,7a*R*)-3,6-DIMETHYL-5,6,7-7a-TETRAHYDRO-2(4*H*)-BENZOFURANONE
E-mintlactone⇒(6*S*,7a*S*)-3,6-DIMETHYL-5,6,7,7a-TETRAHYDRO-2(4*H*)-BENZOFURANONE

MONOETHYL SUCCINATE (ethyl acid succinate) *[1070-34-4]*

Backman (1917)	r	6.4

MORPHOLINE *[110-91-8]*

Hellman & Small (1973,1974)	d	0.04
Hellman & Small (1973,1974)	r	0.25

MOXALONE

Kraft & Fráter (2001)		0.001

muscenone⇒3-METHYLCYCLOPENTADECENONE
muscone⇒3-METHYLCYCLOPENTADECANONE
(3R)-(-)-muscone⇒(3R)-(-)-3-METHYLCYCLOPENTADECANONE
musk ambrette⇒1-*tert*-BUTYL-2-METHOXY-4-METHYL-3,5-DINITROBENZENE
musk C-14⇒ETHYLENE 1,12-DODECANEDIOATE
musk ketone⇒4-*tert*-BUTYL-2,6-DIMETHYL-3,5-DINITROACETOPHENONE
musk R-1⇒16-HYDROXY-11-OXAHEXADECANOIC ACID LACTONE
musk T⇒CYCLO-1,3-ETHYLENEDIOXY-1,3-TRIDECANEDIONE
musk xylene⇒1-*tert*-BUTYL-3,5-DIMETHYL-2,4,6-TRINITROBENZENE
mustard gas⇒2-CHLORO-1-(2-CHLOROETHYLTHIO)ETHANE
myrcene⇒7-METHYL-3-METHYLENE-1,6-OCTADIENE
myristicin⇒5-ALLYL-1-METHOXY-2,3-METHYLENEDIOXYBENZENE

N

1-NAPHTHALDEHYDE *[66-77-3]*

Von Ranson & Belitz (1992b)	d	3.6
Von Ranson & Belitz (1992b)	r	3.4

2-NAPHTHALDEHYDE *[66-99-9]*

Von Ranson & Belitz (1992b)	d	0.19
Von Ranson & Belitz (1992b)	r	0.26

NAPHTHALENE *[91-20-3]*

Backman (1917)	r	0.05 - 0.055
Mitsumoto (1926)	r	4.0 - 4.4
Hesse (1927)	r	0.3
Morimura (1934)	r	3.37 - 5.34
Robbins (1951)		< 1.6
Punter (1983)	d	0.20
Moriguchi *et al.* (1983)	d	0.007
Sävenhed *et al.* (1985)	d	0.01 - 0.04
Nagy (1991)	d	0.45

1-naphthol⇒1-HYDROXYNAPHTHALENE
2-naphthol⇒2-HYDROXYNAPHTHALENE
1-naphthylamine⇒1-AMINONAPHTHALENE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2-naphthylamine⇒2-AMINONAPHTHALENE

2-(1-NAPHTHYL)ETHANAL

Von Ranson & Belitz (1992b)	d	37
Von Ranson & Belitz (1992b)	r	49

neohexane⇒2,2-DIMETHYLBUTANE
neral⇒3,7-DIMETHYL-(*Z*)-2,6-OCTADIENAL
nerol⇒3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-OL
neryl acetate⇒3,7-DIMETHYL-(*Z*)-2,6-OCTADIEN-1-YL ACETATE

NICKEL CARBONYL *[13463-39-3]*

Armit (1907)		3.5
Kincaid *et al.* (1956)		7 – 21

S-(-)-nicotine⇒(*S*)-(-)-3-(1-METHYL-2-PYRROLIDINYL)PYRIDINE

NIRVANOLIDE

Kraft & Fráter (2001)		0.000 1

NITRIC ACID *[7697-37-2]*

Melekhina (1968)	d	0.70

2-NITROBENZALDEHYDE *[552-89-6]*

Backman (1917)	r	4.5

3-NITROBENZALDEHYDE *[99-61-6]*

Backman (1917)	r	3.0

NITROBENZENE *[98-95-3]*

Hermanides (1909)	r	0.0412
Zwaardemaker (1914,1927)	d	0.04 - 0.041
Backman (1917)	r	0.34 - 0.70
Allison & Katz (1919)		146
Henning (1924)	d	0.0065
Katz & Talbert (1930)		9.6
Van Anrooij (1931)	d	0.019
Janiček *et al.* (1960)		19
Andreescheva (1964)		0.0182
Gavaudan & Poussel (1966)		0.15
Leonardos *et al.* (1969)	r	0.024
Randebrock (1971)		0.002
Öztürk (1976)		0.363
Nauš (1982)	d	0.2
Nauš (1982)	r	20
Randebrock (1986)		0.0053

2-nitro-*p*-cresol⇒4-METHYL-2-NITROPHENOL

NITROGEN DIOXIDE *[10544-72-6]*

Beck (1959); Henschler *et al.* (1960)		0.2 - 1.0
Shalamberidze (1967)		0.23
Rumsey & Cesta (1970)		< 1
Knuth (1973)		0.11
Prusakov *et al.* (1976)		0.2 - 0.26
Braker & Mossman (1980)	r	< 9.4
Nagata (2003)	d	0.23

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

NITROMETHANE *[75-52-5]*		
Nagy (1991)	d	124
2-NITROPHENOL *[88-75-5]*		
Backman (1917)	r	0.11 - 0.13
Stuiver (1958)	d	0.0012
3-NITROPHENOL *[554-84-7]*		
Stuiver (1958)	d	3.0
4-NITROPHENOL *[100-02-7]*		
Stuiver (1958)	d	2.3
1-NITROPROPANE *[108-03-2]*		
Dravnieks & Laffort (1972)		28.2
Dravnieks (1974)	d	510
2-NITROPROPANE *[79-46-9]*		
Treon & Dutra (1952)		297 - 1,050
Hine *et al.* (1978)	r	580
Crawford *et al.* (1984)	d	18
NITROSOBENZENE *[586-96-9]*		
Randebrock (1971)		0.15
Randebrock (1986)		0.32
N-NITROSODIMETHYLAMINE *[62-75-9]*		
Prusakov *et al.* (1976)		0.024 - 0.04
3-nitrotoluene⇒1-METHYL-3-NITROBENZENE		
(*E*,*E*)-2,4-NONADIENAL *[5910-87-2]*		
Teranishi *et al.* (1974)		0.000 25
Hall & Andersson (1983)	d	0.000 40
Yang *et al.* (2008)		0.000 2
(*E*,*E*)-2,6-NONADIENAL *[17587-33-6]*		
Von Ranson & Belitz (1992a)	r	0.0014
(*E*,*Z*)-2,6-NONADIENAL *[557-48-2]*		
Teranishi *et al.* (1974)		0.000 06
Ulrich & Grosch (1988a)		0.0080 - 0.022
Ulrich & Grosch (1988b)		0.000 6 - 0.0016
Grosch (1989)		0.000 9 - 0.0025
Von Ranson & Belitz (1992a)	d	0.0011
Von Ranson & Belitz (1992a)	r	0.0039
Guth (1991)		0.000 25
Heiler & Schieberle (1996)		0.000 2
Ömür-Özbek *et al.* (2007)		0.000 11
(*Z*,*E*)-2,6-NONADIENAL		
Von Ranson & Belitz (1992a)	d	0.000 28
Von Ranson & Belitz (1992a)	r	0.0083
(*Z*,*Z*)-2,6-NONADIENAL		
Von Ranson & Belitz (1992a)	d	0.0022
Von Ranson & Belitz (1992a)	r	0.0041

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(*E*,*E*)-3,6-NONADIENAL

Von Ranson & Belitz (1992a)	d	0.36
Von Ranson & Belitz (1992a)	r	0.36

(*E*,*Z*)-3,6-NONADIENAL

Von Ranson & Belitz (1992a)	d	0.0025
Von Ranson & Belitz (1992a)	r	0.094

(*Z*,*E*)-3,6-NONADIENAL

Von Ranson & Belitz (1992a)	d	0.0011
Von Ranson & Belitz (1992a)	r	0.0019

(*Z*,*Z*)-3,6-NONADIENAL *[21944-83-2]*

Von Ranson & Belitz (1992a)	d	0.000 28
Von Ranson & Belitz (1992a)	r	0.000 83

(*E*,*Z*)-2,6-NONADIEN-1-OL *[28069-72-9]*

Heiler (1995); Heiler & Schieberle (1996)	0.000 07

(*Z*)-1,5-NONADIEN-3-OL

Schieberle & Buettner (2001)	0.001 - 0.01

(*Z*)-1,5-NONADIEN-3-ONE

Schieberle & Buettner (2001)	0.000 01 - 0.000 1

γ-nonalactone⇒4-HYDROXYNONANOIC ACID LACTONE
(*R*)-γ-nonalactone ⇒(*R*)-4-HYDROXYNONANOIC ACID LACTONE
(*S*)-γ-nonalactone⇒(*S*)-4-HYDROXYNONANOIC ACID LACTONE
(*R*)-δ-nonalactone⇒(*R*)-5-HYDROXYNONANOIC ACID LACTONE
(*S*)-δ-nonalactone⇒(*S*)-5-HYDROXYNONANOIC ACID LACTONE

NONANAL (nonyl aldehyde, pelargonaldehyde) *[124-19-6]*

Pliška & Janiček (1965)		1.7
Randebrock (1971)		0.000 3
Teranishi *et al.* (1974)		0.030
Hall & Andersson (1983)	d	0.045
Fuller *et al.* (1964)	r	0.0096
Cristoph (1983)	r	0.045 - 0.06
Randebrock (1986)		0.00094
Ulrich & Grosch (1988b)		0.0052 - 0.0121
Lindell (1991)	d	0.23
Khiari *et al.* (1992)		0.01
Von Ranson & Belitz (1992a)	d	0.020
Von Ranson & Belitz (1992a)	r	0.020
Nagata (2003)	d	0.0020
Yang *et al.* (2008)		0.0026
Cometto-Muñiz & Abraham (2010a)	d	0.0031

NONANE *[111-84-2]*

Mullins (1955)	r	108
Laffort & Dravnieks (1973)		60
Nagata (2003)	d	12

2,4-NONADIENE *[34266-16-5]*

Guth & Grosch (1989)	0.0026 - 0.0052

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

NONANOIC ACID (pelargonic acid) *[112-05-0]*

Passy (1893b,1893c)	d	0.02
Backman (1917)	r	0.0016 - 0.0032
Punter (1983)	d	0.12

1-NONANOL (nonyl alcohol) *[143-08-8]*

Pliška (1962)		41
Cristoph (1983)	r	0.015 - 0.02
Nagata (2003)	d	0.0053
Yang *et al.* (2008)		0.018

4-nonalide⇒4-HYDROXYNONANOIC ACID LACTONE

2-NONANONE (methyl heptyl ketone) *[821-55-6]*

Teranishi *et al.* (1974)		0.075
Hall & Andersson (1983)	d	1.7
Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993)	d	5.5
Yang *et al.* (2008)		0.031
Cometto-Muñiz & Abraham (2009a)	d	0.032

(*E*,*E*,*E*)-2,4,6-NONATRIENAL

Schuh & Schieberle (2005)		0.0044

(*E*,*E*,*Z*)-2,4,6-NONATRIENAL

Schuh & Schieberle (2005)		0.000 000 2

(*E*,*Z*,*E*)-2,4,6-NONATRIENAL

Schuh & Schieberle (2005)		0.000 1

(*E*)-2-NONENAL *[18829-56-6]*

Teranishi *et al.* (1974)		0.000 5
Hall & Andersson (1983)	d	0.0036
Koszinowski & Piringer (1986)		0.001
Ulrich & Grosch (1987)		0.0057 - 0.010
Grosch (1989)		0.000 5
Guth (1991); Schieberle & Grosch (1991,1994)		0.000 1 - 0.000 13
Lindell (1991)	d	0.025
Von Ranson & Belitz (1992a)	d	0.000 39
Von Ranson & Belitz (1992a)	r	0.0011
Yang *et al.* (2008)		0.000 09

(*Z*)-2-NONENAL *[60784-31-8]*

Ulrich & Grosch (1988b); Grosch (1989)		0.000 08 - 0.000 23
Guth (1991)		0.000 005
Von Ranson & Belitz (1992a)	d	0.000 022
Von Ranson & Belitz (1992a)	r	0.000 039

(*E*)-3-NONENAL *[109351-28-2]*

Ulrich & Grosch (1988b)		0.0026 - 0.0054
Von Ranson & Belitz (1992a)	d	0.000 28
Von Ranson & Belitz (1992a)	r	0.0014

(*Z*)-3-NONENAL *[31823-43-5]*

Ulrich & Grosch (1988b); Grosch (1989)		0.0023 - 0.0055
Von Ranson & Belitz (1992a)	d	0.000 28
Von Ranson & Belitz (1992a)	r	0.000 56

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(*E*)-4-NONENAL *[2277-16-9]*		
Von Ranson & Belitz (1992a)	d	0.0022
Von Ranson & Belitz (1992a)	r	0.025
(*Z*)-4-NONENAL *[2277-15-8]*		
Von Ranson & Belitz (1992a)	d	0.0045
Von Ranson & Belitz (1992a)	r	0.0045
(*E*)-5-NONENAL		
Von Ranson & Belitz (1992a)	d	0.0070
Von Ranson & Belitz (1992a)	r	0.020
6-NONENAL *[6728-35-4]*		
Lindell (1991)	d	0.023
(*E*)-6-NONENAL *[2277-20-5]*		
Von Ranson & Belitz (1992a)	d	0.000 022
Von Ranson & Belitz (1992a)	r	0.000 14
(*Z*)-6-NONENAL *[2277-19-2]*		
Von Ranson & Belitz (1992a)	d	0.000 14
Von Ranson & Belitz (1992a)	r	0.0036
(*E*)-7-NONENAL		
Von Ranson & Belitz (1992a)	d	0.014
Von Ranson & Belitz (1992a)	r	0.12
8-NONENAL *[39770-04-2]*		
Von Ranson & Belitz (1992a)	d	0.000 20
Von Ranson & Belitz (1992a)	r	0.000 84
1-NONENE *[124-11-8]*		
Koszinowski & Piringer (1983)		37
Nagata (2003)	d	0.0028
1-NONEN-3-OL *[21964-44-3]*		
Koszinowski & Piringer (1983)		0.04
Schieberle & Buettner (2001)		0.001 - 0.01
3-NONEN-2-ONE *[14309-57-0]*		
Koszinowski & Piringer (1986)		0.03
1-NONEN-3-ONE *[24415-26-7]*		
Koszinowski & Piringer (1983,1986)		0.000 02
Schieberle & Buettner (2001)		ca. 0.000 01
4-NONEN-3-ONE		
Koszinowski & Piringer (1986)		0.08
2-NONEN-4-ONE *[32064-72-5]*		
Koszinowski & Piringer (1986)		0.02
(*E*)-2-NONEN-4-ONE *[27743-70-0]*		
Teranishi *et al.* (1974)		0.0027
5-NONEN-4-ONE		

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Koszinowski & Piringer (1986) 0.5

nonyl alcohol⇒1-NONANOL
nonyl aldehyde⇒NONANAL
5-norbornen-2(*exo*)aldehyde⇒BICYCLO[2.2.1]-5-HEPTENE-2(*exo*)-CARBOXALDEHYDE
5-norbornen-2(*endo*)aldehyde⇒BICYCLO[2.2.1]-5-HEPTENE-2(*endo*)-CARBOXALDEHYDE

O

oak lactone⇒4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE
(*E*)-β-ocimene⇒(*E*)-3,7-DIMETHYL-1,3,6-OCTATRIENE
(*Z*)-β-ocimene⇒(*Z*)-3,7-DIMETHYL-1,3,6-OCTATRIENE
(*E*,*Z*)-8-ocimenyl acetate ⇒2,6-DIMETHYL-(*Z*,*E*)-2,5,7-OCTATRIENYL ACETATE
(*Z*,*Z*)-8-ocimenyl acetate ⇒2,6-DIMETHYL-(*Z*,*Z*)-2,5,7-OCTATRIENYL ACETATE

OCTADECANE *[593-45-3]*
Patte (1978); Patte & Punter (1979) d 0.02

cis-9-OCTADECENOIC ACID (oleic acid) *[112-80-1]*
Nagy (1991) d 44

(*E*,*E*)-2,4-OCTADIENAL *[30361-28-5]*
Hall & Andersson (1983) d 0.012

(*Z*)-1,5-OCTADIEN-3-HYDROPEROXIDE
Guth & Grosch (1990); Guth (1991) 0.000 03 - 0.000 06

(*E*)-1,5-OCTADIEN-3-ONE *[359794-78-8]*
Grosch (1989); Guth (1991) 0.000 2 - 0.000 4

(*Z*)-1,5-OCTADIEN-3-OL *[50306-18-8]*
Schieberle & Buettner (2001) 0.0013

(*Z*)-1,5-OCTADIEN-3-ONE *[65767-22-8]*
Ulrich & Grosch (1988a) 0.000 20 - 0.000 51
Ulrich & Grosch (1988b); Grosch (1989) 0.000 01 - 0.000 04
Guth & Grosch (1990);
Schieberle & Buettner (2001) 0.000 003 - 0.000 006

(-)-(4*R*,4a*R*,8a*R*)-1,2,3,4,4a,5,8,8a-OCTAHYDRO-4,8a-DIMETHYLNAPHTHALEN-4a-OL ((-)-dehydrogeosmin)
Huber *et al.* (1993) r 0.000 01

(+)-(4*S*,4a*S*,8a*S*)-1,2,3,4,4a,5,8,8a-OCTAHYDRO-4,8a-DIMETHYLNAPHTHALEN-4a-OL ((+)-dehydrogeosmin)
Huber *et al.* (1993) r 0.000 14

(1*R**,1'*R**,3'*S**)-3,4,5,6,7,8,9,10-OCTAHYDRO-1,3'-DIMETHYL-1*H*-SPIRO(BENZO[8]ANNULENE-2,1'-CYCLOHEXAN)-2'-ONE
Kraft *et al.* (2006a) 0.069

(1*R**,1'*S**,3'*S**)-3,4,5,6,7,8,9,10-OCTAHYDRO-1,3'-DIMETHYL-1*H*-SPIRO(BENZO[8]ANNULENE-2,1'-CYCLOHEXAN)-2'-ONE
Kraft *et al.* (2006a) 0.000 59

1,2,3,4,5,6,7,8-OCTAHYDRO-3,5,5,6,8,8-HEXAMETHYL-2-NAPHTHALENONE (3,5,5,6,8,8-hexamethyl-1,2,3,4,5,6,7,8-octahydro-2-naphthalenone)

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Bachmann & Pesaro (1990) 0.000 13

(1*R**,1'*R**)-3,4,5,6,7,8,9,10-OCTAHYDRO-1-METHYL-1*H*-SPIRO(BENZO[8]ANNULENE-2,1'-CYCLOHEXAN)-2'-ONE
Kraft *et al.* (2006a) 0.213

(1*R**,1'*R**)-3,4,5,6,7,8,9,10-OCTAHYDRO-1-METHYL-1*H*-SPIRO(BENZO[8]ANNULENE-2,1'-CYCLOPENTAN)-2'-ONE
Kraft *et al.* (2006a) 0.0022

(1*R**,1'*S**)-3,4,5,6,7,8,9,10-OCTAHYDRO-1-METHYL-1*H*-SPIRO(BENZO[8]ANNULENE-2,1'-CYCLOPENTAN)-2'-ONE
Kraft *et al.* (2006a) 0.000 80

1,2,3,4,5,6,7,8-OCTAHYDRO-2,3,8,8-TETRAMETHYL-2-ACETOPHENONE
Etzweiler *et al.* (1992) 0.5

(±)-1-[(1*R**,2*R**,8a*S**)-1,2,3,5,6,7,8,8a-OCTAHYDRO-1,2,8,8-TETRAMETHYLNAPHTHALEN-2-YL]-1-ETHANONE
Nussbaumer *et al*. (1999) 0.000 005

(-)-(1*R*,2*S*)-1-(1,2,3,4,5,6,7,8-OCTAHYDRO-1,2,8,8-TETRAMETHYLNAPHTHALEN-2-YL)ETHANONE ((-)-(1*R*,2*S*)-georgywood®)
Fráter *et al*. (2004) 0.000 02

(+)-(1*S*,2*R*)-1-(1,2,3,4,5,6,7,8-OCTAHYDRO-1,2,8,8-TETRAMETHYLNAPHTHALEN-2-YL)ETHANONE ((+)-(1*S*,2*R*)-georgywood®)
Fráter *et al*. (2004) 0.0035

(*R*)-γ-octalactone⇒(*R*)-4-HYDROXYOCTANOIC ACID LACTONE
(*S*)-γ-octalactone⇒(*S*)-4-HYDROXYOCTANOIC ACID LACTONE
(*R*)-δ-octalactone⇒(*R*)-5-HYDROXYOCTANOIC ACID LACTONE
(*S*)-δ-octalactone⇒(*S*)-5-HYDROXYOCTANOIC ACID LACTONE

OCTANAL (caprylaldehyde) *[124-13-0]*

Pliška & Janiček (1965)		5
Teranishi *et al.* (1974)		0.015
Hall & Andersson (1983)	d	0.0078
Cristoph (1983)	r	0.005 - 0.009
Ulrich & Grosch (1988b)		0.0058 - 0.0136
Lindell (1991)	d	0.040
Von Ranson & Belitz (1992a)	d	0.010
Von Ranson & Belitz (1992a)	r	0.010
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	0.021
Nagata (2003)	d	0.000 052
Yang *et al.* (2008)		0.000 4
Cometto-Muñiz & Abraham (2010a)	d	0.000 88
Laska & Ringh (2010)	d	0.17

OCTANE *[111-65-9]*

Jones (1955c)	r	550
May (1966)	d	710
May (1966)	r	1,100
Laffort & Dravnieks (1973)		71
Nagy (1991)	d	61.8
Nagy (1991)	d	90.102

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Nagata (2003)	d	8.0
1-OCTANETHIOL (octyl mercaptan) *[111-88-6]*		
Patte (1978); Patte & Punter (1979)	d	0.000 1
OCTANOIC ACID (caprylic acid) *[124-07-2]*		
Passy (1893b,1893c)	d	0.05
Backman (1917)	r	0.000 3
Jones (1955c)	r	3.2
Punter (1983)	d	0.60
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	0.000 065
Ferreira *et al.* (1998)		0.5
Wise *et al.* (2007); Miyazawa *et al.* (2009a)	d	0.011 - 0.018
Cometto-Muñiz & Abraham (2010b)	d	0.0051
1-OCTANOL (octyl alcohol) *[111-87-5]*		
Backman (1917)	r	0.005 - 0.008
Rocén (1920)	r	0.005
Gavaudan *et al.* (1948)		0.02
Mullins (1955)	r	5.44
Pliška & Janiček (1960)		0.14
Pliška (1962)		9
Stone *et al.* (1967)	d	0.05
Cain (1969)	r	0.5
Punter (1983)	d	0.73
Cristoph (1983)	r	0.03 - 0.05
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	0.037
Nagata (2003)	d	0.014
Cometto-Muñiz & Abraham (2008)	d	0.023
Yang *et al.* (2008)		0.022
2-OCTANOL *[123-96-6]*		
Passy (1892c)	d	0.05
Backman (1917)	r	0.013 - 0.016
Punter (1983)	d	0.14
(*S*)-(+)-2-OCTANOL *[6169-06-8]*		
Stuiver (1958)	d	0.0011
Punter (1983)	d	0.067
(*R*)-(-)-2-OCTANOL *[5978-70-1]*		
Stuiver (1958)	d	0.000 3
Punter (1983)	d	0.22
2-OCTANONE (methyl hexyl ketone) *[111-13-7]*		
Backman (1917)	r	0.06 - 0.1
Stone (1963b,1963c)	d	0.78
Hall & Andersson (1983)	d	0.23
3-OCTANONE (ethyl amyl ketone) *[106-68-3]*		
Appell (1969)		0.0013
(*E*)-2-OCTENAL *[2548-87-0]*		
Teranishi *et al.* (1974)		0.009
Eriksson *et al.* (1976)	d	0.014

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Hall & Andersson (1983)	d	0.047
Lindell (1991)	d	0.012
Von Ranson & Belitz (1992a)	d	0.25
Von Ranson & Belitz (1992a)	r	0.25
Yang *et al.* (2008)		0.0027
(*Z*)-2-OCTENAL *[20664-46-4]*		
Koszinowski & Piringer (1986)		0.007
Ulrich & Grosch (1987)		0.0042 - 0.0083
Ulrich & Grosch (1988b)		0.000 9 - 0.0025
Von Ranson & Belitz (1992a)	d	0.0076
Von Ranson & Belitz (1992a)	r	0.010
(*E*)-3-OCTENAL		
Von Ranson & Belitz (1992a)	d	0.043
Von Ranson & Belitz (1992a)	r	0.088
(*Z*)-3-OCTENAL *[78693-34-2]*		
Von Ranson & Belitz (1992a)	d	0.035
Von Ranson & Belitz (1992a)	r	0.045
(*E*)-4-OCTENAL		
Von Ranson & Belitz (1992a)	d	0.0040
Von Ranson & Belitz (1992a)	r	0.033
(*E*)-5-OCTENAL		
Von Ranson & Belitz (1992a)	d	0.0040
Von Ranson & Belitz (1992a)	r	0.0050
(*Z*)-5-OCTENAL *[41547-22-2]*		
Von Ranson & Belitz (1992a)	d	0.000 50
Von Ranson & Belitz (1992a)	r	0.0075
(*E*)-6-OCTENAL *[63826-25-5]*		
Von Ranson & Belitz (1992a)	d	0.000 18
Von Ranson & Belitz (1992a)	r	0.0020
7-OCTENAL *[21573-31-9]*		
Von Ranson & Belitz (1992a)	d	0.010
Von Ranson & Belitz (1992a)	r	0.018
1-OCTENE *[111-66-0]*		
Dravnieks & Laffort (1972)		0.33
Dravnieks (1974)	d	5
Koszinowski & Piringer (1983)		37
Cometto-Muñiz (1993);		
Cometto-Muñiz & Cain (1994)	d	945
Nagata (2003)	d	0.0046
2-OCTENE *[111-67-1]*		
Dravnieks & Laffort (1972)		0.50
Dravnieks (1974)	d	5
Lindell (1991)	d	2.0
1-OCTEN-3-HYDROPEROXIDE		
Guth & Grosch (1990)		0.000 6 - 0.0012

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1-OCTEN-3-OL *[3391-86-4]*

Hall & Andersson (1983)	d	0.048
Koszinowski & Piringer (1983)		0.11
Ulrich & Grosch (1987)		0.012 - 0.028
Fischer & Grosch (1987)		0.026 - 0.075
Benzo *et al*. (2007)		0.00236
Yang *et al.* (2008)		0.0027
Schieberle & Buettner (2001)		0.001 - 0.01

3-OCTEN-2-ONE *[1669-44-9]*

Koszinowski & Piringer (1986)		0.02
Yang *et al.* (2008)		0.0067

1-OCTEN-3-ONE *[4312-99-6]*

Koszinowski & Piringer (1983,1986)		0.000 1
Fischer & Grosch (1987)		0.000 8 - 0.0022
Ulrich & Grosch (1988b)		0.000 3 - 0.000 6
Grosch (1989)		0.000 4 - 0.0011
Guth & Grosch (1990);		
Schieberle & Buettner (2001)		0.000 03 - 0.000 12

4-OCTEN-3-ONE *[14129-48-7]*

Koszinowski & Piringer (1986)		0.05

2-OCTEN-4-ONE *[4643-27-0]*

Koszinowski & Piringer (1986)		0.2

3-OCTEN-5-ONE (5-octen-4-one)

Koszinowski & Piringer (1986)		0.5

5-octen-4-one⇒3-OCTEN-5-ONE

OCTYL ACETATE *[112-14-1]*

Cristoph (1983)	r	3.1 - 4.7
Cometto-Muñiz & Cain (1991);		
Cometto-Muñiz (1993)	d	2.7
Cometto-Muñiz *et al.* (2008)	d	0.14

2-OCTYL ACETATE (*sec*-octyl acetate) *[2051-50-5]*

Backman (1917)	r	0.022 - 0.038

sec-octyl acetate⇒2-OCTYL ACETATE
octyl alcohol⇒1-OCTANOL

OCTYLBENZENE *[2189-60-8]*

Cometto-Muñiz (1993);		
Cometto-Muñiz & Cain (1994)	d	2.9
Cometto-Muñiz & Abraham (2009b)	d	0.69

7-OCTYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.00405

5-OCTYLDIHYDRO-2(3*H*)-THIOPHENONE (γ-thiododecalactone)

Roling *et al.* (1998); Engel (1999)		0.022

octyl mercaptan⇒1-OCTANETHIOL

1-OCTYNE *[629-05-0]*
Cometto-Muñiz (1993);
Cometto-Muñiz & Cain (1994) d 609

2-OCTYNE *[2809-67-8]*
Dravnieks & Laffort (1972) 0.03
Dravnieks (1974) d 2

oenanthic acid⇒HEPTANOIC ACID
oleic acid⇒*cis*-9-OCTADECENOIC ACID
oryclone⇒4-*tert*-BUTYLCYCLOHEXYL ACETATE
10-oxahexadecanolide⇒16-HYDROXY-10-OXAHEXADECANOIC ACID LACTONE
11-oxahexadecanolide⇒16-HYDROXY-11-OXAHEXADECANOIC ACID LACTONE
12-oxahexadecanolide⇒16-HYDROXY-12-OXAHEXADECANOIC ACID LACTONE
oxalide T⇒16-HYDROXY-10-OXAHEXADECANOIC ACID LACTONE

2-OXAPROPYL SALICYLATE (methoxymethyl salicylate, mesotan)
Backman (1917) r 10

3-OXODECANAL *[56505-80-7]*
Guth & Grosch (1989) 0.000 35 - 0.000 70

5-OXODECANAL
Guth & Grosch (1989) 0.0011 - 0.0022

trans-(2*R*,3*R*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE (*trans*-(2*R*,3*R*)-magnolione)
Stang & Helmchen (2005) 0.0074

cis-(-)-(2*R*,3*S*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE (*cis-(-)*-(2*R*,3*S*)-magnolione)
Stang & Helmchen (2005) 0.071

cis-(+)-(2*S*,3*R*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE (*cis*-(+)-(2*S*,3*R*)-magnolione)
Stang & Helmchen (2005) 0.0029

trans-(2*S*,3*S*)-3-(2-OXOPROPYL)-2-PENTYLCYCLOPENTANONE (*trans*-(2*S*,3*S*)-magnolione)
Stang & Helmchen (2005) 0.006

OXYGEN DIFLUORIDE *[7783-41-7]*
Lester & Adams (1965) 0.22

OZONE *[10028-15-6]*
Witheridge & Yaglou (1939) 0.02 - 0.03
Wilska (1951) < 0.2
Beck (1959); Henschler *et al.* (1960) < 0.04
Buchberg *et al.* (1961) 0.07 - 0.5
Eglite (1968) 0.015
Nagata (2003) d 0.0064
Cain *et al.* (2007a) d 0.014

P

paraldehyde⇒2,4,6-TRIMETHYL-1,3,5-TRIOXANE

(+)-*nor*-PATCHOULENOL
Kraft *et al.* (2005) 0.0028

(-)-PATCHOULOL ((-)-(1*R*,3R,6S,7S,8S)-2,2,6,8-tetramethyltricyclo[5.3.1.0$^{3.8}$]-3-undecanol)

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Kraft *et al.* (2005)		0.000 93

pelargonaldehyde⇒NONANAL
pelargonic acid⇒NONANOIC ACID

PENTABORANE *[19624-22-7]*

Krackow (1953)		2.5

PENTABROMOANISOLE *[1825-26-9]*

Diaz *et al.* (2005)		0.0014

PENTACHLOROANISOLE *[1825-21-4]*

Nijssen (1988)	d	0.000 25
Diaz *et al.* (2005)		0.0016
Strube & Buettner (2010)		0.00219

pentadecalactone⇒15-HYDROXYPENTADECANOIC ACID LACTONE
15-pentadecanolide⇒15-HYDROXYPENTADECANOIC ACID LACTONE

1-PENTADECEN-3-OL *[99814-65-0]*

Koszinowski & Piringer (1983)		150

1-PENTADECEN-3-ONE

Koszinowski & Piringer (1983)		2.9

PENTANAL (valeraldehyde) *[110-62-3]*

Backman (1917)	r	0.009 - 0.010
Teranishi *et al.* (1974)		0.072
Anon. (1980)	d	0.0025
Anon. (1980)	r	0.013
Hall & Andersson (1983)	d	0.034
Cristoph (1983)	r	0.14 - 0.15
Lindell (1991)	d	0.092
Von Ranson & Belitz (1992a)	d	0.12
Von Ranson & Belitz (1992a)	r	0.22
Cometto-Muñiz *et al.* (1998a);		
Cometto-Muñiz (1999)	d	17.5
Nagata (2003)	d	0.0014
Laska & Ringh (2010)	d	0.85

PENTANE *[109-66-0]*

Patty & Yant (1929)		1,450
Mullins (1955)	r	3,090
Laffort & Dravnieks (1973)		350
Nagata (2003)	d	4.1

1,5-PENTANEDIAL (glutaraldehyde) *[111-30-8]*

Colwell (1976)	r	0.16
Cain *et al.* (2007b)	d	0.0015

2,3-PENTANEDIONE *[600-14-6]*

Hall & Andersson (1983)	d	0.063
Blank *et al.* (1992)		0.01 - 0.02

2,4-PENTANEDIONE *[123-54-6]*

Hellman & Small (1974)	d	0.04
Hellman & Small (1974)	r	0.08

1-PENTANETHIOL (amyl mercaptan) *[110-66-7]*

Sales (1958)		0.000 5
Kniebes *et al.* (1969)		0.000 4
Gijs *et al.* (2000)		0.000 7
Nagata (2003)	d	0.000 0034

PENTANOIC ACID (valeric acid) *[109-52-4]*

Passy (1893b,1893c)	d	0.01
Hermanides (1909)	r	0.0021
Zwaardemaker (1914,1927)	d	0.002 - 0.0021
Backman (1917)	r	0.02
Allison & Katz (1919)		29
Henning (1924)	d	0.000 8
Hesse (1926)	r	9.0
Hesse (1927)	r	3.0
Dubrovskaya *et al.* (1961)		0.5 - 1.0
Bocca & Battiston (1964)	d	0.025 - 0.07
Dubrovskaya & Khachaturyan (1973)		0.09 - 1.3
Marshal *et al.* (1981)	d	0.008
Punter (1983)	d	0.03 - 0.12
Neuner-Jehle & Etzweiler (1991)		0.000 15 - 0.000 26
Nagata (2003)	d	0.000 16

1-PENTANOL (amyl alcohol) *[71-41-0]*

Backman (1917)	r	1.0 - 1.2
Allison & Katz (1919)		225
Jung (1936)	d	0.4 - 0.81
Jung (1936)	r	1.62
Janiček *et al.* (1960)		11
Nauš (1962)	d	4
Pliška & Janiček (1965)		1,100
Gavaudan & Poussel (1966)		0.4
May (1966)	d	35
May (1966)	r	80
Stone *et al.* (1972)	d	1.2
Baikov *et al.* (1973)		0.1
Hellman & Small (1974)	d	0.8
Hellman & Small (1974)	r	1.1
Nauš (1982)	d	4
Nauš (1982)	r	30
Punter (1983)	d	2.0
Cristoph (1983)	r	1.0 - 1.1
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	5.0
Lindell (1991)	d	1.3
Ziemer *et al.* (2000)	d	0.020
Nagata (2003)	d	0.36
Yang *et al.* (2008)		0.153

2-PENTANOL (*sec*-pentanol) *[6032-29-7]*

Nagata (2003)	d	1.0

sec-pentanol⇒2-PENTANOL
tert-pentanol⇒2-METHYL-2-BUTANOL

2-PENTANONE (methyl propyl ketone) *[107-87-9]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Backman (1917)	r	11 - 15
May (1966)	d	27
May (1966)	r	48
Hall & Andersson (1983)	d	22
Laska & Hudson (1991)	d	6.7 - 8.3
Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993)	d	30.1
Patterson *et al.* (1993)	d	9.1
Nagata (2003)	d	0.098
Komthong *et al.* (2006)		230
Cometto-Muñiz *et al.* (2008);		
Cometto-Muñiz & Abraham (2009a)	d	0.35

3-PENTANONE (diethyl ketone) *[96-22-0]*

Backman (1917)	r	3.8
May (1966)	d	33
May (1966)	r	49
Dravnieks (1974)	d	3

(*E*)-2-PENTENAL *[1576-87-0]*

Von Ranson & Belitz (1992a)	d	1.4
Von Ranson & Belitz (1992a)	r	2.7

(*Z*)-2-PENTENAL *[1576-86-9]*

Von Ranson & Belitz (1992a)	d	> 1.7
Von Ranson & Belitz (1992a)	r	> 10

(*E*)-3-PENTENAL

Von Ranson & Belitz (1992a)	d	> 0.17
Von Ranson & Belitz (1992a)	r	> 0.17

(*Z*)-3-PENTENAL *[53448-06-9]*

Von Ranson & Belitz (1992a)	d	0.017
Von Ranson & Belitz (1992a)	r	0.017

4-PENTENAL *[2100-17-6]*

Von Ranson & Belitz (1992a)	d	0.017
Von Ranson & Belitz (1992a)	r	0.022

1-PENTENE *[109-67-1]*

Nagata (2003)	d	0.29

1-PENTEN-3-OL *[616-25-1]*

Hall & Andersson (1983)	d	4.3
Koszinowski & Piringer (1983)		0.66
Schieberle & Buettner (2001)		0.01 - 0.1

1-PENTEN-3-ONE (ethyl vinyl ketone) *[1629-58-9]*

Koszinowski & Piringer (1983)		0.02
Ulrich & Grosch (1988a)		0.013 - 0.035
Schieberle & Buettner (2001)		0.000 1 - 0.001

PENTYL ACETATE (amyl acetate) *[628-63-7]*

Grijns (1919)		0.9
Allison & Katz (1919)		39
Jones (1955c)	r	1.6
Gofmekler (1960)		0.6

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Pliška & Janiček (1960)		31
Guadagni (1966)		0.05
Davis (1973)	d	0.04
Hendriks (1979)	d	0.27
Slotnick (1981)		1.3
Laing (1982)	d	0.95
Punter (1983)	d	0.27 - 0.28
Cristoph (1983)	r	0.045 - 0.06
Walker *et al.* (1990)		6.9
Cometto-Muñiz & Cain (1991);		
Cometto-Muñiz (1993)	d	6.3
Walker *et al.* (1996)		0.53 - 5.3
Hoshika *et al.* (1997)	r	41
Ziemer *et al.* (2000)	d	0.049
Walker *et al.* (2003)		0.038 - 0.89
Komthong *et al.* (2006)		10.7 - 230
Olsson & Laska (2010)	d	2.2 - 2.7

PENTYLBENZENE *[538-68-1]*

Cometto-Muñiz (1993);		
Cometto-Muñiz & Cain (1994)	d	6.0

7-PENTYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 013

PENTYL BUTANOATE (amyl butyrate) *[540-18-1]*

Stevens & Cain (1987a)	d	6.5 - 65

2-pentyl-3,5-dimethyl-5,6-dihydropyrazine⇒5,6-DIHYDRO-3,5-DIMETHYL-2-PENTYLPYRAZINE
3-pentyl-2,5-dimethyl-5,6-dihydropyrazine⇒5,6-DIHYDRO-2,5-DIMETHYL-3-PENTYLPYRAZINE
2-pentyldimethyltetrahydropyrazine⇒TETRAHYDRODIMETHYL-2-PENTYLPYRAZINE

2-PENTYLFURAN *[3777-69-3]*

Hall & Andersson (1983)	d	0.27
Yang *et al.* (2008)		0.019

PENTYL 3-METHYLBUTANOATE (amyl isovalerate) *[25415-62-7]*

Allison & Katz (1919)		12

PENTYL PROPANOATE (amyl propionate) *[624-54-4]*

Ziemer *et al.* (2000)	d	0.085

2-PENTYLPYRIDINE *[2294-76-0]*

Schieberle (1993)		0.000 2

PENTYL SALICYLATE (amyl salicylate) *[2050-08-0]*

Backman (1917)	r	0.01
Baldus (1936)	d	0.0075
Baldus (1936)	r	0.016625

6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE (δ-thiodecalactone)

Roling *et al.* (1998); Engel (1999)		0.000 7

(*R*)-6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE ((*R*)-δ-thiodecalactone)

Engel *et al.* (2001)		0.0048

(*S*)-6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE ((*S*)-δ-thiodecalactone)

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Engel *et al.* (2001) 0.0012

PENTYLTHIOPENTANE (diamyl sulphide) *[872-10-6]*
Allison & Katz (1919) 1

pepper pyrazine⇒2-ISOBUTYL-3-METHOXYPYRAZINE
perchloroethylene⇒TETRACHLOROETHENE

PERCHLORYL FLUORIDE (trioxychlorofluoride) *[7616-94-6]*
Braker & Mossman (1980) 42

(3*R*,3a*R*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*R*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) > 0.1

(-)-(3*R*,3a*R*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*R*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 4

(+)-(3*R*,3a*R*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*R*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) > 0.1

(-)-(3*R*,3a*R*,6*S*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*R*,6*S*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 003

(+)-(3*R*,3a*S*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*S*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) > 0.1

(3*R*,3a*S*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*S*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.0075

(-)-(3*R*,3a*S*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*S*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) > 0.1

(+)-(3*R*,3a*S*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*S*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) > 0.1

(3*R*,3a*S*,6*S*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*R*,3a*S*,6*S*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 0075

(3*S*,3a*R*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*S*,3a*R*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.030

(-)-(3*S*,3a*R*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*S*,3a*R*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) > 0.1

(3*S*,3a*R*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*S*,3a*R*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Buhr (2006) > 0.1

(+)-(3*S*,3a*S*,6*R*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*S*,3a*S*,6*R*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 15

(-)-(3*S*,3a*S*,6*R*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*S*,3a*S*,6*R*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 000 001

(+)-(3*S*,3a*S*,6*S*,7a*R*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*S*,3a*S*,6*S*,7a*R*)-3,6-dimethyl-3a,4,5,6,7a-pentahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 5

(3*S*,3a*S*,6*S*,7a*S*)-PERHYDRO-3,6-DIMETHYL-2-BENZO[b]FURANONE ((3*S*,3a*S*,6*S*,7a*S*)-3,6-dimethyl-3a,4,5,6,7a-hexahydro-2(3*H*)-benzofuranone)
Buhr (2006) 0.000 000 007

perillaldehyde⇒4-ISOPROPENYL-1-CYCLOHEXEN-1-CARBOXALDEHYDE
phantolide⇒1-(1,1,2,3,3,6-HEXAMETHYL-1*H*-INDAN-5-YL)ETHANONE
α-phellandrene⇒4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE
(*d*)-α-phellandrene⇒(*S*)-(+)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE
(*l*)-α-phellandrene⇒(*R*)-(-)-4-ISOPROPYL-1-METHYL-1,5-CYCLOHEXADIENE

PHENACYL BROMIDE (α-bromoacetophenone) *[70-11-1]*

Katz & Talbert (1930)		0.12

PHENACYL CHLORIDE (α-chloroacetophenone) *[532-27-4]*

Katz & Talbert (1930)		0.10 - 0.70
Prentiss (1937)		0.2

PHENANTHRENE *[85-01-8]*

Backman (1917)	r	0.055 - 0.06

phenethyl alcohol⇒2-PHENYLETHANOL

PHENOL *[108-95-2]*

Grijns (1906)		2.2 - 6.8
Zwaardemaker (1914,1927)	d	4
Backman (1917)	r	0.13 - 0.26
Henning (1924)	d	1.2
Mukhitov (1962,1963)		0.022
Itskovich & Vinogradova (1962)		3
Pogosyan (1965)		0.022
Korneev (1965)		0.0172
Makhinya (1966)		0.022
Basmadzhieva & Argirova (1968)		0.021
Leonardos *et al.* (1969)	r	0.18
Takhirov (1974)		0.022
Punter (1975,1979)	d	0.8
Makeicheva (1978)		0.027
Anon. (1980)	d	0.046
Anon. (1980)	r	0.22
Nauš (1982)	d	0.2
Nauš (1982)	r	20
Punter (1983)	d	0.23
Moriguchi *et al.* (1983)	d	0.046
Kohler & Homans (1980); Homans (1984)		5.8 - 7.5

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Don (1986); Hoshika *et al.* (1993)	d	0.039
Nagy (1991)	d	0.5
Hoshika *et al.* (1993)	d	0.046
Nagata (2003)	d	0.021

PHENOXYBENZENE (diphenyl ether) *[101-84-8]*

Katz & Talbert (1930)		0.0070
Leonardos *et al.* (1969)	r	0.7
Appell (1969)		0.000 17

phenylacetaldehyde⇒2-PHENYLETHANAL
2-phenylacetaldehyde dimethylacetal⇒(2,2-DIMETHOXYETHYL)BENZENE

PHENYL ACETATE *[122-79-2]*

Jung (1936)	d	0.2
Jung (1936)	r	0.2

PHENYLACETIC ACID *[103-82-2]*

Appell (1969)		0.000 03

phenylbenzene⇒BIPHENYL

4-PHENYLBUTANAL *[18328-11-5]*

Von Ranson & Belitz (1992b)	d	0.050
Von Ranson & Belitz (1992b)	r	0.059

1-PHENYL-3-BUTANOL (β-phenylethylmethyl carbinol) *[2344-70-9]*

Stevens *et al.* (1988)	d	0.0043 - 0.12

(*E*)-2-PHENYLCYCLOPROPAN-1-CARBALDEHYDE

Von Ranson & Belitz (1992b)	d	2.6
Von Ranson & Belitz (1992b)	r	7.9

(*Z*)-2-PHENYLCYCLOPROPAN-1-CARBALDEHYDE

Von Ranson & Belitz (1992b)	d	0.13
Von Ranson & Belitz (1992b)	r	0.69

1-PHENYL-1,2-EPOXYETHANE (styrene oxide) *[96-09-3]*

Hellman & Small (1974)	d	0.3
Hellman & Small (1974)	r	2.0

2-PHENYLETHANAL (phenylacetaldehyde) *[122-78-1]*

Blank *et al.* (1989)		0.000 6 - 0.0012
Von Ranson & Belitz (1992b)	d	0.000 72
Von Ranson & Belitz (1992b)	r	0.0017

(*R*)-1-PHENYLETHANETHIOL

Fischer *et al.* (2008)		0.000 005

(*S*)-1-PHENYLETHANETHIOL

Fischer *et al.* (2008)		0.000 005

1-PHENYLETHANOL (methyl phenyl carbinol, phenylmethylcarbinol, styrallyl alcohol) *[98-85-1]*

Wünsche *et al.* (1995)	d	0.30

2-PHENYLETHANOL (phenethyl alcohol, phenyl ethyl alcohol, β-phenylethyl alcohol) *[60-12-8]*

Appell (1969)		0.000 35

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Blank *et al.* (1989)		0.012 - 0.024
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	0.03
Laska & Hudson (1991)	d	0.0049
McGee *et al.* (1995)	d	0.01 - 0.1
Ferreira *et al.* (1998)		0.16
Dalton *et al.* (2000)	d	0.05 - 0.1
Smeets & Dalton (2002)		d 0.7 - 3.8
Dalton *et al.* (2003)	d	0.014 - 0.016
Tsukatani *et al.* (2003)	d	0.20 - 0.22
Dalton *et al.* (2007)	d	0.012 - 0.021

phenyl ethyl alcohol⇒2-PHENYLETHANOL
β-phenylethyl alcohol⇒2-PHENYLETHANOL

1-PHENYLETHYL ACETATE (styrallyl acetate) *[93-92-5]*

Appell (1969)		0.000 2

2-PHENYLETHYL ACETATE *[103-45-7]*

Schieberle (1998)		0.0067- 0.0268

(β)-phenylethylmethylethyl carbinol⇒3-METHYL-1-PHENYL-3-PENTANOL

6-PHENYLHEXANAL *[16387-61-4]*

Von Ranson & Belitz (1992b)	d	0.014
Von Ranson & Belitz (1992b)	r	0.025

1-phenylhexane⇒HEXYLBENZENE

PHENYL ISOCYANIDE *[931-54-4]*

Allison & Katz (1919)		2
Katz & Talbert (1930)		0.0042
Stone & Pryor (1967)	d	11.0
Stone & Pryor (1967)	r	109.0

PHENYL ISOTHIOCYANATE (phenyl mustard oil) *[103-72-0]*

Katz & Talbert (1930)		0.52
Stone *et al.* (1967)	d	0.033

phenylmethylcarbinol⇒1-PHENYLETHANOL
phenyl mustard oil⇒PHENYL ISOTHIOCYANATE

8-PHENYLOCTANAL

Von Ranson & Belitz (1992b)	d	1.9
Von Ranson & Belitz (1992b)	r	1.8

5-PHENYLPENTANAL

Von Ranson & Belitz (1992b)	d	0.0032
Von Ranson & Belitz (1992b)	r	0.062

(*R*)-2-PHENYLPROPANAL

Bartschat & Mosandl (1996)		> 2.5

(*S*)-2-PHENYLPROPANAL

Bartschat & Mosandl (1996)		0.02

(*R*)-2-PHENYLPROPANAL DIMETHYL ACETAL

Bartschat & Mosandl (1996)		0.25

(*S*)-2-PHENYLPROPANAL DIMETHYL ACETAL

Bartschat & Mosandl (1996)		> 2.5

3-PHENYLPROPANAL (hydrocinnamaldehyde) *[104-53-0]*

Von Ranson & Belitz (1992b)	d	0.12
Von Ranson & Belitz (1992b)	r	0.31

(*R*)-2-PHENYL-1-PROPANOL *[19141-40-3]*

Bartschat & Mosandl (1996)		1

(*S*)-2-PHENYL-1-PROPANOL *[37778-99-7]*

Bartschat & Mosandl (1996)		> 2.5

3-PHENYL-1-PROPANOL (hydrocinnamyl alcohol) *[122-97-4]*

Schwarz (1955)		0.0015
Von Ranson & Belitz (1992b)	d	0.42
Von Ranson & Belitz (1992b)	r	0.90

3-PHENYL-2-PROPENAL (cinnamaldehyde) *[104-55-2]*

Tempelaar (1913)	d	0.098
Backman (1917)	r	0.010 - 0.012
Baldus (1936)	d	0.0201
Baldus (1936)	r	0.05025
Appell (1969)		0.000 03
Randebrock (1971)		0.000 015
Dravnieks *et al.* (1986)	d	0.024
Randebrock (1986)		0.000 081
Von Ranson & Belitz (1992b)	d	0.14
Von Ranson & Belitz (1992b)	r	0.21

3-PHENYL-(*Z*)-2-PROPENAL ((*Z*)-cinnamaldehyde) *[57194-69-1]*

Von Ranson & Belitz (1992b)	d	0.069
Von Ranson & Belitz (1992b)	r	0.37

3-PHENYL-2-PROPEN-1-OL (cinnamic alcohol, cinnamyl alcohol) *[104-54-1]*

Tempelaar (1913)	d	0.19
Appell (1969)		0.000 03
Randebrock (1971)		0.000 013
Randebrock (1986)		0.000 076

PHENYLTHIOBENZENE (diphenyl sulphide) *[139-66-2]*

Katz & Talbert (1930)		0.0026
Leonardos *et al.* (1969)	r	0.036

phenylurethane⇒ETHYL N-PHENYLCARBAMATE
phosgene⇒CARBONYL CHLORIDE

PHOSPHINE *[7803-51-2]*

Valentin (1848)		1.4
Valentin (1850)		0.13
Singh *et al.* (1967)	d	7
Berck (1968)	r	< 2
Leonardos *et al.* (1969)	r	0.03
Dumas & Bond (1974)	d	> 280
Fluck (1976)		0.014 - 2.8

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

PHTHALIC ANHYDRIDE *[85-44-9]*

Slavgorodskiy (1968)		0.32	

2-picoline⇒2-METHYLPYRIDINE
α-picoline⇒2-METHYLPYRIDINE
pinacolone 2,2-DIMETHYL-3-BUTANONE
α-pinene⇒2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(+)-α-pinene⇒(1*R*,5*R*)- (+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(-)-α-pinene⇒(1*S*,5*S*)-(-)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(*R*)-(+)-α-pinene⇒(1*R*,5*R*)-(+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
(*S*)-(-)-α-pinene⇒(1*S*,5*S*)-(-)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE
β-pinene⇒6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE
(-)-β-pinene⇒(1*S*)-(-)-6,6-DIMETHYL-2-METHYLENEBICYCLO[3.1.1]HEPTANE

PIPERIDINE (hexahydroazine) *[110-89-4]*

Geier (1936)	d	0.5	
Geier (1936)	r	2.0	
Nawakowski (1980)		< 7	

cis-PIPERITONE OXIDE *[57130-28-6]*

Benzo *et al.* (2007)		0.022427	

piperonal⇒3,4-METHYLENEDIOXYBENZALDEHYDE
pivaldehyde⇒2,2-DIMETHYLPROPANAL
polywood⇒5,5,8a-TRIMETHYL-2-DECAHYDRONAPHTHYL ACETATE
prenyl mercaptan⇒3-METHYL-2-BUTENE-1-THIOL

PROPANAL (propional, propionaldehyde) *[123-38-6]*

Backman (1917)	r	0.02	
Pliška & Janiček (1965)		240	
Hartung *et al.* (1971)		1.7	
Knuth (1973)		0.026	
Hellman & Small (1974)		d	0.02
Hellman & Small (1974)		r	0.10
Teranishi *et al.* (1974)		0.020	
Bedborough & Trott (1979)	d	0.014	
Anon. (1980)	d	0.0036	
Anon. (1980)	r	0.036	
Hall & Andersson (1983)	d	0.69	
Cristoph (1983)	d	0.33 - 0.40	
Nagy (1991)	d	0.21	
Nagata (2003)	d	0.0024	
Cometto-Muñiz & Abraham (2010a)	d	0.0048	

PROPANE *[74-98-6]*

Patty & Yant (1929)		36,000	
Laffort & Dravnieks (1973)		22,000	
Nagata (2003)	d	2,700	

1,2-PROPANEDIAMINE (1,2-diaminopropane) *[78-90-0]*

Hellman & Small (1974)		d	0.04
Hellman & Small (1974)		r	0.14

1,2-PROPANEDIOL (propylene glycol) *[57-55-6]*

Nagy (1991)	d	16	

1,2-PROPANEDIOL DINITRATE (propylene glycol dinitrate) *[6423-43-4]*

Stewart *et al.* (1974)		1.6	

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

2,2-PROPANEDITHIOL

Mayer & Wittig (1972)		0.000 000 036

1-PROPANETHIOL (propyl mercaptan) *[107-03-9]*

Allison & Katz (1919)		6
Katz & Talbert (1930)		0.005
Blinova (1965)		0.007 - 0.03
Kniebes *et al.* (1969)		0.0016
Wilby (1969)	r	0.0023
Blanchard (1976)		0.018
Nagata (2003)	d	0.000 040

2-PROPANETHIOL (isopropyl mercaptan) *[75-33-2]*

Stuiver (1958)	d	0.000 0025 - 0.000 02
Wilby (1969)	r	0.0014
Blanchard (1976)		0.018
Moschandreas & Jones (1983)	d	0.0081
Moschandreas & Jones (1983)	r	0.0192
Nagata (2003)	d	0.000 019

PROPANOIC ACID (propionic acid) *[79-09-4]*

Passy (1893b,1893c)	d	0.05
Backman (1917)	r	0.5
Grijns (1919)		0.6
Misumoto (1926)	r	1.7 - 2.55
Hesse (1926)	r	4.6
Morimura (1934)	r	1.77 - 2.38
Stone (1963a,1963c)	d	0.39 - 0.68
Stone & Bosley (1965)	d	0.89
Goldenberg (1967)	d	0.003
Hellman & Small (1974)	d	0.08
Hellman & Small (1974)	r	0.10
Anon. (1980)	d	0.0051
Anon. (1980)	r	0.025
Punter (1983)	d	0.44 - 0.58
Dollnick *et al.* (1988)		0.147
Walker *et al.* (1990)		14.1
Nagy (1991)	d	1.2
Walker *et al.* (1996)		0.3 - 3
Nagata (2003)	d	0.017
Van Thriel *et al.* (2006)	d	1.0

(2'*E/Z*)-PROPANOIC ACID (1'',2'',4''-TRIMETHYLPENT-2-ENYLOXY)CARBONYLMETHYL ESTER

Kraft & Eichenberger (2004)		0.016

PROPANOIC ANHYDRIDE (propionic anhydride) *[123-62-6]*

Nagy (1991)	d	3.4

1-PROPANOL (propyl alcohol) *[71-23-8]*

Passy (1892c)	d	5 - 10
Backman (1917)	r	3 - 5
Jung (1936)	d	0.8 - 8
Jung (1936)	r	8 - 24
Jones (1955c)	r	140
Janiček *et al.* (1960)		540
Pliška & Janiček (1965)		25,000
Guadagni (1966)		9

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

May (1966)	d	80
May (1966)	r	150
Khachaturyan *et al.* (1968)		1.25
Cain (1969)	r	660
Corbit & Engen (1971)		46 - 51
Stone *et al.* (1972)	d	2.8
Dravnieks & Laffort (1972)		32.3
Dravnieks (1974)	d	100
Hellman & Small (1974)	d	< 0.075
Hellman & Small (1974)	r	0.2
Laing (1975)	d	100
Nauš (1982)	d	2
Nauš (1982)	r	20
Punter (1983)	d	5.9
Cristoph (1983)	r	2.9 - 3.2
Cometto-Muñiz & Cain (1990,1993);		
Cometto-Muñiz (1993)	d	27.5 - 35
Scharfenberger (1990)		16
Nagata (2003)	d	0.24

2-PROPANOL (isopropyl alcohol, isopropanol) *[67-63-0]*

Passy (1892c)	d	40
Backman (1917)	r	18 - 24
Jung (1936)	d	3.9 - 32.4
Jung (1936)	r	7.8 - 31.2
Scherberger *et al.* (1958)	r	500
Cheesman & Kirkby (1959)	d	43 - 290
May (1966)	d	90
May (1966)	r	120
Gorlova (1970)		2.5
Köster (1968a,1971)	d	64 - 5,400
Dravnieks & Laffort (1972)		57.4
Dravnieks (1974)	d	1,500
Hellman & Small (1974)	d	8.0
Hellman & Small (1974)	r	18.8
Scharfenberger (1990)		491
Nagy (1991)	d	180
Cometto-Muñiz & Cain (1993);		
Cometto-Muñiz (1993,1999)	d	1,245
Smith & Duffy (1995)	d	103
Smith & Duffy (1995)	r	228
Smeets & Dalton (2002)	d	28 - 98
Nagata (2003)	d	65

PROPANONE (acetone) *[67-64-1]*

Zwaardemaker (1914,1927)	d	4 - 7
Backman (1917)	r	4.1 - 4.3
Van Anrooij (1931)	d	1.1
Jung (1936)	d	78
Jung (1936)	r	78
Scherberger *et al.* (1958)	r	1,900
Stuiver (1958)	d	5.8
Feldman (1960)		1.1
Nauš (1962)	d	4
Pogosyan (1965)		1.1
Tkach (1965)		1.1
May (1966)	d	770

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

May (1966)	r	1,660	
Kittel (1968)		11 - 240	
Leonardos *et al.* (1969)	r	240	
Kittel & Wendelstein (1971)	d	75	
Kittel & Wendelstein (1971)	r	121	
Hartung *et al.* (1971)		2.3	
Dravnieks & Laffort (1972)		240	
Artho & Koch (1973)		1,000 - 10,000	
Hellman & Small (1973,1974)	d	48	
Hellman & Small (1973,1974)	r	78	
Dravnieks (1974)	d	1,550	
Takhirov (1974)		1.15	
Makeicheva (1978)		0.94	
Anon. (1980)	d	72	
Anon. (1980)	r	264	
Nauš (1982)	d	1	
Nauš (1982)	r	20	
Punter (1983)	d	8.6	
Nagy (1991)	d	40	
Cometto-Muñiz & Cain (1993);			
Cometto-Muñiz (1993)	d	27,900	
Dalton *et al.* (1997a)	d	199 - 204	
Dalton *et al.* (1997b)	d	626 - 936	
Wysocki *et al.* (1997)	d	97 - 2,026	
Dalton *et al.* (2000)	d	59	
Nagata (2003)	d	101	
Cometto-Muñiz & Abraham (2009a)	d	2	

2-propanoyl-1-pyrroline⇒2-PROPIONYL-1-PYRROLINE
3-(3-propan-2-ylphenyl)butanal⇒3-(3-ISOPROPYLPHENYL)BUTANAL
propargylaldehyde⇒2-PROPYNAL

2-PROPENAL (acrolein) *[107-02-8]*

Katz & Talbert (1930)		4.1	
Plotnikova (1957)		0.8	
Buchberg *et al.* (1961)		0.2 - 0.7	
Leonardos *et al.* (1969)	r	0.48	
Sinkuvene (1970)		0.07	
Knuth (1973)		0.14	
Cormack *et al.* (1974)		0.23	
Teranishi *et al.* (1974)		0.050	
Anon. (1980)	d	0.069	
Anon. (1980)	r	0.32	
Nagata (2003)	d	0.0083	

1-PROPENE (propylene) *[115-07-1]*

Krasovitskaya & Malyarova (1968)		17.3	
Laffort & Dravnieks (1973)		170	
Hellman & Small (1974)		d	38
Hellman & Small (1974)		r	115
Nagata (2003)	d	22	

2-PROPENE-1-THIOL (allyl mercaptan) *[870-23-5]*

Katz & Talbert (1930)		0.0045 - 0.018
Stuiver (1958)	d	0.000 005 - 0.000 04

PROPENOIC ACID (acrylic acid) *[79-10-7]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Hellman & Small (1974)	d	0.27
Hellman & Small (1974)	r	3.0
Piringer & Granzer (1984)		2
Van Thriel *et al.* (2006)	d	1.5

2-PROPEN-1-OL (allyl alcohol) *[107-18-6]*

Katz & Talbert (1930)		3.3
Jones (1955c)	r	83
Dunlap *et al.* (1958)		1.9
Pliška & Janiček (1965)		48
Dravnieks & Laffort (1972)		1.2
Dravnieks (1974)	d	5

2-PROPENYLAMINE (allylamine) *[107-11-9]*

Katz & Talbert (1930)		14
Hine *et al.* (1960)		< 6

(*E*)-1-PROPENYLDITHIO-2-PROPENE

Pino *et al.* (1991)		0.000 048

(*Z*)-1-PROPENYLDITHIO-2-PROPENE

Pino *et al.* (1991)		0.000 018

2-PROPENYLDITHIO-2-PROPENE (diallyl disulphide) *[2179-57-9]*

Katz & Talbert (1930)		0.0072
Gijs *et al.* (2000)		0.07
Nagata (2003)	d	0.0013

2-PROPENYL HEXANOATE (allyl caproate) *[123-68-2]*

Berger *et al.* (1985)	d	0.5 - 1.0

2-PROPENYL ISOCYANIDE (allyl isocyanide) *[2835-21-4]*

Katz & Talbert (1930)		0.049

2-PROPENYL ISOTHIOCYANATE (allyl mustard oil) *[57-06-7]*

Allison & Katz (1919)		8
Katz & Talbert (1930)		0.61
Stone *et al.* (1967)	d	0.19
Stone & Pryor (1967)	d	0.037 - 0.24

(3-(3-propen-2-ylphenyl)butanal) ⇒3-(3-ISOPROPENYLPHENYL)BUTANAL

2-PROPENYLTETRATHIO-2-PROPENE (diallyl tetrasulphide) *[2444-49-7]*

Pino *et al.* (1991)		0.000 0086

2-PROPENYLTHIO-2-PROPENE (diallyl sulphide) *[592-88-1]*

Katz & Talbert (1930)		0.000 65
Gijs *et al.* (2000)		0.07
Nagata (2003)	d	0.0010

propiolaldehyde⇒2-PROPYNAL
propional⇒PROPANAL
propionaldehyde⇒PROPANAL
propionic acid⇒PROPANOIC ACID
propionic anhydride⇒PROPANOIC ANHYDRIDE

2-PROPIONYL-2,3-DIHYDRO-1,4-THIAZINE
Hofmann *et al.* (1995);

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Hofmann & Schieberle (1997)		0.000 1

2-PROPIONYL-1-PYRROLINE (2-propanoyl-1-pyrroline) *[133447-37-7]*

Schieberle (1991); Hofmann *et al.* (1995); Hofmann & Schieberle (1997)		0.000 02

2-PROPIONYL-2-THIAZOLINE *[29926-42-9]*

Hofmann & Schieberle (1997)		0.000 07

PROPYL ACETATE *[109-60-4]*

Backman (1917)	r	12
Jung (1936)	d	0.35
Jung (1936)	r	0.35 - 0.62
May (1966)	d	70
May (1966)	r	110
Hellman & Small (1974)	d	0.2
Hellman & Small (1974)	r	0.6
Cometto-Muñiz & Cain (1991); Cometto-Muñiz (1993)	d	104
Nagata (2003)	d	1.0
Komthong *et al.* (2006)		363

propyl alcohol⇒1-PROPANOL

PROPYLAMINE *[107-10-8]*

Stephens (1971)		0.022
Nagata (2003)	d	0.15

4-PROPYLBENZALDEHYDE *[28785-06-0]*

Von Ranson & Belitz (1992b)	d	1.1
Von Ranson & Belitz (1992b)	r	2.0

PROPYLBENZENE *[103-65-1]*

Cometto-Muñiz (1993); Cometto-Muñiz & Cain (1994)	d	14.4
Nagata (2003)	d	0.019

7-PROPYLBENZO[*b*][1,4]DIOXEPIN-3-ONE

Kraft & Eichenberger (2003)		0.000 1

PROPYL BUTANOATE (propyl butyrate) *[105-66-8]*

Dravnieks & Laffort (1972)		0.25
Nagata (2003)	d	0.058

propyl butyrate⇒PROPYL BUTANOATE
2-propyl-3,5-dimethyl-5,6-dihydropyrazine⇒5,6-DIHYDRO-3,5-DIMETHYL-2-PENTYLPYRAZINE
3-propyl-2,5-dimethyl-5,6-dihydropyrazine⇒5,6-DIHYDRO-2,5-DIMETHYL-3-PROPYLPYRAZINE
2-propyl-3,5-dimethyltetrahydropyrazine⇒TETRAHYDRO-3,5-DIMETHYL-2-PROPYLPYRAZINE
3-propyl-2,5-dimethyltetrahydropyrazine⇒TETRAHYDRO-2,5-DIMETHYL-3-PROPYLPYRAZINE

PROPYLDITHIOPROPANE (dipropyl disulphide, 4,5-dithiaoctane) *[629-19-6]*

Logtenberg (1978)	d	0.13

propylene⇒1-PROPENE
propylene chloride⇒1,2-DICHLOROPROPANE
propylene dichloride⇒1,2-DICHLOROPROPANE
propylene glycol *n*-butyl ether⇒1-BUTOXY-2-PROPANOL
propylene glycol methyl ether acetate⇒1-METHOXY-2-PROPYL ACETATE
propylene glycol monomethyl ether⇒1-METHOXY-2-PROPANOL

propylene glycol dinitrate⇒1,2-PROPANEDIOL DINITRATE
propylene oxide⇒1,2-EPOXYPROPANE

PROPYL FORMATE *[110-74-7]*

Backman (1917)	r	10.5 - 15.0
Nagata (2003)	d	3.5

propyl isobutyrate⇒PROPYL 3-METHYLPROPANOATE
propyl isovalerate⇒PROPYL 3-METHYLBUTANOATE
propyl mercaptan⇒1-PROPANETHIOL

PROPYL 3-METHYLBUTANOATE (propyl isovalerate) *[557-00-6]*

Nagata (2003)	d	0.000 33

PROPYL 3-METHYLPROPANOATE (propyl isobutyrate) *[644-49-5]*

Nagata (2003)	d	0.011

PROPYL PENTANOATE (propyl valerate) *[141-06-0]*

Backman (1917)	r	0.025 - 0.029
Nagata (2003)	d	0.019

2-PROPYLPHENOL *[644-35-9]*

Czerny & Buettner (2009)		0.044

3-PROPYLPHENOL *[621-27-2]*

Czerny & Buettner (2009)		0.000 098

4-PROPYLPHENOL *[645-56-7]*

Czerny & Buettner (2009)		0.011

PROPYL PROPANOATE (propyl propionate) *[106-36-5]*

Backman (1917)	r	0.23 - 0.26
Nagata (2003)	d	0.28

propyl propionate⇒PROPYL PROPANOATE

2-PROPYLPYRIDINE *[622-39-9]*

Schieberle (1993)		0.011

6-PROPYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE (δ-thiooctalactone)

Roling *et al.* (1998); Engel (1999)		0.000 7

(*R*)-6-PROPYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE ((*R*)-δ-thiooctalactone)

Engel *et al.* (2001)		0.0024

(*S*)-6-PROPYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE ((*S*)-δ-thiooctalactone)

Engel *et al.* (2001)		0.0012

S-PROPYL THIOACETATE *[2307-10-0]*

Gijs *et al.* (2000)		0.14

PROPYLTHIOPROPANE (dipropyl sulphide) *[111-47-7]*

Katz & Talbert (1930)		0.053
Wilby (1969)	r	0.11

propyl valerate⇒PROPYL PENTANOATE

2-PROPYNAL (propiolaldehyde, propargylaldehyde) *[123-38-6]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Katz & Talbert (1930)		0.35

prussic acid⇒HYDROGEN CYANIDE
pseudocumene⇒1,2,4-TRIMETHYLBENZENE
pseudocumidine⇒2,4,5-TRIMETHYLANILINE
pulegone⇒4-ISOPROPYLIDENE-1-METHYL-3-CYCLOHEXANONE

PYRIDINE (azine) *[110-86-1]*

Hermanides (1909)	r	0.16
Zwaardemaker (1914,1927)	d	0.04
Backman (1917)	r	0.2
Allison & Katz (1919)		32
Katz & Talbert (1930)		0.74
Van Anrooij (1931)	d	0.078
Geier (1936)	d	0.09
Geier (1936)	r	0.095
Jones (1955c)	r	40
Sales (1958)		0.42
Janiček *et al.* (1960)		4.6
Sutton (1962b)		< 3.2
Kristesashvili (1965)		0.21
Leonardos *et al.* (1969)	r	0.067
Dravnieks & Laffort (1972)		0.33
Laffort & Dravnieks (1973)		0.74
Dravnieks (1974)	d	6
Amoore & Buttery (1978)	d	2.4
Laing *et al.* (1978)	r	2.4
Hangartner (1981)		0.08 - 2.9
Nauš (1982)	d	1
Nauš (1982)	r	10
Moriguchi *et al.* (1983)	d	0.023
Bahnmüller (1983)		0.132 - 1.21
Ahlström *et al.* (1986a)	d	0.124 - 0.146
Amoore (1986a,b)	d	2.1
Don (1986)	d	0.12
Hartigh (1986)	d	0.15 - 0.29
MacLeod *et al.* (1986)		0.054
Cain *et al.* (1987)	d	0.34
Stevens *et al.* (1988)	d	0.13 - 1.2
Cometto-Muñiz & Cain (1990);		
Cometto-Muñiz (1993)	d	4.1
Cain & Gent (1991)	d	0.32
Laska & Hudson (1991)	d	0.039
Nagy (1991)	d	1.5
Berglund & Esfandabad (1992)		0.31 (odor)
Berglund & Esfandabad (1992)		2.5 (irritation)
Nordin *et al.* (1997)		0.34
Nagata (2003)	d	0.20
Cain *et al.* (2010)	d	0.32

PYRROLIDINE (azolidine) *[123-75-1]*

Kendall *et al.* (1968)	r	0.015 - 0.09
Hendriks (1979)	d	1.0

(*R*)-(+)-2-PYRROLIDINECARBOXYLIC ACID (D-proline) *[344-25-2]*

Laska (2010)	d	2.4

(*S*)-(-)-2-PYRROLIDINECARBOXYLIC ACID (L-proline) *[147-85-3]*

Laska (2010)	d	3.1

1-PYRROLINE (3,4-dihydro-2H-azole) *[5724-81-2]*

Amoore (1977)	d	0.0051
Hendriks (1979)	d	0.080

Q

QUINOLINE *[91-22-5]*

Geier (1936)	d	0.03
Geier (1936)	r	0.05 - 0.1
Gundlach & Kenway (1939)	d	28

quinone⇒1,4-BENZOQUINONE

R

cis-rose oxide⇒*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*R*,4*S*)-*cis*-rose oxide⇒(2*R*,4*S*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*S*,4*R*)-*cis*-rose oxide⇒(2*S*,4*R*)-*cis*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
trans-rose oxide⇒*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*R*,4*R*)-*trans*-rose oxide⇒(2*R*,4*R*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN
(2*S*,4*S*)-*trans*-rose oxide⇒(2*S*,4*S*)-*trans*-4-METHYL-2-(2-METHYL-1-PROPENYL)TETRAHYDROPYRAN

S

sabinene⇒1-ISOPROPYL-4-METHYLENEBICYCLO[3.1.0]HEXANE
trans-sabinene hydrate⇒*trans*-5-ISOPROPYL-2-METHYLBICYCLO[3.1.0]-2-HEXANOL
safrole⇒4-ALLYL-1,2-(METHYLENEDIOXY)BENZENE

SALICYL ALCOHOL (α,2-dihydroxytoluene) *[90-01-7]*

Backman (1917)	r	6 - 7

salicylaldehyde⇒2-HYDROXYBENZALDEHYDE
(+)-(*Z*)-α-santalol⇒(+)-(*Z*)-5-(2,3-DIMETHYLTRICYCLO[2.2.1.0^{2,6}]-3-HEPTYL)-2-METHYL-2-PENTEN-1-OL
(-)-(Z)-β-santalol⇒(-)-(Z)-2-METHYL-5-(2-METHYL-3-METHYLENEBICYCLO[2.2.1]-2-HEPTYL-2-PENTEN-1-OL
sedanenolide⇒4-BUTYL-4,5-DIHYDROPHTHALIDE
trans-sedanolide⇒*trans*-3-BUTYLTETRAHYDROPHTHALIDE
sila-bourgeonal⇒3-(4-(2-METHYL-2-SILAPROPANE)PHENYL)PROPANAL
sila-lilial⇒3-(4-(2-METHYL-2-SILAPROPANE)PHENYL)-2-METHYLPROPANAL
sila-linalool⇒3,7-DIMETHYL-3-SILA-1,6-OCTADIEN-3-OL
skatole⇒3-METHYLINDOLE
sorbic aldehyde⇒(*E*,*E*)-2,4-HEXADIENAL
sotolon⇒3-HYDROXY-4,5-DIMETHYL-2(5*H*)-FURANONE
strawberry aldehyde⇒ETHYL 3-METHYL-3-PHENYL-2,3-EPOXYPROPANOATE
styrallyl acetate⇒1-PHENYLETHYL ACETATE
styrallyl alcohol⇒1-PHENYLETHANOL
styrene⇒ETHENYLBENZENE
styrene oxide⇒1-PHENYL-1,2-EPOXYETHANE
sugarone⇒ 2-ETHYL-4-HYDROXY-5-METHYL-3(2*H*)-FURANONE
3-sulfanylhexan-1-ol⇒3-MERCAPTO-1-HEXANOL
(*R*)-3-sulfanylhexan-1-ol⇒(*R*)-3-MERCAPTO-1-HEXANOL
(*S*)-3-sulfanylhexan-1-ol⇒(*S*)-3-MERCAPTO-1-HEXANOL
3-sulfanylhexyl acetate⇒3-MERCAPTOHEXYL ACETATE
(*R*)-3-sulfanylhexyl acetate⇒(*R*)-3-MERCAPTOHEXYL ACETATE
(*S*)-3-sulfanylhexyl acetate⇒(*S*)-3-MERCAPTOHEXYL ACETATE

SULPHUR DICHLORIDE *[10545-99-0]*

Leonardos *et al.* (1969)	r	0.004

SULPHUR DIOXIDE *[7446-09-5]*

Holmes *et al.* (1915)	d	5 - 10
Holmes *et al.* (1915)	r	10 - 13
Smolczyk & Cobler (1930)		< 4
Thomas *et al.* (1943)		1.3 - 1.6
Popov *et al.* (1952)		4 - 6.5
Amdur *et al.* (1953)		2.6 - 21
Dubrovskaya (1957)		2.6 - 3.0
Beck (1959); Henschler *et al.* (1960)		1.3 - 2.6
Bushtueva (1960)		1.5
Bushtueva (1962)		1.6 - 2.6
Makhynia (1966)		0.87 - 0.88
Shalamberidze (1967)		1.6
Leonardos *et al.* (1969)	r	1.2
Nagata (2003)	d	2.3

SULPHUR HEXAFLUORIDE *[2551-62-4]*

Laffort (1968a)		24,000,000

SULPHURIC ACID *[8014-95-7]*

Melekhina (1968)	d	0.60

T

α-terpinene⇒4-ISOPROPYL-1-METHYL-1,3-CYCLOHEXADIENE
γ-terpinene⇒4-ISOPROPYL-1-METHYL-1,4-CYCLOHEXADIENE
1-terpinen-4-ol⇒1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL
(*R*)-terpinen-4-ol⇒(*R*)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL
(*S*)-terpinen-4-ol⇒(*S*)-1-ISOPROPYL-4-METHYL-3-CYCLOHEXEN-1-OL
α-terpineol⇒4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE
α-terpinol⇒4-(2-HYDROXYISOPROPYL)-1-METHYL-1-CYCLOHEXENE
tetra⇒TETRACHLOROMETHANE

1,1,2,2-TETRABROMOETHANE (acetylene tetrabromide) *[79-27-6]*

Hollingsworth *et al.* (1963)	r	< 14

2,3,4,6-TETRACHLOROANISOLE *[938-86-3]*

Strube & Buettner (2010)		0.00574

2,3,4,6-TETRACHLOROANISOLE *[938-22-7]*

Nijssen (1988)	d	0.000 067
Strube & Buettner (2010)		0.000 01

2,3,5,6-TETRACHLOROANISOLE *[6936-40-9]*

Strube & Buettner (2010)		0.00391

1,1,2,2-TETRACHLOROETHANE *[79-34-5]*

Lehmann & Schmidt-Kehl (1936)		20
Dravnieks & Laffort (1972)		1.6
Dravnieks (1974)	d	50

TETRACHLOROETHENE (perchloroethylene, tetrachloroethylene) *[127-18-4]*

Carpenter (1937)		< 340
May (1966)	d	320
May (1966)	r	480

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Leonardos *et al.* (1969)	r	32
Anon. (1980)	d	12
Anon. (1980)	r	55
Torkelson & Rowe (1981)		340
Don (1986); Hoshika *et al.* (1993)	d	8.1 - 8.3
Hoshika *et al.* (1993)	d	12
Nagata (2003)	d	5.2

tetrachloroethylene ⇒TETRACHLOROETHENE

TETRACHLOROMETHANE (tetra, carbon tetrachloride) *[56-23-5]*

Allison & Katz (1919)		4,533
Davis (1934)		500
Lehmann & Schmidt-Kehl (1936)		900
May (1966)	d	1,260
May (1966)	r	1,600
Leonardos *et al.* (1969)	r	135 - 630
Belkov (1969)		11.5 - 58
Nikiforov (1970)		10.58
Dravnieks & Laffort (1972)		280
Dravnieks (1974)	d	3,700
Punter (1983)	d	884
Nagata (2003)	d	29

(*R*)-δ-tetradecalactone⇒(*R*)-4-HYDROXYTETRADECANOIC ACID LACTONE
(*S*)-δ-tetradecalactone⇒(*S*)-4-HYDROXYTETRADECANOIC ACID LACTONE

TETRADECANE *[629-59-4]*

Patte (1978); Patte & Punter (1979)	d	5

tetraethyl orthosilicate⇒ETHYL SILICATE
1,2,3,6-tetrahydrobenzaldehyde⇒3-CYCLOHEXENE-1-CARBOXALDEHYDE

(3*S*,3a*S*,7a*R*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE (wine lactone)
Guth (1996a,b) 0.000 000 01 - 0.000 000 04

(3*R*,3a*R*,7a*S*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE
Guth (1996a,b) > 1

(3*R*,3a*S*,7a*R*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE
Guth (1996a,b) > 1

(3*S*,3a*R*,7a*S*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE
Guth (1996a,b) 0.08 - 0.16

(3*S*,3a*S*,7a*S*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE
Guth (1996a,b) 0.000 07 - 0.000 014

(3*R*,3a*R*,7a*R*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE
Guth (1996a,b) 0.014 - 0.028

(3*R*,3a*S*,7a*S*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE
Guth (1996a,b) 0.008 - 0.016

(3*S*,3a*R*,7a*R*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE
Guth (1996a,b) 0.000 05 - 0.000 2

3a,4,5,7a-tetrahydro-3-ethyl-6-methyl-2(3*H*)-benzofuranone⇒3-ETHYL-6-METHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

BENZOFURANONE

TETRAHYDRODIMETHYL-2-PENTYLPYRAZINE (2-pentyldimethyltetrahydropyrazine)

Kurniadi (2003)	r	> 40

TETRAHYDRO-2,5-DIMETHYL-3-PROPYLPYRAZINE (3-propyl-2,5-dimethyltetrahydropyrazine)

Kurniadi *et al.* (2003)		> 2
Kurniadi (2003)	r	>40

TETRAHYDRO-3,5-DIMETHYL-2-PROPYLPYRAZINE (2-propyl-3,5-dimethyltetrahydropyrazine)

Kurniadi *et al.* (2003)		> 2
Kurniadi (2003)	r	> 40

tetrahydrofuran⇒1,4-EPOXYBUTANE
tetrahydrogeraniol⇒3,7-DIMETHYL-1-OCTANOL

4,5,6,7-TETRAHYDRO-1,1,2,3,3,6-HEXAMETHYL-5-INDANONE (1,1,2,3,3,6-hexamethyl-4,5,6,7-tetrahydro-5-indanone)

Bachmann *et al.* (1992)		0.000 67

3a,4,7,7a-TETRAHYDRO-4,7-METHANOINDENE (dicyclopentadiene, α-dicyclopentadiene) *[77-73-6]*

Kinkead *et al.* (1971b)		0.016
Hellman & Small (1974)	d	0.06
Hellman & Small (1974)	r	0.11
Ventura *et al.* (1997)		0.001

3a,4,5,7a-tetrahydro-6-methyl-2(3*H*)-benzofuranone⇒6-METHYL-3a,4,5,7a-TETRAHYDRO-2(3*H*)-BENZOFURANONE

1,2,3,4-TETRAHYDRONAPHTHALENE *[119-64-2]*

Nagata (2003)	d	0.050

TETRAHYDROTHIOPHENE (THT, thiophane) *[110-01-0]*

Deadman & Prigg (1959)	d	0.0018
Wilby (1964)		0.0025
Nevers & Oister (1965)		0.0037
Kendall *et al.* (1968)	r	0.00014 - 0.0014
Laffort (1968b); Laffort & Dravnieks (1974)		0.01
Vlasek (1968)		0.0036
Kniebes *et al.* (1969)		0.0014
Wilby (1969)	r	0.0028
Blanchard (1976)		0.025
Whisman *et al.* (1978)		0.0014 - 0.0022
Moschandreas & Jones (1983)	d	0.0065
Moschandreas & Jones (1983)	r	0.0155
MacLeod *et al.* (1986)		0.0012
Mannebeck & Mannebeck (2002)	d	0.000 992 - 0.001963
Nagata (2003)	d	0.0022
Maxeiner & Mannebeck (2004)	d	0.00137 - 0.00266
Maxeiner (2006)	d	0.00128 - 0.00146
Maxeiner (2007)	d	0.001831

1,2,3,4-TETRAMETHYLBENZENE *[488-23-3]*

Nagata (2003)	d	0.061

1,2,4,5-TETRAMETHYLBENZENE (durene) *[95-93-2]*

Backman (1917)	r	0.083
Backman (1918)		0.087

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL (2-methylisoborneol, MIB) *[2371-42-8]*
Sävenhed *et al.* (1985) d 0.000 003 - 0.000 05
Khiari *et al.* (1992) 0.000 5
Ömür-Özbek *et al.* (2007) 0.000 48

(1*R*)-1,2,7,7-TETRAMETHYL-*exo*-BICYCLO[2.2.1]-2-HEPTANOL ((-)-2-methylisoborneol)
Belitz & Grosch (1992); Blank & Grosch (2002) 0.000 006 - 0.000 012

(+)-4-((1*S*,5*R*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((+)-*cis*-γ-irone)
Brenna *et al.* (2001a) > 0.1

(-)-4-((1*R*,5*S*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((-)-*cis*-γ-irone)
Brenna *et al.* (2001a) 0.000 75

(+)-4-((1*R*,5*R*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((+)-*trans*-γ-irone)
Brenna *et al.* (2001b) 0.1135

(-)-4-((1*S*,5*S*)-2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((-)-*trans*-γ-irone)
Brenna *et al.* (2001b) 0.02635

4-(*cis*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE (α-irone) *[79-69-6]*
Naves (1953) 0.002

4-(*trans*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE (*iso*-α-irone)
Naves (1953) 0.2

(+)-4-(*trans*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((*d*)-*iso*-α-irone)
Naves (1953) 2.7

(-)-4-(*trans*(1,5):2,5,6,6-TETRAMETHYL-2-CYCLOHEXEN-1-YL)-*cis*-3-BUTEN-2-ONE ((*l*)-*iso*-α-irone)
Naves (1953) 2.7

4,4,6,6-TETRAMETHYL-4,6-DISILAHEPTAN-2-ONE
Büttner *et al.* (2007a) 0.000 005

4,4,6,6-TETRAMETHYL-2-HEPTANONE
Büttner *et al.* (2007a) 0.000 065

(±)-(5'*R**,7'*S**,8'*S**)-1-(4',4',7',8'-TETRAMETHYL-1'-OXASPIRO[4.5]DECAN-7'-YL)ETHAN-1-ONE
Kraft & Popaj (2008) d 0.149

(±)-(5'*R**,7'*S**,8'*S**)-1-(4',4',7',8'-TETRAMETHYL-1'-OXASPIRO[4.5]DECAN-8'-YL)ETHAN-1-ONE
Kraft & Popaj (2008) d 0.346

TETRAMETHYLPYRAZINE *[1124-11-4]*
Fors (1984); Fors & Olofsson (1985) d 0.69
Grosch *et al.* (1996); Wagner *et al.* (1999) > 2

(-)-(1*R*,3R,6S,7S,8S)-2,2,6,8-tetramethyltricyclo[5.3.1.0^{3,8}]-3-undecanol⇒(-)-PATCHOULOL

4,4,6,6-TETRAMETHYL-4-SILAHEPTAN-2-ONE
Büttner *et al.* (2007a) 0.000 01

4,4,6,6-TETRAMETHYL-6-SILAHEPTAN-2-ONE
Büttner *et al.* (2007a) 0.000 025

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(1*R*,2*S*,7*R*,8*R*)-(+)-2,6,9,9-TETRAMETHYLTRICYCLO[5.4.0.0(2.8)]-9-UNDECENE ((+)-α-longipinene) *[5989-02-4]*

Lindell (1991)	d	0.92

nor-TETRAPATCHOULOL

Kraft *et al.* (2005)		0.001 - 0.005

2,4,5,7-TETRATHIAOCTANE ((methylthio)methyl disulphide)

Gijs *et al.* (2000)		0.1

texanol⇒2,2,4-TRIMETHYL-1,3-PENTANEDIOL MONOISOBUTANOATE
2-thenyl mercaptan⇒2-THENYLTHIOL

2-THENYLTHIOL (2-thenyl mercaptan) *[6258-63-5]*

Hofmann & Schieberle (1995)		0.000 003 - 0.000 012

4-thiapentanal⇒3-(METHYLTHIO)PROPANAL

THIAZOLE *[288-47-1]*

Gijs *et al.* (2000)		5.15

thibetolid⇒15-HYDROXYPENTADECANOIC ACID LACTONE
2-thienylthiol⇒2-THIOPHENETHIOL

THIIRANE (ethylene sulphide) *[420-12-2]*

Gijs *et al.* (2000)		2.19

thiobutyl butanoate⇒S-BUTYL THIOBUTANOATE
thiobutyl pentanoate⇒S-BUTYL THIOPENTANOATE
thiobutyl propanoate⇒S-BUTYL THIOPROPANOATE
p-thiocresol⇒4-METHYLBENZENETHIOL
γ-thiodecalactone⇒5-HEXYLDIHYDRO-2(3*H*)-THIOPHENONE
δ-thiodecalactone⇒6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
(*R*)-δ-thiodecalactone⇒(*R*)-6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
(*S*)-δ-thiodecalactone⇒(*S*)-6-PENTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
γ-thiododecalactone⇒5-OCTYLDIHYDRO-2(3*H*)-THIOPHENONE
δ-thiododecalactone⇒6-HEPTYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
thioethyl butanoate⇒S-ETHYL THIOBUTANOATE
thioethyl hexanoate⇒S-ETHYL THIOHEXANOATE
thioethyl isobutanoate⇒S-ETHYL THIO-2-METHYLPROPANOATE
thioethyl isopentanoate⇒S-ETHYL THIO-3-METHYLBUTANOATE
thioglycolic acid⇒MERCAPTOACETIC ACID
γ-thiohexalactone⇒5-ETHYLDIHYDRO-2(3*H*)-THIOPHENONE
δ-thiohexalactone⇒6-METHYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
γ-thiooctalactone⇒5-BUTYLDIHYDRO-2(3*H*)-THIOPHENONE
δ-thiooctalactone⇒6-PROPYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
(*R*)-δ-thiooctalactone⇒(*R*)-6-PROPYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
(*S*)-δ-thiooctalactone⇒(*S*)-6-PROPYLTETRAHYDRO-2*H*-THIOPYRAN-2-ONE
thiophane⇒TETRAHYDROTHIOPHENE

THIOPHENE *[110-02-1]*

Backman (1917)	r	4 - 5
Deadman & Prigg (1959)	d	21
Khikmatullaeva (1967,1968)		1.9 - 2.1
Dravnieks & Laffort (1972)		0.03
Bedborough & Trott (1979)	d	0.078
Moriguchi *et al.* (1983)	d	0.006
Nagata (2003)	d	0.0019

2-THIOPHENETHIOL (2-thienylthiol) *[7774-74-5]*

Burdack-Freitag & Schieberle (2010)		0.000 5

thiophenol⇒BENZENETHIOL
THT⇒TETRAHYDROTHIOPHENE
α-thujone⇒1-ISOPROPYL-4-METHYLBICYCLO[3.1.0]-3-HEXANONE
thymol⇒2-ISOPROPYL-5-METHYLPHENOL
m-tolualdehyde⇒3-METHYLBENZALDEHYDE
o-tolualdehyde⇒2-METHYLBENZALDEHYDE
p-tolualdehyde⇒4-METHYLBENZALDEHYDE

TOLUENE (methylbenzene) *[108-88-3]*

Backman (1917)	r	3.5 - 3.6
Backman (1918)		2.0
Grijns (1919); Zwaardemaker (1927)		170
Zwaardemaker (1927)		2.0
Schley (1934)	d	6
Schley (1934)	r	16
Nader (1958)		0.08 - 19
Deadman & Prigg (1959)	d	5.5
Nauš (1962)	d	2
Stalker (1963)	d	1.0
Gusev (1965)		1.5 - 3.2
May (1966)	d	140
May (1966)	r	260
Leonardos *et al.* (1969)	r	8.1 - 17.8
Dravnieks & O'Donnell (1971)		45
Köster (1971)	d	13.7
Dravnieks & Laffort (1972)		0.53
Artho & Koch (1973)		100 - 1,000
Hellman & Small (1973,1974)	d	0.6
Hellman & Small (1973,1974)	r	7
Dravnieks (1974)	d	60
Winneke & Kastka (1975)		46 - 84
Anon. (1980)	d	3.5
Anon. (1980)	r	18
Nauš (1982)	d	2
Nauš (1982)	r	20
Punter (1983)	d	25.4
Bahnmüller (1983)		5.85 - 29.8
Don (1986); Hoshika *et al.* (1993)	d	3.7 - 3.8
Scharfenberger (1990)		17
Nagy (1991)	d	12
Cometto-Muñiz (1993);		
Cometto-Muñiz & Cain (1994)	d	590
Hoshika *et al.* (1993)	d	3.5
Cometto-Muñiz *et al.* (2002)		0.4
Cometto-Muñiz *et al.* (2003)		0.098
Nagata (2003)	d	1.3
Cometto-Muñiz *et al.* (2004)	d	0.12 - 0.38
Cometto-Muñiz & Abraham (2009b)	d	0.30

α-TOLUENETHIOL (benzyl mercaptan) *[100-53-8]*

Katz & Talbert (1930)		0.013

m-toluidine⇒3-METHYLANILINE
o-toluidine⇒2-METHYLANILINE
p-toluidine⇒4-METHYLANILINE
4-tolyl acetate⇒4-METHYLPHENYL ACETATE
tolylenediisocyanate⇒2,4-DIISOCYANATO-1-METHYLBENZENE
tonalide⇒7-ACETYL-1,1,3,4,4,6-HEXAMETHYL-1,2,3,4-TETRAHYDRONAPHTHALENE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

triallylamine⇒TRI-2-PROPENYLAMINE

2,4,6-TRIBROMOANISOLE *[607-99-8]*

Diaz *et al.* (2005)		0.0014

1,2,4-TRIBROMOBENZENE *[615-54-3]*

Backman (1917)	r	0.0047 - 0.0050

TRIBROMOMETHANE (bromoform) *[75-25-2]*

Passy (1893a)	d	2 - 5
Backman (1917)	r	2.2 - 2.5
Grijns (1919)		150
Rocén (1920)	r	30

TRIBUTYLAMINE *[102-82-9]*

Laing *et al.* (1978)	r	0.53

S,S,S-TRIBUTYL PHOSPHOROTRITHIOATE (butiphos) *[78-48-8]*

Mukhamedova (1968)		0.052

trichlorfon⇒O,O-DIMETHYL(1-HYDROXY-2,2,2-TRICHLOROETHYL)PHOSPHONATE

TRICHLOROACETIC ACID *[76-03-9]*

Backman (1917)	r	1.6 - 2.5

2,3,4-TRICHLOROANISOLE *[54135-80-7]*

Strube & Buettner (2010)		0.03038

2,3,5-TRICHLOROANISOLE *[54135-81-8]*

Strube & Buettner (2010)		0.00177

2,3,6-TRICHLOROANISOLE *[50375-10-5]*

Sävenhed *et al.* (1985)	d	0.004 - 0.01
Diaz *et al.* (2005)		0.0016
Strube & Buettner (2010)		0.000 03

2,4,5-TRICHLOROANISOLE *[6130-75-2]*

Strube & Buettner (2010)		0.0051

2,4,6-TRICHLOROANISOLE *[87-40-1]*

Huber (1984)		< 0.000 051
Nijssen (1988)	d	0.000 0037
Spadone *et al.* (1990)		0.000 004
Khiari *et al.* (1992)		0.01
Diaz *et al.* (2005)		0.0016
Strube & Buettner (2010)		0.000 005

3,4,5-TRICHLOROANISOLE *[54135-82-9]*

Strube & Buettner (2010)		0.03289

1,2,4-TRICHLOROBENZENE *[120-82-1]*

Rowe (1975)		22

1,3,5-TRICHLOROBENZENE *[108-70-3]*

Punter (1983)	d	6.2

2,2,3-TRICHLOROBUTANAL HYDRATE

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Backman (1917)	r	1.7 - 2.3

1,2,3-TRICHLOROBUTANE *[18338-40-4]*

McCawley (1942)		1,980 - 6,600

TRICHLOROETHANAL (chloral) *[75-87-6]*

Leonardos *et al.* (1969)	r	0.28

TRICHLOROETHANAL HYDRATE (chloral hydrate) *[302-17-0]*

Backman (1917)	r	0.035 - 0.050

1,1,1-TRICHLOROETHANE (methylchloroform) *[71-55-6]*

Scherberger *et al.* (1958)	r	1,650
May (1966)	d	2,100
May (1966)	r	3,900
Kendall *et al.* (1968)	r	88
Don (1986)	d	5.3

TRICHLOROETHENE (trichloroethylene) *[79-01-6]*

Lehmann & Schmidt-Kehl (1936)		900
Weitbrecht (1957)		110
Scherberger *et al.* (1958)	r	410
Frantikova (1962)		69
Nauš (1962)	d	3
May (1966)	d	440
May (1966)	r	580
Malyarova (1967)		2.5 - 21
Leonardos *et al.* (1969)	r	115
Torkelson & Rowe (1981)		538
Nauš (1982)	d	3
Nauš (1982)	r	20
Don (1986)	d	3.9
Nagata (2003)	d	21

trichloroethylene⇒TRICHLOROETHENE

TRICHLOROFLUOROMETHANE (fluorotrichloromethane) *[75-69-4]*

Hellman & Small (1974)	d	28
Hellman & Small (1974)	r	760
Braker & Mossman (1980)		1,124,000

TRICHLOROMETHANE (chloroform) *[67-66-3]*

Passy (1893a)	d	30
Tempelaar (1913)	d	3,000
Backman (1917)	r	14.1 - 15.1
Allison & Katz (1919)		3,300
Grijns (1919)		2,350
Rocén (1920)	d	730
Rocén (1920)	r	2,500
Mitsumoto (1926)	r	353.8 - 589.0
Schley (1934)	d	42
Schley (1934)	r	56
Morimura (1934)	r	480 - 622
Lehmann & Schmidt-Kehl (1936)		1,000 - 1,500
Scherberger *et al.* (1958)	r	6,900
Janiček *et al.* (1960)		3,700
Nauš (1962)	d	3

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Dravnieks & Laffort (1972)		150
Dravnieks (1974)	d	1,350
Nauš (1982)	d	0.5
Nauš (1982)	r	20
Punter (1983)	d	650
Nagata (2003)	d	19

TRICHLOROMETHYL CHLOROFORMATE (diphosgene) *[503-38-8]*

Schley (1934)	d	0.008 - 0.01
Schley (1934)	r	0.01 - 0.032
Prentiss (1937)		8.8

1,1,1-TRICHLORO-2-METHYL-3-PROPANOL

Backman (1917)	r	5.7 - 6.5

TRICHLORONITROMETHANE (chloropicrin) *[76-06-2]*

Prentiss (1937)		7.3

2,4,6-TRICHLOROPHENOL *[88-06-2]*

Backman (1917)	r	0.0010 - 0.0016
Kendall *et al.* (1968)	r	0.021
Punter (1983)	d	0.00016

1,1,1-TRICHLORO-2-PROPANOL (isopral) *[76-00-6]*

Backman (1917)	r	16.3 - 18.7

TRICYCLO[3.3.1.1(3.7)]DECANE-1-CARBOXALDEHYDE (1-adamantanaldehyde)

Von Ranson & Belitz (1992b)	d	1.4
Von Ranson & Belitz (1992b)	r	4.4

TRICYCLO(1-HYDROXY-1-METHYLETHYL)SILANE

Sunderkötter *et al.* (2010)		0.000 14

TRIDECANE *[629-50-5]*

Laffort & Dravnieks (1973)		42

(*E*,*Z*,*Z*)-2,4,7-TRIDECATRIENAL *[13552-96-0]*

Blank *et al.* (2001); Lin *et al.* (2001)		0.000 07

1-TRIDECEN-3-OL

Schieberle & Buettner (2001)		> 1

1-TRIDECEN-3-ONE

Schieberle & Buettner (2001)		0.1 - 1

TRIETHANOLAMINE *[102-71-6]*

Nagy (1991)	d	> 61

TRIETHYLAMINE *[121-44-8]*

Tkachev (1970)		0.33
Hellman & Small (1974)	d	< 0.4
Hellman & Small (1974)	r	1.1
Laing *et al.* (1978)	r	11.9
Homans *et al.* (1978)		2.7
Dravnieks *et al.* (1986)	d	1.0
Nagata (2003)	d	0.022

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

triethyl citrate⇒TRIETHYL 2-HYDROXY-1,2,3-PROPANETRICARBOXYLATE

TRIETHYL 2-HYDROXY-1,2,3-PROPANETRICARBOXYLATE (triethyl citrate) *[77-93-0]*

Backman (1917)	r	1.4

TRIETHYLPYRAZINE *[38028-82-9]*

Wagner *et al.* (1999)		0.78

TRIIODOMETHANE (iodoform) *[75-47-8]*

Passy (1893a)	d	0.06 - 0.7
Berthelot (1901)		0.000 3 - 0.03
Backman (1917)	r	0.095
Allison & Katz (1919)		18
Zwaardemaker (1927)		0.03
Cancho *et al.* (2001)		< 0.14

TRIMETHYLAMINE *[75-50-3]*

Tempelaar (1913)	d	2.1
Rotenberg & Mashbits (1967)		2
Sakuma *et al.* (1967)		0.000 7
Leonardos *et al.* (1969)	r	0.000 5
Stephens (1971)		0.0014
Amoore (1977)	d	0.0025
Bedborough & Trott (1979)	d	0.0012
Anon. (1980)	d	0.000 26
Anon. (1980)	r	0.0034
Jensen & Flyger (1983)		0.0031 - 0.027
Langenhove & Schamp (1984)	d	0.002
Homans (1984)		4.4
Nagy (1991)	d	0.0059
Greenman *et al.* (2004)		0.000 041 – 0.0011
Nagata (2003)	d	0.000 077
Van Thriel *et al.* (2006)	d	0.63

2,N,N-TRIMETHYLANILINE *[609-72-3]*

Backman (1917)	r	0.26 - 0.37

4,N,N-TRIMETHYLANILINE *[99-97-8]*

Backman (1917)	r	0.3 - 0.4

2,4,5-TRIMETHYLANILINE (pseudocumidine) *[137-17-7]*

Huijer (1917)	d	8
Backman (1917)	r	0.08 - 0.12

2,4,6-TRIMETHYLANILINE (mesidine) *[88-05-1]*

Tepekina (1968)		0.013

1,2,4-TRIMETHYLBENZENE (pseudocumene) *[95-63-6]*

Backman (1917)	r	0.3 - 0.35
Backman (1918)		0.2
Deadman & Prigg (1959)	d	0.7
Anon. (1980)	d	0.14
Anon. (1980)	r	1.1
Nagata (2003)	d	0.59

1,3,5-TRIMETHYLBENZENE (mesitylene) *[108-67-8]*

Backman (1917)	r	0.35 - 0.4

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Deadman & Prigg (1959)	d	0.2
Dravnieks & Laffort (1972)		2.0
Knuth (1973)		1.2
Dravnieks (1974)	d	12
Anon. (1980)	d	0.18
Anon. (1980)	r	1.4
Punter (1983)	d	10.7
Nagata (2003)	d	0.83

endo-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL (borneol) *[507-70-0]*

Tempelaar (1913)	d	0.052

1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANOL (fenchyl alcohol) *[1632-73-1]*

Schieberle & Grosch (1988)		0.14

1,3,3-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE (fenchone) *[1195-79-5]*

Tempelaar (1913); Zwaardemaker (1927)	d	2.3

1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE (camphor) *[76-22-2]*

Passy (1892a,1892b)	d	5
Zwaardemaker (1914,1927)	d	0.016 - 2
Backman (1917)	r	0.76 - 0.88
Ohma (1922)	d	0.06
Hofmann & Kohlrausch (1925)	r	2 - 33
Mitsumoto (1926)	r	4.4 - 45.0
Tamman & Oelsen (1928)	d	6 - 13
Morimura (1934)	r	1.16 - 32.5
Gundlach & Kenway (1939)	d	0.49
Kleinschmidt (1983)	r	3.35
De Wijk (1989)		2.84

(*S*)-(-)-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTANONE ((*S*)-(-)-camphor) *[464-48-2]*

Facundo *et al.* (2005)		8.3

2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE (α-pinene) *[80-56-8]*

Appell (1969)		0.016
Cristoph (1983)	r	25.0 - 29.0
Randebrock (1986)		0.00036
Laska & Hudson (1991)	d	0.23 - 0.36
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	105
Nagata (2003)	d	0.10

(1*R*,5*R*)-(+)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE ((+)-α-pinene, (*R*)-(+)-α-pinene) *[7785-70-8]*

Lindell (1991)	d	2.1
Jagella & Grosch (1998)		0.018
Molhave *et al.* (2000)	d	23
Buettner & Schieberle (2001a,2001b)		0.0053

(1*S*,5*S*)-(-)-2,6,6-TRIMETHYLBICYCLO[3.1.1]-2-HEPTENE ((-)-α-pinene, (*S*)-(-)-α-pinene) *[7785-26-4]*

Lindell (1991)	d	3.3
Jagella & Grosch (1998)		0.035

3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE (3-carene, δ3-carene) *[13466-78-9]*

Cristoph (1983)	r	8.6 - 9.1
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	9.3

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

(1*S*)-(+)-3,7,7-TRIMETHYLBICYCLO[4.1.0]-3-HEPTENE ((+)-3-carene) *[498-15-7]*

Molhave *et al.* (2000)	d	4

endo-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL ACETATE (bornyl acetate) *[76-49-3]*

Tempelaar (1913)	d	0.44

endo-1,7,7-TRIMETHYLBICYCLO[2.2.1]-2-HEPTYL SALICYLATE (bornyl salicylate) *[560-88-3]*

Baldus (1936)	d	0.03
Baldus (1936)	r	0.04

1-(2,6,6-TRIMETHYL-1,3-CYCLOHEXADIEN-1-YL)-(*E*)-2-BUTEN-1-ONE ((*E*)-β-damascenone) *[23726-93-4]*

Blank *et al.* (1989,1992)		0.000 002 - 0.00 004
Lea & Ford (1991)		0.05
Güntert *et al.* (2001)		0.000 125

3,3,5-TRIMETHYL-2-CYCLOHEXEN-1-ONE (isophorone) *[78-59-1]*

Hellman & Small (1974)	d	1.1
Hellman & Small (1974)	r	3.0
Ziemer *et al.* (2000)	d	0.0017

(1*R*)-(+)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTANONE ((*R*)-(+)-dihydro-α-ionone)

Brenna *et al.* (2002)		0.031

(1*S*)-(-)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-2-BUTANONE ((*S*)-(-)-dihydro-α-ionone)

Brenna *et al.* (2002)		0.1

4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-2-BUTANONE (dihydro-β-ionone) *[17283-81-7]*

Brenna *et al.* (2002)		0.0017

4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE (α-ionone) *[127-41-3]*

Zwaardemaker (1914)	d	0.000 1
Henning (1924)	d	0.000 05
Naves (1947)		0.000 25 - 0.000 5
Neuhaus (1957)	d	0.000 1
Heuhaus (1957)	r	0.000 4 - 0.0024
Goldenberg (1967)	d	0.000 03
Appell (1969)		0.000 01
Moulton & Marshall (1974); Moulton (1977); Marshall & Moulton (1981)	d	0.000 4 - 0.008
Etzweiler *et al.* (1980)		0.0016
Cristoph (1983)	r	0.02 - 0.04

(*R*)-(+)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE ((*R*)-(+)-α-ionone, (*d*)-α-ionone) *[24190-29-2]*

Naves (1947)		0.002 - 0.008
Fuganti *et al.* (2000); Brenna *et al.* (2002)		0.0032

(*S*)-(-)-4-(2,6,6-TRIMETHYL-2-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE ((*S*)-(-)-α-ionone, (*l*)-α-ionone) *[14398-36-8]*

Naves (1947)		0.002 - 0.008
Fuganti *et al.* (2000); Brenna *et al.* (2002)		0.0027

4-(2,6,6-TRIMETHYL-1-CYCLOHEXEN-1-YL)-3-BUTEN-2-ONE (β-ionone) *[14901-07-6]*

Köster (1968a)	d	11,500
Köster (1971)	d	115

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Guadagni (1966)		0.000 07
Cristoph (1983)	r	4.5 - 5.0
Brenna *et al.* (2002)		0.000 12

(+)-(*R*,*E*)-1-(2,6,6-TRIMETHYLCYCLOHEX-2-ENYL)-1-PENTEN-3-ONE

Abate *et al.* (2007)		0.000 53

(-)-(*S*,*E*)-1-(2,6,6-TRIMETHYLCYCLOHEX-2-ENYL)-1-PENTEN-3-ONE

Abate *et al.* (2007)		0.000 24

(4a*R**,8a*R**)-1,1,8a-TRIMETHYLDECAHYDRONAPHTHALENE-4a-OL

Jahnke *et al.* (2009)		0.0131

5,5,8a-TRIMETHYL-2-DECAHYDRONAPHTHYL ACETATE (polywood) *[98676-96-1]*

Wünsche *et al.* (1995)	d	0.000 28

3,7,11-TRIMETHYLDODECA-1,3(*E*),5(*E*),10-TETRAEN-7-OL

Krings *et al.* (2006)		< 1

3,7,11-TRIMETHYLDODECA-1,3(*Z*),5(*E*),10-TETRAEN-7-OL

Krings *et al.* (2006)		< 1

2,2,5-TRIMETHYLHEXANE *[111-84-2]*

Nagata (2003)	d	4.8

(1*R*,2*S*,4*R*,2'*S*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)]⇒(1*R*,2*S*,4*R*,2'*S*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)]
(1*R*,2*S*,4*R*,2'*R*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)]⇒(1*R*,2*S*,4*R*,2'*R*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)]
(1*S*,2*R*,4*S*,2'*R*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)]⇒(1*S*,2*R*,4*S*,2'*R*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)]
(1*S*,2*R*,4*S*,2'*S*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)]⇒(1*S*,2*R*,4*S*,2'*S*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)]
(1*R*,2*R*,4*R*,2'*R*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)]⇒(1*R*,2*R*,4*R*,2'*R*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)]
(1*R*,2*R*,4*R*,2'*S*)-1,7,7-trimethyl-2-*iso*-propylspiro[bicyclo[2.2.1]heptane-2,4'-(1,3-dioxane)]⇒(1*R*,2*R*,4*R*,2'*S*)-2'-ISOPROPYL-1,7,7-TRIMETHYLSPIRO[BICYCLO[2.2.1]HEPTANE-2,4'-(1,3-DIOXANE)]

1,2,6-TRIMETHYL-6-(3-METHYL-2-BUTENYL)CYCLOHEXENOL

Goeke (2006)		0.000 5 / 0.018 (for 2 stereoisomers)

4,11,11-TRIMETHYL-8-METHYLENEBICYCLO[7.2.0]-4-UNDECENE (β-caryophyllene) *[87-44-5]*

Etzweiler *et al.* (1980)		1.5
Cristoph (1983)	r	11.0 - 13.0

(2*E*,5*E*)-5,6,7-TRIMETHYL-2,5-OCTADIEN-4-ONE

Kraft (2004b)		0.5

(2*E*,5*Z*)-5,6,7-TRIMETHYL-2,5-OCTADIEN-4-ONE

Kraft *et al.* (2001); Kraft (2004b)		0.000 5

trimethylolpropane⇒1,1,1-TRIS(HYDROXYMETHYL)PROPANE

1,3,3-TRIMETHYL-2-OXABICYCLO[2.2.2]OCTANE (1,8-cineole, eucalyptol) *[470-82-6]*

Tempelaar (1913); Zwaardemaker (1927)	d	0.055 - 0.2
Grijns (1919); Zwaardemaker (1927)		7
Ohma (1922)	d	0.19
Baldus (1936)	d	0.003
Baldus (1936)	r	0.0035

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Fuller *et al.* (1964)	r	0.000 62
Appell (1969)		0.014
Randebrock (1971)		0.000 2
Herberhold (1975)		0.36
Amoore (1982)	d	0.069
Cristoph (1983)	r	0.03 - 0.05
Dravnieks *et al.* (1986)	d	0.026
Randebrock (1986)		0.000 069
Marin *et al.* (1988)		0.004 - 0.062
De Wijk (1989)		0.52
Laska & Hudson (1991)	d	0.036 - 0.044
Khiari *et al.* (1992)		0.0075
Jagella & Grosch (1998)		0.003
Cometto-Muñiz *et al.* (1998b);		
Cometto-Muñiz (1999)	d	2.0
Güntert *et al.* (2001)		0.0031
Facundo *et al.* (2005)		0.15
Benzo *et al.* (2007)		0.00267 (GC-O technique)
Benzo *et al.* (2007)		0.00508 (dynamic dilution olfactometry)

(±)-(5'*R**,7'*R**)-1-(4',4',7'-TRIMETHYL-1'-OXASPIRO[4.5]DECAN-7'-YL)ETHAN-1-ONE
Kraft & Popaj (2008) d 0.0994

(±)-(5'*R**,8'*R**)-1-(4',4',8'-TRIMETHYL-1'-OXASPIRO[4.5]DECAN-8'-YL)ETHAN-1-ONE
Kraft & Popaj (2008) d 0.0921

2,2,4-TRIMETHYLPENTANE (isooctane) *[540-84-1]*
Nagata (2003) d 3.1

2,2,4-TRIMETHYL-1,3-PENTANEDIOL DIISOBUTANOATE (TXIB) *[6846-50-0]*
Cain *et al.* (2006) d 0.014

2,2,4-TRIMETHYL-1,3-PENTANEDIOL MONOISOBUTANOATE (texanol) *[25265-77-4]*
Ziemer *et al.* (2000) d 0.58

TRIMETHYL PHOSPHITE (methyl phosphite) *[121-45-9]*
Levin & Gabriel (1973) 0.000 5

TRIMETHYLPYRAZINE *[14667-55-1]*
Fors (1984); Fors & Olofsson (1985) d 0.19
Czerny & Wagner (1995) 0.033 - 0.066
Grosch *et al.* (1996); Wagner *et al.* (1999) 0.05 - 0.096

(±)-(5*R**,6*S**)-1,1,6-TRIMETHYLSPIRO[4.5]DECAN-6-OL
Kraft & Bruneau (2007) 0.005

(±)-2,2,6-TRIMETHYLSPIRO[4.5]DECAN-6-OL
Kraft & Bruneau (2007) 0.068

TRIMETHYLTETRAHYDROPYRAZINE
Kurniadi *et al.* (2003) 0.135
Kurniadi (2003) r 2.7

2,4,5-TRIMETHYLTHIAZOLE *[13623-11-5]*
Gasser & Grosch (1990a) 0.0018 - 0.0072
Blank *et al.* (1992) 0.000 5 - 0.001
Gijs *et al.* (2000) 0.022

2,4,6-TRIMETHYL-1,3,5-TRIOXANE (paraldehyde) *[123-63-7]*

Backman (1917)	r	0.02 - 0.025

trioxychlorofluoride⇒PERCHLORYL FLUORIDE

TRI-2-PROPENYLAMINE (triallylamine) *[102-70-5]*

Hine *et al.* (1960)		2.3

1,1,1-TRIS(HYDROXYMETHYL)PROPANE (trimethylolpropane) *[77-99-6]*

Nagy (1991)	d	1.9

3,4,5-trithiaheptane⇒ETHYLTRITHIOETHANE
2,3,5-trithiahexane⇒METHYL(METHYLTHIO)DITHIOMETHANE
2,3,4-trithiapentane⇒METHYLTRITHIOMETHANE

1,2,3-TRITHI-4-ENE

Pino *et al.* (1991)		0.000 017

(*R*)-tropional®⇒(2*R*)-(+)-2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL
(*S*)-tropional®⇒(2*S*)-(-)-2-METHYL-3-(3,4-METHYLENEDIOXYPHENYL)PROPANAL
TXIB⇒2,2,4-TRIMETHYL-1,3-PENTANEDIOL DIISOBUTANOATE

U

γ-undecalactone⇒4-HYDROXYUNDECANOIC ACID LACTONE
(*R*)-γ-undecalactone⇒(*R*)-4-HYDROXYUNDECANOIC ACID LACTONE
(*S*)-γ-undecalactone ⇒(*S*)-4-HYDROXYUNDECANOIC ACID LACTONE
(*R*)-δ-undecalactone⇒(*R*)-5-HYDROXYUNDECANOIC ACID LACTONE
(*S*)-δ-undecalactone⇒(*S*)-5-HYDROXYUNDECANOIC ACID LACTONE

(*Z*)-1,5-UNDECADIEN-3-OL *[56722-23-7]*

Schieberle & Buettner (2001)		0.01 - 0.1

(*Z*)-1,5-UNDECADIEN-3-ONE

Schieberle & Buettner (2001)		0.000 1 - 0.001

UNDECANAL *[112-44-7]*

Pliška & Janiček (1965)		0.7
Randebrock (1971)		0.000 5
Teranishi *et al.* (1974)		0.035
Cristoph (1983)	r	0.009 - 0.012
Randebrock (1986)		0.000 54
Von Ranson & Belitz (1992a)	d	0.14
Von Ranson & Belitz (1992a)	r	0.14

UNDECANE *[1120-21-4]*

Mullins (1955)	r	374
Laffort & Dravnieks (1973)		23
Lindell (1991)	d	9.6
Nagata (2003)	d	5.6

1-UNDECANOL *[112-42-5]*

Pliška (1962)		46

4-undecanolide⇒4-HYDROXYUNDECANOIC ACID LACTONE

2-UNDECANONE (methyl nonyl ketone) *[112-12-9]*

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Tempelaar (1913); Zwaardemaker (1927)	d	1.7
Backman (1917)	r	0.03 - 0.05
Teranishi *et al.* (1974)		0.182
Nagy (1991)	d	3

(*E*,*E*,*Z*)-1,3,5,8-UNDECATETRAENE *[50277-31-1]*

Berger *et al.* (1985)	d	0.2 - 0.3

(*E*,*Z*,*Z*)-1,3,5,8-UNDECATETRAENE *[81717-82-0]*

Berger *et al.* (1985)	d	0.000 02 - 0.000 04

(3*E*,5*E*,9*E*)-1,3,5,9-UNDECATETRAENE

Steinhaus *et al.* (2007)		> 0.2

(3*E*,5*E*,9*Z*)-1,3,5,9-UNDECATETRAENE

Steinhaus *et al.* (2007)		> 0.2

(3*E*,5*Z*,9*E*)-1,3,5,9-UNDECATETRAENE

Steinhaus *et al.* (2007)		0.000 01

(3*E*,5*Z*,9*Z*)-1,3,5,9-UNDECATETRAENE

Steinhaus *et al.* (2007)		0.001

(*E*,*E*)-1,3,5-UNDECATRIENE *[19883-29-5]*

Berger *et al.* (1985)	d	7.5 - 10.0
Steinhaus *et al.* (2007)		0.2

(*E*,*Z*)-1,3,5-UNDECATRIENE *[51447-08-6]*

Berger *et al.* (1985)	d	0.000 01 - 0.000 02
Güntert *et al.* (2001)		0.000 125
Steinhaus *et al.* (2007)		0.000 003

(+)-(*R*)-undecavertol⇒(+)-(5*R*)-4-METHYL-3(*E*)-DECEN-5-OL
(-)-(*S*)-undecavertol⇒(-)-(5*S*)-4-METHYL-3(*E*)-DECEN-5-OL

(*E*)-2-UNDECENAL *[53448-07-0]*

Von Ranson & Belitz (1992a)	d	0.010
Von Ranson & Belitz (1992a)	r	0.044

(*Z*)-2-UNDECENAL *[71277-05-9]*

Von Ranson & Belitz (1992a)	d	0.024
Von Ranson & Belitz (1992a)	r	0.34

(*E*)-3-UNDECENAL

Von Ranson & Belitz (1992a)	d	0.030
Von Ranson & Belitz (1992a)	r	1.3

(*Z*)-3-UNDECENAL

Von Ranson & Belitz (1992a)	d	0.0050
Von Ranson & Belitz (1992a)	r	0.44

(*Z*)-8-UNDECENAL *[147159-49-7]*

Von Ranson & Belitz (1992a)	d	0.0067
Von Ranson & Belitz (1992a)	r	0.0067

9-UNDECENAL (undecylenic aldehyde) *[143-14-6]*

Randebrock (1971)		0.000 2

AIR detection (d) and recogniton (r) odour threshold values in mg/m^3

Randebrock (1986)	0.000 36

1-UNDECEN-3-OL *[35329-42-1]*

Koszinowski & Piringer (1983)	0.37
Schieberle & Buettner (2001)	0.01 - 0.1

1-UNDECEN-3-ONE *[42832-47-3]*

Koszinowski & Piringer (1983)	0.001
Schieberle & Buettner (2001)	0.000 1 - 0.001

undecylenic aldehyde⇒9-UNDECANAL

V

valeraldehyde⇒PENTANAL
valeric acid⇒PENTANOIC ACID
vanillin⇒4-HYDROXY-3-METHOXYBENZALDEHYDE
vinyl acetate⇒ETHENYL ACETATE
vinylbenzene⇒ETHENYLBENZENE
vinyl chloride⇒CHLOROETHENE
vinyl cyanide⇒ETHENYL CYANIDE
p-vinylguaiacol⇒4-ETHENYL-2-METHOXYPHENOL
4-vinylguaiacol⇒4-ETHENYL-2-METHOXYPHENOL
vinylidene chloride⇒1,1-DICHLOROETHENE
vinylpyrazine⇒ETHENYLPYRAZINE
2-vinylpyridine⇒2-ETHENYLPYRIDINE

W-X

whisky lactone⇒4-HYDROXY-3-METHYLOCTANOIC ACID LACTONE
wine lactone⇒(3*S*,3a*S*,7a*R*)-3a,4,5,7a-TETRAHYDRO-3,6-DIMETHYL-2(3*H*)-BENZOFURANONE

XENON *[7440-63-3]*

Laffort (1968a)	2,700,000

m-xylene⇒1,3-DIMETHYLBENZENE
o-xylene⇒1,2-DIMETHYLBENZENE
p-xylene⇒1,4-DIMETHYLBENZENE
2,3-xylenol⇒2,3-DIMETHYLPHENOL
2,4-xylenol⇒2,4-DIMETHYLPHENOL
2,5-xylenol⇒2,5-DIMETHYLPHENOL
2,6-xylenol⇒2,6-DIMETHYLPHENOL
3,4-xylenol⇒3,4-DIMETHYLPHENOL
3,5-xylenol⇒3,5-DIMETHYLPHENOL

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